# Invariant-average kernels and deformation obstructions for a quartic Hamiltonian

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**Abstract**

For the quartic Hamiltonian with potential $Q_0(x, y) = x^4 + 12x^2y^2 + 16y^4$, we study homogeneous angular forces annihilated by every invariant probability on a positive energy shell. A complete criterion combines ordinary torus means, every rational relative phase, and the physical periodic family crossing the origin at zero separation. Abel inversion removes the doubly even component, while a certified resonant coefficient identifies the binary-quartic polynomial intersection with the rotation line. The unrestricted problem has two distinct conclusions. First, derivatives of actual jointly $C^2$ common-mean families must satisfy a common second-force compatibility: pure doubly odd directions are rotational, and a pure even-odd direction with zero vertical value cannot be nonzero. Second, the original $C^1$ even-odd complete-null kernel is finite-dimensional. Its dimension and the existence of a nonzero vertical-trace branch remain undetermined. The finite-dimensional conclusion follows from original-regularity entry, compact observations obtained from complete resonant rows, and exact ordinary endpoint gluing. A quartic Peano value trace and a one-way ambient reconstruction refine the result without asserting a full inverse theorem or a nonpolynomial finite-deformation classification.

**Keywords.** Invariant average; Hamiltonian observation; resonant torus; realizable deformation; finite-dimensional kernel; Peano trace. **Reconstruction convention.** The main text is editable LaTeX. Long technical proofs and the original interval-certificate specification are included as vector PDF source folios in the archive. Original source numbers are explicitly distinguished from new numbers. The archive preserves the only supplied original; it is not a claim that all source calculations have been independently rerun. Finite-dimensionality, not a complete determination of the unrestricted kernel, is the endpoint of this reconstruction.

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# 1 The observation problem and its endpoint

Let

$$Q_0(x,y) = x^4 + 12x^2y^2 + 16y^4, \qquad H(Z,W) = \tfrac{1}{2}|W|^2 + Q_0(Z).$$

For $r \in \{0,1,2\}$ define

$$\mathcal{K}^r = \left\{ \rho^4 a(\theta) : \begin{array}{l} a \in C^r(S^1), \quad \int g \, d\mu = 0 \\ \text{for every invariant probability on } H = 1 \end{array} \right\}. \tag{1}$$

The superscript denotes angular regularity, not homogeneity. Positive homogeneity is used, so no antipodal symmetry is imposed. The shell is compact, and homogeneity transfers the question to any positive energy.

This is an observation problem at one fixed integrable well. It differs from classifying actual families of wells. If a parameter derivative is in $\mathcal{K}^1$, the first-order obstruction has vanished. A family realizing that derivative must still supply one continuous second derivative compatible with all orbit constraints. Conversely, a finite-family theorem in a coefficient neighborhood does not determine the unrestricted angular kernel at the base. We keep these assertions separate.

Reflection parity is denoted EE, EO, OE, OO; the first letter concerns $x$ and the second $y$. Let $\mathcal{K}_{EO} = \mathcal{K}^1 \cap EO$, identify a force with its angular profile, and set

$$\tau(a) = a(\pi/2), \qquad N_0 = \ker(\tau|_{\mathcal{K}_{EO}}).$$

The vertical trace distinguishes two possible finite-dimensional pieces.

**Theorem 1.1** (Finite-dimensional complete EO kernel)**.** *Let $N_E$ be the energy closure of the original space $N_0$ in the positive mechanical energy introduced in Section 5. Then $N_E$ is finite-dimensional and equals $N_0$ algebraically. Its elements retain their original $C^1$ representatives, and*

$$\|a\|_{C^1} \le C_N \sqrt{\mathcal{E}(a)} \qquad (a \in N_0).$$

*The original $C^1$ EO kernel is finite-dimensional. More precisely, either $\mathcal{K}_{EO} = N_0$, or there exists $u_0 \in \mathcal{K}_{EO}$ with $\tau(u_0) = 1$ such that*

$$\mathcal{K}_{EO} = N_0 \oplus \operatorname{span}\{u_0\}.$$

*Neither* $\dim N_0$ *nor the existence of the second alternative is determined.*

This is [source Theorem 10]. The proof appears in Section 6, with its complete mechanical inputs in the archive. The space is the closure of original complete-null inputs; it is not a kernel newly defined by formally extending individual rows.

**Theorem 1.2** (Restrictions on realizable derivatives)**.** *Let $U_s$ be a positive degree-four homogeneous family, jointly $C^2$ in $(s,Z)$ near zero, with $U_0 = Q_0$. Suppose all invariant probabilities on the unit shell of each $U_s$ give the same mean to $\partial_s U_s$. Normalize the phase volume and write the resulting jets as $g = \partial_s V_s|_0$ and $h = \partial_s^2 V_s|_0$. Then:*

1. *If $g$ is pure OO, $g = c g_{\mathrm{rot}}$ and $h_{EE} = c^2 h_{\mathrm{rot}}$.*
2. *If $g$ is pure EO and $g(0,1) = 0$, then $g = 0$.*
3. *If $g$ is odd in $y$ and $g(0,1) = 0$, then $g = c g_{\mathrm{rot}}$ and $h_{EE} = c^2 h_{\mathrm{rot}}$,*

*where*

$$g_{\mathrm{rot}} = 20xy(x^2 + 2y^2), \qquad h_{\mathrm{rot}} = 20x^4 + 60x^2y^2 - 40y^4.$$

This is [source Theorem 9]. It concerns actual jets, not all vectors in the linear kernel. Joint $C^2$ regularity supplies only $g \in C^1$ and $h \in C^0$ in angular variables. No extra derivative is assumed.

### 1.1 Mechanical context

At a common-mean family the centered parameter force is invisible to all invariant statistical states. The question here is how much of that force is detectable when the base admits separated dynamics, including resonances and singular limiting fibers. The parabolic separation of $Q_0$ is known from Rauch-Wojciechowski, Marciniak, and Lundmark [RML99]. The analysis must retain the physical clock: an auxiliary-time average generally differs from the mechanical mean. Centered generators and adiabatic transport provide a related viewpoint [J13], but vanishing invariant means alone is not an existence theorem for a regular generator.

The three main tasks are a complete description of the observations, a distinction between linear nullity and nonlinear realizability, and finite-dimensionality of one unrestricted parity sector. Quantitative reconstruction is a refinement of these results, not a substitute for the missing identification of the remaining finite-dimensional space.

## 2 Separated trajectories and the physical clock

Use the parabolic variables

$$x = uv, \qquad y = (u^2 - v^2)/2, \qquad S = u^2 + v^2, \qquad p_u = vW_x + uW_y, \quad p_v = uW_x - vW_y.$$

Away from the origin they form a canonical two-fold cover. Its deck transformation changes the signs of all four variables. At physical energy one,

$$S(H-1) = \tfrac{1}{2}p_u^2 + U(u) + \tfrac{1}{2}p_v^2 + U(v), \quad U(q) = q^{10} - q^2, \qquad dt = S\,d\tau. \tag{2}$$

The scalar energies are $c$ and $-c$, with $|c| \le c_* = 4\,5^{-5/4}$. The separation integral descends to a polynomial physical integral:

$$C = xW_xW_y - yW_x^2 + 4x^2y(x^2 + 2y^2).$$

Thus the treatment of $C = 0$ must use the original smooth mechanical flow, not discard it as a coordinate singularity.

For $0 < c < c_*$, $u_c$ crosses both wells and $v_c$ librates in the positive well. The negative $v$ well is deck-equivalent. Let $T_u, T_v$ be their auxiliary periods, and choose phases at upper turning points so that

$$u_c(-\alpha) = u_c(\alpha), \quad u_c(\alpha + \pi) = -u_c(\alpha), \quad v_c(-\beta) = v_c(\beta) > 0.$$

The sign $\sigma \in \{1, -1\}$ accounts for the reflected negative-$C$ torus. Put $\overline{S} = \langle u_c^2 \rangle + \langle v_c^2 \rangle$. Physical frequencies satisfy

$$\omega_u = \frac{2\pi}{T_u\overline{S}}, \qquad \omega_v = \frac{2\pi}{T_v\overline{S}}, \qquad \frac{\omega_v}{\omega_u} = \frac{T_u}{T_v}.$$

The phase variables are uniform in $\tau$, not in physical time.

### 2.1 Ordinary means and all rational phases

For a continuous homogeneous force define

$$F_c^\sigma(\alpha, \beta) = S\,g(u_cv_c, \sigma(u_c^2 - v_c^2)/2). \tag{3}$$

Fourier coefficients use normalized Haar measure on the two-torus. The physical torus probability has density $S/\overline{S}$, so its mean is $\widehat{F}_c^\sigma(0,0)/\overline{S}$. This probability is invariant for rational as well as irrational ratios. In the irrational case it is unique.

If $T_u/T_v = m/n$ in lowest terms, the auxiliary orbit has phases $(n\zeta + \alpha_0, m\zeta + \beta_0)$. Put $\chi = m\alpha_0 - n\beta_0$ and choose integers $a_0, b_0$ with $ma_0 - nb_0 = 1$. The actual phase mean is

$$M_c^\sigma(\chi) = \frac{1}{2\pi\overline{S}} \int_0^{2\pi} F_c^\sigma(n\zeta + a_0\chi, m\zeta + b_0\chi)\, d\zeta.$$

The integer coordinate change is a torus automorphism; Haar invariance and Fubini give

$$\widehat{M}_c^\sigma(k) = \frac{\widehat{F}_c^\sigma(km, -kn)}{\overline{S}}. \tag{4}$$

Fejér convergence for continuous $M_c^\sigma$ makes vanishing of all these coefficients equivalent to vanishing at every relative phase. No absolute convergence of a two-dimensional Fourier series is needed. Keeping only the ordinary coefficient would discard precisely this resonant information.

### 2.2 The physical zero-separation family

At $C = 0$, the nonconstant scalar zero-energy solutions are signed translates of

$$h(\tau) = \operatorname{sech}(4\sqrt{2}\tau)^{1/4}.$$

Set $u_d = h(\tau - d/2)$, $v_d = h(\tau + d/2)$ and map them to $(x, y)$. Both infinite auxiliary-time endpoints reach the physical origin in finite time. Matching the physical velocities joins this arc to its $x$ reflection, giving an actual periodic orbit. Let

$$A_0 = \int_{\mathbb{R}} h^2\, d\tau = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\sqrt{2}\,\Gamma(3/4)}, \qquad T_0 = 4A_0.$$

For finite $d$, the joined orbit has primitive period $T_0$. As $|d| \to \infty$ it tends to a double traversal of the vertical orbit, whose primitive period is $T_0/2$. This factor must be retained in second variations.

Writing $F_g(u, v) = (u^2 + v^2)g(uv, (u^2 - v^2)/2)$, its mean is

$$Z_g(d) = \frac{1}{4A_0} \int_{\mathbb{R}} [F_g(u_d, v_d) + F_g(u_d, -v_d)]\, d\tau. \tag{5}$$

The integral is absolutely convergent. The two arcs cancel $x$-odd inputs, and $\tau \mapsto -\tau$ cancels $y$-odd inputs. Thus only the EE component is observed by this family.

## 3 The complete criterion and the polynomial intersection

**Lemma 3.1** (Abel injectivity at zero separation)**.** *For a continuous homogeneous force, $Z_g(d) = 0$ for every finite $d$ if and only if $g_{EE} = 0$.*

*Proof.* It remains to treat an EE profile $a$. For $d > 0$ write $b_0 = 4\sqrt{2}$, $D = b_0 d/2$, $k = \tanh D$, and

$$A(\xi) = \tfrac{1}{2}[(1 - \xi)^{-1/2} + (1 + \xi)^{-1/2}], \qquad \vartheta(\xi) = \arctan\Big(\sinh(\tfrac{1}{2}\operatorname{arctanh}\xi)\Big),$$

and $B(\xi) = A(\xi)^5 a(\vartheta(\xi))$, $c_D = (\cosh D)^{-1/2}$. Substitution in the pulse formula, including the clock, gives

$$Z_g(d) = \frac{2c_D^5}{b_0 A_0 k^{3/2}} \int_0^k (k^2 - \xi^2)^{1/4} B(\xi)\, d\xi.$$

With $K = k^2$ and $f(t) = t^{-1/2} B(\sqrt{t})$, the last integral is $\frac{1}{2}\int_0^K (K - t)^{1/4} f(t) dt$. The input is locally integrable including zero. Convolution with $K^{-1/4}$ and the beta integral convert a vanishing output into a positive multiple of $\int_0^K (K - t) f(t) dt$. Two distributional derivatives give $f = 0$. Since $\vartheta : (0, 1) \to (0, \pi/2)$ is onto, continuity and parity imply $a = 0$. The converse is the reflection cancellation in (5). □

**Theorem 3.2** (Complete invariant-average criterion)**.** *A continuous homogeneous degree-four force belongs to* $\mathcal{K}^0$ *if and only if* $g_{EE} = 0$ *and, for both* $\sigma = \pm 1$*,*

1. $\widehat{F}_c^{\sigma}(0,0) = 0$ *for every* $0 < c < c_*$*;*
2. $\widehat{F}_c^{\sigma}(km, -kn) = 0$ *for every rational* $T_u(c)/T_v(c) = m/n$ *in lowest terms and every integer* $k \geq 1$*.*

*Proof.* Necessity follows from actual torus and periodic probabilities and Lemma 3.1. For sufficiency, disintegrate any invariant probability over the continuous integral $C$. Invariance of the conditionals holds almost everywhere, first at rational times for a countable dense family of test functions and then at all times by continuity. On regular irrational tori the conditional is the clock-weighted Haar probability. On rational tori it is a mixture of periodic phase probabilities. The two conditions annihilate these measures by (4). Endpoint brakes are limits of torus means. On $C = 0$, the scalar quadratures give exactly the joined-pulse family and the vertical limiting orbit; their means vanish by Lemma 3.1. These cases exhaust the conditionals. □

Both physical reflections preserve the Hamiltonian, so each parity projection preserves $\mathcal{K}^r$. On an $x$-odd sector the ordinary means and zero-level means vanish automatically; the remaining resonant rows are those with $m$ odd and $k$ odd. A countable family of smooth observations need not be injective on an infinite-dimensional angular space.

**Proposition 3.3** (Binary-quartic kernel)**.** *The intersection of* $\mathcal{K}^0$ *with binary quartic polynomials is exactly* $\operatorname{span}\{g_{\mathrm{rot}}\}$*.*

*Proof.* Lemma 3.1 removes $x^4, x^2y^2, y^4$. The remaining space is spanned by $x^3y, xy^3$. For angular momentum $\ell = xW_y - yW_x$, $L_H\ell = -g_{\mathrm{rot}}$, so $g_{\mathrm{rot}} \in \mathcal{K}^0$. The source interval certificate constructs a $9:4$ torus with

$$\mathcal{A}_{9,4}[x^3y] = \tfrac{1}{2}(U_7V_3 - U_3V_7) < -9/10^7,$$

where $U_j = 2\langle u^j \cos 9\alpha\rangle$ and $V_j = 2\langle v^j \cos 4\beta\rangle$. The exact rotation identity gives $20\mathcal{A}[x^3y] + 40\mathcal{A}[xy^3] = 0$. Hence the resonant coefficient of $bx^3y + dxy^3$ vanishes only when $d = 2b$, which is precisely the rotation line. The complete certificate specification is retained in Archive A.14. □

Every binary quartic has only EE and OO components, so this proposition contains no EO classification. For an unnormalized quartic first force with common unit-energy mean $b$, subtract $3bQ_0$ before applying the centered criterion, because $\int Q_0 d\mu = 1/3$.

## 4 Original regularity and one common second force

For the family in Theorem 1.2, define

$$D_U(s) = \int_0^{2\pi} U_s(e_\theta)^{-1/2} d\theta, \qquad V_s = \left(\frac{D_U(s)}{D_U(0)}\right)^2 U_s.$$

Microcanonical coarea and the virial identity show that every invariant mean of $\partial_s V_s$ is zero. The normalization is jointly $C^2$ and preserves positivity and homogeneity; it need not preserve joint convexity. In particular

$$g = \partial_s V_s|_0 \in C^1, \qquad h = \partial_s^2 V_s|_0 \in C^0.$$

**Proposition 4.1** (Necessary second-force compatibility)**.** *The first jet $g$ belongs to $\mathcal{K}^1$. The single continuous second jet $h$ satisfies*

$$T_0 Z_h(d) + \mathcal{Q}_d(g,g) = 0, \tag{6}$$

$$\int_0^{2\pi} h(e_\theta) Q(\theta)^{-3/2} d\theta = \tfrac{3}{2} \int_0^{2\pi} g(e_\theta)^2 Q(\theta)^{-5/2} d\theta, \tag{7}$$

$$\int_0^{T_c} h(q_{c,\phi}(t)) dt + \mathcal{Q}_{c,\phi}(g,g) = 0. \tag{8}$$

*Here $Q(\theta) = Q_0(e_\theta)$. Equation (6) holds on every finite zero-level orbit; equation (8) holds at every relative phase of each regular rational torus with nonzero isoenergetic twist. The quadratic data are the actual free-period mechanical Jacobi data, with the period variable retained.*

The full proof is [source Theorem 8 and Appendix F.1–F.4]. On a fixed regular rational torus, nonzero isoenergetic twist identifies the kernel of the free-period Hessian with the initial-angle directions. Normal reduction and a compact-envelope argument make the reduced action constant, and twice differentiating gives (8). Two ordinary derivatives of the constant volume give (7).

The zero-level cylinder is not compact in its relative-shift parameter. Its identity cannot be inferred by applying the compact-torus argument uniformly. The source proves (6) separately using the actual physical monodromy, the vertical primitive orbit at both ends, and the cancellation of the EO soft mode. The factor two in the vertical limit agrees with Section 2.

By Lemma 3.1, (6) determines $h_{EE}$ uniquely. An inverse applied to an arbitrary quadratic datum need not be continuous; continuity here is inherited from the actual $C^2$ family. Thus the same $h$, not a different compensator for each orbit, is a substantive compatibility requirement.

### 4.1 Entry at the vertical endpoint

**Proposition 4.2** (Original ordinary-null entry)**.** *Let $g(\rho e_\theta) = \rho^4 a(\theta)$ be EO with $a \in C^1(S^1)$ and suppose the ordinary means vanish on a fixed interval $0 < c < c_0$. Put $A = a(\pi/2)$ and $\delta = \pi/2 - \theta$. Then*

$$a(\pi/2 - \delta) - A \sin^3(\pi/2 - \delta) = o(\delta^2).$$

*On a fixed collar the left side is bounded in absolute value by $C\|a\|_{C^1}\delta^2$.*

This is [source Theorem 18], proved in Archive A.4 from the ordinary scalar-measure equation. It is a value statement and asserts no limit of a derivative quotient. For $A = 0$, the resulting physical-window entry legitimizes the second-variation forms and the energy completion. No compactness or finite-dimensionality theorem is used to prove that entry. The profile $\sin^3\theta$ is a subtraction reference, not a claimed full-null vector.

## 5 Positive energy and realizability obstructions

The exact positive mixture of zero-level probabilities eliminates $h$ between (6) and (7). It yields the necessary condition

$$\mathcal{B}(g,g) = 0, \tag{9}$$

where, with the source normalization,

$$\mathcal{B}(g_1,g_2) = -\frac{1}{T_0}\int_0^1 \mu(K)\mathcal{Q}_{d(K)}(g_1,g_2)dK - \frac{3}{2}\int_{S^1} a_1 a_2 Q^{-5/2} d\theta.$$

The density $\mu$ is positive and is explicitly defined by the beta-series mixture in [source Appendix F.5]. All orbit data are full physical-period data.

Let $I = (0, \pi/2)$ and $R_*(\theta) = Q'(\theta) = 20 \cos\theta \sin\theta(1 + \sin^2\theta) > 0$ on $I$. The source Green calculation yields a strictly positive off-diagonal kernel $\kappa$ and the ground-state form

$$\mathcal{D}(a) = \frac{1}{2} \int_{I^2} \kappa(s,t) R_*(s) R_*(t) \left| \frac{a(s)}{R_*(s)} - \frac{a(t)}{R_*(t)} \right|^2 ds dt. \tag{10}$$

For every original $C^1$ OO profile, $\mathcal{B}(a,a) = -\mathcal{D}(a) \le 0$, and equality holds exactly on the rotation line. The statement includes the diagonal distribution and endpoint limits: off-diagonal positivity alone would not justify it. The source computes the multiplication delta from the Green derivative jump, rules out derivative-supported diagonal terms, and passes from compactly supported profiles by the rotation radical identity.

For a zero-vertical EO force in the physical-window domain, the sign flip $\mathrm{sign}(x)g$ is a legitimate $C^1$ OO force. The difference of EO and OO physical boundary conditions gives a rank correction:

$$\mathcal{B}(g,g) = -\mathcal{E}(a), \qquad \mathcal{E}(a) = \mathcal{D}(a) + \frac{32}{3\pi} \int_0^1 \mu(K) |I_a(K)|^2 dK. \tag{11}$$

Here $I_a(K)$ is the pairing with the right Jacobi field, including its physical-origin boundary term; the exact formula is [source (356)]. The energy is positive definite on the stated zero-vertical EO domain. The source definitions, sign proof, origin term, and domain extensions are retained in Archive A.5; the notation in (11) abbreviates those precise operators rather than introducing formal replacements.

### 5.1 Proof of the realizability theorem

For pure OO, (9) and (10) force $g = c g_{\mathrm{rot}}$. For pure EO of zero vertical value, Proposition 4.2 puts $g$ in the legitimate domain of (11), whose strict positivity forces $g = 0$.

For a $y$-odd jet of zero vertical value, write $g = g_{EO} + g_{OO}$. The physical Jacobi operator and its kernel projection commute with the half-period reflection; the two forcings lie in opposite eigenspaces. The mixed term in $\mathcal{B}$ therefore vanishes. The angular mixed term also vanishes by parity. Both remaining terms are nonpositive; the EO term is strictly negative unless $g_{EO} = 0$. Hence $g = c g_{\mathrm{rot}}$. Finally the exact rotation family supplies second force $c^2 h_{\mathrm{rot}}$. Subtract its zero-level identity from (6); Abel injectivity gives $h_{EE} = c^2 h_{\mathrm{rot}}$. This proves Theorem 1.2.

**Corollary 5.1** (Tangent image of one family)**.** *For one normalized jointly $C^2$ multiparameter common-mean family, let $T$ be its derivative image at $Q_0$. If $T$ is entirely EO, then* $\dim T \le 1$. *If every element of $T$ is odd in $y$, then*

$$\ker(g \mapsto g(0,1)|_T) \subset \mathbb{R} g_{\mathrm{rot}}, \qquad \dim T \le 2.$$

*Proof.* Each parameter line is an actual one-parameter family, so Theorem 1.2 applies to every vector of its derivative image. In the EO case vertical evaluation is injective. In the $y$-odd case its kernel has dimension at most one. Rank–nullity gives the claims. □

The corollary does not subtract realizable directions from unrelated families, and it does not assert that a parity projection of a realizable mixed jet is independently realizable.

### 5.2 Rotation is attained even with joint convexity

For any $c \in \mathbb{R}$, let $P_s(Z) = Q_0(e^{csJ} Z)$ and $U_s = e^{Ms^2} P_s$. The normal Hessian of $Q_0$ is uniformly positive on the unit circle. For sufficiently large $M$ and then sufficiently small $|s|$, its Schur complement in the full $(s, Z)$ Hessian is positive away from zero; homogeneity

extends the Hessian continuously by zero at the origin. Thus $U$ is jointly convex on a strip. Since $L_H \ell = -\partial_\theta U_s$ and quartic virial gives $\langle U_s \rangle = 1/3$, every invariant mean of $\partial_s U_s$ equals $2Ms/3$. Volume normalization removes $e^{Ms^2}$ and leaves the exact rotation family. Hence $cg_{\mathrm{rot}}$ is attained. The detailed constant choices are preserved in [source Proposition 6.1, p. 43]. This example addresses realizability, not trapping of all dissipative trajectories.

# 6 Complete resonant rows and finite-dimensionality

The difficult part of finite-dimensionality is not the final Hilbert-space argument. It is establishing energy entry and compact observation bounds at the original regularity, with the exact endpoint and sampling terms of every mechanical resonant row.

Let $\mathcal{H}_P$ be the completed Peano energy space from [source Theorem 23], with injective angular realization $J$. Its original domain consists of zero-vertical $C^1$ EO profiles satisfying the physical-window condition and the $o(\delta^2)$ value entry. Proposition 4.2 ensures $N_0$ lies in this domain before completion. Let $N_E$ be its energy closure. Choose a smooth horizontal cutoff $\chi_H$.

**Proposition 6.1** (The three mechanical inputs)**.** *On $N_E$ the following hold.*

1. *Original-domain entry and angular realization are valid, with an energy coercivity bound controlling angular $L^2$.*
2. *The maps $a \mapsto Ja$ into $L^2(I)$ and $a \mapsto \chi_H Ja$ into $L^2(I, \theta^{-1/2} d\theta)$ are compact.*
3. *Exact ordinary endpoint gluing gives*

$$\mathcal{E}(a) \le C \left( \|Ja\|_{L^2}^2 + \|\chi_H Ja\|_{L^2(\theta^{-1/2} d\theta)}^2 \right). \tag{12}$$

These statements correspond to [source Theorems 18, 22–24, Corollary J.2, Lemma 85]. The archive includes their full proofs in dependency order. The first input is independent of compactness. For the second, complete integer-row analysis gives interior regularity and the signed horizontal boundary estimates; endpoint tails and local Rellich compactness then give compact maps. A finite list of Fourier rows cannot replace this step. For the third, the ordinary equation is differentiated as a raw physical coarea integral, then split into its exact vertical, horizontal, and interior pieces. The output restriction is applied after the distributional derivative. Non-nested leakage is bounded on the ambient spaces. No cutoff profile is assumed to remain null.

*Proof of Theorem 1.1.* If $N_E$ were infinite-dimensional, choose an energy-orthonormal sequence $a_j$. It is weakly zero. Compactness of the two maps in Proposition 6.1 makes both observation norms tend to zero. The gluing inequality forces $\mathcal{E}(a_j) \to 0$, contradicting normalization. Thus $N_E$ is finite-dimensional.

The original linear space $N_0$ is dense in $N_E$ by construction. A linear subspace of a finite-dimensional space is closed, hence $N_0 = N_E$. Choose an energy-orthonormal basis $e_1, \dots, e_d$ from $N_0$. For $a = \sum c_j e_j$,

$$\|a\|_{C^1} \le \left( \sum_j \|e_j\|_{C^1}^2 \right)^{1/2} \left( \sum_j |c_j|^2 \right)^{1/2} = C_N \sqrt{\mathcal{E}(a)}.$$

This gives a qualitative bound, not a computed constant or explicit basis. Finally vertical evaluation has scalar range and kernel exactly $N_0$. If its range is nonzero, choose and normalize a full-null element $u_0$ of nonzero trace. Subtracting $\tau(a) u_0$ from any $a \in \mathcal{K}_{EO}$ yields the direct sum in the theorem. Otherwise the whole EO kernel is $N_0$. □

Finite-dimensionality does not imply zero dimension. It also does not imply equality between the static $C^2$ EO kernel and its $C^1$ counterpart. If the unit-trace branch exists it is the affine fiber $u_0 + N_0$, not the fiber formed by adding an arbitrary convenient reference profile.

# 7 Value traces and one-way ambient reconstruction

## 7.1 A fourth-order value trace

**Theorem 7.1** (Quartic Peano value)**.** *Every original $C^1$ EO profile of zero vertical value whose ordinary means vanish on a fixed interval $0 < c < c_0$ has*

$$a(\pi/2 - \delta) = \lambda(a)\delta^4 + o(\delta^4).$$

*The coefficient is a bounded $C^1$ linear functional; on $N_0$ it is also energy-continuous.*

This is [source Theorems 11 and 25]. The proof uses the ordinary equation to obtain subquartic growth, then a convergent critical forcing primitive. The inverse at a simple Mellin zero retains its homogeneous constant and converts that primitive into the value coefficient. It is logically independent of finite-dimensionality except for the final energy-continuity statement. Its exact cutoff-independent formula is in Archive A.12.

This is not $C^4$ endpoint regularity. It does not prove $\lambda(a) = 0$, injectivity of the trace, or absence of flat null profiles. The nonzero-vertical-trace branch remains a different affine problem.

## 7.2 Reconstruction on known null inputs

Define the observation space and map

$$\mathcal{Z} = L^2(I) \oplus L^2((0, \theta_H), \theta^{-1/2} d\theta), \qquad Ka = (Ja, \chi_H Ja).$$

The exact ordinary equation gives a supported inverse on the vertical coordinate space. With the operators defined explicitly in [source Appendix N], let $D_v$ be the signed mechanical remainder, $P_V$ the support projection, $S_v$ the full-line normal inverse, and $B$ the leakage-plus-observation operator. Then

$$\mathcal{R}(g, h) = U_0 g + U_V (\mathrm{Id} + D_v)^{-1} P_V S_v B(g, h) \tag{13}$$

is a bounded operator on all of $\mathcal{Z}$, and

$$\mathcal{R}Ka = a \quad (a \in N_0). \tag{14}$$

The physical lifts $U_0, U_V$ are defined without division by a cutoff. The support projection is applied after the full-line inverse. The source gives explicit compatible vertical collars with $\|D_v\| < 2^{-12}$; this is not a numerical bound for the horizontal observation tail.

Let $\Pi_m$ be a specified finite-rank orthogonal observation projection and suppose

$$\|(\mathrm{Id} - \Pi_m)Ka\|_{\mathcal{Z}} \le \eta_m \|a\|_E \qquad (a \in N_0).$$

The ambient comparison space $V_m = \mathcal{R} \operatorname{Ran} \Pi_m$ and its energy-orthogonal projection $P_m$ satisfy

$$\|(\mathrm{Id} - P_m)a\|_E \le \|a - \mathcal{R}\Pi_m Ka\|_E \le \|\mathcal{R}\| \eta_m \|a\|_E.$$

This argument is legitimate even though the approximation remainder need not be null. In contrast, applying (12) directly to an arbitrary projection remainder would be invalid.

If the inverse in (13) is truncated after $M$ terms, its additional operator error is bounded by $vsbq^{M+1}/(1 - q)$ in the source notation. The small $q$ controls this Neumann-series error only. It does not replace the observation error $\eta_m$, finite representation errors, or an ambient bound for later testing.

### 7.3 What a successful separation certificate would require

**Proposition 7.2** (Sufficient finite-carrier criterion)**.** *Let $P$ be the exact energy-orthogonal projection onto a specified finite-dimensional ambient space $V$. Suppose that, on $N_0$, $\|(\mathrm{Id} - P)a\|_E \le \epsilon\|a\|_E$ with $0 \le \epsilon < 1$. Let an ambient observation $\mathcal{O}$ satisfy $\|\mathcal{O}\| \le B$, $\mathcal{O}a = 0$ on $N_0$, and $\|\mathcal{O}v\| \ge s_*\|v\|_E$ on $V$. If*

$$s_*^2(1 - \epsilon^2) > B^2\epsilon^2,$$

*then $N_0 = \{0\}$.*

*Proof.* For $a \in N_0$, $\mathcal{O}Pa = -\mathcal{O}(\mathrm{Id} - P)a$. Orthogonality gives $\|Pa\|_E^2 \ge (1 - \epsilon^2)\|a\|_E^2$. The two observation estimates contradict the strict inequality unless $a = 0$. □

This reproduces [source Proposition N.1] as a conditional criterion, not a successful separation. The source does not provide all numerical horizontal-tail, retained-interval, and continuous-output Gram bounds needed to instantiate it. Even $N_0 = 0$ would leave the full EO kernel at most one-dimensional; the existence of a unit-trace vector, the OE and OO linear kernels, and general finite deformations would still be separate questions.

## 8 Contribution boundary and proof architecture

The completed endpoint is finite-dimensionality together with realizability obstructions at the original regularity. The result is not presented as an unrestricted nonpolynomial rigidity classification. The logical order is

$$\text{physical trajectories and clock} \Longrightarrow \text{complete observation criterion},$$
$$\text{ordinary-null endpoint entry} \Longrightarrow \text{legitimate mechanical energy},$$
$$\text{complete rows and endpoint gluing} \Longrightarrow \text{compact energy control} \Longrightarrow \dim N_0 < \infty.$$

Actual two-jets form a separate branch: their one common second force gives $\mathcal{B}(g, g) = 0$, and the signed energy identities then restrict realizability. The quartic value trace and ambient reconstruction refine these statements without changing their endpoint.

The remaining finite-dimensional uncertainty is explicit. Determining its dimension or finding a nonzero direction would change the identification result. Improving an isolated contraction constant without closing the ambient observation and separation bounds would not. No claim about finite families at nonzero parameters follows solely from an initially rotational derivative.

## References


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# A Proof archive and source-number concordance

The archived source folios are included in this PDF, preserving equations, labels, and complete technical proofs. The main text gives the new order and self-contained principal arguments. Old section numbers in the folios are provenance labels. Source references retain their original bibliography at the end. This hybrid format is a reconstruction draft, not a fully retyped submission source. Boundary folios may retain adjacent context so no proof lines are cropped away.

| New task | Source location | Role |
|---|---|---|
| Physical criterion | Section 5, pp. 35–40 | All phases, clock, zero-separation family |
| Actual two-jets | Section 6, pp. 40–43 | One continuous common second force |
| Original entry | Appendix E, pp. 132–138 | Prior to energy completion |
| Signed forms | Appendix F, pp. 138–156 | Monodromy, mixture, diagonal and origin terms |
| Complete-row analysis | G–I, pp. 156–207 | Endpoint, sampling, Mellin and local regularity |
| Finite-dimensionality | J–K, pp. 207–226; source Theorem 10 | Compactness and exact ordinary gluing |
| Quantitative vertical bound | L, pp. 226–233 | Separate from horizontal observation bounds |
| Value trace | M, pp. 233–237 | Fourth-order value, not derivative regularity |
| Ambient reconstruction | N, pp. 237–240 | One-way identity and conditional separation |
| Polynomial coefficient | O, pp. 241–247 | Reported rational $9 : 4$ certificate |
| Shared foundation | pp. 9–14 | Volume and compact normal reduction |

domains with $C^9$ boundaries sufficiently close to a circle, using their convention $r = 8$. The invariant-distribution theorem of Avila and Kocsard [7] concerns orientation-preserving $C^\infty$ circle diffeomorphisms with irrational rotation number; the invariant probability spans all invariant distributions, while smooth cohomological stability requires a Diophantine rotation number. Neither result is a theorem for the natural Hamiltonians considered here. They provide context for why periodic tests and regular solvability must be distinguished. The mechanical content proved below is the classification of allowed potential deformations, followed by a controlled corrector and an actual critical-flow argument.

The sextic action argument uses the classical minimal-average action of twist maps. We use Mather's rational differentiability criterion in the $C^1$ twist-map formulation recorded by Siburg [19, 22]; the $C^1$ map and $C^2$ generating-function framework is also explicit in [18]. The parameter invariance of the physical minimum actions is proved here from the actual force condition. It is distinct from the classical existence and differentiability results used after that invariance has been established.

For vanishing damping, energy convergence, objective-value decay and point convergence have different hypotheses. Cabot, Engler and Gadat [13] distinguish energy convergence from convergence to a minimizer and identify a necessary damping condition for the latter. Attouch, Chbani and Riahi [4] prove value decay $O(t^{-2\alpha/3})$ for $0 < \alpha \leq 3$ in the general convex setting; that estimate alone does not select one point of a minimizing valley. Aujol, Dossal and Rondepierre [5] obtain finite-length convergence under coercivity, matched flatness and growth of order $\gamma > 2$, and $\alpha \geq (\gamma + 2)/(\gamma - 2)$. The continuous-time result of Aujol, Dossal, Labarrière and Rondepierre [6] permits nonunique minimizers under growth and second-order regularity of the minimizing set, with $\alpha > 9/2 + 6/(\gamma - 2)$. The latter two finite-length convergence regimes do not contain the damping $\alpha = 1 + 2/p < 2$ treated here. Moreover, homogeneity only in the normal variables is not the full-state flatness hypothesis of [5]. Our conclusion uses the mechanical common-mean structure and an actual compact tail; it asserts neither a general finite-length theorem nor a selection-coordinate rate.

# 2 Frozen action identities and volume normalization

**Lemma 1** (Fixed-energy action variation)**.** *Let a differentiable family of periodic orbits of $H_s = |W|^2/2 + P(s, Z)$ have energy $h$, period $T$, and unnormalized action $J = \oint W \cdot \mathrm{d}Z$. Then*

$$J_h = T, \qquad J_{s_j} = -\int_0^T P_{s_j}(s, Z(t))\,\mathrm{d}t. \tag{6}$$

*For a homogeneous potential, $J(s, h) = h^\nu J(s, 1)$, where $\nu = 1/2 + 1/p$. In particular, the mean force along the branch is*

$$-\frac{J_s}{J_h} = -\frac{h}{\nu}\nabla_s \log J(s, 1).$$

**Proof.** Parametrize the loop on a fixed circle. Variation and integration by parts give $\delta J = \int (\delta W \cdot \dot{Z} - \dot{W} \cdot \delta Z)\,\mathrm{d}t$. Hamilton's equations and variation of $H_s = h$ identify the integrand with $\delta h - P_s \cdot \delta s$. Periodicity removes the boundary term. Finally, $Z = h^{1/p} z$, $W = h^{1/2} w$, and $t = h^{1/p-1/2}\tau$ give the claimed scaling. These are the standard action identities; see [11]. ∎

For any invariant probability measure on $H_s = h$, the virial identity follows by integrating $\mathcal{L}_s(Z \cdot W) = 2H_s - (p+2)P$:

$$\int P \,\mathrm{d}\mu = \frac{2h}{p+2}. \tag{7}$$

This applies to all invariant measures, including measures supported on periodic orbits.

Let $n$ be the normal dimension and define

$$\Omega(s,h) = \mathrm{vol}_{2n}\{H_s \le h\}, \qquad \Omega_1(s) = \Omega(s,1), \quad D(s) = \Omega_1(s)^{1/n}.$$

The power $1/n$ is the normal-dimensional root, not the phase-dimensional root $1/(2n)$.

**Lemma 2** (Microcanonical identification and normalization)**.** *The positive-energy shells are regular, and*

$$\Omega(s,h) = h^{n\nu}\Omega_1(s), \qquad b(s) = -\frac{1}{n\nu}\nabla \log \Omega_1(s) = -\frac{1}{\nu}\nabla \log D(s) \tag{8}$$

*whenever* (CM) *holds. The functions $\Omega_1, D, b$ are respectively $C^2, C^2, C^1$. For $n = 2, p = 4$,*

$$\Omega(s,h) = \frac{2\pi}{3} h^{3/2} \mathcal{I}(P(s,\cdot)), \qquad b = -\frac{2}{3}\nabla \log \mathcal{I}(P). \tag{9}$$

*The normalized family* (1) *has $\mathcal{I}(V) = \mathcal{I}_0$ and every invariant force mean equal to zero. Its differentiable periodic branches have constant fixed-energy action.*

**Proof.** If the normal gradient of $H_s$ vanished on $H_s = h > 0$, then $W = 0$, $P_Z = 0$, and Euler's identity $Z \cdot P_Z = pP$ would give $h = 0$. Coarea and variation of a compact regular sublevel therefore give

$$\Omega_h = \int_{H_s = h} \frac{\mathrm{d}\Sigma}{|\nabla H_s|}, \qquad \Omega_{s_j} = -\int_{H_s = h} \frac{P_{s_j}\,\mathrm{d}\Sigma}{|\nabla H_s|}.$$

The probability $\Omega_h^{-1}|\nabla H_s|^{-1}\,\mathrm{d}\Sigma$ is invariant, since the Hamiltonian flow preserves both Lebesgue volume and energy. It is the microcanonical measure, rather than normalized Euclidean surface area. The homogeneous coordinate change has Jacobian $h^{n\nu}$, so its mean force is $-h\nabla \log \Omega_1/(n\nu)$, proving (8).

If $Q(s) = \mathrm{vol}_n\{P(s,Z) \le 1\}$, polar integration gives

$$Q(s) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} P(s,\omega)^{-n/p}\,\mathrm{d}\omega, \qquad \Omega_1(s) = \frac{(2\pi)^{n/2}\Gamma(1+n/p)}{\Gamma(1+n\nu)} Q(s). \tag{10}$$

Local uniform coercivity permits two parameter derivatives under the integral. For $n = 2, p = 4$, $Q = \mathcal{I}/2$ and the constant in (10) is $4\pi/3$, yielding (9).

Write $V = c(s)P$. The common mean of $V_{s_j}$ on $H_V = 1$ is $b_j + (1/3)\partial_j \log c$. Indeed the term $cP_{s_j}$ has mean $b_j$ after transport from the $P$ shell of energy $1/c$, while $c_{s_j}P$ has mean $\partial_j \log c/3$ by (7). For $c = (\mathcal{I}(P)/\mathcal{I}_0)^2$, this sum vanishes by (9). The action conclusion follows from Lemma 1. ∎

An equality of invariant means alone does not reverse the implication obtained by integrating $\mathcal{L}_s A = P_s - H_s b$: it gives necessary obstruction identities for $A$, not its existence or a bound for $A_s$. The classifications below resolve this issue in two particular mechanical classes.

## 2.1 Convex geometry of the effective energy

**Lemma 3.** *Let $U \subset \mathbb{R}^m$ be open and convex and $P \in C^2(U \times \mathbb{R}^n)$ jointly convex and $p$-homogeneous, $p > 1$. If $P$ is coercive at one parameter, it is positive away from zero at every parameter, locally uniformly coercive, and $D = \Omega_1^{1/n}$ is positive, $C^2$ and concave on $U$. Hence*

$$U_I(s) = (I/D(s))^{1/\nu} \quad (I > 0)$$

*is convex and $C^2$ on $U$. If $D$ is affine on any nontrivial selected segment, then $P(s, \cdot)$ is constant on that segment. In particular all frozen potentials agree on $M = \operatorname{argmax}_U D$ when this set is nonempty.*

**Proof.** Convexity and homogeneity give $P(s, 0) = 0$, $P_Z(s, 0) = 0$, and $P \geq 0$. If $P(s_0, z) = 0$ for $z \neq 0$, write an arbitrary $s \in U$ as $s = \theta s_0 + (1 - \theta)s_1$ with $s_1 \in U$ and $0 < \theta < 1$. Joint convexity at $(s_0, z/\theta)$ and $(s_1, 0)$ gives $P(s, z) \leq 0$. Thus that direction would vanish at every parameter, contradicting coercivity at one. Compactness of the sphere then gives local uniform coercivity.

The configuration bodies $K_s = \{P(s, \cdot) \leq 1\}$ obey $(1 - \theta)K_{s_0} + \theta K_{s_1} \subset K_{s_\theta}$. The Brunn–Minkowski inequality [10] and (10) make $D$ concave. Composition with the decreasing convex function $x \mapsto I^{1/\nu} x^{-1/\nu}$ gives convexity of $U_I$.

For completeness, the strict equality consequence uses additional homogeneous structure. For $w \neq 0$ let $h_s(w)$ be the support function of $K_s$. The partial Legendre transform in $Z$ is concave in $s$ by joint convexity, and radial maximization gives

$$\sup_Z (w \cdot Z - P(s, Z)) = (p - 1)p^{-p/(p-1)} h_s(w)^{p/(p-1)}.$$

If $D(s_\theta) = (1 - \theta)D(s_0) + \theta D(s_1)$, equality of volumes in the inclusion and Brunn–Minkowski imply $K_{s_\theta} = (1 - \theta)K_{s_0} + \theta K_{s_1}$. Nested full-dimensional compact convex bodies of equal volume coincide. Thus the support at $s_\theta$ is the affine average of its endpoint supports. Concavity of its $p/(p - 1)$ power and strict convexity of that scalar power force $h_{s_0}(w) = h_{s_1}(w)$ for every $w$. The endpoint bodies and potentials coincide. A

concave function meeting its chord at an interior point agrees with the chord throughout the segment; repeating the support argument proves constancy throughout. Applying this to segments in the convex maximizing set $M$ proves the last assertion. ∎

At every $u \in M$, if $T = \operatorname{span}(M - M)$, joint Hessian positivity also gives

$$\Pi_T P_s(u, Z) = 0, \qquad P_{ss}(u, Z)v = P_{Zs}(u, Z)v = 0 \quad (v \in T). \tag{11}$$

Indeed restriction to each segment in $M$ is constant, its second derivative vanishes, and a vector with zero quadratic form for a positive-semidefinite matrix belongs to its kernel. Continuity includes segment endpoints. This geometric fact does not by itself supply a force corrector off $M$.

# 3 Exact action reduction and finite radial rigidity

The normal configuration space in this section is $\mathbb{R}^2$. The parameter space may have any finite dimension. Homogeneity always means positive homogeneity in the normal variable; neither antipodal symmetry nor polynomial dependence is assumed. Convexity is not needed for the frozen rigidity statements below. In particular, amplitude normalization will not be assumed to preserve joint convexity.

## 3.1 Compact reduction for the mechanical action

We first give the reduction argument that supplies the periodic continuations used in the proof. This also specifies the function spaces and the role of the period variable.

**Lemma 4** (Compact envelope)**.** *Let $N$ be a compact boundaryless $C^1$ manifold and let $a \in C^1((-\varepsilon, \varepsilon) \times N)$. Suppose*

$$d_m a(r, m) = 0 \quad \Longrightarrow \quad \partial_r a(r, m) = 0, \qquad a(0, m) = a_0.$$

*Then $a(r, m) = a_0$ everywhere. The same conclusion holds in a sufficiently small convex finite-dimensional parameter neighborhood if the implication holds for every parameter derivative.*

**Proof.** Put $M(r) = \max_N a(r, \cdot)$ and $m_*(r) = \min_N a(r, \cdot)$. Both are locally Lipschitz, since $\partial_r a$ is bounded on compact parameter intervals times $N$. Every extremizer is a critical point, because $N$ has no boundary. If $x_0$ maximizes at $r$ and $x_h$ maximizes at $r + h$, then, for $h > 0$,

$$a(r + h, x_0) - a(r, x_0) \le M(r + h) - M(r) \le a(r + h, x_h) - a(r, x_h).$$

Every accumulation point of $x_h$ as $h \to 0$ maximizes at $r$. Divide by $h$, write the outer differences as integrals of $\partial_r a$, and use compactness and continuity. Both bounds tend to

zero. The argument for $h < 0$, and then for the minimum, is identical. Thus $M' = m'_* = 0$ everywhere. Their initial values coincide, so $m_* = a = M = a_0$. For several parameters apply the interval assertion on each segment from the base parameter in a small convex neighborhood. ∎

**Lemma 5** (Normal reduction with zero critical parameter derivative)**.** *Let $E$ be a real Hilbert space, $\mathcal{O} \subset E$ open, and $F \in C^2(U \times \mathcal{O})$, where $U$ is a neighborhood of $0$ in a finite-dimensional parameter space. Let $N \subset \mathcal{O}$ be a compact, connected, boundaryless, finite-dimensional $C^2$ embedded manifold of critical points of $F(0, \cdot)$. Suppose its Riesz Hessian $L_m$ is self-adjoint Fredholm and*

$$\ker L_m = T_m N \qquad (m \in N).$$

*Suppose also that every nearby critical point of $F(s, \cdot)$ satisfies $\partial_{s_j} F(s, y) = 0$ for every parameter component. After restricting $U$, there is a unique small $C^1$ normal graph*

$$Y(s, m) = m + \psi(s, m), \qquad \psi(s, m) \in E_m := (T_m N)^{\perp}, \qquad \psi(0, m) = 0,$$

*whose points are all critical and satisfy $F(s, Y(s, m)) = F(0, N)$. Every sufficiently close critical point belongs to this graph.*

**Proof.** Self-adjointness and the closed range property imply $\operatorname{ran} L_m = (\ker L_m)^{\perp} = E_m$. The restriction of $L_m$ to $E_m$ is therefore boundedly invertible. In local normal frames the Hessians and tangent projections vary continuously. Compactness gives a uniform inverse bound. The $C^1$ implicit function theorem applied to

$$\Pi_{E_m} \nabla_y F(s, m + \psi) = 0$$

produces the normal graph. Uniqueness glues the local constructions and compactness gives one parameter neighborhood and one tube. The reduced function $a(s, m) = F(s, Y(s, m))$ is $C^1$. Its tangent graph, together with $E_m$, spans $E$, since it is $C^1$ close to the zero section. Consequently

$$d_m a(s, m) = 0 \quad \Longleftrightarrow \quad d_y F(s, Y(s, m)) = 0.$$

Moreover, $\partial_{s_j} \psi \in E_m$, so the normal equation gives $\partial_{s_j} a = \partial_{s_j} F(s, Y)$. The base critical value is constant on connected $N$. Lemma 4 now makes $a$ identically that value, and the equivalence just proved makes all graph points full critical points. Normal uniqueness includes every critical point in a sufficiently small common tube. ∎

For a jointly $C^2$ natural potential $V(s, Z)$ define, on $E = H^1(\mathbb{R}/\mathbb{Z}; \mathbb{R}^2) \times \mathbb{R}$ with $T > 0$,

$$J_s(q, T) = \int_0^1 \left( \frac{|\dot{q}|^2}{2T} - TV(s, q) + T \right) du. \tag{12}$$

A dot in this formula denotes differentiation in $u$. Physical time is $t = Tu$. On a tube about a compact family of loops and positive periods, $H^1 \hookrightarrow C^0$ confines all loop ranges to one compact ball. The ordinary first and second derivative formulas for the potential, together with uniform continuity of its second derivatives on that ball, show that $J$ is twice continuously Fréchet differentiable. The kinetic term is smooth for $T > 0$. Only quadratic velocity dependence is used. The first variations are

$$d_q J[\eta] = T^{-1} \int_0^1 \dot{q} \cdot \dot{\eta}\, du - T \int_0^1 V_Z(s,q) \cdot \eta\, du,$$
$$\partial_T J = \int_0^1 \left( 1 - \frac{|\dot{q}|^2}{2T^2} - V(s,q) \right) du.$$

The first equation gives $\ddot{q} + T^2 V_Z(s,q) = 0$. A weak solution is classical by this equation. Its energy is constant, and the second equation sets it equal to one. Thus critical points are exactly parametrized unit-energy periodic orbits, with the specified period (which need not be primitive). At a critical point,

$$\partial_{s_j} J_s = -\int_0^T V_{s_j}(s, Q(t))\, dt. \tag{13}$$

The full Hessian, evaluated at a critical point, is

$$D^2 J[(\eta,\tau),(\xi,\sigma)] = T^{-1} \int_0^1 \dot{\eta} \cdot \dot{\xi}\, du - T \int_0^1 \eta \cdot V_{ZZ} \xi\, du$$
$$- 2\tau \int_0^1 V_Z \cdot \xi\, du - 2\sigma \int_0^1 V_Z \cdot \eta\, du + \frac{\tau\sigma}{T^3} \int_0^1 |\dot{q}|^2\, du. \tag{14}$$

The leading loop form becomes coercive on adding $T^{-1} \int \eta \cdot \xi$ and a positive period form. All corrections are compact: loop terms factor through the compact embedding $H^1 \hookrightarrow L^2$, and period terms are finite rank. The Riesz Hessian is therefore self-adjoint Fredholm. For a $p$-homogeneous potential, the virial identity and unit energy give, at every critical point,

$$J_s = \int_0^T |\dot{Q}|^2\, dt = \frac{T}{\nu}, \qquad \nu = \frac{1}{2} + \frac{1}{p}. \tag{15}$$

In particular, the conclusion of Lemma 5 gives constant periods as well as constant actions in the homogeneous case.

## 3.2 Centering and the two radial-base critical manifolds

For a positive $p$-homogeneous planar potential set

$$\mathscr{I}(s) = \int_0^{2\pi} P(s, e_\theta)^{-2/p}\, d\theta, \qquad e_\theta = (\cos\theta, \sin\theta).$$

Direct integration first in momentum and then in the radial position variable gives the phase volume

$$\Omega(s,E) = \frac{\pi p}{p+2} E^{1+2/p} \mathscr{I}(s). \tag{16}$$

The two critical values of $Q$ are distinct, so a rotation preserving $Q$ must preserve each labeled critical line. Its angle is 0 or $\pi$ modulo $2\pi$. Conversely every binary quartic is invariant under $-I$. Thus the stabilizer is exactly $\{I, -I\}$. The orbit is a smooth circle parametrized by $\theta$ modulo $\pi$, with nonzero derivative $\mathcal{R}Q$. Local inversion gives $C^2$ angles; on a simply connected $S$, the circle-valued map lifts through $\mathbb{R} \to \mathbb{R}/(\pi\mathbb{Z})$. Undoing (1) gives $\alpha = (\mathcal{I}_0/\mathcal{I}(P))^2$. On a nonsimply-connected region even a global $SO(2)$-valued lift may fail, since an odd multiple of $\pi$ preserves the potential but changes the rotation matrix. This does not affect the orbit-valued statement. ∎

The neighborhood contains exactly two real projective critical lines, throughout. It is therefore outside the maximal-four-critical-line hypothesis for binary quartics. No nearby potential is assumed to retain parabolic separation, coordinate-reflection symmetry, or integrability.

# 5 The complete invariant-average kernel at the non-radial quartic

The coefficient-neighborhood theorem of Section 4 and a classification of all centered forces at one well are different assertions. We now study the latter question at

$$Q_0(x, y) = P_0(x, y) = x^4 + 12x^2y^2 + 16y^4, \qquad H = \tfrac{1}{2}|W|^2 + Q_0.$$

For $r \in \{0, 1, 2\}$ let $\mathcal{K}^r$ consist of the forces $g(\rho e_\theta) = \rho^4 a(\theta)$, $a \in C^r(\mathbb{S}^1)$, whose mean is zero for every invariant probability on $H = 1$. No central symmetry is imposed. The superscript records angular regularity, not a homogeneity degree. The continuous class is sufficient for the exact criterion below. For a jointly $C^2$ parameter family the first force is in $C^1$, and its second force is only continuous; neither will be given an extra spatial derivative.

## 5.1 Separation and the physical clock

Put

$$\begin{gathered} x = uv, \qquad y = (u^2 - v^2)/2, \qquad S = u^2 + v^2, \\ p_u = vW_x + uW_y, \qquad p_v = uW_x - vW_y. \end{gathered} \tag{75}$$

Away from the origin this is a canonical two-fold covering, with deck transformation $(u, v, p_u, p_v) \mapsto (-u, -v, -p_u, -p_v)$. Direct substitution gives

$$H = \frac{p_u^2 + p_v^2}{2S} + \frac{u^{10} + v^{10}}{S}.$$

At physical energy $E > 0$, the regularized Hamiltonian and clock are

$$S(H - E) = \tfrac{1}{2}p_u^2 + U_E(u) + \tfrac{1}{2}p_v^2 + U_E(v), \qquad U_E(q) = q^{10} - Eq^2, \qquad \mathrm{d}t = S\,\mathrm{d}\tau. \tag{76}$$

The scalar energies are $c$ and $-c$, where $|c| \le c_*(E) = 4\,5^{-5/4}E^{5/4}$. We write $c_* = c_*(1)$ and $U(q) = q^{10} - q^2$. The separation integral descends to the physical variables and extends polynomially through the origin:

$$C = xW_xW_y - yW_x^2 + 4x^2y(x^2 + 2y^2). \tag{77}$$

Indeed $C = p_u^2/2 + u^{10} - Hu^2$ in the covering, and expansion yields (77). The separated equations prove $\{C, H\} = 0$ away from the origin, hence everywhere by continuity. The level $C = 0$ need not be a regular level: the vertical straight orbit is a critical part of it.

For $0 < c < c_*$, the $u$ oscillator crosses both wells, while $v$ librates in the positive well. The negative $v$ well is deck-equivalent. This parametrizes one physical two-torus; the negative-$C$ torus is its reflection in $y$. Denote the auxiliary periods by $T_u(c), T_v(c)$ and choose normalized phases $\alpha, \beta$ starting at upper turning points. Thus

$$u_c(-\alpha) = u_c(\alpha), \quad u_c(\alpha + \pi) = -u_c(\alpha), \quad v_c(-\beta) = v_c(\beta) > 0.$$

Let $\bar{S} = \langle u_c^2\rangle + \langle v_c^2\rangle$, with normalized auxiliary-phase averages. The canonical scalar actions satisfy

$$\begin{aligned} \partial_c I_u &= T_u/(2\pi), & \partial_c I_v &= -T_v/(2\pi), \\ \partial_E I_u &= T_u\langle u^2\rangle/(2\pi), & \partial_E I_v &= T_v\langle v^2\rangle/(2\pi). \end{aligned}$$

Their Jacobian is $-T_uT_v\bar{S}/(4\pi^2) \neq 0$. Inverting this matrix therefore gives the physical action frequencies

$$\omega_u = \frac{2\pi}{T_u\bar{S}}, \qquad \omega_v = \frac{2\pi}{T_v\bar{S}}, \qquad \frac{\omega_v}{\omega_u} = \frac{T_u}{T_v}. \tag{78}$$

The scalar phases are uniform in $\tau$, not in $t$. This distinction accounts for the factor $S$ in every force integral below.

## 5.2 All ordinary means and all resonant phases

For $\sigma = \pm 1$ define

$$\begin{aligned} F_c^\sigma(\alpha, \beta) &= S\, g\big(u_cv_c, \sigma(u_c^2 - v_c^2)/2\big) \\ &= \frac{S^5}{16}\, a\big(\arg[2u_cv_c + i\sigma(u_c^2 - v_c^2)]\big)\,. \end{aligned} \tag{79}$$

The argument is a point of the unit circle. Fourier coefficients use normalized Haar measure on $\mathbb{T}^2$. The clock-weighted torus probability has density $S/\bar{S}$ relative to Haar measure, so its mean is

$$\langle g\rangle_{c,\sigma} = \widehat{F}_c^\sigma(0, 0)/\bar{S}. \tag{80}$$

This measure is invariant even when the period ratio is rational. When that ratio is irrational it is the unique invariant probability on the torus: invariant measures for a positive time change correspond by multiplication by the clock to invariant measures of the linear flow.

If $T_u/T_v = m/n$, with $(m,n) = 1$, the auxiliary orbit has phases $(\alpha, \beta) = (ns+\alpha_0, ms+\beta_0)$. Its auxiliary period is $nT_u = mT_v$ and its physical period is $nT_u\bar{S}$, independent of the relative phase, since $S = u^2 + v^2$ is additive. Put $\chi = m\alpha_0 - n\beta_0$. Choose integers $a_0, b_0$ with $ma_0 - nb_0 = 1$. The actual phase mean is the continuous function

$$M_c^\sigma(\chi) = \frac{1}{2\pi\bar{S}} \int_0^{2\pi} F_c^\sigma(ns + a_0\chi, ms + b_0\chi)\,\mathrm{d}s.$$

The change $(s,\chi) \mapsto (ns + a_0\chi, ms + b_0\chi)$ is a torus automorphism. Haar invariance and Fubini give, without differentiating the input,

$$\widehat{M_c^\sigma}(k) = \widehat{F_c^\sigma}(km, -kn)/\bar{S}. \tag{81}$$

Separate evenness in $\alpha, \beta$ makes these coefficients real. The Fejér sums of the continuous function $M_c^\sigma$ converge uniformly; thus all actual phase means vanish if and only if its constant coefficient and all the coefficients in (81) vanish. Absolute convergence of a two-dimensional Fourier series is unnecessary.

For reference, each coefficient is an explicit scalar quadrature. If $A > 1$ is the positive turning point at energy $c$, and $b < B$ are the positive turning points at energy $-c$, set $p_e(q) = \sqrt{2(e + q^2 - q^{10})}$. Then

$$\widehat{F_c^\sigma}(0,0) = \frac{4}{T_uT_v} \int_{-A}^{A} \int_b^B \frac{(s^2+t^2)g(st, \sigma(s^2-t^2)/2)}{p_c(s)p_{-c}(t)}\,\mathrm{d}t\,\mathrm{d}s.$$

The $(j,k)$ coefficient has the additional factors $\cos(j\alpha(s))\cos(k\beta(t))$, where $\alpha(s) = (2\pi/T_u)\int_s^A p_c^{-1}$ and $\beta(t) = (2\pi/T_v)\int_t^B p_{-c}^{-1}$.

At $c = c_*$ the $v$ oscillator is fixed at $v_0 = 5^{-1/8}$, giving the curved brake $y = x^2/(2v_0^2) - v_0^2/2$. Its reflected partner occurs at $-c_*$. Their physical means are limits of (80); continuity follows from the regularized scalar turning-point integrals. The only straight directions are the axes, and on either one

$$\langle g \rangle_e = \frac{g(e) + g(-e)}{6Q_0(e)}. \tag{82}$$

The virial identity gives $\langle Q_0 \rangle = 1/3$ and the two half-oscillations have equal duration. Consequently the unrestricted angular class imposes an opposite-ray sum, not two separate ray values.

## 5.3 The physical zero-separation family and Abel inversion

The nonconstant zero-energy scalar solutions are signed translates of

$$h(\tau) = \operatorname{sech}(4\sqrt{2}\tau)^{1/4}.$$

For $d \in \mathbb{R}$ put $u_d = h(\tau - d/2)$, $v_d = h(\tau + d/2)$, and $q_d = (u_dv_d, (u_d^2 - v_d^2)/2)$. Since $\mathrm{d}t = (u_d^2 + v_d^2)\,\mathrm{d}\tau$, both infinite ends of this auxiliary trajectory reach the physical origin in

finite time. Direct differentiation of the displayed pulse gives the outgoing and incoming physical velocities

$$(\sqrt{2}\,\mathrm{sech}(\sqrt{2}d), -\sqrt{2}\tanh(\sqrt{2}d)), \qquad (-\sqrt{2}\,\mathrm{sech}(\sqrt{2}d), -\sqrt{2}\tanh(\sqrt{2}d)).$$

The $x$ reflection of the first arc begins with the latter velocity and returns with the former. Uniqueness of the smooth physical flow therefore joins the two arcs into an actual periodic orbit. Put

$$A_0 = \int_{\mathbb{R}} h(\tau)^2 \,\mathrm{d}\tau = \frac{\sqrt{\pi}\Gamma(1/4)}{4\sqrt{2}\Gamma(3/4)}, \qquad T_0 = 4A_0. \tag{83}$$

Each arc lasts $2A_0$. For finite $d$ the two-arc orbit has primitive period $T_0$: its two interior arcs lie in opposite open half-planes and there are exactly two origin crossings. At $d = 0$ this is the horizontal straight oscillator. As $|d| \to \infty$ the orbit tends to a double traversal of the vertical straight orbit, whose primitive period is $T_0/2$.

With $F_g(u, v) = (u^2 + v^2)g(uv, (u^2 - v^2)/2)$ the mean is

$$Z_g(d) = \frac{1}{4A_0} \int_{\mathbb{R}} [F_g(u_d, v_d) + F_g(u_d, -v_d)] \,\mathrm{d}\tau. \tag{84}$$

The integral is absolutely convergent, since $|F_g| \le C(u^2 + v^2)^5$. It kills every $x$-odd input by reflection of the arcs and every $y$-odd input by $\tau \mapsto -\tau$. Only the doubly even component is seen.

**Theorem 6** (Zero-level Abel injectivity)**.** *For every continuous angular force $g$, $Z_g(d) = 0$ for all finite $d$ if and only if its component even in both $x$ and $y$ is zero.*

**Proof.** Write the doubly even component as $\rho^4 a(\theta)$. For $d > 0$ put $b_0 = 4\sqrt{2}$, $D = b_0 d/2$, $k = \tanh D$, $z = \tanh(b_0\tau)$ and $\xi = kz$. Set

$$\begin{aligned} A(\xi) &= \tfrac{1}{2}[(1-\xi)^{-1/2} + (1+\xi)^{-1/2}], \\ \vartheta(\xi) &= \arctan\bigl(\sinh(\tfrac{1}{2}\operatorname{arctanh}\xi)\bigr), \quad c_D = (\cosh D)^{-1/2}, \\ B(\xi) &= A(\xi)^5 a(\vartheta(\xi)). \end{aligned}$$

Substitution in the pulse formulas gives physical radius $c_D(1-z^2)^{1/4}A(\xi)$ and angle $\vartheta(\xi)$. Including the physical clock in (84) yields

$$Z_g(d) = \frac{2c_D^5}{b_0 A_0 k^{3/2}} \int_0^k (k^2 - \xi^2)^{1/4} B(\xi) \,\mathrm{d}\xi.$$

With $K = k^2$, $t = \xi^2$ and $f(t) = t^{-1/2}B(\sqrt{t})$, the integral is $\frac{1}{2}\int_0^K (K-t)^{1/4} f(t) \,\mathrm{d}t$. The function $f$ is locally integrable, including at zero. Convolution in $K$ with $K^{-1/4}$, followed by the beta integral and Fubini, gives a positive constant times $\int_0^K (K-t)f(t) \,\mathrm{d}t$. If the original integral vanishes, two distributional derivatives give $f = 0$. The map $\vartheta : (0,1) \to (0, \pi/2)$ is onto, so continuity and reflection give $a = 0$ everywhere. The converse follows directly from the two reflection cancellations. ∎

**Theorem 7** (Complete invariant-average criterion)**.** *A continuous degree-four homogeneous force $g$ belongs to $\mathcal{K}^0$ if and only if its doubly even component is zero and, for both $\sigma = \pm 1$,*

1. *$\widehat{F}_c^{\sigma}(0,0) = 0$ for every $0 < c < c_*$;*
2. *$\widehat{F}_c^{\sigma}(km, -kn) = 0$ for every rational torus $T_u(c)/T_v(c) = m/n$ in lowest terms and every integer $k \geq 1$.*

**Proof.** Necessity follows from the actual torus and periodic measures and Theorem 6. Conversely, disintegrate an invariant probability on the compact energy shell over the continuous integral $C$. The conditional probabilities are invariant almost everywhere: apply invariance at rational times to a countable dense collection of continuous functions, then use continuity to obtain all times and functions on one full-measure set of levels. A regular nonendpoint level is the physical torus described above. Its irrational flow has the unique clock-weighted Haar measure; at rational ratio all invariant probabilities are mixtures of the periodic phase measures. Equations (80) and (81) annihilate both cases. The endpoint brakes are limits of torus means. On $C = 0$, the scalar zero-energy quadrature shows that every nonconstant coordinate is a translated signed pulse or zero. The resulting physical orbits are precisely the concatenations in (84) and the exceptional vertical orbit. Their means vanish by Theorem 6, including the vertical limit. These cases exhaust the conditional invariant measures and hence the original probability. ∎

## 5.4 Parity and the polynomial intersection

Denote parity in $(x, y)$ by EE, EO, OE and OO. Both physical reflections preserve $Q_0$ and carry invariant measures to invariant measures, so each parity projection preserves $\mathcal{K}^r$. The EE component is zero. On either $x$-odd sector, the ordinary means, the endpoint brakes and (84) vanish automatically. Since $F(\alpha+\pi, \beta) = -F(\alpha, \beta)$, only odd first Fourier indices remain. Thus its exact remaining condition is

$$\widehat{F}_c^{\sigma}(km, -kn) = 0 \quad \text{whenever } T_u/T_v = m/n, \quad m \text{ odd}, \quad k \text{ odd}. \tag{85}$$

Within one fixed $y$-parity sector the two signs $\sigma$ are redundant up to that sign. Neither countability nor smoothness of these rows proves injectivity on an infinite-dimensional angular space.

The rotation force

$$g_{\mathrm{rot}} = 20xy(x^2 + 2y^2), \qquad R_*(\theta) = 20 \cos\theta \sin\theta (1 + \sin^2\theta) \tag{86}$$

belongs to the OO kernel. Indeed for $L = xW_y - yW_x$, $\mathcal{L}L = -g_{\mathrm{rot}}$, and an invariant probability on the compact shell annihilates this bounded coboundary.

**Proposition 5.1** (The binary-quartic linear kernel)**.** *At $Q_0$, the intersection of $\mathcal{K}^0$ with the space of binary quartic polynomials is exactly* $\operatorname{span}\{g_{\mathrm{rot}}\}$.

**Proof.** Theorem 6 eliminates $x^4, x^2y^2, y^4$. For $x^3y$ one has $F = (u^7v^3 - u^3v^7)/2$. On the certified $9:4$ torus of Appendix O, set $U_j = 2\langle u^j \cos 9\alpha\rangle$ and $V_j = 2\langle v^j \cos 4\beta\rangle$. Its double-cosine coefficient is

$$A_{9,4}[x^3y] = \tfrac{1}{2}(U_7V_3 - U_3V_7) < -9/10^7.$$

The exact rotation identity gives $20A_{9,4}[x^3y] + 40A_{9,4}[xy^3] = 0$. Thus the coefficient of $bx^3y + dxy^3$ vanishes only if $d = 2b$. Conversely this line is the rotation coboundary. The appendix supplies the validated orbit and Fourier-tail bounds for the strict inequality. ∎

This is a linear statement at the single integrable well. It neither replaces the nonlinear coefficient-neighborhood theorem nor identifies any nonpolynomial parity sector. In particular every binary quartic has only EE and OO components, so Proposition 5.1 contains no EO information. For an unnormalized first force with common mean $b$, the centered force is $g - 3bQ_0$, by the virial identity. The criterion therefore applies exactly to the first-order question produced by the volume normalization.

# 6 Realizable jets at the nonradial well

The linear observation problem and the deformation problem have different logical scopes. We now impose the common-mean condition on an actual family, not merely on its derivative at one parameter value. Throughout this section

$$P_0(x,y) = x^4 + 12x^2y^2 + 16y^4, \qquad Q(\theta) = P_0(\cos\theta, \sin\theta).$$

The first letter in a parity designation refers to reflection of $x$, and the second to reflection of $y$.

Let $U_s$ be a positive, degree-four homogeneous family, jointly $C^2$ in $(s,q)$ near $s = 0$, with $U_0 = P_0$. Suppose that every invariant probability on its unit-energy shell gives the same mean to $\partial_s U_s$. Its phase-volume normalization is

$$D_U(s) = \int_0^{2\pi} U_s(e_\theta)^{-1/2}\,\mathrm{d}\theta, \qquad V_s = \left(\frac{D_U(s)}{D_U(0)}\right)^2 U_s. \tag{87}$$

Every invariant mean of $\partial_s V_s$ is zero. The normalization is jointly $C^2$; it need not preserve joint convexity. In particular the actual jets have precisely the regularity

$$g = \partial_s V_s|_0 \in C^1, \qquad h = \partial_s^2 V_s|_0 \in C^0. \tag{88}$$

Both are homogeneous of degree four. No angular second derivative of $g$ is assumed below. The normalization and the mechanical second-variation identities are proved in Appendix F.

**Proposition 5.1** (The binary-quartic linear kernel)**.** *At $Q_0$, the intersection of $\mathcal{K}^0$ with the space of binary quartic polynomials is exactly* $\operatorname{span}\{g_{\mathrm{rot}}\}$.

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Both are homogeneous of degree four. No angular second derivative of $g$ is assumed below. The normalization and the mechanical second-variation identities are proved in Appendix F.

**Theorem 8** (Necessary common second force)**.** *For the family above, $g$ belongs to the complete original invariant-average kernel of Theorem 7, and $g_{\mathrm{EE}} = 0$. There is one continuous function $h$, the actual second derivative in* (88)*, with the following properties.*

1. *On every member $q_d$ of the physical zero-separation family,*

$$T_0 Z_h(d) + \mathcal{Q}_d(g, g) = 0. \tag{89}$$

   *Here $Z_h$ is its physical mean and $\mathcal{Q}_d$ is the mechanical Jacobi quadratic datum defined in Appendix F. It includes the full period and has the correct double-to-primitive limit at the vertical ends.*

2. *The normalized volume identity is*

$$\int_0^{2\pi} h(e_\theta) Q(\theta)^{-3/2}\,\mathrm{d}\theta = \frac{3}{2}\int_0^{2\pi} g(e_\theta)^2 Q(\theta)^{-5/2}\,\mathrm{d}\theta. \tag{90}$$

3. *At every fixed regular rational torus with nonzero isoenergetic twist, the same $h$ satisfies*

$$\int_0^{T_c} h(q_{c,\varphi}(t))\,\mathrm{d}t + \mathcal{Q}_{c,\varphi}(g, g) = 0 \quad \textit{for every relative phase } \varphi. \tag{91}$$

   *The datum in this formula uses the normal free-period Hessian, including its period variable.*

*The zero-separation equation determines $h_{\mathrm{EE}}$ uniquely through the continuous EE Abel transform. A distributional inverse of an arbitrarily prescribed quadratic datum need not be continuous; continuity here comes from the actual family.*

**Proof.** At $s = 0$ the normalized common-mean condition is exactly the complete first-order observation criterion. Its EE part vanishes by Theorem 6. Proposition F.1 proves the regular-torus statement at joint $C^2$ regularity. The zero-level cylinder is noncompact in its relative-shift parameter, so its identity does not follow from that compact-torus argument. Proposition F.2 proves it separately, using the primitive vertical orbit to anchor both ends and the actual EO soft-mode cancellation. Differentiating the constant normalized volume twice gives (90). Finally the injectivity of the EE transform gives the stated uniqueness. ∎

Put

$$R(\theta) = Q'(\theta) = 20\cos\theta\sin\theta(1 + \sin^2\theta), \qquad g_{\mathrm{rot}} = 20xy(x^2 + 2y^2), \tag{92}$$

and

$$h_{\mathrm{rot}} = 20x^4 + 60x^2y^2 - 40y^4. \tag{93}$$

The positive zero-level mixture eliminates $h$ from (89)–(90); it gives a quadratic necessary condition $\mathcal{B}(g, g) = 0$. Its sign is proved below only on the domains stated in the next theorem.

**Theorem 9** (Parity restrictions on actual two-jets)**.** *Let $g, h$ be the normalized jets of an actual family as above.*

1. *If $g$ is pure OO, then $g = cg_{\mathrm{rot}}$ for some $c \in \mathbb{R}$, and*

$$h_{\mathrm{EE}} = c^2 h_{\mathrm{rot}}.$$

2. *If $g$ is pure EO and $g(0,1) = 0$, then $g = 0$.*

3. *More generally, if $g$ is odd in $y$ and $g(0,1) = 0$, then $g = cg_{\mathrm{rot}}$ and $h_{\mathrm{EE}} = c^2 h_{\mathrm{rot}}$.*

*These are restrictions on jets of actual families at $P_0$. They do not identify the unrestricted OO or EO linear observation kernels.*

**Proof.** The exact positive mixture and volume identity give $\mathcal{B}(g,g) = 0$ by Proposition F.3. Theorem 19 proves, for every original $C^1$ OO profile, that $\mathcal{B} \le 0$ and that its zero set is precisely $\mathbb{R}R$. This proves the first-order conclusion in part (1) without assuming that arbitrary linear nullvectors have second jets.

For an EO ordinary-null profile with zero vertical value, the original Peano theorem and Lemma 53 place the profile in the finite-window domain. Proposition F.4 then gives $\mathcal{B}(g,g) = -E(g) < 0$ unless $g = 0$. This proves the second assertion.

For the third, write $g = g_{\mathrm{EO}} + g_{\mathrm{OO}}$. Reflection preserves the complete linear kernel. The OO ordinary means vanish by symmetry, so $g_{\mathrm{EO}}$ is ordinary-null; its vertical value is zero because every OO force vanishes there. The actual Jacobi operator and its kernel projection commute with the half-period reflection. The EO and OO forcings lie in opposite eigenspaces, and therefore

$$\mathcal{B}(g,g) = \mathcal{B}_{\mathrm{EO}}(g_{\mathrm{EO}}, g_{\mathrm{EO}}) + \mathcal{B}_{\mathrm{OO}}(g_{\mathrm{OO}}, g_{\mathrm{OO}}).$$

The angular cross term also vanishes by reflection. Both terms are nonpositive; the first is strictly negative unless its input is zero. Thus $g_{\mathrm{EO}} = 0$ and $g_{\mathrm{OO}} = cg_{\mathrm{rot}}$.

For rotation, the exact response is $-Jq_d$, where $J(x,y) = (-y,x)$, and $\mathcal{Q}_d(g_{\mathrm{rot}}, g_{\mathrm{rot}}) = -\int h_{\mathrm{rot}}(q_d)\,\mathrm{d}t$. Subtracting $c^2 h_{\mathrm{rot}}$ from the actual second force in (89) and applying EE Abel injectivity proves the stated second-order conclusions. No conclusion about the other parity components of $h$ is required. ∎

**Corollary 6.1** (The tangent image of one family)**.** *For one normalized jointly $C^2$ multi-parameter common-mean family, let $\mathcal{T}$ be its first-derivative image at $P_0$. If $\mathcal{T}$ is entirely EO, then* $\dim \mathcal{T} \le 1$. *If every element of $\mathcal{T}$ is odd in $y$, then*

$$\ker\bigl(g \mapsto g(0,1)\,\big|_{\mathcal{T}}\bigr) \subseteq \mathbb{R}g_{\mathrm{rot}}, \qquad \dim \mathcal{T} \le 2.$$

**Proof.** Every parameter line is an actual one-parameter family. In the pure EO case vertical evaluation is injective by Theorem 9; in the odd-$y$ case its kernel has dimension at most one. Rank–nullity proves the assertions. The argument concerns the image of the derivative of the same family; it does not subtract directions arising from unrelated families or assert realizability of parity projections. ∎

**Proposition 6.1** (Attainment in the joint-convex class)**.** *Every $cg_{\mathrm{rot}}$ is attained as a normalized first direction of a local positive homogeneous common-mean family which is jointly convex in parameter and position.*

**Proof.** Set $P_s(q) = P_0(e^{csJ}q)$ and $U_s(q) = e^{Ms^2}P_s(q)$. Strict positivity of $P_0$ and its spatial Hessian on the unit circle gives $m, \lambda > 0$, uniformly under rotation, such that

$$P_s \geq m|q|^4, \qquad (P_s)_{qq} \geq \lambda|q|^2 \,\mathrm{Id}\,.$$

Choose finite constants $A, B, D, E$ for $|\nabla P_s| \leq A|q|^3$, $|\nabla\partial_\theta P_s| \leq B|q|^3$, $|\partial_\theta P_s| \leq D|q|^4$ and $|\partial_\theta^2 P_s| \leq E|q|^4$. First choose $M$ with

$$2Mm > c^2E + (|c|B + 1)^2/\lambda + 2,$$

then choose $\delta > 0$ with $2M\delta A \leq 1$ and $4M\delta|c|D \leq 1$. For $|s| < \delta$, the Schur complement of the spatial Hessian of $U$ is bounded below, at $q \neq 0$, by

$$e^{Ms^2}|q|^4\left(2Mm - 4M\delta|c|D - c^2E - \frac{(2M\delta A + |c|B)^2}{\lambda}\right) > 0.$$

At $q = 0$ all Hessian blocks vanish continuously, with orders $2, 3, 4$. Thus the whole Hessian is positive semidefinite on the convex strip.

For $\ell = xp_y - yp_x$, Hamilton's equation gives $X_{H_s}\ell = -\partial_\theta U_s$. Its integral under every invariant probability is zero, while quartic virial gives $\langle U_s\rangle = 1/3$. Consequently

$$\langle\partial_s U_s\rangle = \langle 2MsU_s + c\partial_\theta U_s\rangle = 2Ms/3.$$

This is the original common-mean condition. Moreover $D_U(s) = e^{-Ms^2/2}D_U(0)$, so (87) gives exactly $V_s = P_s$. Its first and second jets are $cg_{\mathrm{rot}}$ and $c^2h_{\mathrm{rot}}$. ∎

A nonzero pure EO realizable first direction, if one exists, must therefore have nonzero vertical value. It remains subject to every original first-order observation and to the same continuous second force in Theorem 8. Neither the preceding theorems nor the finite-dimensional result for the zero-vertical EO kernel in the next section decides existence of that affine fiber. They also give no classification of OE-containing mixtures or of families at nonzero parameter merely from an initially rotational derivative.

# 7 Finite-dimensional EO structure and exact reconstruction

The complete first-order problem has a further compactness consequence in the EO sector. It is useful to state precisely both its domain and its limits. Define

$$\mathcal{K}_{\mathrm{EO}} = \mathcal{K}^1 \cap \{g \text{ even in } x,\ \text{odd in } y\}, \qquad \tau(a) = a(\pi/2), \qquad \mathcal{N}_0 = \ker(\tau|_{\mathcal{K}_{\mathrm{EO}}}).$$

We identify a force with its angular profile. The energy $E$ is the positive Dirichlet-plus-rank form constructed in Appendix F; its exact normalization is recorded in Appendix J. Every element of $\mathcal{N}_0$ belongs to its original finite-energy domain, by Theorem 18 and the physical window argument of Lemma 53. Let $\mathcal{N}_E$ be its energy closure under the injective angular realization $J$. It is the closure of original complete-null inputs, not a larger kernel defined by formal completed rows.

**Theorem 10** (EO finite-dimensionality)**.** *The space $\mathcal{N}_E$ is finite-dimensional and $\mathcal{N}_E = \mathcal{N}_0$ algebraically. Its elements have their original $C^1$ representatives and*

$$\|a\|_{C^1} \le C_{\mathcal{N}} \sqrt{E(a)} \qquad (a \in \mathcal{N}_0).$$

*The entire original $C^1$ EO kernel is finite-dimensional. More precisely, if $d = \dim \mathcal{N}_0$, then either*

$$\mathcal{K}_{\mathrm{EO}} = \mathcal{N}_0,$$

*or there exists an original full-null $u_0$ with $\tau(u_0) = 1$ and*

$$\mathcal{K}_{\mathrm{EO}} = \mathcal{N}_0 \oplus \operatorname{span}\{u_0\}.$$

*The dimension $d$ and the existence of the second alternative are not determined here.*

**Proof.** The original entry theorem is independent of any energy compactness. The closed energy realization and the complete resonant-row analysis then give two compact maps on $\mathcal{N}_E$: ordinary angular $L^2$ inclusion (Corollary J.2) and the horizontal inclusion into $L^2(\theta^{-1/2}\,\mathrm{d}\theta)$ (Theorem 24). The latter uses all integer rows, including their exact endpoint and sampling terms. It cannot be replaced by a fixed finite-frequency family. The exact ordinary gluing inequality (497) controls the energy by the squares of these two compact observation norms. An energy-bounded weakly zero sequence therefore converges strongly in energy. An infinite-dimensional Hilbert space would have an orthonormal weakly zero sequence of norm one, a contradiction. Thus $\mathcal{N}_E$ is finite-dimensional. The dense linear subspace $\mathcal{N}_0$ of a finite-dimensional space is closed, so it equals $\mathcal{N}_E$. Choose an energy-orthonormal basis $e_1, \dots, e_d$ from $\mathcal{N}_0$. Then $\|\sum c_j e_j\|_{C^1} \le (\sum \|e_j\|_{C^1}^2)^{1/2} (\sum |c_j|^2)^{1/2}$, proving the qualitative norm bound. Finally the evaluation map $\tau$ has scalar range. Its kernel is exactly $\mathcal{N}_0$, by the original ordinary-null entry theorem. If its range is

nonzero, divide any nonzero-trace element by its trace and subtract its multiple from every other kernel member. This gives the displayed direct sum; otherwise the first alternative holds. ∎

The $C^1$ estimate is a consequence of finite dimension and supplies no computed constant or explicit basis. The static $C^2$ EO kernel is a subspace of this finite-dimensional $C^1$ kernel; equality of those regularity classes is not asserted. On any fixed translate $v_0 + \mathcal{N}_E$ in the ambient energy space, weak convergence is strong convergence, so no energy defect remains there. If a unit-trace full-null $u_0$ exists, the unit-trace kernel fiber is $u_0 + \mathcal{N}_0$. The convenient reference $a_*(\theta) = \sin^3\theta$ must not be substituted for $u_0$: it is not asserted to be full-null.

**Theorem 11** (Vertical value trace)**.** *Every original $C^1$ EO profile with zero vertical value whose ordinary means vanish on a fixed interval $0 < c < c_0$ has*

$$a(\pi/2 - \delta) = \lambda(a)\delta^4 + o(\delta^4).$$

*This fourth-order Peano value coefficient is the bounded $C^1$ linear functional given in* (523) *and* (525)*. On $\mathcal{N}_0$ it is also energy-continuous.*

**Proof.** The ordinary equation first gives subquartic growth, then a convergent critical forcing primitive. The exact inverse at the simple Mellin zero converts that primitive into the value coefficient; this is Theorem 25. It uses neither complete-null compactness nor finite dimension. Only the final energy-continuity assertion composes its $C^1$ bound with Theorem 10. ∎

Exact finite-dimensional approximation has an additional requirement. An inequality controlling null inputs by compact observations does not permit the same inequality to be applied to an arbitrary projection remainder, since such a remainder need not be null. The ordinary mechanical equation supplies the needed ambient reconstruction instead.

**Proposition 7.1** (Exact reconstruction and ambient approximation)**.** *There is an explicitly defined bounded operator $\mathcal{R} : \mathcal{Z} \to \mathcal{H}_P$, given by* (527)*, such that $\mathcal{R}\mathcal{K}a = a$ on $\mathcal{N}_0$, where $\mathcal{K}a = (Ja, \chi_H Ja)$. For any finite-rank orthogonal observation projection satisfying* (529)*, the known ambient space $\mathcal{R}\operatorname{Ran}\Pi_m$ approximates the nullspace with the energy error* (530)*. Truncating the mechanical Neumann series gives* (531)*, with an explicit geometric factor $q < 2^{-12}$ at the collars in* (500)*.*

**Proof.** The exact supported ordinary equation, its physical lift and the projection estimate are proved in Appendix N. The stated contraction is for the full signed vertical mechanical remainder; the scalar normal inverse has the separate positive-convolution approximation of Proposition K.1. ∎

These results give finite-dimensionality, endpoint values and a constructive reduction of the EO identification problem. They do not determine its dimension, show that the

quartic trace vanishes, or prove that the trace is injective. The exact reconstruction is one-way on known null inputs; its values on arbitrary data need not be ordinary-null. A numerical observation-tail bound and a separating Gram inequality are still required for the finite comparison of Proposition N.1. The OE and OO linear kernels, general mixed-parity realizability and a nonpolynomial finite-neighborhood classification remain separate questions.

# 8 Controlled correctors and actual point selection

We first prove a dynamical implication in arbitrary finite normal and selected dimensions. Let $U \subset \mathbb{R}^m$ be open and convex and $P \in C^2(U \times \mathbb{R}^n)$ be jointly convex, $p$-homogeneous in $Z$, $p > 2$, and coercive near a compact selected tail $K \Subset U$. Set

$$a = 2/p, \qquad \nu = 1/2 + 1/p, \qquad \sigma = \log t, \qquad \varepsilon = e^{-a\sigma}.$$

The physical equations have damping $(1 + a)/t$. Substitution of $q = \varepsilon Z$, $W = \varepsilon Z_\sigma$ and $k = s_\sigma$ gives exactly

$$Z' = \varepsilon^{-1} W, \qquad W' = -\varepsilon^{-1} P_Z, \qquad s' = k, \qquad k' = -ak - P_s. \tag{94}$$

Here and below primes in this section denote $\sigma$ derivatives. In particular,

$$H = \tfrac{1}{2}|W|^2 + P(s, Z), \quad E = H + \tfrac{1}{2}|k|^2, \qquad H' = k \cdot P_s, \quad E' = -a|k|^2. \tag{95}$$

The formulas are exact, not an averaged model.

**Lemma 11** (Compact-tail and relative-energy estimates)**.** *On such a tail, all rescaled states are bounded, $k \in L^2$, $k \to 0$, and $H \to E_\infty \geq 0$. If the normal state is not identically zero, then $H > 0$ at every finite time and, eventually, $|k| \leq CH$. In the case $E_\infty = 0$,*

$$E = H + O(H^2), \qquad |H'| \leq CH^2, \qquad H(\sigma) \geq \frac{1}{H(S)^{-1} + C(\sigma - S)}.$$

**Proof.** The nonnegative decreasing energy bounds $k, W, Z$ by coercivity. Its dissipation makes $k \in L^2$. The selected force and hence $k'$ are bounded on the compact state region, so uniform continuity and $L^2$ integrability give $k \to 0$. Equation (95) then gives $H \to E_\infty$. Local uniqueness for the $C^1$ vector field makes the zero normal state invariant and prevents an excited solution from reaching it at a finite time.

Homogeneity and compactness give $|P_s| \leq C|Z|^p \leq CH$. Thus $|(logH)'| \leq C|k| \to 0$. For each $0 < d < a$, at sufficiently late $S$ and $S \leq u \leq \sigma$,

$$H(u) \leq H(\sigma) e^{d(\sigma - u)}, \qquad H(\sigma) \geq H(S) e^{-d(\sigma - S)}.$$

**Proof.** Normalize as in Section A, so the radial member is $|Z|^6$ and the angular volume is fixed. On a small parameter neighborhood write $V(s,\rho e_\theta)=\rho^6(1+f(s,\theta))$. All action and brake hypotheses used above follow from the original common-mean assumption by that section. Theorem 16 and Lemma 49 give $\sum_{n\ge1,\ n\ \mathrm{odd}} n^3 w_n \le C_2 a^2$, where $w_n = 4|\widehat{f}(n)|^2$ for positive odd $n$. Lemma 22 gives

$$ca^2 \le \sum_{n\ge5,\ n\ \mathrm{odd}} n^2 w_n, \qquad w_n \le C_1 a^4 \quad (n\ge5,\ n\ \mathrm{odd}).$$

For every fixed integer $M\ge5$,

$$ca^2 \le C_1' a^4 M^3 + C_2 a^2/M.$$

Choose $M$ once with $C_2/M \le c/2$. If $a>0$, this forces $a^2 \ge c/(2C_1'M^3) > 0$, whereas $a \le C\delta$. For sufficiently small $\delta$ this is impossible. If $a=0$, $\|f\|_{H^1}\le Ca$ gives $f=0$ directly. This proves local finite isolation, including arbitrarily high angular frequencies.

The normalized radial locus is nonempty and relatively closed by angular $C^2$ continuity. Apply the same local argument at any radial member to see that it is relatively open. Connectedness makes it all of $S$. Undoing the fixed volume normalization gives the stated positive amplitude; $\kappa(s)=P(s,e_0)$ is $C^2$. Only the frozen action, mechanical and transport arguments were used. In particular no convexity of the joint parameter–position Hessian or of the normalized family was used. Joint convexity of the original potential remains a separate hypothesis of the actual trajectory selection theorem. ∎

# E Ordinary means and the original vertical Peano class

This appendix supplies the endpoint entry used in the physical second variation and in the energy completion. It precedes both arguments logically and uses only the original angular $C^1$ norm. Write $I=(0,\pi/2)$ and $\delta=\pi/2-\theta$.

**Theorem 18** (Original ordinary-null entry)**.** *Let $g(\rho e_\theta)=\rho^4 a(\theta)$ be EO, with $a\in C^1(\mathbb{S}^1)$, and suppose its ordinary torus means vanish for all sufficiently small positive $c$. Put $A=a(\pi/2)$. Then*

$$a(\pi/2-\delta) - A\sin^3(\pi/2-\delta) = o(\delta^2). \tag{312}$$

*If the interval of ordinary nullity is fixed, the left side has modulus at most $C\|a\|_{C^1}\delta^2$ on a fixed collar. This is a value statement; no derivative quotient or second angular derivative is asserted.*

## E.1 The complete scalar-measure reduction

Subtract $Ag_{\mathrm{ref}}$, where $g_{\mathrm{ref}}(x,y)=y^3\sqrt{x^2+y^2}$, and call the remaining force $g_0$. Its clock-weighted parabolic observable $F_0(u,v)=(u^2+v^2)g_0(uv,(u^2-v^2)/2)$ is $C^1$, homo-

geneous of degree ten, antisymmetric in $u, v$, and zero on the axes. Choose a fixed small ratio cutoff $\chi$, equal to one near zero, supported in $[0, \kappa)$, and set

$$\beta(r) = \chi(r)F_0(r, 1) = -\frac{\chi(r)(1 + r^2)^5}{16}[a - A\sin^3](\pi/2 - 2\arctan r).$$

Then $\beta \in C_c^1([0, \infty))$ and $\beta(0) = \beta'(0) = 0$. For $U(q) = q^{10} - q^2$ define the positive-half scalar measures

$$\nu_c^+(\,\mathrm{d}q) = \frac{\mathbf{1}_{[0,q_+(c)]}\,\mathrm{d}q}{\sqrt{2(c - U(q))}}, \qquad \nu_c^-(\,\mathrm{d}q) = \frac{\mathbf{1}_{[q_l(-c),q_+(-c)]}\,\mathrm{d}q}{\sqrt{2(-c - U(q))}}, \quad \nu^0(\,\mathrm{d}q) = \frac{\mathbf{1}_{(0,1)}\,\mathrm{d}q}{\sqrt{2}q\sqrt{1 - q^8}},$$

and $\Delta\nu_c = \nu_c^+ - \nu_c^-$. All finite-$c$ supports lie in a fixed compact interval. Although $\nu^0$ has infinite mass, every integral against it below has a positive-order zero at the origin. Put

$$k(z) = (z^2 + 1)^{-1/2} - \mathbf{1}_{z>1}(z^2 - 1)^{-1/2}, \qquad D_\epsilon\beta = \frac{1}{\sqrt{2}}\int_0^\infty \beta(\epsilon z)k(z)\,\mathrm{d}z,$$

$$B\beta(c) = \int_0^1 v^{10} D_{\sqrt{c}/v}\beta\,\frac{\mathrm{d}v}{\sqrt{2}v\sqrt{1 - v^8}}. \tag{313}$$

These are convergent signed integrals. The raw physical numerator is $T_g(c) = \int g\,\delta(H - 1)\delta(C - c)\,\mathrm{d}q\,\mathrm{d}W$. The exact cone decomposition is obtained from

$$G(u, v) = v^{10}\beta(u/v), \qquad R_0(u, v) = F_0(u, v) - G(u, v) + G(v, u).$$

The function $R_0$ is supported where $u, v$ are comparable. The physical covering, clock and parity give $T_{g_0}/8 = \iint F_0\,\mathrm{d}\Delta\nu_c(u)\,\mathrm{d}\nu_c^-(v)$. Define

$$\begin{aligned} P_c(v) &= v^{10} D_{\sqrt{c}/v}\beta, \quad A_c(v) = \Delta\nu_c[G(\cdot, v)] - P_c(v), \\ \Theta_c(u) &= \nu_c^-[G(\cdot, u)], \quad \Psi_c(u) = \nu_c^-[R_0(u, \cdot)]. \end{aligned}$$

Thus, with no discarded angular or scalar piece,

$$\begin{aligned} T_g(c) &= 8B\beta(c) + E_g(c), \\ E_g(c) &= AT_{\mathrm{ref}}(c) + 8\{\nu_c^-(A_c) + (\nu_c^- - \nu^0)(P_c) - \Delta\nu_c(\Theta_c) + \Delta\nu_c(\Psi_c)\}. \end{aligned} \tag{314}$$

**Lemma 50** (Endpoint remainder)**.** *The same remainder in* (314) *extends to* $C^1([0, c_1])$*, with* $E_g(0) = 0$ *and* $\|E_g'\|_{C^0([0,c_1])} \le C\|a\|_{C^1}$*, for a fixed sufficiently small* $c_1 > 0$*.*

**Proof.** We give the scalar estimates and the endpoint closure, since boundedness of $E_g'$ on $(0, c_1]$ alone would not give its limit. For $j > 2$ let $X_j$ be the weighted $C^1$ test space with norm

$$\|\psi\|_{X_j} = \sup_{q>0}(q^{-j}|\psi(q)| + q^{1-j}|\psi'(q)|).$$

On the fixed scalar interval one has

$$|\Delta\nu_c(\psi)| + |(\nu_c^- - \nu^0)(\psi)| \le C_j c\|\psi\|_{X_j}, \tag{315}$$

$$|(\nu_c^+)'(\psi)| + |(\nu_c^-)'(\psi)| \le C_j \|\psi\|_{X_j}, \tag{316}$$

$$\int q^j \,\mathrm{d}|\Delta\nu_c| + \int q^j \,\mathrm{d}|\nu_c^- - \nu^0| \le C_j\sqrt{c}. \tag{317}$$

For $\psi(0) = 0$ with $\|\psi'\|_\infty \le M$, the second estimate is replaced by $CMc^{-1/2}$. For tests of quadratic order it is $CM(1 + |\log c|)$. To prove these assertions, use $w = q\sqrt{1-q^8}$ near the scalar saddle. Its inverse satisfies $q(w) = w + O(w^9)$. The transformed minus integral is $\int_{\sqrt{c}}^\infty f(w)(w^2-c)^{-1/2}\,\mathrm{d}w$ and its derivative is

$$\frac{1}{2}\int_{\sqrt{c}}^\infty \frac{f'(w)/w - f(w)/w^2}{\sqrt{w^2-c}}\,\mathrm{d}w. \tag{318}$$

The harmless factor $1/\sqrt{2}$ is restored in the scalar measure. Split at $2\sqrt{c}$. Below it scaling applies; above it the difference of quadratic kernels is bounded by $Cc/w^3$. Formula (318) is uniformly integrable for $j > 2$; for the weaker test use $\int_{\sqrt{c}}^\infty [w\sqrt{w^2-c}]^{-1}\,\mathrm{d}w = \pi/(2\sqrt{c})$. At the outer root the substitution $q = q_+(e) - z^2$ gives a bounded functional on $C^1$ tests. The $L^1$ difference of two square-root endpoint densities whose roots differ by $O(c)$ is $O(\sqrt{c})$: split within distance $O(c)$ of the roots and use $Cc\,d^{-3/2}$ outside. These facts prove (315)–(317). Moreover the differentiated functionals have limits $L_+, L_-$ on each $X_j$. On $w < \epsilon$, their modulus is $C\epsilon^{j-2}$ uniformly in small $c$, while dominated convergence applies on $w \ge \epsilon$. This also proves convergence on test families convergent in $X_j$.

We first establish the uniform derivative bound. Write $M = \|a\|_{C^1}$. The reflected and comparable-coordinate functions $Z_c = \Theta_c, \Psi_c$ satisfy

$$|Z_c(u)| \le CMu^{10}, \quad |\partial_u Z_c(u)| \le CMu^9, \quad |\partial_c Z_c(u)| \le CMc^{-1/2}u^9.$$

For the reflected cone these follow from $|\beta(r)| + |10\beta - r\beta'| \le CMr$ and $u^{-1}\int_{\sqrt{c}}^{C\kappa u} w(w^2 - c)^{-1/2}\,\mathrm{d}w \le C$. For the comparable cone the scalar mass on a fixed-ratio interval is bounded; near the saddle this is the difference of two arcosh values. The product rule and (316)–(317) show that $\partial_c\Delta\nu_c(Z_c) = O(M)$ without a mixed derivative of $Z_c$. The same reasoning applies to the outer-measure correction: $P_c$ obeys $|P_c| \le CMv^{10}$, $|P_c'| \le CMv^9$ and $|\partial_c P_c| \le CMv^9/\sqrt{c}$, so $\partial_c[(\nu_c^- - \nu^0)(P_c)] = O(M)$.

For the geometric correction one must not estimate a difference of unknown values of $\beta'$. In the original scalar coordinate put $W(u) = u\sqrt{1-u^8}$ and compare

$$j_\pm^e(u) = (W(u)^2 \pm c)_+^{-1/2}, \qquad j_\pm^0(u) = (u^2 \pm c)_+^{-1/2},$$

where the plus case has no positive-part restriction. For $0 < L \le L_0$,

$$\|j_\pm^e - j_\pm^0\|_{L^1(0,L)} \le CL^4, \qquad \int_0^L |j_-^e - j_-^0|\frac{\mathrm{d}u}{u} \le CL^3. \tag{319}$$

Above $2\sqrt{c}$ the difference is $O(u^7)$. Below it rescale $u = \sqrt{c}z$: the two simple roots differ by $O(c^4)$ and their square-root density difference has $L^1$ norm $O(c^2)$, also for a truncated

upper endpoint. Division by $u$ costs $c^{-1/2}$. This proves (319); smooth factors $1 + O(u^8)$ preserve it. It follows that $|A_c(v)| \le CMv^{14}$ and $|\partial_v A_c(v)| \le CMv^{13}$. For its $c$ derivative, transform (318) back to $u = q(w)$ before comparing. The terms with $\beta'$ become

$$v^9 \int \beta'(u/v) \left\{ \frac{q'(W(u))j_c(W(u))}{W(u)} - \frac{j_c(u)}{u} \right\} \mathrm{d}u, \qquad j_c(w) = \frac{\mathbf{1}_{w>\sqrt{c}}}{\sqrt{w^2 - c}}.$$

The two unknown derivatives now have the same argument, and (319) bounds this by $CMv^{12}$. The terms with $\beta$ have the same weighted comparison. The additional geometric factor satisfies

$$q''(W(u))/(q'(W(u))W(u)) = O(u^6),$$

and gives the same bound $CMv^{12}$. Direct differentiation of the plus kernel costs at most $CMv^{16}$. Thus $|\partial_c A_c(v)| \le CMv^{12}$ and $\partial_c \nu_c^-(A_c) = O(M)$. The undifferentiated estimates give $E_g = O(c)M$ by (315); in the direct cone the geometric error is $O(cMv^{16})$, obtained by the same split at $2\sqrt{c}$.

The reference cannot be omitted. Its observable is

$$F_{\mathrm{ref}}(u, v) = \frac{(u^{10} - v^{10}) - (u^8 v^2 - u^2 v^8) - 2(u^6 v^4 - u^4 v^6)}{16}.$$

Writing $I_j^{\pm} = \nu_c^{\pm}(q^j)$ and $\Delta I_j = I_j^+ - I_j^-$ gives

$$T_{\mathrm{ref}} = \tfrac{1}{2}[\Delta I_{10} I_0^- - \Delta I_0 I_{10}^- - \Delta I_8 I_2^- + \Delta I_2 I_8^- - 2\Delta I_6 I_4^- + 2\Delta I_4 I_6^-]. \tag{320}$$

With $L = \log(1/c)$, $a_0 = 1/(2\sqrt{2})$ and $d_0 = 1/(4\sqrt{2})$,

$$\begin{array}{ll} I_0^- = a_0 L + b_0 + o(1), & (I_0^-)' = -a_0/c + O(1), \\ \Delta I_2 = -a_0 c L + b_2 c + o(c), & (\Delta I_2)' = -a_0(L-1) + b_2 + o(1), \\ I_8^- = d_0 + O(c), & \Delta I_{10} = d_0 c + O(c^2), \quad (\Delta I_{10})' = d_0 + O(c). \end{array}$$

Here the constants and derivative limits follow directly by subtracting the constant and quadratic saddle numerators. Their primitives are $\operatorname{arcosh}(\epsilon/\sqrt{c})$ and

$$\tfrac{1}{2}[\epsilon\sqrt{\epsilon^2 + c} - c \operatorname{arsinh}(\epsilon/\sqrt{c})], \quad \tfrac{1}{2}[\epsilon\sqrt{\epsilon^2 - c} + c \operatorname{arcosh}(\epsilon/\sqrt{c})].$$

The remaining transformed densities have order at least eight and fall under the weighted derivative estimates. The degree-ten remainder has a bounded second scalar derivative. Also $(\Delta I_0)'$ has a finite limit, all degree $4, 6, 8, 10$ moments have continuous first derivatives, and $(I_2^-)' = O(1+L)$. In the differentiated sum $\Delta I_{10} I_0^- + \Delta I_2 I_8^-$ the terms $d_0 a_0 L$ and $-d_0 a_0 L$ cancel, leaving the finite limit $d_0(b_0 + b_2)$. Every other term of (320) has a finite derivative limit. This proves the required reference bound and its endpoint continuity, rather than differentiating an unspecified little-oh term.

It remains to establish the endpoint limit for arbitrary input. First take a smooth EO input. Then $\beta = O(r^2)$ and $\beta' = O(r)$. The reflected and comparable $Z_c$ converge in $X_8$ to $Z_0$; on a fixed interval away from zero this follows by scalar endpoint regularization, while

their common $u^{10}, u^9$ bounds make the $u < \epsilon$ part negligible. Their $c$ derivatives have the sharper bound $Cu^8(1+L)$, so the second product-rule term tends to zero by (317), and the first tends to $(L_+ - L_-)(Z_0)$. Likewise $P_c \to 0$ in $X_8$ and $|\partial_c P_c(v)| \le Cv^8(1+L)$, so its outer correction has derivative tending to zero. For $A_c$ use

$$h_v(w) = v^{10}\{\beta(q(w)/v)q'(w) - \beta(w/v)\}.$$

Smoothness gives $|h_v| \le Cv^8w^{10}$, $|\partial_w h_v| \le Cv^8w^9$. Therefore

$$\partial_c A_c(v) \longrightarrow -\frac{1}{\sqrt{2}} \int h_v(w) w^{-3}\,\mathrm{d}w,$$

with uniform bound $Cv^{16}$; also $A_c \to 0$ in $X_8$. Dominated convergence for the outer scalar measure proves the last derivative limit. Thus $E_g'$ extends continuously for smooth inputs. Finally approximate any original $C^1$ EO profile by smooth EO profiles in $C^1$. Linearity of every term in (314) and the uniform bound already proved make the extended derivatives uniformly Cauchy on $[0, c_1]$. Their uniform limit is continuous and agrees with $E_g'$ on $(0, c_1]$. Since all values at zero are zero, integration gives the claimed $C^1$ extension. No bound on a second angular derivative is used in this closure. ∎

## E.2 A critical inverse acting on the original growth class

For $f(t) = \beta(\sqrt{t})$ define

$$\mathcal{A}_{\mathrm{ax}} f(c) = 2\int_0^\infty f(t) d(c/t) \frac{\mathrm{d}t}{t},$$

where

$$d(\rho) = \frac{1}{32}\left[\mathbf{1}_{\rho<1}\int_\rho^1 \frac{z^{9/2}\,\mathrm{d}z}{\sqrt{z-\rho}\sqrt{1-z^4}} - \int_0^1 \frac{z^{9/2}\,\mathrm{d}z}{\sqrt{z+\rho}\sqrt{1-z^4}}\right].$$

Reversing the convergent signed integrals in (313) gives $B\beta = -2\mathcal{A}_{\mathrm{ax}} f$. Its Mellin transform is

$$D(\alpha) = \frac{\sqrt{\pi}}{128} B\left(\frac{5-\alpha}{4}, \frac{1}{2}\right) \frac{\Gamma(\alpha+1/2)}{\Gamma(\alpha+1)} \tan\frac{\pi\alpha}{2}. \tag{321}$$

For example, evaluate the inner signed power integral by the beta substitutions $z = \rho + t$ and $z = \rho - t$, then the outer integral by $z^4 = t$; initially $-1/2 < \Re\alpha < 1$ ensures convergence. The result elsewhere is its meromorphic continuation.

Set $K(x) = 2e^{-x} d(e^x)$. Direct endpoint regularization gives

$$\begin{aligned} &K(x) = 1/32 + O(e^{2x}), \quad K'(x) = O(e^{2x}) && (x \to -\infty),\\ &K(x), K'(x) = O(e^{-3x/2}) && (x \to +\infty),\\ &K(0+) - K(0-) = -\pi/32. \end{aligned}$$

For the first line transfer up to four derivatives in the lower Abel integral to $z^{9/2}$ times a smooth cutoff; its $j$th derivative is $O(z^{9/2-j})$. Integration by parts shows cancellation

of the even jets and gives $d(\rho) = \rho/64 + O(\rho^3)$ and $d'(\rho) = 1/64 + O(\rho^2)$. At $\rho = 1$, the substitution $z = \rho + (1-\rho)s$ gives the beta mass $\pi/64$ for the positive branch. The negative branch is smooth. At infinity differentiate that branch directly. Thus $\mu = K' = -(\pi/32)\delta_0 + k(x)\,\mathrm{d}x$ is a finite measure with exponential moments of every order below one. Its bilateral transform is

$$M(z) = 2zD(1+z), \qquad |\Re z| < 1, \quad M(0) = -1/32.$$

It has no zero in this strip: the gamma factors have no zero or pole, the tangent zeros lie at the boundary, and the pole at $z = 0$ cancels with the displayed nonzero value.

For completeness, a genuine inverse measure follows from this formula. On each smaller closed strip the gamma-ratio expansion gives $M(z) = -\pi/32 + O(|\Im z|^{-1})$ and $M'(z) = O(|\Im z|^{-2})$. These bounds can be obtained from

$$\frac{\Gamma(w)}{\Gamma(w+1/2)} = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-wt}(1-e^{-t})^{-1/2}\,\mathrm{d}t$$

by rotating the integration ray into the right half-plane and expanding $[t/(1-e^{-t})]^{1/2} = 1 + t/4 + O(t^2)$; the remainder and its derivative are integrable after multiplication by the exponential. The two square root factors have opposite vertical arguments, fixing the principal limit $-\pi/32$. Nonvanishing on a compact frequency rectangle and this limit bound $1/M$ on every smaller strip. The inverse Fourier transform $r$ of $1/M(i\xi) + 32/\pi$ satisfies $r, xr \in L^2$, hence $r \in L^1$. Integrating the Fourier integral once by parts and shifting its contour to $\Re z = -\eta$ for $x > 0$ and $\Re z = \eta$ for $x < 0$ gives $|r(x)| \le C_\eta e^{-\eta|x|}/|x|$ for $|x| \ge 1$, $0 < \eta < 1$. Consequently

$$\nu = -\frac{32}{\pi}\delta_0 + r(x)\,\mathrm{d}x, \qquad \mu * \nu = \delta_0, \qquad \int e^{\eta|x|}\,\mathrm{d}|\nu|(x) < \infty \quad (\eta < 1). \tag{322}$$

Fourier uniqueness proves the convolution identity, and exponential moments allow its use in the weighted spaces below.

To prove Theorem 18, set $F(x) = e^{-x}f(e^x)$ and $G(x) = e^{-x}\mathcal{A}_{\mathrm{ax}}f(e^x)$. Then $G' = \mu * F$ in distributions. By (314) and ordinary nullity,

$$G'(x) = \frac{1}{16}\left[E_g'(e^x) - e^{-x}E_g(e^x)\right] \longrightarrow 0 \quad (x \to -\infty).$$

This function is bounded globally: away from the small output interval use the original cutoff $C^1$ input and the regular kernel; at $+\infty$ it is $O(e^{-3x/2})$. Crucially, the input has only the original bound $F(x) = o(e^{-x/2})$ at $-\infty$ and vanishes for large $x$. Therefore $e^{\eta x}F, e^{\eta x}G' \in L^2$ for any $1/2 < \eta < 1$. The inverse (322) acts on this weighted $L^2$ space, so $F = \nu * G'$. Finite total variation first gives $\|F\|_\infty \le C\|a\|_{C^1}$; dominated convergence then gives $F(x) \to 0$ at $-\infty$. This is exactly $\beta(r) = o(r^2)$. The smooth nonvanishing amplitude and $\delta = 2\arctan r$ yield (312) and its uniform quadratic bound. No Peano or finite-energy assumption was made to obtain them.

# F Actual second variations and the signed Green forms

This appendix proves the two-jet statements of Section 6 and the exact positive energy representation used subsequently. Write $P = P_0$, $Q(\theta) = P(e_\theta)$ and

$$I = (0, \pi/2), \qquad R = Q' = 20 \sin\theta \cos\theta (1 + \sin^2\theta),$$
$$b_{\mathrm{p}} = 4\sqrt{2}, \qquad A_0 = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\sqrt{2}\,\Gamma(3/4)}, \qquad T_0 = 4A_0.$$

All orbit integrals below use physical time. A quadratic datum on the vertical orbit repeated twice is twice its primitive-period datum.

## F.1 Normalization and the free-period second variation

The shell mass of $|p|^2/2 + U(q) = 1$ is $\pi D_U$, where $D_U = \int U(e_\theta)^{-1/2}\,\mathrm{d}\theta$. Momentum polar integration and then radial position integration give

$$\int \delta(H_U - 1)\partial_s U\,\mathrm{d}q\,\mathrm{d}p = \frac{\pi}{3}\int_0^{2\pi} \partial_s U(e_\theta) U(e_\theta)^{-3/2}\,\mathrm{d}\theta.$$

Hence its common mean is $-(2/3)D_U'/D_U$. Scaling the energy and using quartic homogeneity shows that multiplication by $\lambda = (D_U/D_U(0))^2$ changes that mean by $\lambda'/(3\lambda)$; the two terms cancel. This proves (87). The normalized $D_V$ is constant, and two ordinary derivatives, valid under joint $C^2$, give (90).

For the normalized family use the action on $H^1_{\mathrm{per}}([0,1];\mathbb{R}^2) \times (0,\infty)$,

$$\mathscr{A}_s(q,T) = \int_0^1 \left(\frac{|q'|^2}{2T} - TV_s(q) + T\right)\mathrm{d}u. \tag{323}$$

It is jointly $C^2$: in one dimension $H^1$ embeds in $C^0$, so the first two derivatives of the Nemytskii terms use only the continuous derivatives of $V$ on a fixed compact set. Its critical points are exactly the energy-one periodic orbits. There

$$\mathscr{A}_s = \oint p\cdot\,\mathrm{d}q, \qquad \partial_s\mathscr{A}_s = -\int_0^T \partial_s V_s(q(t))\,\mathrm{d}t = 0.$$

The $T$ equation imposes energy one, rather than an unconstrained energy-varying periodic problem.

**Proposition F.1** (A fixed regular resonant torus)**.** *On a fixed regular rational torus with nonzero isoenergetic twist, the nearby periodic-loop family of a normalized common-mean deformation persists, with the same action and period. Its second variation is* (91)*, with one and the same continuous $h$.*

# F Actual second variations and the signed Green forms

This appendix proves the two-jet statements of Section 6 and the exact positive energy representation used subsequently. Write $P = P_0$, $Q(\theta) = P(e_\theta)$ and

$$I = (0, \pi/2), \qquad R = Q' = 20 \sin\theta \cos\theta (1 + \sin^2\theta),$$
$$b_{\mathrm{p}} = 4\sqrt{2}, \qquad A_0 = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\sqrt{2}\,\Gamma(3/4)}, \qquad T_0 = 4A_0.$$

All orbit integrals below use physical time. A quadratic datum on the vertical orbit repeated twice is twice its primitive-period datum.

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Hence its common mean is $-(2/3)D_U'/D_U$. Scaling the energy and using quartic homogeneity shows that multiplication by $\lambda = (D_U/D_U(0))^2$ changes that mean by $\lambda'/(3\lambda)$; the two terms cancel. This proves (87). The normalized $D_V$ is constant, and two ordinary derivatives, valid under joint $C^2$, give (90).

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$$\mathscr{A}_s = \oint p\cdot\,\mathrm{d}q, \qquad \partial_s\mathscr{A}_s = -\int_0^T \partial_s V_s(q(t))\,\mathrm{d}t = 0.$$

The $T$ equation imposes energy one, rather than an unconstrained energy-varying periodic problem.

**Proposition F.1** (A fixed regular resonant torus)**.** *On a fixed regular rational torus with nonzero isoenergetic twist, the nearby periodic-loop family of a normalized common-mean deformation persists, with the same action and period. Its second variation is* (91)*, with one and the same continuous $h$.*

**Proof.** In action–angle coordinates, a vector in the free-period Hessian kernel has action variation $\delta I$ and period variation $\delta T$ satisfying

$$\omega \cdot \delta I = 0, \qquad TD\omega \cdot \delta I + \omega \delta T = 0.$$

The nonzero isoenergetic derivative of the frequency ratio forces $\delta I = 0$, then $\delta T = 0$. Thus the kernel is precisely the two initial-angle directions of the compact critical torus. The Hessian is symmetric Fredholm of index zero: its leading loop block is the derivative quadratic form, the potential is a compact perturbation, and the constant and period variables are finite dimensional. It is invertible on the normal complement, uniformly on this one torus.

The normal implicit-function theorem gives a coherent $C^1$ graph $v_s(z)$ over the torus. Its reduced action $\mathscr{R}(s,z) = \mathscr{A}_s(z, v_s(z))$ is $C^2$ by the envelope identities $\mathscr{R}_s = \mathscr{A}_s$ and $\mathscr{R}_z = \mathscr{A}_z$ on the normal graph. At every critical point of $\mathscr{R}(s,\cdot)$, the actual common-mean condition gives $\mathscr{R}_s = 0$. Compactness and the uniform first-order expansion in $s$ therefore give zero derivative for both its maximum and minimum. For example, compare a maximizer at $s$ with one at $s+\epsilon$ and pass to a convergent subsequence; every limiting maximizer is critical and has zero $s$ derivative. Both envelopes are constant. They agree at $s = 0$, so $\mathscr{R}$ is constant on the whole normal graph. No differentiable choice of a maximizer is needed.

Quartic virial gives $\oint p \cdot \mathrm{d}q = 4T/3$, so the periods too remain constant. Let $\mathscr{H}_z$ be the normal Hessian and

$$\mathfrak{b}_{g,z}(\eta,\tau) = -T\int_0^1 \nabla g(q_z)\cdot\eta\,\mathrm{d}u - \tau\int_0^1 g(q_z)\,\mathrm{d}u.$$

It annihilates the tangent kernel. Differentiating the normal equation once gives $v_s'|_0 = -\mathscr{H}_z^{-1}\mathfrak{b}_{g,z}$, whence

$$\mathscr{R}_{ss}(0,z) = -\int_0^T h(q_z)\,\mathrm{d}t - \mathfrak{b}_{g,z}(\mathscr{H}_z^{-1}\mathfrak{b}_{g,z}).$$

This is zero and proves the result, with the displayed pairing as $\mathcal{Q}_{c,\varphi}$. Its sign is not presumed positive. The normal graph is equivariant under each fixed loop time shift by uniqueness. If initial states are desired, the identity $p(0) = T\int_0^1 (1-u)\nabla V_s(q(u))\,\mathrm{d}u$ supplies their $C^1$ dependence; it avoids taking a pointwise derivative of an arbitrary $H^1$ variation. ∎

## F.2 The exact zero-level monodromy

For the zero-level family of Section 5.3, start at the origin with

$$q_d(0) = 0, \quad q_d'(0) = \sqrt{2}n, \qquad n = (\epsilon, -h), \quad m = (h,\epsilon),$$
$$\epsilon = \operatorname{sech}(\sqrt{2}d), \qquad h = \tanh(\sqrt{2}d).$$

Its half-period relation is $q_d(t+T_0/2) = R_x q_d(t)$, where $R_x = \operatorname{diag}(-1,1)$. Set $\kappa_0 = 6\pi/T_0$.

**Lemma 51.** *In the order $(\delta q, \delta p)$ the physical monodromy is*

$$M_d = \begin{pmatrix} \mathrm{Id} & (T_0/2)nn^t \\ \kappa_0\epsilon^2 mm^t & \mathrm{Id} \end{pmatrix}. \tag{324}$$

*For finite $d$, the fixed-period and energy-one free-period kernels consist exactly of time translation and relative-shift variation. The primitive vertical transverse monodromy is $-\mathrm{Id}$.*

**Proof.** At $d = 0$, $q = (x(t), 0)$, $x'' = -4x^3$, $x'^2 = 2(1 - x^4)$. The longitudinal Jacobi fields $x'/\sqrt{2}$ and $(\sqrt{2}/4)(x + tx')$ give the block $\left(\begin{smallmatrix} 1 & T_0/2 \\ 0 & 1 \end{smallmatrix}\right)$. For the transverse equation $Y'' + 24x^2Y = 0$, on the outgoing quarter set

$$Y_1(x) = x\sqrt{1 - x^4}, \quad J_0(x) = \int_0^x \frac{t^2}{\sqrt{1 - t^4}}\,\mathrm{d}t, \quad Y_0(x) = 1 - \tfrac{3}{2}x^4 + \tfrac{3}{2}Y_1(x)J_0(x).$$

Substitution in $2(1-x^4)Y_{xx} - 4x^3Y_x + 24x^2Y = 0$ verifies these fields. Put $B_0 = \frac{3}{2}J_0(1) = \frac{3}{2}\sqrt{\pi}\Gamma(3/4)/\Gamma(1/4)$. Near the turn, the part of $Y_0 - B_0Y_1$ is analytic in $1 - x^4$, whereas $Y_1 = 2\sqrt{2}|t - A_0| + O(|t - A_0|^3)$. Smooth continuation across the turn therefore changes $Y_0$ to $Y_0 - 2B_0Y_1$ on the returning arc. Its data at the next origin are $(1, 2\sqrt{2}B_0)$. The field with initial data $(0, 1)$ is $xx'/2$ and is half-periodic. Repeating the half-transfer gives the transverse block $\left(\begin{smallmatrix} 1 & 0 \\ 4\sqrt{2}B_0 & 1 \end{smallmatrix}\right)$; the beta integrals give $4\sqrt{2}B_0 = 6\pi/T_0$.

The invariant

$$C = xp_xp_y - yp_x^2 + 4x^2y(x^2 + 2y^2)$$

commutes with $H$. Its flow at $q = 0$ has $X_C p = (-p_xp_y, p_x^2)$, so $z_d = \Phi_C^{-d}(z_0)$. It conjugates the physical monodromies. Its differential on this origin section is block diagonal $\mathrm{diag}(F, F^{-t})$; transport of $X_H$ gives $Fe_1 = n$, and transport of $X_C$ gives $F^{-t}e_2 = \epsilon m$. Conjugation gives (324). Writing $\delta q = un + vm$, $\delta p = \alpha n + \beta m$ shows that fixed-period closure requires $\alpha = v = 0$. In the free-period problem energy first gives $\alpha = 0$, and $(M_d - \mathrm{Id})\xi + X_H\delta T = 0$ then gives $\delta T = v = 0$. The two remaining directions are precisely the phase and $d$ variations.

On a positive vertical half-excursion put $h_0(\sigma) = \mathrm{sech}(b_{\mathrm{p}}\sigma)^{1/4}$, $y = h_0^2/2$, $\mathrm{d}t = h_0^2\,\mathrm{d}\sigma$. The transverse fields $h_0e^{\pm\sqrt{2}\sigma}$ have endpoint data

$$(0, 2\sqrt{2}/2^{1/4}) \longmapsto (2^{1/4}, 0), \qquad (2^{1/4}, 0) \longmapsto (0, -2\sqrt{2}/2^{1/4}).$$

One half-excursion has transfer $\left(\begin{smallmatrix} 0 & 1/2 \\ -2 & 0 \end{smallmatrix}\right)$; its square is $-\mathrm{Id}$. The longitudinal primitive block has the nonzero homogeneity shear $T_y/2$, where $T_y = T_0/2$. Thus the primitive energy-one orbit is nondegenerate modulo time. ∎

On $T_0$-periodic variations let $\mathsf{S}\eta(t) = R_x\eta(t + T_0/2)$ and $L_d = -\partial_t^2 - D^2P(q_d(t))$. It commutes with $\mathsf{S}$. Both finite-$d$ kernel fields are $\mathsf{S}$-even. Compactifying by the initial velocity angle makes the coefficients smooth on a closed interval. At a vertical double

traversal there are three kernel fields: the longitudinal time field and two primitive-antiperiodic transverse fields. All three are again $\mathsf{S}$-even. Consequently the fixed-domain $\mathsf{S}$-odd inverse obeys

$$\sup_d \|(L_d^-)^{-1}\|_{L^2\to H^2} < \infty. \tag{325}$$

Indeed each operator is self-adjoint Fredholm with zero kernel on this block; inverse continuity and a finite compact cover give the assertion. For $x$-odd $g$, $F_d = \nabla g(q_d)$ lies in this block, so

$$\mathcal{Q}_d^-(g,g) = \int_0^{T_0} F_d \cdot (L_d^-)^{-1} F_d \,\mathrm{d}t$$

is continuous through both ends and bounded by $C\|g\|_{C^1}^2$. At either end the forcing is transverse and primitive-periodic; the response is the primitive response repeated twice. Hence $\mathcal{Q}^-_{\text{double}} = 2\mathcal{Q}^-_{\text{primitive}}$.

## F.3 The physical-window bridge at the vertical crossing

Let $\Phi_t$ denote the physical flow on $H = 1$ and put

$$\mathcal{R}_T g(z) = T_0^{-1} \int_0^{T_0} g(q(\Phi_t z))\,\mathrm{d}t, \qquad h_T[g] = g - g\circ\Phi_{T_0} + \mathcal{R}_T g.$$

The return maps have the exact local factorizations

$$\Phi_{T_0/2} - R_x = CY_{1/2}, \qquad \Phi_{T_0} - \mathrm{Id} = CY_1, \tag{326}$$

with analytic bounded coefficients near the zero level. To check the singular crossing rather than merely regular points, use the exact section

$$z(a,b) = (a,0,b,\sqrt{2-b^2-2a^4}), \qquad C(z) = ab\sqrt{2-b^2-2a^4}.$$

The coordinates $u = a$, $v = b\sqrt{2-b^2-2a^4}$ give $C = uv$. Every component of either return difference vanishes on both axes by the physical half-period relation. Double Hadamard division gives $f(u,v) = uv\int_0^1\int_0^1 f_{uv}(su,tv)\,\mathrm{d}s\,\mathrm{d}t$. Transport along the flow covers the vertical orbit; ordinary division by $C$ covers the remaining regular zero-level points. Quotients agree on overlaps because they agree where $C \neq 0$.

We need more than zero-level centering for an EO input. The following estimate uses only its values to pay for the critical crossing.

**Lemma 52** (A quadratic value bound)**.** *For an angular $C^1$ EO force $f$ with zero vertical value put*

$$M(f) = \sup_{x\neq 0, q\neq 0} \frac{|f(x,y)|}{x^2(x^2+y^2)}.$$

*On a fixed neighborhood of the critical vertical orbit,*

$$\left\|\frac{\mathcal{R}_T f}{C}\right\|_\infty + \left\|\frac{h_T[f]}{C}\right\|_\infty \le C_0\big(\|f\|_{C^1(\mathbb{S}^1)} + M(f)\big), \qquad C \neq 0. \tag{327}$$

**Proof.** Bounded physical phase changes alter $\mathcal{R}_T f$ by at most $C|C|\|f\|_{C^1}$, by the full-return identity (326). Time reversal has the same property. Flow to the upward section $z(a,d) = (a, 0, 2d, \sqrt{2-4d^2-2a^4})$, where $|C| \asymp |ad|$. Set $t_* = T_0/4$. Pairing the two half-periods, then the times $t, -t$, reduces the mean, up to that controlled error, to

$$\int_0^{t_*} \big[f(q(t;a,d)) - f(q(t;a,-d))\big]\,\mathrm{d}t. \tag{328}$$

If $|a| \asymp |d|$, smooth dependence gives $|x| \le C(|a|+|d|)$ and the value bound directly gives $CM|ad|$. The two highly unbalanced sectors are equivalent: the nearby downward $y=0$ crossing, followed by time reversal, has coordinates $a' = dA_1(a,d)$, $d' = aA_2(a,d)$ with analytic positive factors equal to one at the origin. Indeed on $d=0$ that crossing is the physical origin, and on $a=0$ its transverse velocity vanishes, by the exact half-period symmetry; the first transverse variations give the two linear coefficients. Its time differs from $t_*$ by $O(ad)$. It therefore suffices to treat $0 < d/a = \lambda \ll 1$, $a > 0$.

Retain the exact axis orbit $q(t;a,0) = (aA_a(t), Y_a(t))$, rather than replacing it by a linearized vertical trajectory. The vertical transverse fields $z_1 = X_1$, $z_2 = 2X_2$ obey

$$z_1' = -2z_2^3, \quad z_2' = 2z_1^3, \quad z_1^4 + z_2^4 = 1, \quad Y_0 = z_1 z_2/\sqrt{2},$$

with $(z_1,z_2)(0) = (1,0)$ and $(z_1,z_2)(t_*) = (0,1)$. Taylor expansion of the analytic physical flow gives, uniformly in time,

$$q(t;a,\pm d) = (aA_a(t) \pm dz_2(t), Y_a(t)) + O(ad). \tag{329}$$

The error is $O(ad+d^2) = O(ad)$: the first $d$ derivative differs from its value at $a=0$ by $O(a)$. Thus a $C^1$ force pays this replacement with $C|ad|\|f\|_{C^1}$.

On $(0,t_*)$ set

$$U_a = A_a/Y_a, \quad V_a = z_2/Y_a, \quad J_a = Y_a^4 V_a/U_a', \quad L_a = V_a'/U_a'.$$

They satisfy $U_a > 0$, $U_a' < 0$. At $t=0$ the bounds are

$$U_a \asymp t^{-1}, \quad U_a' \asymp -t^{-2}, \quad J_a = O(t^6),\ J_a' = O(t^5),\ L_a = O(t^3),\ L_a' = O(t^2).$$

At the other end, $s = t_* - t$, they are

$$U_a \asymp 1, \quad U_a' \asymp -s^3, \quad J_a, J_a' = O(1), \quad L_a = O(s^{-5}), \quad L_a' = O(s^{-6}), \quad |V_a/U_a| \le C/s.$$

These are bounds for the exact axis orbit, uniform for small $a$. For verification at the delicate endpoint, cubic homogeneity of the acceleration gives $q_a(t_* - s) = v_a s + O(s^5)$. Dividing the two components gives $U_a = u_a + k_a s^4 + O(s^5)$; at $a=0$, $U_0 = \sqrt{2}/z_2$ and $U_0' \asymp -16\sqrt{2}s^3$. Thus $k_a > 0$, $U_a'/s^3$ has a nonzero analytic limit, and $J_a = Y_a^3 z_2/U_a'$ extends regularly. At the initial endpoint even/odd Taylor expansions give the first list. On the compact middle the strict inequality follows from $U_0' = -2\sqrt{2}z_1^3/z_2^2 < 0$.

Take $\delta = L\lambda^{1/5}$ with a fixed sufficiently large $L$. On $s \le \delta$, the value estimate alone gives

$$\int_{t_*-\delta}^{t_*} |f(aA_a \pm dz_2, Y_a)|\,\mathrm{d}t \le CM\big[(a^2+a^4)\delta^5 + d^2\delta^3 + d^4\delta\big] \le C_L Mad.$$

This includes any ratio fold in the endpoint interval. On the remaining interval write $f(x,y) = y^4 b(x/y)$ for $y > 0$; then $|b(r)| \le M(r^2+r^4)$. For $|k| \le d$ put $r_k = aU_a + kV_a$ and $\eta = k/a$. Choosing $L$ large gives $|\eta L_a| \le 1/2$ and $|\eta V_a/U_a| \le 1/2$, so $r_k > 0$ and $r_k' < 0$. The preceding endpoint bounds imply, for $m = 2, 4$,

$$\left[U_a^m \left|\frac{J_a}{1+\eta L_a}\right|\right]_\partial + \int U_a^m \left|\left(\frac{J_a}{1+\eta L_a}\right)'\right| \mathrm{d}t \le C.$$

At the initial end the relevant powers are $t^{6-m}$, $t^{5-m}$ and $t^{8-m}$; at the truncated final end the only extra cost is $|\eta| \int_\delta s^{-6}\,\mathrm{d}s \le C\lambda\delta^{-5}$. With $F_k = a^{-1}J_a/(1+\eta L_a)$, integration by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}k}\int Y_a^4 b(r_k)\,\mathrm{d}t = [F_k b(r_k)]_\partial - \int F_k' b(r_k)\,\mathrm{d}t, \qquad \left|\frac{\mathrm{d}}{\mathrm{d}k}\int Y_a^4 b(r_k)\,\mathrm{d}t\right| \le CM(a+a^3).$$

One first truncates away from $t = 0$; the displayed powers make the boundary vanish and justify the limit. Thus no bound for $b'$ at infinity is used. Integrating in $k$ from $-d$ to $d$ proves the required bound for (328). Finally $f \circ \Phi_{T_0} - f$ is $C$ times a bounded continuous coefficient directly from (326). ∎

**Lemma 53** (Peano values and the continuous window quotient)**.** *Let $g$ be angular $C^1$ and EO, $A = g(0,1)$, and $g_* = y^3\sqrt{x^2+y^2}$. If*

$$g - Ag_* = o\big(|q|^4\cos^2\theta\big) \quad \textit{at the vertical directions},$$

*then $\mathcal{R}_T g/C$ and $h_T[g]/C$ extend continuously through the critical vertical orbit, with value zero there. In particular this holds for every original EO ordinary-null input, by Theorem 18.*

**Proof.** First the analytic coefficients $Y_1, Y_{1/2}$ in (326) vanish on the vertical orbit. At the upward section let $X_1, X_2$ be transverse variations with data $(1,0), (0,1)$. Their Wronskian and the conserved transverse quadratic form give $X_1X_2 = y/\sqrt{2}$. Explicitly $y'XX' - y(X')^2 + 8y^3X^2$ is constant; for each field it is zero, so the sum of their distinct logarithmic derivatives is $y'/y$, and the initial values determine the constant. The mixed longitudinal variation is

$$Z'' + 192y^2 Z = -48yX_1X_2 = -24\sqrt{2}y^2, \qquad Z = (y' - \sqrt{2})/8.$$

Both $Z, Z'$ vanish at $T_0/2$ and $T_0$. The mixed transverse variation vanishes identically. Since $\partial_a\partial_b C = \sqrt{2}$, the divided return coefficients vanish at the crossing, hence along its whole orbit by flow commutation.

The reference $g_*$ is globally $C^2$. Its window vanishes on both section axes, so double division gives a continuous quotient. On the vertical line $(g_*)_{xx} = \operatorname{sign}(y)y^2$ and $(g_*)_y = 4|y|^3$. Its mixed second variation is exactly

$$(g_*)_{xx}X_1X_2 + (g_*)_y Z = \tfrac{1}{2}|y|^3 y',$$

whose full-period integral is zero. The quotient consequently vanishes at the crossing. Phase shifts preserve this zero value because their difference is an integral of the full-return coefficient just proved to vanish. This deals with the possibly nonzero vertical value $A$.

Put $g_0 = g - Ag_*$. Cut it off inside vertical cones of width $\epsilon$. The removed part tends to zero in $\|\cdot\|_{C^1} + M(\cdot)$: Peano values control $M$ and the cutoff term $\epsilon^{-1}\sup|g_0|$, while the angular derivative tends to zero at vertical by even-$x$ parity and $C^1$ regularity. Smooth angular convolution outside those cones, preserving parity, therefore gives smooth EO forces $f_n$ which vanish in vertical cones and converge to $g_0$ in that norm. Each window quotient extends continuously by double division, and its mixed variation is zero at vertical because the derivatives of $f_n$ there vanish. Estimate (327) makes these quotients uniformly Cauchy, including their continuous extensions by density of $C \neq 0$. Their limits are the claimed continuous quotients, with zero vertical trace. Add the reference term. At regular zero-level points ordinary $C^1$ division gives the same extension. No estimate of an angular derivative divided by the distance to the axis has been assumed. ∎

## F.4 The EO soft mode and the noncompact second-order anchor

For an EO force the zero-level mean vanishes by the physical reflection and reversal symmetries. Thus $F_d = \nabla g(q_d)$ is orthogonal both to $q_d'$ and to the relative-shift variation. Denote its kernel-orthogonal quadratic datum by $\mathcal{Q}_d^{\mathrm{EO}} = \langle F_d, L_d^\dagger F_d\rangle$. The inverse itself need not remain bounded at a vertical end.

**Lemma 54.** *If the EO force is $C^1$ and $\mathcal{R}_T g = Cb_g$ with $b_g$ continuous and zero on the vertical orbit, then $\mathcal{Q}_d^{\mathrm{EO}}$ has continuous endpoint limits*

$$\mathcal{Q}_{\pm\infty}^{\mathrm{EO}} = 2\mathcal{Q}_{\mathrm{vertical,primitive}}^{\mathrm{EO}}.$$

*It is bounded by $C(\|g\|_{C^1} + \|b_g\|_\infty)^2$.*

**Proof.** Let $X_d$ be the homogeneous Jacobi field with initial data $(m, 0)$. Equation (324) gives $X_d(T_0) = m$, $X_d'(T_0) = \kappa_0\epsilon^2 m$. It is periodic in $H^1$, and its exact form defect is

$$\mathfrak{l}_d(X_d, v) = \kappa_0\epsilon^2\ell_d(v), \qquad \ell_d(v) = m \cdot v(0).$$

Compactify by the velocity angle $\psi$. The time field and $\partial_\psi q_\psi$ remain linearly independent at the endpoint, are orthogonal by time parity, and give a smooth rank-two projection $P_d$. The trace $\ell_d$ annihilates both. Therefore $\phi_d = (\mathrm{Id} - P_d)X_d$ satisfies

$$L_d\phi_d = \kappa_0\epsilon^2\ell_d \text{ in } H^{-1}, \quad \ell_d(\phi_d) = 1, \quad \mathfrak{l}_d(\phi_d, \phi_d) = \kappa_0\epsilon^2.$$

At the endpoint $\phi_d$ tends to the extra transverse time-even zero mode $X_*$. A small spectral circle about zero contains the two exact zeros and one simple soft eigenvalue $\lambda_d$. Write $\phi_d = \alpha_d e_d + \chi_d$, where $e_d$ is its unit eigenvector and $\chi_d$ is in the uniformly gapped complement. The outer inverse is uniformly bounded $H^{-1} \to H^1$, since the common form norm is equivalent to that of $(\mathrm{Id} + |L_d|)^{1/2}$ for uniformly bounded matrix potentials. Hence

$$\|\chi_d\|_{H^1} = O(\epsilon^2), \qquad \lambda_d \alpha_d^2 = \kappa_0 \epsilon^2 \big(1 - \ell_d(\chi_d)\big), \qquad \alpha_d^2 \longrightarrow \|X_*\|_2^2 > 0. \tag{330}$$

This determines the sign and scale of the soft eigenvalue from an actual boundary defect, not from a monodromy eigenvalue alone.

At $z_d$, $\mathrm{d}C(m,0) = -2\epsilon$. Differentiating the fixed physical window along that energy-tangent variation gives exactly

$$\langle F_d, X_d \rangle = T_0 \, \mathrm{d}(\mathcal{R}_T g)_{z_d}(m,0) = -2\epsilon T_0 b_g(z_d).$$

Only continuity of $b_g$ is needed when differentiating $C b_g$ at $C = 0$. Since $F_d \perp \operatorname{Ran} P_d$, its soft quadratic term is

$$\frac{|\langle F_d, e_d \rangle|^2}{\lambda_d} = \frac{[-2\epsilon T_0 b_g(z_d) - \langle F_d, \chi_d \rangle]^2}{\kappa_0 \epsilon^2 (1 - \ell_d(\chi_d))} \le C\big(|b_g(z_d)|^2 + \epsilon^2 \|g\|_{C^1}^2\big). \tag{331}$$

It tends to zero. This does not bound the soft response itself. The outer inverses converge in $L^2 \to H^2$ and $F_d$ converges uniformly. At vertical the EO force is $(0, 4A|y(t)|^3)$, where $A = g(0,1)$; it need not vanish. It is longitudinal, primitive-periodic and orthogonal to the time field. The limiting outer response is therefore the primitive longitudinal response repeated twice. This proves the assertion and its normalization. ∎

**Proposition F.2** (The actual zero-level second-order anchor)**.** *For every normalized jointly $C^2$ common-mean family at $P$, the full first force satisfies* (89) *for every finite $d$.*

**Proof.** EE Abel injectivity gives $g_{\mathrm{EE}} = 0$. Decompose $g = g_{\mathrm{EO}} + g_{x\text{-odd}}$. The ordinary means of the second part vanish by symmetry; hence the first part is ordinary-null and Lemma 53 applies, including its nonzero-vertical reference case. The two forces lie in opposite eigenspaces of $\mathsf{S}$, so their Green cross pairing is zero. Lemma 54 and (325) show that the total quadratic datum is continuous at each end and has twice the primitive value. At vertical the two forcings are longitudinal and transverse, so the cross pairing is zero there as well.

The primitive vertical orbit has transverse Poincare derivative $-\mathrm{Id}$, so the ordinary implicit-function theorem gives a nearby energy-one periodic branch. Common means make its action constant, and quartic virial makes its period exactly $T_y = T_0/2$. Its genuine second derivative is

$$T_y Z_h(\text{vertical}) + \mathcal{Q}_{\text{vertical,primitive}} = 0.$$

Thus $r_2(d) := -T_0 Z_h(d) - \mathcal{Q}_d(g,g)$ tends to zero at both ends.

On any slightly enlarged compact $d$ interval, the kernel calculation in Lemma 51 gives a uniformly invertible normal free-period Hessian on the compact phase cylinder. Its coherent normal graph is equivariant under fixed time shifts, so the reduced $C^2$ action is $\mathscr{R}(s,d)$, independent of phase. At a critical $d$ it is a genuine orbit and satisfies $\mathscr{R}_s = 0$. The fixed-period response used above has energy variation zero: from $L_d q_d = -2\nabla P$ and $q \cdot \nabla g = 4g$ one gets $\int \nabla P \cdot \eta_d = 0$; the constant linearized energy $q_d' \cdot \eta_d' + \nabla P \cdot \eta_d + g(q_d)$ has integral zero. Thus this is the same free-period response with $\delta T = 0$, modulo the tangent kernel. The action calculation gives

$$\mathscr{R}(s,d) = J_0 + \tfrac{1}{2} s^2 r_2(d) + o(s^2), \qquad \mathscr{R}_s(s,d) = s r_2(d) + o(s)$$

uniformly on each compact slab. If $r_2(d_*) > 0$, choose finite endpoints where it is smaller, using its zero end limits. For small $s > 0$, a maximum on this slab occurs in its interior and has $r_2$ bounded positively away from zero. There $\mathscr{R}_d = 0$ but $\mathscr{R}_s > 0$, a contradiction. A negative value is excluded by a minimum. Hence $r_2 \equiv 0$. This argument assumes no uniform parameter interval for the whole infinite cylinder and no persistence of that cylinder at nonzero parameter. ∎

## F.5 The positive volume mixture and the quadratic necessity

Define

$$\mathcal{A}(\xi) = \frac{(1-\xi)^{-1/2} + (1+\xi)^{-1/2}}{2}, \quad \vartheta(\xi) = \arctan\left(\sinh \frac{\operatorname{arctanh} \xi}{2}\right).$$

For $K = \tanh^2(2\sqrt{2}d)$, the zero-level Abel formula of Section 6 is

$$b_{\mathrm{p}} A_0 K^{3/4}(1-K)^{-5/4} Z_b(d(K)) = \int_0^K (K-t)^{1/4} f_b(t)\,\mathrm{d}t, \qquad f_b(t) = t^{-1/2}\mathcal{A}(\sqrt{t})^5 b(\vartheta(\sqrt{t})).$$

Changing angle to $t$ gives $\int_{\mathbb{S}^1} bQ^{-3/2} = 4\int_0^1 w(t) f_b(t)\,\mathrm{d}t$, where

$$w(t) = \frac{1}{32}(1-t)^{7/4}\left(1 - \frac{\sqrt{1-t}}{2} - \frac{1-t}{4}\right)^{-3/2}.$$

Define $a_j > 0$ by $(1 - z/2 - z^2/4)^{-3/2} = \sum_{j\ge 0} a_j z^j$ and set

$$\begin{aligned} c_j &= \frac{a_j}{32\,\mathrm{Beta}(5/4, 3/2 + j/2)}, \\ \lambda(K) &= \sum_{j\ge 0} c_j (1-K)^{1/2 + j/2}, \\ \mu(K) &= b_{\mathrm{p}} T_0 \lambda(K) K^{3/4}(1-K)^{-5/4}. \end{aligned} \tag{332}$$

The series has radius greater than one after the displayed half-power is removed; reciprocal beta factors grow only polynomially. Termwise beta integration gives $w(t) = \int_t^1 (K - t)^{1/4} \lambda(K)\,\mathrm{d}K$. Thus, for every continuous homogeneous degree-four position observable,

$$\int_{\mathbb{S}^1} bQ^{-3/2}\,\mathrm{d}\theta = \int_0^1 \mu(K) Z_b(d(K))\,\mathrm{d}K. \tag{333}$$

For the interchange, Tonelli applied to $|f_b|$ gives the finite integral $\int w|f_b|$; signed Fubini then follows, even if $f_b$ alone is not integrable at one. Both sides see only the EE part. The measure is positive, $\mu = O(K^{3/4})$ at zero and $\mu \asymp (1-K)^{-3/4}$ at one. Taking $b = Q$ and using virial gives $\int \mu = 3\int Q^{-1/2}$. We shall also use the exact normalization

$$b_{\mathrm{p}} T_0 c_0 = 21/16, \tag{334}$$

which follows from $b_{\mathrm{p}} T_0 = 4\,\mathrm{Beta}(1/4, 1/2)$ and $\mathrm{Beta}(5/4, 3/2) = \frac{2}{21}\,\mathrm{Beta}(1/4, 1/2)$.

Define, on the domains with finite continuous quadratic data just proved,

$$\mathcal{B}(g_1, g_2) = -\frac{1}{T_0}\int_0^1 \mu(K)\mathcal{Q}_{d(K)}(g_1, g_2)\,\mathrm{d}K - \frac{3}{2}\int_{\mathbb{S}^1} a_1 a_2 Q^{-5/2}\,\mathrm{d}\theta. \tag{335}$$

This includes all original $C^1$ $x$-odd forces and the EO window domain.

**Proposition F.3.** *The actual normalized first force satisfies $\mathcal{B}(g, g) = 0$. For every original $C^1$ $x$-odd $g$, $\mathcal{B}(g_{\mathrm{rot}}, g) = 0$.*

**Proof.** Insert the actual continuous $h$ in (333), then use (89) and (90). For the second assertion the exact rotation response is $-Jq_d$, whence

$$\mathcal{Q}_d(g_{\mathrm{rot}}, g) = -\int_0^{T_0} \partial_\theta g(q_d)\,\mathrm{d}t.$$

Apply (333) to this continuous position observable. Integration by parts on the angular circle gives $\int a' Q^{-3/2} = \frac{3}{2}\int aRQ^{-5/2}$, cancelling the other term of (335). Only one angular derivative is used. ∎

## F.6 The complete OO off-diagonal kernel

On the first physical quarter of $q_d$, $d > 0$, write $q = r(\cos s, -\sin s)$, with $s$ decreasing from $\alpha(K) = \vartheta(\sqrt{K})$ to zero. Put

$$\mathfrak{q}(s) = \frac{4\sin^2 s}{(1+\sin^2 s)^2}, \qquad K_d = b_{\mathrm{p}}\sqrt{K}(1-K), \qquad T_K'(s) = -J_K(s).$$

The angle is strictly monotone: in the separated pulses the logarithmic velocity difference is $\sqrt{2}[\tanh(b_{\mathrm{p}}\sigma + 2\sqrt{2}d) - \tanh(b_{\mathrm{p}}\sigma - 2\sqrt{2}d)] > 0$. The exact radial and time densities are

$$r_K(s) = \mathcal{A}(\sqrt{\mathfrak{q}})[(1-K)(K-\mathfrak{q})/K]^{1/4},$$

$$J_K(s) = \frac{2\mathcal{A}(\sqrt{\mathfrak{q}})}{b_{\mathrm{p}}\vartheta'(\sqrt{\mathfrak{q}})}(1-K)^{1/4}K^{1/4}(K-\mathfrak{q})^{-3/4}, \tag{336}$$

$$H_K(s) := r_K^3 J_K = \frac{2\mathcal{A}(\sqrt{\mathfrak{q}})^4}{b_{\mathrm{p}}\vartheta'(\sqrt{\mathfrak{q}})}\frac{1-K}{\sqrt{K}},$$

$$G(K,s) := r_K^4 J_K = \frac{2\mathcal{A}(\sqrt{\mathfrak{q}})^5}{b_{\mathrm{p}}\vartheta'(\sqrt{\mathfrak{q}})}(1-K)^{5/4}K^{-3/4}(K-\mathfrak{q})^{1/4}.$$

In particular $H_K$ has no moving-edge singularity. The OO quarter problem is the coupled operator $L_d$ with

$$\eta(0) = 0, \qquad \eta_x(A_0) = 0, \qquad \eta_y'(A_0) = 0.$$

Its inverse is uniformly bounded up to both ends, by the time-odd part of (325).

Let $V_d = \partial_d q_d$ and $E_d = (q_d + tq_d')/4$ be the left Dirichlet Jacobi fields. Define the right field by

$$Y_C(A_0) = 0, \qquad Y_C'(A_0) = ((av)^{-1}, 0), \quad a = (1-K)^{1/4}, \quad v = \sqrt{2K}. \tag{337}$$

It has energy variation zero and $C$ variation one. With $W(Y,Z) = Y \cdot Z' - Y' \cdot Z$,

$$W(E_d, q_d') = -1, \quad W(V_d, q_d') = 0, \quad W(V_d, Y_C) = -1.$$

The ordered Green matrix is $-Y_{\mathrm{left}}(t)W^{-t}Y_{\mathrm{right}}(t')^t$ for $t < t'$. Its term not involving $q_d'(t')$ is $V_d(t)Y_C(t')^t$. For angular inputs with strictly ordered disjoint supports, the later input pairs to zero with $q_d'$. Therefore the full-period datum is

$$\mathcal{Q}_d(g_f, g_h) = 4\left(\int_0^{A_0} \nabla g_f \cdot V_d \,\mathrm{d}t\right)\left(\int_0^{A_0} \nabla g_h \cdot Y_C \,\mathrm{d}t\right),$$

where the larger-angle input $f$ is earlier in physical time. The factor four is the full physical period, including all branches. Moreover

$$\int \nabla g_f \cdot V_d \,\mathrm{d}t = -K_d \int f(s)\partial_K G(K,s)\,\mathrm{d}s.$$

There is no lower-edge atom here because $G$ vanishes at $K = \mathfrak{q}(s)$.

Here is a closed formula for the interior density in the other factor. For $0 < q < K < 1$ set

$$b = \sqrt{1-q}, \quad \Delta = (1-K)(K-q), \quad D = \operatorname{arctanh}\sqrt{K}, \quad x = \operatorname{arctanh}\sqrt{q/K},$$

$$J(z) = \int_0^z \sqrt{\cosh w}\,\mathrm{d}w, \qquad L_q(K) = J(D+x) - J(D-x),$$

$$A_q = 3K + 2(1-q) - 3q/K, \qquad H_q(K) = \frac{A_q L_q - 4\Delta L_q'}{b^2}. \tag{338}$$

Every prime fixes $q$. Then

$$\Lambda_C(K,s) = \frac{3(1+b)^3}{32 b_{\mathrm{p}} b^{3/2}}(1-K)^{5/4}(K-q)^{-3/4}K^{-1/4}H_q(K), \qquad q = \mathfrak{q}(s). \tag{339}$$

To verify that this is the physical right-field density, rather than a formal separated expression, use the even scalar energy variation

$$e(z) = \frac{\cosh(z)^{1/4} - (3/2)\operatorname{sech}(z)^{1/4}\tanh z\, J(z)}{8}.$$

For $z_- = -(D+x)$ and $z_+ = D - x$, put $u = \operatorname{sech}(z_-)^{1/4}$, $v_1 = \operatorname{sech}(z_+)^{1/4}$, $T_\pm = \tanh z_\pm$ and $\Delta_T = T_+ - T_- > 0$. The separated variations of energies $+1, -1$ are $e(z_-), -e(z_+)$ plus one common auxiliary time shift, chosen to impose (337). This common shift is a total derivative of the occupation integral for compact interior angular supports. With $E_\pm = e(z_\pm)/\operatorname{sech}(z_\pm)^{1/4}$ one has

$$E'_\pm = -T_\pm \operatorname{sech}(z_\pm)^{-1/2}/8 - 3(1 - T_\pm^2)J(z_\pm)/16,$$

$$r = (u^2 + v_1^2)/2, \quad \theta_X = uv_1\Delta_T/(4r), \quad \delta r = u^2 E_- - v_1^2 E_+, \quad \delta\theta/\theta_X = 4(E_- + E_+)/\Delta_T.$$

The angular integration by parts has density $[-10r^4\delta r + \partial_X(2r^5\delta\theta/\theta_X)]/(b_{\mathrm{p}}\theta_X)$. Collecting its terms gives

$$\frac{3r^6}{b_{\mathrm{p}}uv_1\Delta_T^2}\left[(3T_-T_+ - 2)(J(z_-) + J(z_+)) - 2(T_+u^{-2} + T_-v_1^{-2})\right],$$

which simplifies to (339). These formulas are regular when an individual scalar velocity changes sign. The origin boundary, absent for these compact later supports, will be retained explicitly in (356). A useful check, also fixing $L_q$ without branch ambiguity, is

$$L_q(K) = K^{3/4}(1-K)^{-1/4}\int_0^{\sqrt{q}} [\sqrt{1+y} + \sqrt{1-y}](K - y^2)^{-5/4}\,\mathrm{d}y. \tag{340}$$

For $s < t$, the off-diagonal kernel of $\mathcal{B}$ is therefore

$$\kappa(s,t) = -\frac{4}{T_0}\int_{\mathfrak{q}(t)}^1 G(K,t)W'_{\mathfrak{q}(s)}(K)\,\mathrm{d}K, \qquad W_q(K) = \mu(K)K_d\Lambda_C(K,s). \tag{341}$$

Use symmetry for $t < s$. In deriving this expression by integration by parts in $K$, the lower boundary vanishes because $G = 0$, and the upper one vanishes because $W_q = O_q((1-K)^{5/4})$.

## F.7 An elementary strict scalar sign

Put

$$F_q = K(1-K)^{3/2}(K-q)^{-3/4}H_q, \qquad S(\beta) = \sum_{j\geq 0} c_j\beta^j, \quad \beta = \sqrt{1-K}.$$

All prefactors in the exact identity

$$W_q(K) = \frac{3b_{\mathrm{p}}T_0(1+b)^3}{32b^{3/2}}S(\beta)F_q(K) \tag{342}$$

are positive. We prove $F_q > 0$ and $F_q' < 0$ on $q < K < 1$.

Let $d_q = \sqrt{2-2b}$ and define

$$T_q = 2d_q K^{3/4}\Delta^{-1/4}, \qquad E_q = \frac{2d_q}{b}(1+b-K)K^{-1/4}\Delta^{-1/4}.$$

For $f(z) = \sqrt{\cosh z}$, $(2f)' = f\tanh z$ and $(2f\tanh z)' = f(1+\operatorname{sech}^2 z)$. Integrating between $D-x$ and $D+x$ gives the strict sandwich

$$T_q < T_q/\sqrt{K} < L_q < E_q. \tag{343}$$

It remains valid if $D - x < 0$. For example, writing $\eta = \sqrt{q}$ and $\alpha_\pm = \sqrt{1\pm\eta}$ gives $\alpha_+ - \alpha_- = d_q$, $\alpha_+\alpha_- = b$, $\eta(\alpha_+ + \alpha_-) = d_q(1+b)$, and

$$f(D\pm x) = K^{1/4}\Delta^{-1/4}\alpha_\pm, \qquad \tanh(D\pm x) = \frac{K\pm\eta}{\sqrt{K}(1\pm\eta)}.$$

These identities give both endpoint expressions in (343) directly.

Define positive factors

$$\mathfrak{G} = K(1-K)^{3/2}(K-q)^{-3/4}/b^2, \qquad \mathfrak{U} = (d_q/2)K^{-1/4}\Delta^{-5/4},$$

and

$$p = \frac{1}{K} - \frac{3}{2(1-K)} - \frac{3}{4(K-q)}, \quad N = (2+b)K - (2+b-b^2), \quad C_q = pA_q + A_q',$$

$$B_q = [A_q - 4p\Delta + \Delta' + \Delta/K]N - 4\Delta(2+b).$$

Direct endpoint differentiation gives $L_q' = \mathfrak{U}N$, and hence

$$F_q'/\mathfrak{G} = C_q L_q + \mathfrak{U}B_q. \tag{344}$$

For a fully algebraic sign check put $u = (1-K)/b^2$, $w = 1-u$. Then

$$C_q = \frac{33b^2u^3 - 56b^2u^2 + 20b^2u - 42u^2 + 75u - 30}{4uwK}, \quad B_q = b^4[11(b+2)u^2 - 7(2b+5)u + 10].$$

The exact identities are

$$\frac{C_qT_q + \mathfrak{U}B_q}{\mathfrak{U}} = -b^4P_0, \qquad \frac{C_qE_q + \mathfrak{U}B_q}{\mathfrak{U}} = -b^4wP_1/K, \tag{345}$$

where the following expansions exhibit strict positivity:

$$\begin{aligned} P_0 &= b^2(6u^3 + 33u^2w + 14uw^2 + 20w^3) \\ &\quad + b(1-b)(3u^3 + 17u^2w + 54uw^2 + 40w^3) + 20(1-b)^2w^2, \\ P_1 &= b^3(2u^3 + 23u^2w + 74uw^2 + 20w^3) \\ &\quad + b^2(1-b)(5u^3 + 143u^2w + 198uw^2 + 60w^3) \end{aligned}$$

$$+ b(1-b)^2(2u^3 + 106u^2w + 164uw^2 + 60w^3) + 20(1-b)^3w.$$

These identities are checked by expansion using $w = 1 - u$; equivalently $P_0 = 20w^2 + bu(14 - 11u) - b^2u(33u^2 - 56u + 20)$ and

$$P_1 = 33b^3u^3 - 34b^3u^2 + 11b^2u^2 - 10b^2u - 42bu^2 + 44bu - 20u + 20.$$

If $C_q < 0$, use the lower comparator in (343); if $C_q > 0$, use the upper comparator; if it is zero either identity suffices. This proves, including the quantitative version needed later,

$$-F_q' \geq \mathfrak{G}\mathfrak{U}b^4 \min\{P_0, wP_1/K\} > 0. \tag{346}$$

For fixed $q$, the sandwich and $L_q' = \mathfrak{U}N$ give $H_q = O_q((1-K)^{-1/4})$, so $F_q \to 0$ at one. Strict decrease now implies $F_q > 0$. In (342), $S$ has positive coefficients and $\beta$ decreases with $K$, so $W_q' < 0$. Equation (341) consequently proves

$$\kappa(s,t) > 0 \qquad (0 < s \neq t < \pi/2). \tag{347}$$

For later absolute bounds it is useful to record an elementary decomposition of the same, positive density. Put $M = \int_{D-x}^{D+x} \cosh(z)^{-3/2}\,\mathrm{d}z$ and $P_s = 3(1-K^2) + b(K+3)$. The identity $L_q = E_q - M$ gives

$$\begin{aligned} \Lambda_E &= \frac{3}{64}\frac{(1+b)^{5/2}}{b^{9/2}}\sqrt{q}(1-K)K^{-3/2}P_s, \\ \Lambda_M &= -\frac{3(1+b)^3}{128\sqrt{2}b^{7/2}}\beta^{5/2}K^{-1/4}A_qM(K-q)^{-3/4}, \\ \Lambda_C &= \Lambda_E + \Lambda_M \geq 0. \end{aligned} \tag{348}$$

No term has been dropped when its sign is negative.

## F.8 The full diagonal and the original $C^1$ ground-state identity

The off-diagonal calculation alone does not rule out derivative-supported diagonal terms. Let $\mathbf{G}_K$ be the complete quarter Green matrix, $e(s) = (\cos s, -\sin s)$, $n(s) = (\sin s, \cos s)$, $A_K = H_K n$ and $D_K = -4H_K e$. For OO profiles set $\mathcal{F}_{K,f} = A_K f' + D_K f$. The exact full-period formula is

$$\mathcal{Q}_K(f,h) = 4\iint \mathcal{F}_{K,f}(s)^t \mathbf{G}_K(T_K(s), T_K(t))\mathcal{F}_{K,h}(t)\,\mathrm{d}s\,\mathrm{d}t.$$

The Green matrix is continuous and its first physical-time derivative has jump $-\mathrm{Id}$. Consequently

$$\partial_t\partial_{t'}\mathbf{G}_K = \text{piecewise bounded kernel} + \mathrm{Id}\ \delta(t-t').$$

Two angular integrations by parts produce the multiplication delta

$$4H_K(s)^2J_K(s)\delta(s-t).$$

No derivative of a delta occurs. The remaining kernel is

$$\partial_s\partial_t(A_s^t\mathbf{G}A_t) - \partial_s(A_s^t\mathbf{G}D_t) - \partial_t(D_s^t\mathbf{G}A_t) + D_s^t\mathbf{G}D_t$$

away from the diagonal. At the moving angle $\alpha(K)$, $\mathbf{G}_K(0,t) = \mathbf{G}_K(t,0) = 0$, so the complete kernel has no moving-edge boundary term. Near its corner it vanishes at least like the minimum of the two distances to that edge to power $1/4$. This uses the complete Green matrix; an isolated right Jacobi factor need not vanish at the physical origin.

On each compact angular interval, $H_K$ and its first angular derivative are $O(1-K)$, while $J_K(s) \le C(1-K)^{1/4}(K-\mathfrak{q}(s))^{-3/4}$. The uniform quarter inverse bounds the physical Green matrix and its piecewise first and mixed derivatives. After integration against $\mu$, the worst regular angular kernel is bounded by

$$C\int_{\max(q_s,q_t)}^1 (1-K)^{7/4}(K-q_s)^{-3/4}(K-q_t)^{-3/4}\,\mathrm{d}K \le C|q_s-q_t|^{-1/2}.$$

The comparison integral is $\mathrm{Beta}(1/4,1/2)|q_s-q_t|^{-1/2}$. The diagonal density is bounded locally by $C\int_{q_s}^1(1-K)^{3/2}(K-q_s)^{-3/4}\,\mathrm{d}K$. Thus, on compact tests,

$$\begin{aligned}\mathcal{B}(f,h) &= \int_I V(s)f(s)h(s)\,\mathrm{d}s + \iint_{I^2}\kappa(s,t)f(s)h(t)\,\mathrm{d}s\,\mathrm{d}t,\\ V(s) &= -\frac{4}{T_0}\int_{q_s}^1 \mu H_K^2 J_K\,\mathrm{d}K - 6Q(s)^{-5/2}.\end{aligned} \tag{349}$$

The local kernel is integrable, bounded by $C(1+|s-t|^{-1/2})$. The displayed bounds justify Fubini through the whole parameter integral; no finite-part prescription or additional diagonal distribution occurs.

**Theorem 19** (OO sign and exact equality case)**.** *For every original $C^1$ OO profile $f$,*

$$-\mathcal{B}_{\mathrm{OO}}(f,f) = D(f) = \frac{1}{2}\iint_{I^2}\kappa(s,t)R(s)R(t)\left[\frac{f(s)}{R(s)} - \frac{f(t)}{R(t)}\right]^2\,\mathrm{d}s\,\mathrm{d}t. \tag{350}$$

*The integral is finite. Its zero set and the bilinear radical are exactly $\mathbb{R}R$.*

**Proof.** Choose smooth increasing interior cutoffs $\chi_n \uparrow 1$. For a $C^1$ OO profile, $f_n = \chi_n f$ has uniformly bounded $C^1$ norm, since $f$ vanishes linearly at both axes. The finite forcing seminorm

$$N(f)^2 = \frac{1}{T_0}\int_0^1 \mu(K)\|\nabla g_f(q_K)\|^2_{L^2(0,T_0)}\,\mathrm{d}K + \int_{\mathbb{S}^1} f^2 Q^{-5/2}\,\mathrm{d}\theta$$

controls $|\mathcal{B}(f,h)|$ by $CN(f)N(h)$. The forcing differences converge almost everywhere in $(K,t)$ and are uniformly bounded. Since $\mu$ has finite mass and no endpoint atoms, dominated convergence gives $N(f-f_n)\to 0$. This does not require convergence in $C^1$ at the axes.

Apply the rotation radical identity to $R_n = \chi_n R$ and a nonnegative compact test. Positivity of $\kappa$ and monotone convergence in (349) give the exact row identity

$$\int_I \kappa(s,t)R(s)\,\mathrm{d}s = -V(t)R(t). \tag{351}$$

It also proves the required weighted tails are finite. Use this identity with the compact test $f^2/R$. The inequality $2|f(s)f(t)| \le R(s)f(t)^2/R(t) + R(t)f(s)^2/R(s)$ justifies the ground-state algebra absolutely and proves (350) for compact profiles.

For general $C^1$ OO $f$, $\psi = f/R$ is bounded and continuous up to the axes. Fatou first gives finiteness of the nonnegative difference-square integral, since $D(f_n) = -\mathcal{B}(f_n, f_n)$ converges. Also $D(R_n) \to -\mathcal{B}(R,R) = 0$. The quotient difference of $f - f_n$ is

$$(1-\chi_n(s))[\psi(s)-\psi(t)] + \psi(t)[\chi_n(t)-\chi_n(s)].$$

Its first squared contribution tends to zero by dominated convergence in the already finite difference-square measure, and the second is bounded by $C\|\psi\|_\infty^2 D(R_n)$. Thus $D(f - f_n) \to 0$; the triangle inequality for pairwise differences proves equality in the Fatou passage. There is no surviving axis functional. Finally (347) and $R > 0$ imply that $D(f) = 0$ forces $f/R$ to be constant almost everywhere, hence everywhere in the interior and then on the full circle by $C^1$ parity. The rotation radical proves the converse and the bilinear assertion. ∎

**Lemma 55** (The complete positive row)**.** *With $\delta = \pi/2 - s$, the full row in* (351) *is*

$$a_0(s) := -V(s) = \frac{1}{R(s)}\int_I \kappa(s,t)R(t)\,\mathrm{d}t = \frac{6Q(s)^{-3/2}}{1-\sin^4 s} + 36Q(s)^{-5/2} \le C(1+\delta^{-2}). \tag{352}$$

*In particular $a_0(0) = 42$ and $D(f) \le 2\int_I a_0(s)f(s)^2\,\mathrm{d}s$ whenever the latter is finite.*

**Proof.** Here is a direct evaluation of the multiplication part, retaining its normalization. Put $x = \cos s$, $q = \mathfrak{q}(s)$ and $m_0 = Q^{-3/2}$. From (336),

$$H_K = \frac{x^{-5}(1-K)}{\sqrt{2}\sqrt{K}}, \qquad J_K = \frac{x(1-K)^{1/4}K^{1/4}}{\sqrt{2}(2-x^2)^{3/2}(K-q)^{3/4}}.$$

Since $w(q) = \int_q^1 (K-q)^{1/4}\lambda(K)\,\mathrm{d}K$,

$$\int_q^1 (1-K)(K-q)^{-3/4}\lambda(K)\,\mathrm{d}K = -4(1-q)w'(q) - w(q).$$

Substitution in $d_0 = (4/T_0)\int \mu H_K^2 J_K$ gives

$$d_0 = \frac{\sqrt{2}b_{\mathrm{p}}x^{-9}}{(2-x^2)^{3/2}}[-4(1-q)w'(q) - w(q)] = \frac{6m_0}{1-\sin^4 s} - \frac{20m_0'}{R}.$$

The last equality follows by differentiating the explicit $w$ above and $q = 4\sin^2 s/(1+\sin^2 s)^2$; all its factors are displayed. As $m_0'/R = -(3/2)Q^{-5/2}$, adding the local angular term $6Q^{-5/2}$ in (349) proves (352). Its endpoint bound is immediate from $1 - \sin^4 s \asymp \delta^2$ near vertical. The last inequality follows from $(u-v)^2 \le 2u^2 + 2v^2$ and the positive row identity, so it includes cross pairs as well as diagonal neighborhoods. ∎

## F.9 The exact EO rank correction and its origin boundary

For a zero-vertical original $C^1$ EO force $g$, $\widetilde{g} = \operatorname{sign}(x)g$ is a genuine $C^1$ OO force: $x$-evenness makes its transverse derivative vanish at the vertical axis, and zero vertical value makes both the value and longitudinal derivative there vanish. The two forces agree on the first quarter. The EO quarter response has left Neumann conditions, while the OO response has left Dirichlet conditions; their common right conditions are $x$-Dirichlet and $y$-Neumann. The EO quarter kernel is $\mathbb{R}q_d'$. Its compatibility is exactly $\int_0^{A_0} \nabla g \cdot q_d' \,\mathrm{d}t = g(q_d(A_0)) - g(q_d(0)) = 0$.

The right field (337) has the precise origin data

$$m \cdot Y_C(0) = -\frac{1}{2\epsilon}, \qquad n \cdot Y_C'(0) = 0, \qquad Y_C'(0) = \frac{\kappa_0 \epsilon}{8} m. \tag{353}$$

The first two identities are its unit $C$ variation and zero energy variation. For the third, the reflected physical half-return satisfies

$$\operatorname{diag}(R_x, R_x) M_{d,\mathrm{half}} = \mathrm{Id} + \tfrac{1}{2}(M_d - \mathrm{Id}).$$

At $d = 0$ this is the longitudinal half-transfer $\left(\begin{smallmatrix} -1 & -T_0/4 \\ 0 & -1 \end{smallmatrix}\right)$ and transverse half-shear $\kappa_0/2$ already computed; the commuting $C$ flow and its equivariance under $R_x$ prove it for finite $d$. Reflection about the quarter endpoint gives $Y_C(T_0/2) = R_x Y_C(0)$ and $Y_C'(T_0/2) = -R_x Y_C'(0)$. The half-return formula gives instead $Y_C'(T_0/2) = R_x[Y_C'(0) - \kappa_0 \epsilon m/4]$. Equating them proves the factor eight in (353).

Define the actual quarter trace

$$I_f(K) = \int_0^{A_0} \nabla g_f(q_d(t)) \cdot Y_C(t) \,\mathrm{d}t. \tag{354}$$

If $\eta_D$ is the Dirichlet-left response, Green's identity with $q_d'$ gives $\eta_D'(0) = r_0 m$, and that with $Y_C$ gives $I_f = -r_0/(2\epsilon)$. Hence the Neumann-left response is

$$\eta_N = \eta_D + \gamma q_d' + \frac{16}{\kappa_0} I_f Y_C.$$

The phase term pairs to zero. Restoring the full period yields the exact positive rank-one difference

$$\mathcal{Q}_d^{\mathrm{EO}}(g, g) - \mathcal{Q}_d^{\mathrm{OO}}(\widetilde{g}, \widetilde{g}) = \frac{64}{\kappa_0} I_f(K)^2. \tag{355}$$

Polarization gives the corresponding bilinear identity.

The complete angular trace, including its boundary subtraction, is

$$I_f(K) = -c(K) f(\alpha(K)) + \int_0^{\alpha(K)} \Lambda_C(K, s) f(s) \,\mathrm{d}s, \qquad c(K) = \frac{(1 + \sqrt{1-K})^3}{16\sqrt{2}\sqrt{K(1-K)}}. \tag{356}$$

For clarity, the physical origin does contribute to this formula. The actual expansions are

$$q_d(t) = \sqrt{2} n t - \frac{\sqrt{2}}{10} \nabla P(n) t^5 + O(t^9),$$

$$s(t) = \alpha - \frac{R(\alpha)}{10} t^4 + O(t^8), \qquad H_K(s) \longrightarrow H_0 = 5\sqrt{2}/R(\alpha).$$

Writing $B(s) = H_K(s) n(s) \cdot Y_C(T_K(s))$ and $D(s) = -4H_K(s) e(s) \cdot Y_C(T_K(s))$, the pairing is $\int_0^\alpha [Bf' + Df]\,\mathrm{d}s$. Here $B(0) = 0$, while $B(\alpha) = -5\sqrt{2}/[2\epsilon R(\alpha)] = -c(K)$. Furthermore

$$B(s) = -c(K) + b_1(K)(\alpha - s)^{1/4} + O(\alpha - s), \quad b_1 = H_0 \frac{\kappa_0 \epsilon}{8} \left( \frac{10}{R(\alpha)} \right)^{1/4} > 0.$$

These are expansions of analytic physical solutions, so the differentiated remainders have the claimed orders. Thus $B \in W^{1,1}$ and $D - B' = \frac{1}{4} b_1 (\alpha - s)^{-3/4} + O(1)$ is integrable. It agrees with (339) on compact tests and hence throughout the open interval. Integration by parts proves (356) for the original $C^1$ input. Dropping its first term would change the operator.

**Lemma 56** (The entire low-$K$ rank map)**.** *For each $k < 1$,*

$$\int_0^k \mu(K) |I_f(K)|^2 \,\mathrm{d}K \le C_k \|f\|^2_{L^2(I)}. \tag{357}$$

*The expression* (356) *has its resulting bounded extension on angular $L^2$; it does not evaluate a fixed point trace of an $L^2$ input.*

**Proof.** On a compact subinterval of $0 < K < 1$, $K \mapsto \alpha(K)$ is a smooth diffeomorphism and the multiplication term in (356) is bounded by change of variables. The history kernel has the just proved integrable order $(\alpha - s)^{-3/4}$ with smooth local amplitudes. Young's inequality treats it. The separated history is bounded, including $s = 0$, where (340) gives $\Lambda_C = O(s)$.

It remains to include $K = 0$. Write $K = u^2$, $q = v^2 = u^2 r^2$ and $b_f(v) = f(\vartheta(v))$. The exact decomposition (348) gives, uniformly for small $u$,

$$|\Lambda_C(u^2, \vartheta(ur))| \le C u^{-2} r (1 - r^2)^{-3/4}.$$

Indeed its elementary term is bounded by $Cr/u^2$; the other term has $|A_q| \le C$ and $M \le Cr$. For $r \le 1/2$ the latter follows from the length $2 \operatorname{arctanh} r$, and for $r > 1/2$ from $\int_{\mathbb{R}} \operatorname{sech}^{3/2} z \,\mathrm{d}z < \infty$. Since $\vartheta'$ is bounded above and below here,

$$|I_f(u^2)| \le C u^{-1} \big( |b_f(u)| + \mathcal{T}|b_f|(u) \big), \qquad \mathcal{T} b(u) = \int_0^1 r (1 - r^2)^{-3/4} b(ur) \,\mathrm{d}r.$$

The origin multiplication has been retained in this inequality. Extend $b_f$ by zero above a fixed small $u_0$. Minkowski and dilation give $\|\mathcal{T} b\|_2 \le \frac{1}{2} \operatorname{Beta}(3/4, 1/4) \|b\|_2$. Also $\mu(u^2) 2u \le C u^{5/2}$. The low-$K$ integral is therefore at most $C \int_0^{u_0} u^{1/2} (|b_f| + \mathcal{T}|b_f|)^2 \,\mathrm{d}u \le C \|f\|_2^2$. Combine with the compact part. ∎

**Proposition F.4** (The positive EO energy on its original core)**.** *Let $\mathcal{W}_0$ be the original $C^1$ EO zero-vertical class for which $\mathcal{R}_T g = Cb_g$ with a bounded continuous quotient vanishing at the vertical orbit. For its first-quadrant profile $f$,*

$$E(f) := -\mathcal{B}_{\mathrm{EO}}(f,f) = D(f) + \frac{32}{3\pi}\int_0^1 \mu(K) I_f(K)^2 \,\mathrm{d}K. \tag{358}$$

*Here $c_E = 32/(3\pi)$. Both terms are finite, and $E(f) > 0$ for $f \neq 0$. The subcore $\mathcal{W}_P$ defined by $f = o(\delta^2)$ contains every original zero-vertical ordinary-null input by Theorem 18 and Lemma 53.*

**Proof.** The EO datum is bounded and continuous by Lemma 54. The sign-flipped input is genuinely $C^1$ OO, so its datum is bounded by (325). Equation (355) then proves integrability of the rank square against the finite positive measure $\mu$; this integrability was not assumed in advance. The angular squares of the two inputs are identical, so (335) and $64/(\kappa_0 T_0) = 32/(3\pi)$ prove (358). If it is zero, the OO ground-state theorem makes the common quadrant profile $cR$. Even-$x$ $C^1$ parity gives $f'(\pi/2) = 0$, whereas $R'(\pi/2) = -40$. Thus $c = 0$. The last assertion follows from the independent value theorem and the physical-window bridge; it asserts no nullity of arbitrary cutoff profiles. ∎

The nonzero-vertical reference $\sin^3\theta$ is never sign-flipped into this OO domain. It has a legitimate finite EO window datum by Lemma 53; its affine subtraction requires the complete boundary term and all original first-order data. Equation (358) therefore proves the stated zero-vertical obstruction, without excluding the unresolved nonzero-vertical affine class or identifying a general mixed-parity linear kernel.

# G Uniform calculus of the actual resonant rows

This appendix proves the estimates for resonant observations that enter the energy argument. All angular restrictions below are restrictions of the same input. No restricted input is assumed to satisfy a null equation. Constants may depend on fixed compact intervals and smooth cutoffs, but never on an integer frequency or on a scale tending to zero.

## G.1 The complete density, clocks, and denominator cells

We use the parabolic coordinates and physical clock of Section 5.1. For positive separation $c$, put

$$p_e(q) = \sqrt{2(e + q^2 - q^{10})}, \qquad \tau_e(q) = \int_q^{q_+(e)} \frac{\mathrm{d}v}{p_e(v)}, \qquad \Delta T = T_u - 2T_v, \qquad U = T_u \overline{S}.$$

Here $q_+(-c)$ and $q_\ell(-c)$ are the upper and lower positive roots of $q^2 - q^{10} = c$. All root-dependent integrals are evaluated on their physical intervals. Write $r(c) = T_u(c)/T_v(c)$

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and

$$\kappa = \frac{3}{4}B(3/4, 1/2), \qquad \lambda_\epsilon(\theta) = (\sec\theta + \tan\theta)^\epsilon, \quad \epsilon \in \{+1, -1\}.$$

The sign $\epsilon$ distinguishes reflected physical angles; it is not a new choice of separation sign.

**Lemma 57** (Complete angular density)**.** *For an actual resonance $m = 2n + 2k$, $k, n \geq 1$, the EO observation in Section 5.2, with the original physical normalization, is*

$$R_{k,n}[a] = \int_0^{\pi/2} a(\theta) b_{k,n}(\theta)\,\mathrm{d}\theta, \qquad b_{k,n} = B_+ - B_-, \tag{359}$$

$$C_\epsilon = \frac{\lambda_\epsilon \sec\theta (1 + \lambda_\epsilon^2)^5}{2T_v U}, \tag{360}$$

$$B_\epsilon = C_\epsilon \int_{q_\ell(-c)}^{\min(q_+(-c), q_+(c)/\lambda_\epsilon)} \frac{v^{11}\cos(\omega\tau_c(\lambda_\epsilon v))\cos(\omega\tau_{-c}(v))}{p_c(\lambda_\epsilon v) p_{-c}(v)}\,\mathrm{d}v, \tag{361}$$

$$\omega = \frac{4\pi k}{\Delta T(c)}. \tag{362}$$

*An empty integration interval contributes zero. All positive integer $k$, including nonprimitive harmonics, occur in these formulas.*

**Proof.** On a positive angular ray, $u = \lambda_+ v$ and $|\partial_\theta u| = v\lambda_+ \sec\theta$. The homogeneous force times the physical clock is $S^5 a/16$, where $S = u^2 + v^2$. Folding by $x$-evenness in the full double-cosine Fourier formula supplies $8/(T_u T_v \overline{S})$. The resulting Jacobian and force give exactly (361); the reflected physical angle has $\lambda_-$. The EO extension subtracts that term. Resonance gives $n\Delta T = 2kT_v$, hence $\omega = n(2\pi/T_v) = 4\pi k/\Delta T$. In particular, both sum and difference phases in the product of cosines must be retained. ∎

**Lemma 58** (Conormal clocks and exact cells)**.** *For every fixed integer $j \geq 0$,*

$$c^j \partial_c^j \left( \frac{\Delta T(c)}{c} - \frac{\kappa}{\sqrt{2}} \right) = O_j(c), \qquad T_v(c) = \frac{\log(1/c)}{\sqrt{2}} + O(1), \qquad U(c) \longrightarrow T_0 > 0.$$

*The last two assertions have the corresponding finite conormal derivative bounds. On a fixed positive compact $s$ interval, with $c = hs$ and $L = \log(1/h)$, define*

$$\nu_h(s) = \frac{2}{r(hs) - 2}, \quad n_{k,h}(s) = k\nu_h(s), \quad w_h(s) = \frac{h}{L}|\nu_h'(s)| = \frac{2h^2 r'(hs)}{L(r(hs) - 2)^2}.$$

*For all small $h$, $r' > 0$, $n_{k,h}$ is decreasing, and its half-integer cells $C_{k,n} = n_{k,h}^{-1}([n - 1/2, n + 1/2))$ satisfy*

$$|C_{k,n}| \asymp \frac{h}{kL}, \qquad \int_{C_{k,n}} w_h(s)\,\mathrm{d}s = \frac{h}{kL}, \qquad w_h \longrightarrow \frac{2}{\kappa s^2} \quad \text{in each fixed finite } C^j \text{ norm.} \tag{363}$$

*The actual node $s_{k,n} = c_{k,n}/h$ belongs to its cell; it need not be its Euclidean midpoint.*

**Proof.** Near the scalar saddle use $w = q\sqrt{1-q^8}$. Its inverse has $\mathrm{d}q/\,\mathrm{d}w = 1 + w^8 G(w^2)$ with $G$ analytic. If $I_+ = T_u/4$, $I_- = T_v/2$, the singular difference is

$$I_+ - I_- = \frac{1}{\sqrt{2}} \left\{ \operatorname{arsinh} \frac{\delta}{\sqrt{c}} - \operatorname{arcosh} \frac{\delta}{\sqrt{c}} + N_+(c) - N_-(c) \right\} + J(c) - J(-c),$$

where $J$ is the simple-upper-root regularized outer integral and

$$N_+ = \int_0^\delta \frac{w^8 G(w^2)}{\sqrt{w^2 + c}}\,\mathrm{d}w, \qquad N_- = \int_{\sqrt{c}}^\delta \frac{w^8 G(w^2)}{\sqrt{w^2 - c}}\,\mathrm{d}w.$$

The elementary difference and $J(c) - J(-c)$ are analytic at zero. In $N_-$, put $z = \sqrt{w^2 - c}$; the integrand becomes $(z^2 + c)^{7/2} G(z^2 + c)$. Replacing its upper endpoint by $\delta$ changes it by an analytic function. Splitting the differentiated integrals at $\sqrt{c}$ gives, for $j \geq 2$, bounds for $c^j N_\pm^{(j)}$ of order $c^j$ if $j < 4$, $c^4(1 + |\log c|)$ if $j = 4$, and $c^4$ if $j > 4$. The first two derivatives are integrable directly. For the coefficient, let $P(q) = q^2 - q^{10}$. The saddle subtraction and simple-root regularization give $(I_+ - I_-)'(0) = -(1/\sqrt{2})\,\mathrm{FP} \int_0^1 P(q)^{-3/2}\,\mathrm{d}q$, where the two counterterms are $1/(2\eta^2)$ at zero and $2 \cdot 8^{-3/2} \zeta^{-1/2}$ at $1 - \zeta$. For $H(q) = (-4 + 6q^8)/(q^2\sqrt{1-q^8})$, direct differentiation gives

$$H'(q) = 8q^{-3}(1-q^8)^{-3/2} + 12q^5(1-q^8)^{-1/2}.$$

Its endpoint expansions are $-4q^{-2} + O(q^6)$ and $\zeta^{-1/2}/\sqrt{2} + O(\sqrt{\zeta})$, with no constant terms. Integration and the two counterterm subtractions therefore give $\mathrm{FP} \int_0^1 P^{-3/2} = -3B(3/4, 1/2)/16$. Multiplying by the exact factor four in $\Delta T$ proves $\Delta T'(0) = \kappa/\sqrt{2}$. Thus the remainder after the linear term has $c^j \partial_c^j$ size $O_j(c^2)$, proving the asserted difference estimate without differentiating a bare asymptotic. The individual periods and the moment defining $U$ follow from the identical decomposition, with an extra factor $q^2$ for the moment.

It follows that

$$r(c) - 2 = \frac{\kappa c}{\log(1/c)}(1 + O(1/\log(1/c))), \qquad r'(c) = \frac{\kappa}{\log(1/c)}(1 + O(1/\log(1/c))),$$

with finite conormal versions. Their positive leading terms establish monotonicity. The exact cell mass follows by the monotone substitution $n = k\nu_h(s)$. The remaining claims follow by differentiation on fixed compact $s$ intervals. ∎

We shall repeatedly use two elementary consequences. For $g \in H^1(C)$, $x_0 \in \overline{C}$, and $\ell = |C|$,

$$\ell |g(x_0)|^2 \leq C \int_C (|g|^2 + \ell^2 |g'|^2). \tag{364}$$

This is the fundamental theorem of calculus followed by averaging. Moreover, if $S$ is monotone and $0 < c \leq |S'| \leq C$, changing variable $t = S(x)$ and periodizing with period $2\pi$ gives

$$\sum_{j \in \mathbb{Z}} \left| \int f(x) A(x) e^{ijS(x)}\,\mathrm{d}x \right|^2 \leq 2\pi \sup_t \sum_{S(x_\ell) = t + 2\pi\ell} \frac{|A(x_\ell)|^2}{|S'(x_\ell)|}\,\|f\|_2^2. \tag{365}$$

Indeed Cauchy–Schwarz on each set of inverse images bounds the square of the periodized function by the product of its two weighted sums. Integration of the input sum and Parseval prove the formula. For phase images of length less than $2\pi$ there is at most one inverse image, and Parseval is exact.

## G.2 Endpoint reflection and the buffered stationary expansion

A buffered product means fixed compact input and output intervals with $\xi > s/4$ by a fixed positive margin, where $\theta = h\xi$ and $c = hs$. The endpoint estimates must be proved before interpolation in a row index.

**Lemma 59** (Complete endpoint bounds)**.** *On a buffered product, choose fixed smooth radial endpoint caps disjoint from the positive stationary point. The complete endpoint density satisfies*

$$|Lb_{\mathrm{end},k,n}(h\xi)| \le C_M k^{-M} \qquad (M \ge 1)$$

*at every actual node, uniformly in all $k, n$ and small $h$. The same conclusion holds on $0 \le \xi \le \varepsilon_0 < s_{\min}/8$, with the correct upper-turn chart for that product. No derivative of the lower integer carrier in a noninteger continuation is asserted.*

**Proof.** Write $P = p_c(\lambda v)$, $Q = p_{-c}(v)$. The exact difference is

$$\Delta = P^2 - \lambda^2 Q^2 = 2c(1+\lambda^2) + 2\lambda^2(1-\lambda^8)v^{10}. \tag{366}$$

At the lower turn let $s_\ell$ be ascending negative-energy time. The actual integer identities $\omega T_u/4 = \pi(n+k)$ and $\omega T_v/2 = \pi n$ turn the complete cosine product into $(-1)^k \cos(\omega\zeta_c(\lambda v))\cos(\omega s_\ell)$, where $\zeta_c(u) = \int_0^u p_c^{-1}$. Reflect this full even product across zero. With $z = \operatorname{sgn}(s_\ell)\sqrt{v^2 - \ell_c^2}$,

$$v = \sqrt{\ell_c^2 + z^2}, \quad Q^2 = z^2 g_c(v)^2, \quad g_c(v)^2 = 2\left(1 - \sum_{j=0}^{4} v^{8-2j}\ell_c^{2j}\right).$$

The transformed amplitude and inverse phase derivative are

$$a_\ell = \frac{\chi_\ell(v)v^{10}}{Pg_c(v)}, \qquad H_\ell = \frac{c}{-\partial_z(\zeta_c(\lambda v) - s_\ell)} = \frac{vPg_c(v)(P + \lambda z g_c(v))}{\Delta/c}.$$

The lower cap can be fixed so that $\Delta/c$ is positive and bounded away from zero. In the weight $v$, $a_\ell$ has degree nine and $H_\ell$ degree three: the $j$th derivative of a degree-$d$ function is bounded by $C_j v^{d-j}$. The recurrence $a_{j+1} = \partial_z(H_\ell a_j)$ has degree $9+2j$ and bounded $L^1$ norm. Integration by parts on the full line gives $C_j(\omega c)^{-j} \le C_j k^{-j}$. The small numerator $P + \lambda z g_c$ causes no difficulty; no lower bound on that numerator is needed.

At an upper endpoint use the signed time $t$ of the oscillator that turns first. The other upper time satisfies the exact analytic identity

$$s(t)^2 = t^2 + \varepsilon d(\varepsilon, t^2, \text{parameters}), \qquad d \ge d_0 > 0.$$

Here $\varepsilon = h$ on fixed scaled products, and the first-turn gap fixes the positive sign of $d$. For the small-input product both physical signs turn first at $q_+(-c)$, since $q_+(c) - \lambda q_+(-c) = h(s/4 - \epsilon\xi) + O(h^2) > 0$. The complete reflected amplitude is $A(t^2)/s(t)$ with smooth compact support. For $\phi = t - s(t)$ the exact rationalization is

$$\frac{\varepsilon}{\phi'} = \frac{s(s+t)}{d - t(s+t)d_T}, \qquad T = t^2.$$

The denominator stays positive after fixing a sufficiently small cap. With weight $(\varepsilon+t^2)^{1/2}$, the inverse derivative has degree two and the amplitude degree minus one. The repeated integration-by-parts amplitude has degree $j-1$, hence bounded $L^1$ norm for $j \geq 1$. This proves $C_j(\omega\varepsilon)^{-j}$. Reflection eliminates a boundary at the actual turn; smooth cutoffs eliminate artificial outer boundaries. On a middle compact the two difference derivatives are rationalized using (366), and the sum derivative is $-\lambda/P - 1/Q$. The latter is treated with its actual parameter $k/h$. This proves the claim for both physical signs and both cosine phases. ∎

**Lemma 60** (Interior expansion and all-integer Bessel bound)**.** *On a buffered compact product, put $x = \log\xi$ and $a_{h,F}(he^x) = h^{-1/2}e^{-x/2}F(x)$. Choose the radial interior cutoff equal to one near all positive critical points. Then, for any fixed sufficiently large $M$,*

$$u_k(s) := \int e^{x/2}\sqrt{k}\,Lb_{\mathrm{int},h,k}(s, he^x)F(x)\,\mathrm{d}x = v_k(s) + r_k(s), \quad (367)$$

$$v_k = e^{-i\pi/4}I_k + e^{i\pi/4}I_{-k}, \qquad I_j = \int A_h(s,x)F(x)e^{ijS_h(s,x)}\,\mathrm{d}x, \quad (368)$$

$$\sum_{k\geq 1}\int \left(|u_k|^2 + |k^{-1}\partial_s u_k|^2\right)\mathrm{d}s \leq C\|F\|_2^2, \quad (369)$$

$$\sum_{k>K}\int \left(|r_k|^2 + |k^{-1}\partial_s r_k|^2\right)\mathrm{d}s \leq CK^{-2}\|F\|_2^2. \quad (370)$$

*The positive amplitude $A_h$ and phase $S_h$ have uniform required finite derivatives. In particular $-S_{h,x}$ and $S_{h,s}$ are positive and bounded away from zero. The complete actual analysis $B_h a = (\sqrt{L}\,\rho(c_{k,n}/h)R_{k,n}[a])_{k,n}$ on this input window has uniformly bounded norm.*

**Proof.** For the positive difference phase

$$\Phi_h(v) = \frac{4\pi}{\Delta T}[\tau_c(\lambda_+ v) - \tau_{-c}(v)], \qquad v_h^{10} = \frac{c(1+\lambda_+^2)}{\lambda_+^2(\lambda_+^8 - 1)}.$$

There is one critical point, tending to $(s/(4\xi))^{1/10}$, strictly inside the radial compact. Its Hessian is negative and nonzero. The negative physical difference and both sum phases are nonstationary there. The clock lemma and the analytic regularized upper-time integrals

give finite-order smooth convergence of the difference phase before exponentiation. Its limit is

$$S_0(s,\xi) = A_1(v_*) - (\xi/s)B_1(v_*), \quad B_1(v) = \frac{4\pi}{\kappa\sqrt{1-v^8}}.$$

The envelope identities are $S_{0,x} = -(\xi/s)B_1(v_*) < 0$ and $S_{0,s} = (\xi/s^2)B_1(v_*) > 0$.

The one-dimensional Morse coordinate at the exact critical point reduces the phase to its exact critical value minus a positive square. Taylor expansion of the transformed smooth amplitude, integration of its even Gaussian moments, and integration by parts of the Taylor remainder give an expansion in $k^{-j}$ after multiplication by $\sqrt{k}$. All coefficients are independent of $k$ and have bounded finite derivatives. Differentiating once in $s$, then multiplying by $k^{-1}$, preserves these bounds. Applying enough integrations on the compact nonstationary pieces, using $k/h$ for the sums, gives a summable final remainder. The original cosine expansion assigns $1/4$ to each stationary exponential; the Hessian sign gives the factors in (368).

Apply (365) to each coefficient of the finite expansion, using the monotone exact $S_h$. The differentiated amplitudes are $iS_{h,s}A_h$ and $k^{-1}\partial_s A_h$; neither contains a growing integer factor. This proves (369). Extracting $K^{-1}$ from the first higher coefficient on $k > K$ proves (370); later terms and the sufficiently long final remainder are smaller. No condition $kh \to 0$ is used.

By (363) and (364), the actual interior array has squared norm bounded by

$$C \int \sum_k \bigl(|u_k|^2 + (h/L)^2 |k^{-1}u_k'|^2\bigr)\,\mathrm{d}s.$$

For the endpoints, combine Lemma 59 with the count

$$\#\{n : c_{k,n}/h \in I\} \le CkL/h.$$

The resulting squared Hilbert–Schmidt tail is $C\sum_{k>K} k^{-3} \le CK^{-2}$. This also establishes the full target bound and retains the endpoint histories without differentiating their integer carriers. ∎

**Lemma 61** (Local normal modulo compact terms)**.** *Shrink the fixed log-input window so that every phase image has length less than $2\pi$. The complete local actual normal has the form*

$$B_h^* B_h = D_h + K_h + E_h, \qquad \|E_h\| \longrightarrow 0,$$

*where $D_h$ is positive multiplication, uniformly bounded below, and $K_h$ converges in norm to a compact operator. In particular its compact terms obey, after fixed logarithmic coordinate changes,*

$$|\langle K_h u, v\rangle| \le \varepsilon \|u\|_2 \|v\|_2 + C_\varepsilon \|u\|_{H^{-1}(\mathbb{R})} \|v\|_2 + o(1)\|u\|_2\|v\|_2.$$

*The exact positive multiplier is*

$$D_h(x) = 2\pi \int w_h(s)\rho(s)^2 \frac{A_h(s,x)^2}{-S_{h,x}(s,x)}\,\mathrm{d}s. \tag{371}$$

**Proof.** For the two separate stationary signs, integer Parseval gives $2\pi \int A_h^2 |F|^2/(-S_{h,x})$ minus the zero-mode square. The zero-mode kernels converge smoothly on compact rectangles, hence in Hilbert–Schmidt norm. The opposite-sign kernel has phase $k(S_h(s,x) + S_h(s,y))$, whose $s$ derivative is positive. Repeated integration by parts gives $C_N k^{-N}$ with the required finite input derivatives; its series is collectively compact. The high-$k$ difference between full and leading analyses has norm $O(K^{-1})$ by (370) and the endpoint tail. Every fixed finite-$k$ part converges in Hilbert–Schmidt norm by Proposition H.1, including the moving logarithmic brake. These facts give norm convergence of the compact part by first choosing $K$ and then $h$.

Sampling is paid once over the whole integer sum. For $V_k = \rho^2 u_k \overline{v_k}$, the exact cell mass and the fundamental theorem of calculus bound the product-cell error by

$$C\frac{h}{L}\int \sum_k k^{-1}|V_k'(s)|\,\mathrm{d}s \le C\frac{h}{L}\|F\|_2\|G\|_2.$$

The last step is Cauchy–Schwarz in the joint variables $(s,k)$ followed by (369). It is not a uniform band error summed infinitely many times. Finally approximate a compact limit by finite-rank operators with smooth compactly supported source vectors. Their pairings are bounded by the $H^{-1}$ norm through $H^1$ duality, which proves the last assertion. ∎

## G.3 The reflected upper source and actual fractional sampling

The source interval in a mixed normal may cross the moving inner brake. Absolute logarithmic estimates at that brake do not give the all-integer bound needed here. The following construction keeps the full time chart.

**Lemma 62** (Whole-line reflected upper-source symbol)**.** *Let $\chi$ be a fixed even smooth time cutoff and let $\tau > 0$ be its radius. There is an exact upper component, acting on the whole signed scaled-angle line, whose Fourier kernel on $|\eta| \ge C_0 k$ is*

$$\widehat{K}_{h,k,\tau}(s,\eta) = e^{i\Psi_h(s,\eta;k)}\big(|\eta|^{-1}a_{0,h,\tau}(s,k/\eta) + a_{-2,h,k,\tau}(s,\eta)\big), \tag{372}$$

$$\Psi_h = -\eta\beta_h(s) + \frac{k^2}{\eta}E_h(s,(k/\eta)^2), \qquad \beta_h(s) = \theta_b(hs)/h, \tag{373}$$

$$|\partial_s\partial_\eta\Psi_h| \ge c > 0. \tag{374}$$

*The principal symbol has uniform finite $s$ and weighted-$\eta$ seminorms, independent of $\tau$, and is supported where $|\eta| \ge ck/\tau$. The remainder is of order minus two, with constants allowed to depend on fixed $\tau$. On $|\eta| \le C_0 k$, the exact complementary kernel and any fixed number of $s$ derivatives are bounded by $C_{N,\tau}h^{N-1}k^{-N}$. For $0 \le t \le 1$, the high principal and remainder satisfy, respectively,*

$$\|T_0 f\|_{H_s^t}^2 \le C_t \int_{|\eta|\ge ck/\tau} |\eta|^{2t-2}|\widehat{f}(\eta)|^2\,\mathrm{d}\eta, \tag{375}$$

$$\|T_{-2}f\|_{H^t_s}^2 \le C_{t,\tau}\int_{|\eta|\ge C_0k}|\eta|^{2t-4}|\widehat{f}(\eta)|^2\,\mathrm{d}\eta. \tag{376}$$

*A fixed smooth output cutoff is understood. The signed EO upper source is the difference of this density at $\xi$ and $-\xi$.*

**Proof.** Let $u = U_c(t)$ and $v = U_{-c}(s_1)$ be the even trajectories from their upper turns, and let $\Theta(c,t,s_1) = \arg(2uv + i(u^2 - v^2))$. Reflect both time variables before any integration. The exact full-plane density is

$$K(s,\xi) = \frac{L}{8T_vU}\int_{\mathbb{R}^2}(u^2+v^2)^5\chi(t/\tau)\chi(s_1/\tau)\delta(h\xi-\Theta)e^{i\omega(t-s_1)}\,\mathrm{d}t\,\mathrm{d}s_1.$$

The full-plane exponential is four times the quadrant cosine product, which accounts for its coefficient. Set $q = t + s_1$, $p = (t - s_1)/h$. Fourier transformation of the delta contributes $1/h$, while the time Jacobian is $h/2$. Thus the Fourier phase is $kD_h(s)p - \eta\Xi_h(s,q,p)$, with $D_h = 4\pi h/\Delta T$ and $\Xi_h = \Theta(hs,q,hp)/h$; the amplitude is $L(u^2 + v^2)^5\chi((q+hp)/(2\tau))\chi((q-hp)/(2\tau))/(16T_vU)$. The full $p$ support has length $O(1/h)$.

Analyticity of the even scalar trajectories and the zero-energy interchange symmetry give

$$\Theta(c,q,d) = qdH_0(q,d) + cG(c,q,d), \quad H_0(0,0) = -4, \quad G(0,0,0) = 1/4.$$

Hence $\Xi_h = qpH_0(q,hp) + sG(hs,q,hp)$. On the long-$p$ part, $|\partial_q\Xi_h| \ge c|p|$ after choosing the maximum time chart sufficiently small. Every higher $q$ derivative is $O(1+|p|)$. Repeated $q$ integrations therefore give $C_{N,\tau}|\eta|^{-N}\int(1+|p|)^{-N}\,\mathrm{d}p$; this controls the actual long chart and all fixed parameter derivatives.

On the bounded chart put $r_1 = k/\eta$. The Hessian at $r_1 = 0$ is the negative of

$$\mathrm{Hess}_{q,p}\,\Xi_h(s,0,0) \longrightarrow \begin{pmatrix} -5s & -4 \\ -4 & 0 \end{pmatrix},$$

with determinant $-16$. The implicit-function theorem gives one critical point, odd in $r_1$, with both critical times equal to $r_1$ times smooth nonzero factors. Its critical value is even in $r_1$. For the essential cancellation, work also in the original compact time chart with phase $\lambda(t-s_1) - \Theta(c,t,s_1)$. At $c = 0$ its critical point lies exactly on $t = s_1$, so its critical value is zero for all small $\lambda$. Time reversal makes that value even in $\lambda$; interchange of the oscillators makes it odd in $c$. At $\lambda = 0$ it is $-\theta_b(c)$. Analytic division thus gives

$$U_{\mathrm{crit}}(c,\lambda) + \theta_b(c) = c\lambda^2 G_*(c^2,\lambda^2).$$

Substitute $c = hs$, $\lambda = (k/\eta)D_h$. This proves (373), with no $k^2/(h\eta)$ term. Since $\beta_h' \to 1/4$, its mixed derivative has a fixed sign for a fixed sufficiently small $|k/\eta|$. The compact Morse chart has signature zero. Two-dimensional stationary phase at parameter

$|\eta|/h$, including the exact prefactor $1/h$, gives the order-minus-one principal and an order-minus-two remainder. The compact nonstationary part and long chart are included by further integrations. Differentiation takes place in these exact parameter charts, so it gives genuine symbol bounds.

The critical times have the form $r_1 T(s, r_1^2)$ and $r_1 V(s, r_1^2)$, with $T, V$ bounded away from zero. Thus the evaluated cutoff vanishes unless $|r_1| \le C\tau$. On every cutoff derivative, $|r_1|/\tau$ is bounded; this proves uniform principal $s$ and $r_1\partial_{r_1}$ bounds. The latter are precisely weighted Fourier derivative bounds. No such uniformity is needed for the order-minus-two constants.

For completeness, the graph estimate is elementary. On one frequency half-line, the phase difference at $\eta, \zeta$ has $s$ derivative of magnitude at least $c|\eta - \zeta|$, and its higher normalized derivatives are bounded. Two or more integrations in the compact $s$ variable bound the frequency-side $T^*T$ kernel by $C_N(1 + |\eta - \zeta|)^{-N}$. Schur's test proves the order-zero $L^2$ estimate. Absorb $|\eta|^{-1}$ or $|\eta|^{-2}$ into the input. One $s$ derivative of the output raises the symbol order by one because $\Psi_s = O(|\eta|)$. Interpolation of these weighted Hilbert-space estimates proves (375)–(376). The pairing uses $\widehat{K}(s, -\eta)$, a harmless reversal of frequency.

Finally choose the maximum time cap after fixing $C_0$ so that $C_0|\Theta_t| \le D_{\min}/2$. In the low-frequency region the original compact phase $(k/h)[D_h(t - s_1) - (\eta/k)\Theta]$ has a nonzero $t$ derivative. Arbitrarily many full-line integrations give $h^{N-1}k^{-N}$, including each prescribed $s$ derivative. This also retains the complete cosine product at the original time origin. ∎

**Lemma 63** (Fractional sampling and cap-smallness)**.** *For $1/2 < t < 1$, the entire upper-time component on the actual rows satisfies*

$$\|B_{U,\tau;k\ge K}\| \le C_t\left[\tau + (h/L)^t \tau^{1-t}\right] + C_\tau K^{-1} + C_{N,\tau} h^{N-1}. \tag{377}$$

*The input space is the common $L^2$ space. The actual output entries are*

$$\sqrt{h/L}\, T_{h,k,\tau} f(s_{k,n}).$$

*Bounded diagonal row weights are permitted.*

**Proof.** For a cell of length $\ell$, the one-dimensional $H^t$ embedding, after scaling, gives

$$\ell|g(x_0)|^2 \le C_t\left(\|g\|^2_{L^2(C)} + \ell^{2t}\iint_{C^2} \frac{|g(x) - g(y)|^2}{|x - y|^{1+2t}}\,\mathrm{d}x\,\mathrm{d}y\right).$$

The embedding follows from Fourier Cauchy–Schwarz because $\int (1 + \eta^2)^{-t}\,\mathrm{d}\eta < \infty$. Sum disjoint actual cells and use (363); writing $A = L/h$ gives

$$\frac{h}{L}\sum_n |g(s_{k,n})|^2 \le C_t k\big(\|g\|_2^2 + (kA)^{-2t}\|g\|^2_{H^t}\big).$$

For the principal term keep its common input frequency and support before summing $k$. At a fixed $\eta$ only $k \le C\tau|\eta|$ occur, so $\sum k \le C\tau^2\eta^2$ and $\sum k^{1-2t} \le C_t(\tau|\eta|)^{2-2t}$. Equations (375) and these sums give squared norm $C_t[\tau^2 + A^{-2t}\tau^{2-2t}]$. For the remainder restricted to $k \ge K$, the same calculation has an additional $|\eta|^{-2}$ and $|\eta| \ge cK$, giving squared norm $C_\tau K^{-2}$. The low-frequency kernel has input frequency length $O(k)$; its output $H^t$ norm is at most $C_{N,\tau}h^{N-1}k^{1/2-N}$. The remaining sum is $\sum k^{2-2N} < \infty$ for $N \ge 2$. Tonelli applies to these nonnegative finite-truncation bounds. The square-root triangle inequality proves (377). In particular no sum of unrelated alias norms, and no borderline $H^1$ harmonic sum, occurs. ∎

## G.4 Separated mixed normals, including the inner brake

**Lemma 64** (Separated buffered windows)**.** *Let $X, Y$ be separated compact logarithmic input intervals, both buffered above the positive inner brake on a fixed output interval. For a smooth compact frequency weight $\chi(k/K)$ away from zero, the complete mixed form satisfies*

$$\left| L \sum_{k,n} \chi(k/K)\rho(c_{k,n}/h)^2 R_{k,n}[a_{h,F}]\overline{R_{k,n}[a_{h,G}]} \right| \le C(K^{-1/2} + h/L)\|F\|_2\|G\|_2. \tag{378}$$

*For a fixed smooth finite-overlap dyadic exhaustion, its all-integer mixed normal converges in norm, as $h \downarrow 0$, to a compact operator.*

**Proof.** The phase geometry has two branches, so a globally injective canonical relation cannot be assumed. Put $z = \xi/s$, $v = (4z)^{-1/10}$ and

$$q(z) = -\partial_{\log z} S_0(z) = \frac{\pi}{\kappa v^{10}\sqrt{1 - v^8}}.$$

It has one minimum, at $v^8 = 5/7$. Suppose $\xi > \xi'$ and put $r_1 = \xi'/\xi \in (0, 1)$, uniformly away from one by separation. The equation $\partial_s[S_0(\xi/s) - S_0(\xi'/s)] = 0$ is exactly

$$u = \frac{1 - r_1^2}{r_1^{-4/5} - r_1^2}, \qquad s_*(\xi, \xi') = 4\xi u^{5/4}.$$

It has only one possible root. At that root the two arguments lie on opposite sides of the minimum, and

$$\partial_s^2(S_0(\xi/s) - S_0(\xi'/s)) = -\frac{zq'(z) - z'q'(z')}{s^2} < 0.$$

The compact separated geometry makes this Hessian uniformly nonzero. Finite-order convergence of the exact phases then gives a unique exact critical graph wherever the limiting root lies near the output support; a uniform nonstationary gap on its compact complement excludes extra roots. Smooth output cutoffs and a finite parameter partition handle roots entering or leaving that support.

Let $\Phi_h(x,y)$ be the reduced exact phase at the critical output. The envelope identities give $\Phi_{h,x} \le -c_0$ and $\Phi_{h,y} \ge c_0$ on each parameter rectangle. Output stationary phase gives the same-sign kernel $k^{-1/2}b_h(x,y)e^{ik\Phi_h(x,y)} + O(k^{-3/2})$. For opposite signs the output phase is $\pm(S_h(s,x) + S_h(s,y))$ and is nonstationary. The smooth integer sum is evaluated by Poisson summation:

$$\sum_{k\ge 1} \chi(k/K)k^{-1/2}e^{ik\Phi} = K^{1/2}\sum_{\ell\in\mathbb{Z}} \widehat{t^{-1/2}\chi(t)}\big(K(\Phi - 2\pi\ell)\big).$$

It is bounded by $C_N K^{1/2}(1 + K\,\mathrm{dist}(\Phi, 2\pi\mathbb{Z}))^{-N}$. Only finitely many aliases meet the compact phase range. Integrating in $y$ using $|\Phi_{h,y}| \ge c_0$ gives $CK^{-1/2}$, and integrating in $x$ gives the same bound. These two Schur estimates retain every alias; pointwise peak size alone would not suffice. The sum of the stationary remainders is also $O(K^{-1/2})$.

The exact product-cell comparison in the proof of Lemma 61, with the all-integer Bessel bounds on both inputs, costs $Ch/L$ over the entire sum. Endpoint pieces have actual Hilbert–Schmidt tails; higher stationary coefficients have tails from (370). Thus (378) concerns the complete rows. For the all-integer operator sum the continuum smooth bands geometrically, and pay the cell error only once over the whole sum. Finite-$k$ pieces converge in Hilbert–Schmidt norm by Proposition H.1. Their limit is compact, and the uniform tail makes the entire mixed limit compact. ∎

**Lemma 65** (Separated source crossing the inner brake)**.** *Fix a buffered target interval and a positive compact source interval separated from it. The source may cross the moving inner brake. With the same actual rows, smooth output weight, and fixed smooth dyadic exhaustion, the complete mixed normal converges in operator norm to a compact limit. It has the $H^{-1}$ compact-defect bound of Lemma 61.*

**Proof.** Remove the entire two-time upper component of Lemma 62 before subdividing the angular source. Its high-$k$ mixed norm is bounded by (377) times the uniform complete target norm. A row-dependent angular strip must not be inserted into that estimate.

For fixed $\tau$, the remaining source near the upper endpoint is bounded away from both upper turns. Indeed the difference of the squared upper times is $O(h)$; if either time is outside the removed cap then the other is outside a smaller fixed cap for sufficiently small $h$. The remaining radial support satisfies $v \le 1 - c_\tau$. Choose a separate fixed lower cap and retain its entire integer history as in Lemma 59. On $\xi \le s/4 + \delta_\tau$, choose $\delta_\tau > 0$ so small that the possible stationary point is outside the residual radial support. This follows explicitly from

$$\Delta/h = 4s - 16\xi v^{10} + O(h) \ge 4s_{\min}[1 - (1 - c_\tau)^{10}] - 16\delta_\tau + O(h) > 0.$$

On the whole bounded source product the lower cap can also be fixed with $v_\ell^{10} \le s_{\min}/(16\xi_{\max})$, making its divided gap positive. Full reflected lower integration and regular

nonstationary integration give $|Lb_{\mathrm{res}}| \le C_{M,\tau} k^{-M}$ on the nonstationary part. Counting actual denominators gives squared high-pass analysis norm $C_{M,\tau} \sum_{k \ge K} k^{1-2M}$.

The rest of the compact source/output product has a finite cover by buffered rectangles. Subdivide only this residual using smooth product cutoffs. Its regular radial part has exactly the stationary phases of Lemma 60, with modified smooth amplitudes. If a critical point lies in the removed time cap, its amplitude is zero; the phase itself remains smoothly defined. Therefore the Bessel, differentiated Bessel, and separated-phase arguments repeat with constants depending on fixed $\tau$. The lower cap is still estimated pointwise, without an output derivative. The residual mixed tail is $C_\tau K^{-1/2} + C_\tau h/L$.

Combining all parts gives

$$\|C_{h,\ge K}\| \le C_t[\tau + (h/L)^t \tau^{1-t}] + C_\tau K^{-1} + C_{N,\tau} h^{N-1} + C_\tau K^{-1/2} + C_\tau h/L.$$

For fixed $\tau$ the full source analysis is bounded: use (377) at $K = 1$, the residual Bessel bound, and the nonstationary Hilbert–Schmidt bound. Thus finite row sums define the same bounded mixed operator by analysis and synthesis. Choose $\tau$ first, then $K$, then $h$. Every fixed finite-$k$ complete piece converges in Hilbert–Schmidt norm, including its logarithmic brake. Its limit is compact, and the displayed tail proves norm convergence of the full mixed operator. Smooth finite-rank approximation gives the $H^{-1}$ form. ∎

## G.5 The two unbounded relative-scale tails

**Lemma 66** (Small and large original-input scales)**.** *On a fixed positive output collar there are constants $\varepsilon_0, \theta_0, M_0, h_0 > 0$ such that the complete actual analysis obeys*

$$\|B_h 1_{(0,\varepsilon h)}\|^2 \le C\varepsilon, \qquad 0 < \varepsilon \le \varepsilon_0, \tag{379}$$

$$\|B_h 1_{(Mh,hZ')}\|^2 \le C\left[M^{-6/5} + (h/L)^2 (Z')^{4/5}\right], \qquad M_0 \le M \le Z' \le \theta_0/h. \tag{380}$$

*The input norm is the original angular $L^2$ norm. In particular the full large tail has squared bound $C[M^{-6/5} + h^{6/5}/L^2]$.*

**Proof.** For the small scale the exact gap (366) is $\Delta/h = 4s - 16\epsilon\xi v^{10} + O(h) > 0$ on $0 \le \xi \le \varepsilon_0 < s_{\min}/8$. The correct first-turn and reflection calculation in Lemma 59, together with the regular middle integrations, proves $|Lb_{k,n}(h\xi)| \le C_M k^{-M}$ for both physical branches separately. With $M = 2$, the row norm squared on an angular interval of length $\varepsilon h$ is at most $C\varepsilon h/(L^2 k^4)$. Counting $CkL/h$ denominators and multiplying by $L$ gives (379).

For the large scale put

$$d(\theta) = \frac{\lambda_+^2(\lambda_+^8 - 1)}{1 + \lambda_+^2}, \qquad V = (c/d(\theta))^{1/10}, \qquad \varrho = V^8.$$

Here $d(\theta)/\theta \to 4$, and the exact positive stationary point is $v = V$. For $\theta \geq M_0 h$, $V$ is uniformly small while the lower root is much smaller, since $\ell_c/V \leq C\sqrt{d(\theta)}V^4$. Set $v = Vw$ near the critical point. The normalized momenta are

$$\widetilde{P}^2 = 2[\lambda^2 w^2 + \varrho(d - \lambda^{10} w^{10})], \quad \widetilde{Q}^2 = 2[w^2 - \varrho(d + w^{10})].$$

They are uniformly positive on a fixed $w$ chart. The centered difference phase is constructed by integrating

$$\Phi_w = \frac{2(1+\lambda^2)(1-w^{10})}{\widetilde{P}\widetilde{Q}(\widetilde{P} + \lambda\widetilde{Q})}, \qquad \Phi_{ww}(1) = -\frac{10(1+\lambda^2)}{\lambda^2 \widetilde{Q}(1)^3} < 0.$$

The radial amplitude is $V^{10} w^{11}/(\widetilde{P}\widetilde{Q})$ and its oscillatory parameter is $k\gamma/V^2$, where $\gamma = 4\pi c/\Delta T$. Stationary phase therefore gives at actual nodes

$$Lb_{k,n}(\theta) = k^{-1/2} V^{11} \sum_{\epsilon=\pm 1} A_\epsilon(c, \theta; h) e^{i\epsilon k S(c,\theta)} + E_{k,n}(\theta), \qquad |E_{k,n}| \leq C k^{-3/2} V^{13}. \tag{381}$$

The leading amplitudes and one $s$ derivative are bounded; no derivative of $E_{k,n}$ is required.

Here are the uniform endpoint and regular-complement checks behind that remainder. On a lower cap $v \leq 2aV$, the reflected recurrence from Lemma 59 has degree $9 + 2j$ and support length $O(V)$, so its $L^1$ norm is $O(V^{10+2j})$. This gives $C_j k^{-j} V^{10+2j}$ with the exact $(-1)^k$ carrier retained. At the upper turn the other time satisfies $s(t)^2 = t^2 + \theta d_u(\theta, t^2, c/\theta)$, with a positive uniform $d_u$. The reflected recurrence gives $C_j(\omega\theta)^{-j} \leq C_j k^{-j} V^{10j}$. On regular dyads $v \asymp a \geq BV$, the difference derivative has size $\theta a^8$ after scaling the dyad, and the amplitude has size $a^{10}$. The bound is $C_j k^{-j} V^{10j} a^{10-8j}$; its dyadic sum is dominated by $a \asymp V$ for $j \geq 2$. On $v \asymp V$ outside the saddle the parameter is $k/V^2$. Both sum phases have nonzero normalized derivative and give $C_j k^{-j} c^j$. Thus every omitted nonstationary part is $C_j k^{-j} V^{10+2j}$, proving (381).

Write $z = \theta/h$, so $V^{10} \asymp z^{-1}$. The actual remainder has squared Hilbert–Schmidt norm at most $C\sum_{k\geq 1} k^{-2} \int_M^{Z'} z^{-13/5}\,\mathrm{d}z \leq CM^{-8/5}$. For the leading term, the exact envelope derivative is

$$\partial_z S_h = -\frac{\gamma(c)}{s}\frac{\sec\theta}{\widetilde{Q}(1)}, \qquad -C \leq \partial_z S_h \leq -c < 0, \quad |\partial_s S_h| \leq Cz.$$

To verify the last bound uniformly as the saddle approaches zero, split $\tau_e(q)$ at a fixed $q_0 > V$ and differentiate the integral below $q_0$. There $p_e(q) \asymp q$, so $|\partial_e \tau_e(q)| \leq Cq^{-2}$. The difference of the two times is $O(\theta + c/V^2) = O(\theta)$; its $c$ derivative at the critical point is $O(V^{-2})$, with no moving-critical-point term. The conormal bounds on $\gamma$ give the claimed derivative. Consequently, with $B_\epsilon = V^{11} A_\epsilon$,

$$|B_\epsilon| + |\partial_s B_\epsilon| \leq Cz^{-11/10}, \qquad |S_{h,s} B_\epsilon| \leq Cz^{-1/10}.$$

The inverse images of integer phase translates are uniformly separated. In (365) their two relevant sums are bounded by $CM^{-6/5}$ and $C(Z')^{4/5}$, since $\sum z_\ell^{-11/5} \leq CM^{-6/5}$ and

$\sum_{z_\ell \le Z'} z_\ell^{-1/5} \le C(Z')^{4/5}$. Thus the undifferentiated leading Bessel sum has the first bound and the $k^{-1}$ differentiated sum the second. Applying exact actual-cell sampling proves (380). The scale-resolved $Z'$ term is retained before any subsequent bin summation. ∎

For later use, let $h_j = e^{j\ell}$ be a logarithmic lattice and let the source lie in a fixed-width log bin at scale $h_i$. After composing with the buffered target, Lemma 66 gives the analysis bounds

$$t_{ji} \le Ce^{(i-j)\ell/2}, \qquad i \ll j, \tag{382}$$

$$t_{ji} \le C\left[e^{-3(i-j)\ell/5} + \frac{e^{3j\ell/5+2i\ell/5}}{1+|j|}\right], \qquad i \gg j. \tag{383}$$

These are genuine all-integer estimates, not repeated finite-channel bounds. The second term in (383) is the actual-cell correction and cannot be erased before a weighted matrix sum.

## G.6 The remaining physical source and its common row exhaustion

The off-cap estimate is a mixed estimate. In particular a source analysis bound over all integers is neither asserted nor needed on the entire physical off-cap interval. Fix a smooth output weight $W_h(k,n)$ supported in a compact $c/h$ collar and satisfying, at fixed $k \asymp K$,

$$|\partial_n^j W_h(k,n)| \le C_j N^{-j}, \qquad N = KL/h.$$

A fixed smooth function of $c/h$, times a bounded $k$-only weight, has this property. Fix one smooth nonnegative finite-overlap dyadic partition in $k$ and its increasing partial sums. All mixed limits in this subsection use that same partition and row weight.

**Lemma 67** (Complete physical off-cap estimate)**.** *Let $a$ have finite weighted norm*

$$\|a\|_W^2 = \int_0^{\pi/2} (1+\delta^{-4})|a(\phi)|^2 \,\mathrm{d}\phi, \qquad \delta = \pi/2 - \phi.$$

*Fix a physical horizontal cap $0 < \theta < \theta_0$, and a buffered target at angular scale $h$. For $a_{\mathrm{off}} = 1_{(\theta_0,\pi/2)}a$, the complete actual mixed sum converges in target $L^2$ and satisfies*

$$\left| L \sum_{k,n} W_h(k,n)\rho(c_{k,n}/h)^2 R_{k,n}[a_{\mathrm{off}}]\overline{R_{k,n}[v_h]} \right| \le Ch^{3/5}\|a\|_W \|v_h\|_2. \tag{384}$$

*For a fixed finite set $F$ of positive $k$,*

$$L \sum_{k\in F,\, c_{k,n}/h \in I} |R_{k,n}[a_{\mathrm{off}}]|^2 \le C_F h^{6/5}\|a\|_W^2, \tag{385}$$

$$\sum_{k\in F,\, 0<c_{k,n}<d} c_{k,n}^{-\gamma}\log(1/c_{k,n})|R_{k,n}[a_{\mathrm{off}}]|^2 \le C_{F,\gamma} d^{6/5-\gamma}\|a\|_W^2, \quad 0 \le \gamma < 6/5. \tag{386}$$

*Thus on the Peano energy closure, where the vertical coercivity theorem gives* $\|a\|_W^2 \le CE(a)$*, these are the exponents* $3/5$ *for the mixed norm and* $6/5$ *for the squared fixed-frequency data norm.*

**Proof.** We give the source estimates and their summation, keeping the three physical regions distinct. First

$$\|1_{\delta<d}a\|_2 \le d^2\|a\|_W. \tag{387}$$

This is a tail of $a$, not an assertion about the energy of its cutoff. Fix a small physical vertical neighborhood and a large fixed $M$. Its cap $\delta < M\sqrt{h}$, its middle $M\sqrt{h} < \delta < \delta_V$, and the remaining fixed core cover the entire off-cap interval.

**The fixed rescaled vertical cap.** Put $\mu = \tan(\delta/2)$ and $D = q_+(c) - q_\ell(-c)/\mu$. On the visible positive branch use one smooth partition $\chi_{\rm in}(L^2D) + \chi_{\rm out}(L^2D) = 1$; the first is supported in $0 \le D \le 2C_0/L^2$. Both positive parts vanish on the invisible side. The complete negative term is retained once on the entire cap.

Here is the joint analysis bound on the positive collar. Let $U_c(t)$ start at the positive upper turn and $V_c(s_1)$ at the negative lower turn. The exact constraint $V_c(s_1) = \mu U_c(t)$ has an even solution $c = C_\phi(t, s_1)$ near the brake $c_b(\phi)$. Since $\partial_c(V_c - \mu U_c) \asymp h^{-1/2}$, its expansion is

$$c_b - C_\phi = c_b(8t^2 + 2s_1^2) + O(c_b^2(t^2 + s_1^2) + c_b(t^2 + s_1^2)^2).$$

The Jacobian identity

$$\frac{\mathrm{d}r\,\mathrm{d}u}{p_c(u)p_{-c}(\mu u)} = \frac{r'(c)}{\partial_c(V_c - \mu U_c)}\,\mathrm{d}t\,\mathrm{d}s_1$$

cancels both turning square roots. Actual integer parity supplies $(-1)^n\cos(n\Omega_v s_1)$ at the lower turn. Reflect both times and Fourier transform in the ratio $r$. With $u_b = (r(c_b) - 2)L/h$, $z = L(t, s_1)$, $\lambda = \xi h/L$, the exact transform has the form

$$C(-1)^n\frac{\sqrt{h}}{L^4}e^{-2i\xi - iu_b\lambda}\int A_h(u_b, z)e^{i(\lambda/L^2)q_h(u_b,z)+i(n/L^2)\ell_h(u_b,z)}\,\mathrm{d}z,$$

where the amplitude has fixed compact support and uniform finite derivatives, and

$$q_h \to u_b(8z_1^2 + 2z_2^2), \qquad \ell_h \to 2\pi\sqrt{2}(z_1 + z_2)$$

in finite $C^j$ norms. On $|\lambda| \le c_0 n$, full-plane integration by parts gives arbitrarily high powers of $(n/L^2)^{-1}$. On $|\lambda| \ge c_0 n$, the phase $q_h + (n/\lambda)\ell_h$ has one uniformly nondegenerate critical point on a buffered parameter rectangle. Morse reduction, including the nonstationary complement in its exact symbol, yields the common input transform

$$\widehat{H}_n^{\rm hi}(\xi) = (-1)^n e^{-2i\xi}\frac{h^{-1/4}}{L\xi}\mathcal{U}_{n,h}F(\xi h/L), \quad \mathcal{U}_{n,h}F(\lambda) = \int e^{-iu_b\lambda + i\lambda L^{-2}\mathcal{W}_h(u_b,n/\lambda)}b_{n,h}F\,\mathrm{d}u_b. \tag{388}$$

Here $F = a(\phi_h(u_b))\sqrt{|\phi_h'(u_b)|}$ is the same input for all $n$, $|\phi_h'| = \sqrt{h}\, j_h$, and $\mathcal{W}_h, b_{n,h}$ have uniform finite derivatives, with weighted derivatives in $\lambda$ for the symbol. The critical value tends to $-5\pi^2(n/\lambda)^2/(4u_b)$. This construction is valid also when the critical point is outside the smaller amplitude support, and through $n/\lambda = 0$.

For clarity the joint bound implicit in (388) can be proved without assuming orthogonality of aliases. In its quadratic form, factor the difference phase as

$$-(u_b - v_b)\eta, \quad \eta = \lambda[1 - L^{-2}\mathcal{D}(u_b, v_b, n/\lambda)], \quad \mathcal{D} = \int_0^1 \mathcal{W}_{h,u}(v_b + t(u_b - v_b), \nu)\,\mathrm{d}t.$$

Both $\eta/\lambda$ and $\partial_\lambda \eta$ lie between $1/2$ and $3/2$ for small $h$. Its inverse is $\lambda = \eta H(u_b, v_b, n/\eta)$ with bounded finite derivatives. After this change, the quadratic form

$$\sum_n n^{-1/2} \int |\lambda|^{-1/2} |\mathcal{U}_{n,h} F(\lambda)|^2 \,\mathrm{d}\lambda$$

has a compact double symbol with mixed derivatives through two in each input variable uniformly bounded, because its support has $n \le C|\eta|$ and

$$|\eta|^{-1/2} \sum_{n \le C|\eta|} n^{-1/2} \le C.$$

Fourier-expanding that double symbol in the two compact spatial variables gives coefficients bounded by $C(1 + j^2)^{-1}(1 + \ell^2)^{-1}$. Each coefficient is a bounded Fourier multiplier between modulations; absolute summation and Plancherel prove the joint bound by $C\|F\|_2^2$. Smooth high cutoffs give legitimate zero extensions; finite $n$ and frequency truncations justify the calculation before monotone passage.

At the actual samples $r = 2 + 2k/n$, the reciprocal cell has length $\pi n$. Weighted alias Cauchy–Schwarz gives

$$\left| \sum_p \xi_p^{-1} \mathcal{U}_n F(\xi_p h/L) \right|^2 \le \left( \sum_{|\xi_p| \ge cnL/h} |\xi_p|^{-3/2} \right) \left( \sum_p |\xi_p|^{-1/2} |\mathcal{U}_n F(\xi_p h/L)|^2 \right).$$

The first factor is at most $C/[n\sqrt{nL/h}]$. Tiling the frequency line and using the preceding joint bound gives

$$\|B_{P,\mathrm{in}} a\|_{\ell^2}^2 \le C h^{-1/2} L^{-1} \|1_{\delta < M\sqrt{h}} a\|_2^2.$$

The low transform gives a convergent sum after sufficiently many integrations, since every relevant $n \ge cL/h$. The complete negative cap, reflected at both turns with its nonstationary phase, gives squared analysis norm $Ch^{5/2}\|1_{\delta < M\sqrt{h}} a\|_2^2$.

On the positive exterior use the exact circle substitution $u = b + D\sin^2 t$, $b = q_\ell(-c)/\mu$. Since $b$ stays in a positive compact interval, its two critical points are uniformly nondegenerate, also at $D = 0$ after dividing the phase by $\sqrt{D}$. The exact two-wave symbols are

$$P_{k,n} = \frac{D^{-1/4}}{L\sqrt{k}} \sum_{\epsilon = \pm 1} e^{i\epsilon n \psi} A_\epsilon, \qquad \psi = \pi - \Omega_v \sqrt{D}, F_0(c, D),$$

with smooth positive $F_0$. At fixed $k$, conormal inversion gives $\partial_n c = -(r-2)/(nr')$. Consequently $|\partial_n(n\psi) - \pi| \le C/(L\sqrt{D}) + C/L$ and the $j$th symbol derivative is bounded by $C_j L^{1/2}(L^2/N)^j$ on $D \ge C_0/L^2$. For the middle-plus-upper target charts, their time phases and normalized $n$ derivatives are $O(1/L)$. Choose $C_0$ large first; the source-target phase increments then have distance at least $\pi/2$ from $2\pi\mathbb{Z}$. Repeated discrete summation by parts, with the smooth compact $n$ window, gives a summable band norm. The lower target is kept separately with its pointwise reflected estimate. More explicitly, two discrete differences cost $(L^2/N)^2$. The target radial majorant has integral $CL$, so the middle-plus-upper pointwise mixed kernel is at most $ChL^{7/2}K^{-1/2}$ after summing $O(N)$ denominators and $O(K)$ numerators. Its target and source lengths are $O(h)$ and $O(\sqrt{h})$, which gives $Ch^{7/4}L^{7/2}K^{-1/2}$ in operator norm. The full circle stationary bound is the uniform bound by $\min(1, (\omega\sqrt{D})^{-1/2})$ before division into its two waves. Integrating its square in the cap, including the central interval $\omega\sqrt{D} \le 1$, gives $\|Lb_{P,k,n}\|_{L^2(V_h)} \le Ch^{1/4}k^{-1/2}$. Against the lower target row norm $Ch^{1/2}L^{-1}k^{-2}$, the exact count $CkL/h$ gives a band norm $Ch^{-1/4}K^{-1/2}$. Together these estimates give

$$\|C_{\mathrm{out,K}}\| \le Ch^{-1/4}K^{-1/2}.$$

The upper and middle part in fact has the smaller bound $Ch^{7/4}L^{7/2}K^{-1/2}$; the complete lower target determines the displayed coarser scale. The circle derivative bounds are uniform on the entire fixed rescaled cap, since its visible lower endpoint stays positive. The dyadic $K^{-1/2}$ sum is absolutely convergent. Compose the positive collar and complete negative analyses with the complete bounded target, and add this exterior series. The result is the entire cap mixed bound

$$\|C_{V,h}a\|_2 \le Ch^{-1/4}\|1_{\delta<M\sqrt{h}}a\|_2 \le Ch^{3/4}\|a\|_W.$$

The positive partition is used only once and the complete negative source is subtracted only once.

**The middle source and the fixed physical core.** For $M\sqrt{h} < \delta < \delta_V$ put $\varepsilon = c/\mu^2$, $a_*^{\mathrm{sc}} = \varepsilon^{1/10}$ and $\varrho = \varepsilon^{4/5}$. To avoid confusing this scale with an angular reference, it is used only in this paragraph. The positive coarea integral is exactly

$$\frac{(1+\mu^2)^6}{4T_vU}\int_{q_\ell(-c)/\mu}^{q_+(c)} \frac{u^{11}\cos(\omega\tau_c(u))\cos(\omega\tau_{-c}(\mu u))}{p_c(u)p_{-c}(\mu u)}\,\mathrm{d}u.$$

Under $u = a_*^{\mathrm{sc}}w$ its two normalized squared momenta are $F = 1 + \varrho(\mu^2/w^2 - w^8)$ and $Q = 1 - \varrho(w^{-2} + \mu^8 w^8)$. For the difference $G = \tau_{-c}(\mu u) - \tau_c(u)$,

$$\frac{G_w}{\varrho} = \frac{(1-\mu^8)w^{10} - (1+\mu^2)}{\sqrt{2}w^3\sqrt{FQ}(\sqrt{F}+\sqrt{Q})}.$$

Its unique critical point is $w_* = [(1+\mu^2)/(1-\mu^8)]^{1/10}$, with positive nonzero second derivative $(5/\sqrt{2})(1-\mu^8)w_*^6 Q(w_*)^{-3/2}$. The large parameter is $\lambda_* = \omega\varrho$. The exact $1/4$ cosine factors, radial measure, and stationary factor give

$$LP = k^{-1/2}\varepsilon^{11/10}\sum_{\epsilon=\pm1} e^{i\epsilon n\psi}A_\epsilon + L\mathcal{R},$$
$$|L\mathcal{R}| \le Ck^{-3/2}\varepsilon^{11/10}h^{1/5}, \qquad |LN| \le Ch^{3/10}k^{-2}\varepsilon^{11/10}.$$

The complete remainders follow by the same signed-time recurrences: at the lower turn use $u = \sqrt{\varepsilon}Y_c(s_1)$, where $Y_c$ is even and solves $Y_c'' = 2Y_c - 10c^4Y_c^9$; the difference of the two squared time speeds has a fixed positive numerator $1+\mu^2+O(\varepsilon^4 Y_c^{10})$. At the upper turn reflect the even positive trajectory; on a fixed upper cap the other speed has a strict gap from one. On intervening dyads the difference derivative has sizes $\varepsilon/u^2$ below the critical scale and $u^8$ above it, while the sum derivative stays nonzero. Their geometric sums are dominated by the critical scale. The negative physical branch has no critical point and two integrations give the stated bound. This treats the entire histories, without differentiating either remainder.

For the fixed core, $\mu$ lies in a compact subset of $(0,1)$. The same normalized chart remains uniform because $w_*$ is bounded and $1-\mu^8 > 0$; the positive upper endpoint is indeed $q_+(c)$, since $q_+(-c)/\mu > q_+(c)$ there for small $c$. Both endpoint gaps are strict on this compact set. The resulting complete pointwise bounds are

$$|LP| \le C[h^{11/10}k^{-1/2} + h^{13/10}k^{-3/2}], \qquad |LN| \le Ch^{7/5}k^{-2}.$$

Thus this use on the core is checked directly and does not extrapolate a small-$\mu$ estimate beyond its domain.

The weighted source integral has the decisive improved exponent

$$\int_{M\sqrt{h}<\delta<\delta_V} (h/\mu^2)^{11/10}|a|\,\mathrm{d}\phi \le Ch^{11/10}\left(\int_0^{\mu_0}\mu^{-2/5}\,\mathrm{d}\mu\right)^{1/2}\|a\|_W \le Ch^{11/10}\|a\|_W.$$

Consequently the source leading, remainder, and negative pairings have sizes, respectively,

$$L^{-1}h^{11/10}k^{-1/2}\|a\|_W, \quad L^{-1}h^{13/10}k^{-3/2}\|a\|_W, \quad L^{-1}h^{7/5}k^{-2}\|a\|_W.$$

Actual counting and complete target synthesis give mixed remainder bounds $Ch^{4/5}\|a\|_W$ and $Ch^{9/10}\|a\|_W$. The complete target endpoint row norm is at most $Ch^{1/2}L^{-1}k^{-2}$; its individual-row sum against the leading source is $Ch^{3/5}\sum k^{-3/2}\|a\|_W$.

It remains to pair the regular difference target with the leading source. The source critical phase has the exact subtraction

$$G_* = \frac{\log(1/\mu)}{\sqrt{2}} + O_{\mathrm{con}}(\varepsilon^{4/5}),$$

obtained by splitting the scalar times at a fixed $q_0$ and integrating the subtracted momentum $1/p_e - 1/(\sqrt{2}q)$. At fixed $k$ this implies

$$\partial_n(n\psi) = \frac{2\pi \log(1/\mu)}{L} + O(L^{-2} + \varepsilon^{4/5}/L).$$

The difference target on a regular radial compact has phase and conormal derivatives of size $O(h/L)$. Thus its integer increment has a nonzero gap from $2\pi\mathbb{Z}$, of size $c\log(1/\mu)/L$ on the middle and $c/L$ on the fixed core. For $q(n) = (e^{i(\Phi(n+1)-\Phi(n))} - 1)^{-1}$, $|\partial_n^j q| \le C_j L[\log(1/\mu)]^{-1} N^{-j}$. Two transfers of a discrete difference, with no boundary terms from the smooth $n$ window, give the band kernel bound

$$|C_{DQ,K}(\theta,\phi)| \le ChK^{-1/2}(h/\mu^2)^{11/10}[\log(1/\mu)]^{-2}.$$

After the displayed weighted source integration and target length $O(h)$, its norm is $Ch^{13/5}K^{-1/2}\|a\|_W$. On the fixed core the same calculation uses only its original $L^2$ norm. The target sum phases are not assigned this gap: on the regular radial compact their exact derivative is $-\lambda/P - 1/Q$, so their actual $k/h$ parameter gives the pointwise bound $C_J(h/k)^J/L$. Their individual-row sum is $O(h^{13/5})$ instead. This is why both target endpoints, both differences, and both sums have all been counted. The complete middle and core bounds are therefore $C(h^{3/5} + h^{4/5} + h^{9/10} + h^{13/5})\|a\|_W$.

Every finite row sum has exactly the decomposition just used. The two source error arrays converge in $\ell^2$; the endpoint and target-sum terms converge absolutely over individual rows; the difference-target terms and positive-cap exterior converge absolutely over the fixed smooth bands. Their recombination proves (384) with the stated common exhaustion.

For fixed finite $F$, the complete cap row bound $\|1_{\delta<M\sqrt{h}} L b_{k,n}\|_2^2 \le C\sqrt{h}/k$ and (387) give its squared data bound $C_F h^{3/2}\|a\|_W^2$. The middle and core pointwise bounds give $C_F h^{6/5}\|a\|_W^2$ by the exact denominator count. Add the three pairings before squaring, using $|u+v+w|^2 \le 3(|u|^2+|v|^2+|w|^2)$. This proves (385); the factor depending on $|F|$ is not summable over all $k$. Finally partition the actual nodes into $2^{-j-1}d < c \le 2^{-j}d$. Applying the band estimate and summing $\sum_j (2^{-j}d)^{6/5-\gamma}$ proves (386). Every node is included exactly once. ∎

## G.7 An explicit principal coefficient and the remaining effective gap

**Lemma 68** (An elementary numerical scalar collar)**.** *For $c \le e^{-512}$, writing $L_c = \log(1/c)$, one has*

$$\tfrac{3}{2} \le \kappa < 3, \quad \tfrac{1}{2} < T_0 < 4, \quad \tfrac{3}{4} \le \Delta T/c < 4, \quad \frac{L_c}{2\sqrt{2}} \le T_v \le L_c, \quad T_0/2 \le U \le 3T_0/2,$$

*and $3/4 \le \kappa r' c^2/[L_c(r-2)^2] \le 5/4$.*

**Proof.** Here are explicit regularized estimates proving one common collar. Set $w_0 = 1/8$ and initially $|e| \le 2^{-16}$. In the saddle inverse write $q_s' = 1 + w^8 G_p(w^2)$ and $q_s^2 q_s' = w^2[1 + w^8 G_m(w^2)]$. For $u = q_s^8 \le 2^{-16}$, their exact formulas are

$$G_p = \frac{5 - (1+\sqrt{1-u})^{-1}}{(1-5u)(1-u)^4}, \qquad G_m = \frac{(1+\sqrt{1-u})^{-1} + 5\sqrt{1-u}}{(1-5u)(1-u)^{9/2}}.$$

They give $|G_p| \le 6$, $|G_m| \le 7$, $|G_p'|, |G_m'| \le 1$ and $|G_p''| \le 16$, where primes are in $w^2$. For example $x = w^2 = t - t^5$, $t = q_s^2$, gives $|u_x| \le 1/512$, $|u_{xx}| \le 1/8$; the $u$ derivatives of $G_p$ are bounded by 102 and 4096. These rational bounds prove the claimed analytic derivative estimates even at the origin.

The upper root $Q(e)$ lies in $[99/100, 101/100]$, with $|Q'| \le 1/7$ and $|Q''| \le 1/3$, by its polynomial equation. Factor $e + q^2 - q^{10} = (Q - q)R(Q, q)$. Here

$$R = (Q+q)(Q^8 - 1) + \sum_{j=2}^{9} Q^{9-j} q^j.$$

For $q \ge 1/2$, retaining the $Q^7 q^2$ term gives $R \ge 709/10000 > 1/16$. Below $1/2$, $p_e(q) \ge q$ on the outer interval, and $R \ge 1/256$. The two pieces of the outer period are bounded by $\log 4 < 2$ and $2\sqrt{8}\sqrt{51/100} < 5$, so $|J_{\mathrm{out}}| \le 7$. Use $q = Q - (Q - q_s(w_0))t^2$ to obtain the exact fixed-domain formula

$$J_{\mathrm{out}}(e) = \sqrt{2(Q - q_s(w_0))} \int_0^1 R(Q, q(t))^{-1/2} \, \mathrm{d}t.$$

The directional $Q$ derivatives of $R$ are at most 128 and 1024; thus $|R_e| \le 32$, $|R_{ee}| \le 128$. Differentiating the fixed-domain formula gives $|J_{\mathrm{out}}'| \le 2^{18}$ and $|J_{\mathrm{out}}''| \le 2^{32}$. Inserting $q(t)^2$ gives outer moment derivative at most $2^{20}$.

In the negative saddle term put $a = w_0^2$, $l = \sqrt{a - c}$ and $H(x) = x^{7/2} G_p(x)$. Its exact derivatives are

$$N_-' = \int_0^l H'(z^2 + c) \, \mathrm{d}z - \frac{H(a)}{2l}, \qquad N_-'' = \int_0^l H''(z^2 + c) \, \mathrm{d}z - \frac{H'(a)}{2l} - \frac{H(a)}{4l^3}.$$

The preceding bounds give $|N_-'|, |N_-''| < 1$; both moving-endpoint terms are included. The elementary logarithm in $T_v$ has derivative at most 64, and the elementary difference in $I_+ - I_-$ has second derivative at most $2^{15}$. Consequently

$$T_v = L_c/\sqrt{2} + b_v, \quad |b_v| \le 19, \quad |b_v'| \le 2^{20}, \quad |(I_+ - I_-)''| \le 2^{34}.$$

For $d(c) = \Delta T/c$, the exact identity $c(I_+ - I_-)' - (I_+ - I_-) = \int_0^c t(I_+ - I_-)''(t) \, \mathrm{d}t$ gives $|d - \kappa/\sqrt{2}| \le 2^{35} c$ and $|d'| \le 2^{35}$. The moment identities $U = 4J_+ + 2rJ_-$, $J_\pm(0) = T_0/8$, together with the elementary saddle moment primitives, give $|J_\pm - T_0/8| \le 2^{21} c(1 + L_c)$ and $|U - T_0| \le 2^{25} c(1 + L_c)$ when $r \le 3$. The beta integral gives $3/2 \le \kappa < 3$; the physical period formula $T_0 = 2^{-1/2} \int_{\mathbb{R}} \operatorname{sech}(z)^{1/2} \, \mathrm{d}z$ gives $1/2 < T_0 < 4$.

For $c \le 2^{-37}$ one already has $3/4 \le d(c) < 4$. For $L_c \ge 64$, the period lower bound gives $r \le 3$, and $2^{25}e^{-64}(1+64) < 1/4 \le T_0/2$ proves the moment bounds. Finally differentiate the signed counting function before taking absolute values. Its factor is

$$\widetilde{\rho}(c) = \frac{\kappa}{L_c d(c)}[T_v - cT_v' + cT_v d'/d].$$

Subtracting one and using the displayed estimates gives, uniformly on $0 < c \le d_*$,

$$|\widetilde{\rho} - 1| \le \frac{2^{35}d_*}{3/4} + \frac{3(1/\sqrt{2} + 19 + d_* 2^{20})}{(3/4)\log(1/d_*)} + \frac{3d_* 2^{35}(1/\sqrt{2} + 19/\log(1/d_*))}{(3/4)^2} \le \frac{128}{\log(1/d_*)}.$$

The last inequality holds for $\log(1/d_*) \ge 64$ using monotonicity of $Le^{-L}$. At 512 it is at most $1/4$, which proves the counting sign and its asserted interval simultaneously. This factor equals $\kappa r' c^2/[L_c(r-2)^2]$ once the sign is known. ∎

**Proposition G.1** (Explicit finite-scale principal bounds)**.** *Let $X = [-1/64, 1/64]$, $I = [1, 17/16]$, and $p(x) = e^{3x/8}$. Choose $0 \le \rho \le 1$ smooth, supported in $I$, equal to one on $[65/64, 67/64]$, and a radial cutoff equal to one on $[1/2, 15/16]$. For every $0 < h \le e^{-513}$,*

$$2^{-27} < p(x)^2 D_h(x) < 2^{15}.$$

*The two-sign continuum leading analysis multiplied by $p$ has norm less than 256. These bounds concern the exact principal expression; they do not bound the complete sampled normal by the same constants.*

**Proof.** Put $\theta = he^x$, $c = hs$, $\lambda = \sec\theta + \tan\theta$ and $Q = p_{-c}(v_h)$. At the critical point $P = \lambda Q$,

$$H_h = |\Phi_h''(v_h)| = \frac{40\pi(\lambda^8 - 1)v_h^9}{\Delta T Q^3}, \qquad q_h = -S_{h,x} = \frac{4\pi\theta\sec\theta, v_h}{\Delta T Q}.$$

The exact leading amplitude including the exponential factor $1/4$ is

$$A_h = e^{x/2}\frac{L\sec\theta(1+\lambda^2)^5 v_h^{11}}{8T_v U Q^2}\sqrt{\frac{2\pi}{H_h}}.$$

Substitution into (371) cancels $Q$, $\Delta T$ and $T_v$ exactly, leaving

$$D_h(x) = \int_I \rho(s)^2 \frac{hLr'(c)\sec\theta(1+\lambda^2)^{10}v_h^{12}}{1280\,U(c)^2(\lambda^8 - 1)}\,\mathrm{d}s. \tag{389}$$

Lemma 68 gives on $c \le e^{-512}$

$$\tfrac{3}{2} \le \kappa < 3, \quad \tfrac{1}{2} < T_0 < 4, \quad \tfrac{3}{4} \le \Delta T/c < 4, \quad \frac{L_c}{2\sqrt{2}} \le T_v \le L_c, \quad T_0/2 \le U \le 3T_0/2,$$

and the exact counting factor $\kappa r' c^2/[L_c(r-2)^2]$ lies in $[3/4, 5/4]$. Since $L/L_c < 2$, they imply $1/8 < Lr' < 512$ and $1/4 < U < 8$. Also $1 \le \lambda \le 33/32$ and $8\theta \le \lambda^8 - 1 < 16\theta$.

The exact critical equation gives $1/32 < v_h^{10} < 17/63 < 1/3$, hence $1/2 < v_h < 15/16$ and $v_h^{12} > 1/64$. Moreover $\lambda v_h < 495/512 < 99/100$, so both scalar upper-root constraints are satisfied. The lower root is below $v_h$. Since $v_h^8 < 1/2$ and $v_h^2 > 1/2$, one has $v_h/Q < 2$ and $q_h < 512/3$. The phase image of $X$ has length less than $16/3 < 2\pi$.

The integrand in (389), without $\rho^2$, is larger than $1/(5\,2^{18})$. The plateau has length $1/32$ and $p^2 > 1/2$, so the lower result follows from $1/(5\,2^{24}) > 2^{-27}$. For the upper result use $h/(\lambda^8 - 1) < 1/4$, $(1+\lambda^2)^{10} < 3^{10} < 2^{16}$ and $U^{-2} < 16$; the integrand is below $2^{20}/5 < 2^{18}$. The interval has length $1/16$ and $p^2 < 2$, giving the upper result. Finally exact integer Parseval and $|I_k + e^{i\alpha} I_{-k}|^2 \le 2(|I_k|^2 + |I_{-k}|^2)$ give squared continuum norm at most $2 \sup p^2 D_h < 2^{16}$. ∎

The preceding numerical principal bound is useful but does not itself make the weighted compactness argument a numerical kernel certificate. The latter still requires one uniform absorbable packet of the complete all-integer compact-normal, inner-brake, separated, and cell-error constants, and an enclosed finite-channel off-cap coefficient. Their analytic decay and uniformity have been proved above; numerical values satisfying a common absorption inequality have not been obtained here. A fixed finite family of boundary Mellin multipliers cannot replace that packet: each multiplier tends to zero at large Mellin frequency because its logarithmic kernel is integrable. Thus finite compact-frequency coverage alone does not control arbitrary high Mellin oscillations.

# H The signed horizontal boundary family and its Mellin transform

This section retains the clock-normalized Fourier rows of Section 5.2. In particular, the two physical angular branches are subtracted, and the two time phases in each branch are kept until they have been estimated. The integer channel is fixed throughout all passages from an actual row to a boundary operator. These passages will also be proved in operator norm, on the complete shrinking output half-line, with a horizontal weight.

## H.1 Actual channels, phases, and the boundary kernel

For the separated potential $U(q) = q^{10} - q^2$, write

$$p_e(q) = \sqrt{2(e + q^2 - q^{10})}, \qquad \tau_e(v) = \int_v^{q_+(e)} \frac{\mathrm{d}q}{p_e(q)},$$
$$T_0 = 8 \int_0^1 \frac{q^2\,\mathrm{d}q}{p_0(q)}, \qquad \kappa = \frac{3}{4} B\left(\frac{3}{4}, \frac{1}{2}\right).$$

Here $q_+(e)$ is the upper positive root near 1; for $e = -c < 0$, $\ell_c$ denotes the lower positive root. All roots and times in this section have the branches fixed by the real oscillators.

The exact critical equation gives $1/32 < v_h^{10} < 17/63 < 1/3$, hence $1/2 < v_h < 15/16$ and $v_h^{12} > 1/64$. Moreover $\lambda v_h < 495/512 < 99/100$, so both scalar upper-root constraints are satisfied. The lower root is below $v_h$. Since $v_h^8 < 1/2$ and $v_h^2 > 1/2$, one has $v_h/Q < 2$ and $q_h < 512/3$. The phase image of $X$ has length less than $16/3 < 2\pi$.

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For the separated potential $U(q) = q^{10} - q^2$, write

$$p_e(q) = \sqrt{2(e + q^2 - q^{10})}, \qquad \tau_e(v) = \int_v^{q_+(e)} \frac{\mathrm{d}q}{p_e(q)},$$
$$T_0 = 8 \int_0^1 \frac{q^2\,\mathrm{d}q}{p_0(q)}, \qquad \kappa = \frac{3}{4} B\left(\frac{3}{4}, \frac{1}{2}\right).$$

Here $q_+(e)$ is the upper positive root near 1; for $e = -c < 0$, $\ell_c$ denotes the lower positive root. All roots and times in this section have the branches fixed by the real oscillators.

Put $r(c) = T_u(c)/T_v(c)$. For $k \geq 1$, the actual channel consists of

$$m = 2n + 2k, \qquad r(c_{k,n}) = 2 + \frac{2k}{n}, \qquad \omega_{k,n} = m\Omega_u(c_{k,n}) = n\Omega_v(c_{k,n}). \tag{390}$$

No coprimality restriction is imposed: a nonprimitive pair is an actual higher Fourier harmonic on the same resonant torus. Here $\Omega_u = 2\pi/T_u$ and $\Omega_v = 2\pi/T_v$ are the auxiliary-time frequencies. The small-$c$ period and clock estimates of Lemma 58 give

$$\begin{aligned} r(c) - 2 &= \frac{\kappa c}{\log(1/c)}(1 + o(1)), \qquad & r'(c) &= \frac{\kappa}{\log(1/c)}(1 + o(1)), \\ T_u(c)\overline{S}(c) &\longrightarrow T_0, & \frac{\log(1/c)}{T_v(c)} &\longrightarrow \sqrt{2}. \end{aligned} \tag{391}$$

In particular, if $c_{k,n} = hs_n$ and $s_n$ stays in a positive compact interval, then, with $L = \log(1/h)$,

$$\frac{nh}{L} \longrightarrow \frac{2k}{\kappa s_n}, \qquad h\omega_{k,n} - \frac{4\pi\sqrt{2}k}{\kappa s_n} \longrightarrow 0, \tag{392}$$

uniformly on that interval. As throughout the row calculus, the asymptotic assertions here and below concern the actual integer nodes. The real function $2k/(r(c) - 2)$ will be used only to describe their cells.

Here is the exact density from Lemma 57 from which the limit is taken. For $\epsilon \in \{1, -1\}$ set

$$\lambda_\epsilon(\theta) = \sec\theta + \epsilon\tan\theta, \qquad H = \min\{q_+(-c), q_+(c)/\lambda_\epsilon\},$$
$$P(v) = p_c(\lambda_\epsilon v), \qquad Q(v) = p_{-c}(v).$$

The contribution is zero if its permitted interval is empty. Otherwise

$$\begin{aligned} B_\epsilon(c, n, \theta) &= C_\epsilon(c, \theta) \int_{\ell_c}^{H} \frac{v^{11}}{P(v)Q(v)} \cos\big(\omega\tau_c(\lambda_\epsilon v)\big) \cos\big(\omega\tau_{-c}(v)\big)\ \mathrm{d}v, \\ C_\epsilon(c, \theta) &= \frac{\lambda_\epsilon \sec\theta (1 + \lambda_\epsilon^2)^5}{2T_u T_v \overline{S}}, \qquad b_{k,n}(\theta) = B_+(c_{k,n}, n, \theta) - B_-(c_{k,n}, n, \theta), \\ R_{k,n}[a] &= \int_0^{\pi/2} a(\theta) b_{k,n}(\theta)\ \mathrm{d}\theta. \end{aligned} \tag{393}$$

Indeed $u = \lambda_\epsilon v$ has angular Jacobian $v\lambda_\epsilon \sec\theta$. Combining it with $S^5 a(\theta)/16$ and the positive-half coarea factor $8/(T_u T_v \overline{S})$ yields exactly (393). Thus its normalization is that of the complex Fourier row divided by $\overline{S}$, rather than an unnormalized time integral.

Define the moving-root derivative and the boundary phases by

$$J(v) = \partial_e \tau_e(v)|_{e=0}, \qquad A_k(v) = \frac{8\pi\sqrt{2}k}{\kappa} J(v), \qquad B_k(v) = \frac{4\pi k}{\kappa\sqrt{1 - v^8}}. \tag{394}$$

The derivative of the turning time is taken after regularizing its upper root. In particular it includes the root's displacement. On compact subsets of $(0, 1)$,

$$J'(v) = \frac{1}{2\sqrt{2}\, v^3 (1 - v^8)^{3/2}}, \quad A_k'(v) = \frac{4\pi k}{\kappa v^3 (1 - v^8)^{3/2}}, \quad \frac{A_k'(v)}{B_k'(v)} = \frac{1}{4v^{10}}. \tag{395}$$

The analytic upper-turn coordinate gives

$$\tau_e(v) = \sqrt{q_+(e) - v}\, T(e, q_+(e) - v), \qquad T(0,0) = \tfrac{1}{2}, \quad q_+'(0) = \tfrac{1}{8}.$$

Consequently

$$J(v) = \frac{1}{32}(1-v)^{-1/2} + O((1-v)^{1/2}), \qquad A_k(v) - \frac{1}{4}B_k(v) \longrightarrow 0. \tag{396}$$

This fixes the constant in $A_k$; specifying only its derivative would not suffice for the signed kernel.

For $z > 0$, $z \neq 1/4$, put

$$\begin{aligned} P_k(z) &= \frac{4\sqrt{2}}{T_0} \int_0^1 \frac{v^9}{1-v^8} \cos(A_k(v) - B_k(v)z)\, \mathrm{d}v, \\ N_k(z) &= \frac{4\sqrt{2}}{T_0} \int_0^1 \frac{v^9}{1-v^8} \cos(A_k(v) + B_k(v)z)\, \mathrm{d}v, \\ K_k(z) &= P_k(z) - N_k(z) = \frac{8\sqrt{2}}{T_0} \int_0^1 \frac{v^9}{1-v^8} \sin A_k(v) \sin(B_k(v)z)\, \mathrm{d}v. \end{aligned} \tag{397}$$

The upper endpoints are improper oscillatory integrals. At $z = 1/4$ we use their locally integrable extensions; the value assigned at the single point does not affect any operator.

A particularly useful exact representation follows from

$$X = (1-v^8)^{-1/2}, \qquad a(X) = (1-X^{-2})^{1/4}, \qquad F(X) = X - \int_X^\infty \big((1-t^{-2})^{-5/4} - 1\big)\, \mathrm{d}t. \tag{398}$$

Indeed $F'(X) = (1-X^{-2})^{-5/4}$, (395) and (396) imply $A_k(v) = (\pi k/\kappa)F(X)$, and $v^9\, \mathrm{d}v/(1-v^8) = a(X)\, \mathrm{d}X/(4X)$. With $c_k = \pi k/\kappa$ and $d_k = 4c_k$,

$$K_k(z) = \frac{2\sqrt{2}}{T_0} \int_1^\infty a(X) \sin(c_k F(X)) \sin(d_k X z)\, \frac{\mathrm{d}X}{X}. \tag{399}$$

Since $a(X) = 1 + O(X^{-2})$ and $F(X) = X - 5/(4X) + O(X^{-3})$, the upper integrand differs from $\sin(c_k X)\sin(d_k X z)/X$ by an $L^1(\,\mathrm{d}X)$ function, uniformly in $z$. The elementary estimate

$$\left| \int_2^R \frac{\cos(tX)}{X}\, \mathrm{d}X \right| \leq C \left(1 + \log_+ \frac{1}{|t|}\right), \qquad t \neq 0,$$

follows by splitting at $1/|t|$ and integrating the remaining tail by parts. Applied to $t = c_k \mp d_k z$, it proves convergence of (399) off $1/4$, the uniform truncated-kernel bound

$$|K_{k,R}(z)| \leq C_k \left(1 + \log_+ \frac{1}{|z - 1/4|}\right), \qquad K_k(z) = -\frac{\sqrt{2}}{T_0} \log|z - 1/4| + O_k(1). \tag{400}$$

The corresponding remainder is continuous at $1/4$: the difference from the leading cosine integral has an absolutely integrable tail, uniformly near that point. The negative branch is continuous there.

## H.2 Compact actual-row convergence, including the moving brake

**Proposition H.1** (Fixed-channel compact transfer)**.** *Fix $k \geq 1$ and positive compact intervals $J$ and $I$. For actual nodes $c_{k,n} = hs_n$ with $s_n \in I$, set*

$$D_{k,h}f(n) = \frac{L}{\sqrt{h}} R_{k,n}[h^{-1/2} f(\cdot/h)] = \int_J f(\xi) L b_{k,n}(h\xi)\, \mathrm{d}\xi.$$

*There is an isometric step-function embedding of this row vector, with squared norm $(h/L)\sum_{s_n \in I} |D_{k,h}f(n)|^2$, into $L^2(I^+, \mathrm{d}\mu_k)$, where $I^+$ is a fixed compact enlargement of $I$ and*

$$\mathrm{d}\mu_k(s) = \frac{2k}{\kappa} s^{-2}\, \mathrm{d}s.$$

*The embedded operators converge in Hilbert–Schmidt norm to*

$$f \longmapsto \mathbf{1}_I(s) \int_J K_k(\xi/s) f(\xi)\, \mathrm{d}\xi.$$

*The assertion holds also for any fixed finite set of channels.*

**Proof.** We give the uniform estimate needed to pass varying inputs through the limit. First, on $\delta \leq v \leq 1 - \delta$,

$$\tau_c(\lambda_\epsilon v) - \tau_{-c}(v) = 2cJ(v) - \epsilon h\xi \frac{v}{p_0(v)} + o(h).$$

Using (392), its phase tends uniformly to $A_k(v) - \epsilon B_k(v)\xi/s_n$. The sum phase is nonstationary on this compact interval, with derivative comparable to $h^{-1}$, so one integration by parts makes its integral tend to zero. The half-product identity

$$\cos\alpha \cos\beta = \tfrac{1}{2}\cos(\alpha - \beta) + \tfrac{1}{2}\cos(\alpha + \beta)$$

is used before this estimate. The clock limits give the coefficient $4\sqrt{2}/T_0$ in each limiting cosine integral.

At the lower end, uniformly for these parameters,

$$P(v) \geq C^{-1}v, \qquad Q(v) \geq C^{-1}\sqrt{v^2 - \ell_c^2}, \qquad \int_{\ell_c}^{\delta} \frac{v^{11}}{P(v)Q(v)}\, \mathrm{d}v \leq C\delta^{10}. \tag{401}$$

For the last inequality put $t = \sqrt{v^2 - \ell_c^2}$; the resulting integrand is bounded by $C\delta^9\, \mathrm{d}t$. This estimate covers the lower turn and the origin together.

For the upper end retain the exact two phases

$$\Psi_d = \omega(\tau_c(\lambda v) - \tau_{-c}(v)), \qquad \Psi_s = \omega(\tau_c(\lambda v) + \tau_{-c}(v)),$$

and the polynomial

$$\Delta = P^2 - \lambda^2 Q^2 = 2c(1 + \lambda^2) + 2\lambda^2(1 - \lambda^8)v^{10}. \tag{402}$$

The integration-by-parts quotients, with their signs, are exactly

$$\frac{v^{11}/(PQ)}{\Psi_d'} = \frac{v^{11}(P+\lambda Q)}{\omega\Delta}, \qquad \frac{v^{11}/(PQ)}{-\Psi_s'} = \frac{v^{11}}{\omega(P+\lambda Q)}. \tag{403}$$

Off $\xi = s_n/4$, a sufficiently small fixed upper cap has $|\Delta| \geq C^{-1}h$ in the positive branch. On that cap the first quotient has endpoint values and total variation $O(\sqrt{\delta}+\sqrt{h})$: $P+\lambda Q$ has that variation, and $1/(\omega\Delta)$ and its derivative are bounded. The second quotient is monotone near the upper turn, and its endpoint and variation bounds tend to zero, of size $O(\sqrt{h})$ at the moving upper endpoint and a vanishing bound at the cap's lower endpoint. In the negative branch $\Delta/h$ tends to $4s_n(1+4(\xi/s_n)v^{10})$, so the same argument has no positive-angle brake. Improper integration by parts means first stopping short of a turning point and then taking the limit; none of its endpoint terms is set to zero. Together with the compact bulk and (401), this proves

$$Lb_{k,n}(h\xi) - K_k(\xi/s_n) \longrightarrow 0 \tag{404}$$

uniformly on compact sets separated from $\xi = s_n/4$.

It remains to control the moving logarithm, rather than deducing an operator limit from (404) alone. The exact positive brake satisfies

$$\lambda_+(\theta_b(c)) = \frac{q_+(c)}{q_+(-c)}, \qquad \theta_b(c) = \frac{c}{4} + O(c^3), \qquad \xi_b(h,s) = \frac{\theta_b(hs)}{h} = \frac{s}{4} + O(h^2).$$

The cubic remainder follows because the logarithm of the root ratio is odd in $c$, as is $\theta = \arctan(\sinh(\log\lambda_+))$. Set $e = |\xi - \xi_b(h, s_n)|$ and $g = q_+(c)/\lambda_+(h\xi) - q_+(-c)$. Then $C^{-1}he \leq |g| \leq Che$. With $r = \sqrt{q_+(-c) - v}$, simple-root factorization bounds the upper coarea amplitude by

$$C\frac{\mathrm{d}r}{\sqrt{r^2+g}}, \qquad r \geq \sqrt{\max(0,-g)}.$$

If $e \leq C_0 h$, then $|g| \geq C^{-1}e^2/C_0$; the arsinh or arcosh primitive bounds the complete cap by $C(1+\log_+(1/e))$.

If $e > C_0 h$, choose $C_0$ so large that $|g| \leq e^2/4$. The part $r \geq e$ has the same logarithmic absolute bound. On the inner part, of $v$-length $O(e^2)$, the polynomial (402) satisfies

$$|\Delta| \asymp he, \quad |1/(\omega\Delta)| \leq C/e, \quad |\partial_v(1/(\omega\Delta))| \leq C/e^2.$$

Indeed its value at $q_+(-c)$ has size $he$, its derivative is $O(h)$, and its variation on this part is $O(he^2)$. Here $P, Q$ decrease and $P+\lambda Q$ has supremum and variation $O(e)$. The first quotient in (403) therefore has bounded endpoints and variation: its differentiated terms are bounded by $C$, $Ce^2$, and $Ce$. The second is increasing, and its final value is at most

$$\frac{C}{\omega\sqrt{|g|}} \leq C\sqrt{h/e} \leq C.$$

Integration by parts in both phases bounds the inner integral by $C$. For $e$ bounded away from zero use the off-brake upper-cap argument already given. The negative branch is bounded on the whole compact parameter set by that same nonstationary argument. As $LC_\epsilon$ is bounded, we obtain the complete-row estimate

$$|Lb_{k,n}(h\xi)| \le C\left(1+\log_+\frac{1}{|\xi-\xi_b(h,s_n)|}\right). \tag{405}$$

In particular these kernels have uniformly bounded $L^p(J)$ norms for every finite $p$, uniformly in the actual nodes in $I^+$. This also rules out a contact mass at the moving brake.

For completeness, the embedding is exact. Define

$$n_{k,h}(s)=\frac{2k}{r(hs)-2}, \qquad C_{n,h}=\{s : n-\tfrac12 \le n_{k,h}(s) < n+\tfrac12\}.$$

The differentiable estimate (391) gives

$$\frac{h}{L}|n'_{k,h}(s)| \longrightarrow \frac{2k}{\kappa}s^{-2}, \quad \operatorname{diam} C_{n,h}=O(h/L), \quad \mu_k(C_{n,h})=\frac{h}{L}(1+o(1))$$

uniformly on $I^+$. Use all full cells with centers $s_n \in I$, and define

$$\alpha_{n,h}=\left(\frac{h/L}{\mu_k(C_{n,h})}\right)^{1/2}, \qquad A_{k,h}(s,\xi)=\alpha_{n,h}Lb_{k,n}(h\xi) \quad (s\in C_{n,h},\ s_n\in I),$$

with value zero elsewhere. Then

$$\left\|\int_J A_{k,h}(s,\xi)f(\xi)\,\mathrm{d}\xi\right\|^2_{L^2(\mu_k)} = \frac{h}{L}\sum_{s_n\in I}|D_{k,h}f(n)|^2 = L\sum_{s_n\in I}|R_{k,n}[h^{-1/2}f(\cdot/h)]|^2. \tag{406}$$

The union of these full cells differs from $I$ only in endpoint bands of width $O(h/L)$. Thus (404) gives almost-everywhere convergence of $A_{k,h}$ to $\mathbf{1}_I(s)K_k(\xi/s)$. The exceptional brake graph and output endpoints have product measure zero. Estimate (405), with any $p>2$, gives uniform $L^p$ bounds on the fixed finite product-measure space. Uniform integrability therefore gives $L^2$ convergence of the kernels, precisely the claimed Hilbert–Schmidt convergence. A finite sum proves the finite-channel assertion. ∎

**Lemma 69** (Complete actual-row scale tails)**.** *For a fixed $k$, there are a horizontal cap $\theta_0>0$, constants $\varepsilon_0>0$, $M_0>1$, and a sufficiently small actual-node collar such that, with $L_n=\log(1/c_{k,n})$,*

$$\begin{aligned} &L_n(|B_+|+|B_-|)\le C_k, && 0<\theta\le \varepsilon_0 c_{k,n},\\ &L_n(|B_+|+|B_-|)\le C_k\frac{c_{k,n}}{\theta}, && M_0c_{k,n}\le\theta<\theta_0. \end{aligned} \tag{407}$$

*The bounds hold separately for the complete double-cosine integral of each physical sign, with both phases included.*

**Proof.** Use (393)–(403), now with $h = c = c_{k,n}$. The prefactor multiplied by $L_n$ is bounded and $1/\omega \le C_k c$. Each momentum has at most one maximum, as follows directly from

$$P' = (2\lambda^2 v - 10\lambda^{10} v^9)/P, \qquad Q' = (2v - 10v^9)/Q.$$

Their suprema and total variations are uniformly bounded. The lower-turn estimate (401) holds with any small upper endpoint $V$ in place of $\delta$.

For $\theta \ge M_0 c$ put $V = D(c/\theta)^{1/10}$. Choose $D$ large, then $M_0$ large, and finally the collar and $\theta_0$ small. These choices ensure $\ell_c < V < \delta < H$ and $V^{10} \ge c/4$. The interval below $V$ contributes absolutely at most $CV^{10} = Cc/\theta$. Above it write

$$\mathcal{D}(v) = \Delta(v)/v^{10} = 2c(1+\lambda^2)v^{-10} + 2\lambda^2(1-\lambda^8).$$

Since $|2\lambda^2(1-\lambda^8)| \asymp \theta$, the function $\mathcal{D}$ is monotone, has a fixed nonzero sign, and

$$\|1/\mathcal{D}\|_\infty + \mathrm{Var}(1/\mathcal{D}) \le C/\theta.$$

The difference quotient becomes $v(P+\lambda Q)/(\omega\mathcal{D})$; its endpoints and variation are bounded by $C_k c/\theta$.

For the sum quotient observe the exact identities

$$P^2 - vPP' = 2c + 8\lambda^{10} v^{10} > 0, \qquad Q^2 - vQQ' = -2c + 8v^{10}.$$

Thus $v^{11}/(P+\lambda Q)$ is increasing when $v^{10} \ge c/4$. The two upper-root gaps are

$$q_+(c)/\lambda_\pm - q_+(-c) = c/4 \mp \theta + O(c^2 + \theta^2).$$

Their absolute sizes are comparable to $\theta$ in the present range. At $H$ one momentum vanishes and the other is comparable to $\sqrt{\theta}$. The sum quotient consequently has endpoint and variation bound $C_k c/\sqrt{\theta} \le C_k c/\theta$. This proves the second bound, including all cutoff and turning endpoint terms.

If $\theta \le \varepsilon_0 c$, choose $\varepsilon_0$ small. Then $\Delta \asymp c$ on the whole permitted fiber, both upper gaps are positive and comparable to $c$, and

$$\|1/\Delta\|_\infty + \mathrm{Var}(1/\Delta) \le C/c.$$

The first quotient in (403) has bounded endpoints and variation. For the sum phase split at $V_0 = D_0 c^{1/10}$ with $V_0^{10} \ge c/4$. The lower part is $O(c)$ absolutely; the upper increasing quotient has final value at most $C_k c/\sqrt{c}$. These bounds prove the first assertion. No positive-minus-negative cancellation is required in either estimate. ∎

## H.3 Ordinary Mellin transform and the weighted logarithmic line

**Lemma 70** (Boundary kernel tails)**.** *For every fixed positive integer $k$,*

$$K_k(z) = O_k(z) \quad (z \downarrow 0), \qquad K_k(z) = O_k\left(\frac{\log(2+z)}{z}\right) \quad (z \to \infty). \tag{408}$$

*Together with* (400)*, these estimates imply*

$$\int_0^\infty z^{\alpha-1}|K_k(z)|\,\mathrm{d}z < \infty \qquad (0 < \alpha < 1).$$

**Proof.** On $1 < X \le 2$ let $g(X) = a(X)\sin(c_kF(X))/X$. Direct differentiation in (398) gives

$$|g(X)| \le C_k(X-1)^{1/4}, \qquad |g'(X)| \le C_k\big((X-1)^{-3/4} + (X-1)^{-1}\big).$$

For small $z$, $|\sin(d_kXz)| \le d_kXz$ gives $O_k(z)$. For $z \ge 2$ split at $1 + 1/z$. The initial interval has absolute integral $O_k(z^{-5/4})$; integration by parts on the rest gives $O_k((1+\log z)/z)$, including both boundary terms. The singular derivative has thus never been integrated through $X = 1$.

For $X \ge 2$ put

$$H_k(X) = \frac{a(X)}{X}\exp\big(ic_k(F(X) - X)\big), \qquad I_k(t) = \int_2^\infty H_k(X)e^{itX}\,\mathrm{d}X \quad (t \ne 0).$$

The exact expressions give $H_k^{(j)}(X) = O_k(X^{-1-j})$ for $j = 0, 1, 2$. Integration by parts proves convergence and $|I_k(t)| \le C_k/|t|$. The upper contribution to $K_k$ is

$$\frac{\sqrt{2}}{T_0}\,\mathrm{Re}\big(I_k(c_k - d_kz) - I_k(c_k + d_kz)\big),$$

which is $O_k(1/z)$ for $z \ge 1/2$. If $0 \le z \le 1/8$, both arguments lie in $[c_k/2, 3c_k/2]$. Integrate by parts twice:

$$I_k(t) = -\frac{H_k(2)e^{2it}}{it} + \frac{H_k'(2)e^{2it}}{(it)^2} + \frac{1}{(it)^2}\int_2^\infty H_k''(X)e^{itX}\,\mathrm{d}X.$$

The last integral can be differentiated because $XH_k'' \in L^1$. Thus $I_k'$ is bounded on that compact frequency interval, and the difference is $O_k(z)$. This justifies the small-$z$ estimate without differentiating a conditionally convergent integral in its original form. Integrability at $1/4$ follows from (400). ∎

**Proposition H.2** (Ordinary Mellin formula)**.** *For $0 < \mathrm{Re}\,w < 1$, the absolutely convergent Mellin transform of the signed kernel is*

$$\begin{aligned} M_k(w) &= \int_0^\infty z^{w-1}K_k(z)\,\mathrm{d}z \\ &= \frac{8\sqrt{2}}{T_0}\Gamma(w)\sin\frac{\pi w}{2}\int_0^1 \frac{v^9}{1-v^8}\sin A_k(v)B_k(v)^{-w}\,\mathrm{d}v \\ &= \frac{2\sqrt{2}}{T_0}\Gamma(w)\sin\frac{\pi w}{2}\left(\frac{4\pi k}{\kappa}\right)^{-w}\int_1^\infty a(X)X^{-w-1}\sin\left(\frac{\pi k}{\kappa}F(X)\right)\,\mathrm{d}X. \end{aligned} \tag{409}$$

*All positive real powers in this formula use the real logarithm. Both integrals on the right are absolutely convergent.*

**Proof.** An Abel factor supplies a justified interchange before it is removed. For $\varepsilon > 0$, first truncate the $X$ integral at $R$. Fubini is then valid, and

$$\int_0^\infty z^{w-1}e^{-\varepsilon z}\sin(Bz)\,\mathrm{d}z = \frac{\Gamma(w)}{2i}\big((\varepsilon - iB)^{-w} - (\varepsilon + iB)^{-w}\big).$$

These are principal complex powers. For nonreal $w$ this is a difference of powers, not an imaginary-part formula. If $\alpha = \operatorname{Re} w > 0$, then

$$|(\varepsilon \pm iB)^{-w}| \le e^{\pi|\operatorname{Im} w|/2}B^{-\alpha}, \qquad \int_0^1 \frac{v^9}{1-v^8}B_k(v)^{-\alpha}\,\mathrm{d}v < \infty.$$

The latter integrand is $O(v^9)$ at zero and $O((1-v)^{-1+\alpha/2})$ at one. The uniform truncated-kernel bound (400) allows $R \to \infty$ on the original damped $z$ side; the displayed radial majorant permits the same limit on the transformed side. Next let $\varepsilon \downarrow 0$. On the radial side dominated convergence gives

$$(-iB)^{-w} - (iB)^{-w} = 2i\sin(\pi w/2)B^{-w}.$$

On the original side Lemma 70 supplies absolute Mellin integrability and hence dominated convergence. This proves the first two lines of (409); the exact change of variables (398) proves the last. ∎

For $0 \le \gamma < 1$ define the boundary operator, initially on compactly supported functions, by

$$(T_k f)(c) = \int_0^\infty K_k(\theta/c)f(\theta)\,\mathrm{d}\theta. \tag{410}$$

The unitary logarithmic coordinate maps for its natural weighted spaces are

$$\begin{aligned} F_\gamma(x) &= e^{(1-\gamma)x/2}f(e^x), & \|F_\gamma\|_2^2 &= \int_0^\infty \theta^{-\gamma}|f(\theta)|^2\,\mathrm{d}\theta, \\ G_\gamma(y) &= e^{-(1+\gamma)y/2}(T_k f)(e^y), & \|G_\gamma\|_2^2 &= \int_0^\infty c^{-2-\gamma}|T_k f(c)|^2\,\mathrm{d}c. \end{aligned} \tag{411}$$

Direct substitution gives

$$G_\gamma(y) = \int_{\mathbb{R}} F_\gamma(y+t)H_{k,\gamma}(t)\,\mathrm{d}t, \qquad H_{k,\gamma}(t) = e^{(1+\gamma)t/2}K_k(e^t). \tag{412}$$

The tails from Lemma 70 are respectively $O_k(e^{(3+\gamma)t/2})$ at $-\infty$ and $O_k((1+t)e^{-(1-\gamma)t/2})$ at $+\infty$. The finite logarithmic singularity is integrable. Thus $H_{k,\gamma} \in L^1(\mathbb{R})$ and Young's inequality extends $T_k$ boundedly between the spaces in (411). With the unitary convention $\widehat{F}(\eta) = (2\pi)^{-1/2}\int e^{-i\eta x}F(x)\,\mathrm{d}x$, its multiplier is exactly

$$m_{k,\gamma}(\eta) = \int_{\mathbb{R}} H_{k,\gamma}(t)e^{i\eta t}\,\mathrm{d}t = M_k\left(\frac{1+\gamma}{2} + i\eta\right). \tag{413}$$

There is no extra Fourier-normalization factor. The shifted line is $(1+\gamma)/2$, as is also forced by the output weight.

## H.4 The inverse strip of the model elliptic primitive

We now establish the complex-domain facts needed to exclude a common Mellin zero. They concern the actual inverse of the real function $F$ in (398), not an arbitrarily chosen preimage of an Abelian integral.

**Lemma 71** (Holomorphic inverse strip)**.** *The real inverse $X = X(u)$ of $F : (1, \infty) \to \mathbb{R}$ extends holomorphically to*

$$\mathcal{S} = \{u \in \mathbb{C} : |\operatorname{Im} u| < \kappa\}.$$

*There are holomorphic nonvanishing functions $q(u), y(u)$ on $\mathcal{S}$, matching $q, y > 0$ on the real axis, such that*

$$y^2 = 1 - q^4, \qquad X = 1/y, \qquad (1 - X^{-2})^{3/2} = q^6.$$

*The logarithms of $q, y, X$ are fixed by continuation from those positive real values.*

**Proof.** Consider the smooth compactification $E$ of $y^2 = 1 - q^4$. It is a genus-one curve, since the double cover of the $q$ sphere is branched at $1, -1, i, -i$. Its two infinite points $I_\pm$ are distinguished by $y/q^2 \to \pm i$. Set

$$\omega = \frac{2\,\mathrm{d}q}{q^2 y^3}, \qquad I = \int_0^1 \frac{q^2\,\mathrm{d}q}{\sqrt{1-q^4}} = \frac{\kappa}{3}, \qquad R(q,y) = \frac{3q^4 - 2}{qy}.$$

On the real arc $0 < q < 1$ (so $q = v^2$ in the boundary integral), with $y = \sqrt{1-q^4}$ and $X = 1/y$, we have $\mathrm{d}F = \omega$. Direct differentiation gives the exact identity and its normalized primitive

$$\omega = \mathrm{d}R - 3\frac{q^2\,\mathrm{d}q}{y}, \qquad F(q,y) = \kappa + R(q,y) - 3\int_0^q \frac{t^2\,\mathrm{d}t}{\sqrt{1-t^4}}. \tag{414}$$

The integral begins on the positive-$y$ sheet. Since $R - X \to 0$ at $q = 1$ and $3I = \kappa$, this is precisely the normalization $F(X) - X \to 0$ at $X = \infty$. At the origin chart it gives

$$F(q,y) = \kappa - \frac{2}{q} + O(q^3). \tag{415}$$

We compute all finite critical values of the lifted primitive. At $(q, y) = (0, \pm 1)$ the differential is $\pm 2q^{-2}(1 + O(q^4))\,\mathrm{d}q$. At a branch point $q = a$, $a^4 = 1$, use $y$ as local coordinate:

$$\omega = -q^{-5}y^{-2}\,\mathrm{d}y, \qquad q = a(1-y^2)^{1/4}.$$

All six poles are double with zero residue. At infinity, with $t = 1/q$, we have

$$y = \pm i t^{-2}(1 + O(t^4)), \qquad \omega = \mp 2it^6(1 + O(t^4))\,\mathrm{d}t.$$

Thus the only zeros of $\omega$ are order six at $I_+$ and $I_-$; its divisor is

$$(\omega) = 6I_+ + 6I_- - 2P_{0,+} - 2P_{0,-} - 2P_1 - 2P_{-1} - 2P_i - 2P_{-i}. \tag{416}$$

A primitive has simple poles at the six pole points and order-seven critical points at the two infinite points.

The full period lattice must be computed, rather than inferred from a real periodization length. Let $A$ be the closed lift of the real arc from $-1$ to 1 and back on the other sheet. Let $B$ be the closed lift of an arc from 1 to $i$ in the first quadrant, with interior disjoint from $A$. They intersect once in the square-root coordinate at 1, so, after an orientation choice, they are an integral homology basis. Displace these cycles off the poles when integrating $\omega$; zero residues make this displacement harmless. By (414),

$$\int_A \frac{q^2\,\mathrm{d}q}{y} = 4I, \qquad \int_A \omega = -4\kappa.$$

For $B$ deform the arc to $1 \to 0 \to i$. On $q = it$ the integrand $q^2\,\mathrm{d}q/y = -it^2\,\mathrm{d}t/\sqrt{1-t^4}$, so

$$\int_B \frac{q^2\,\mathrm{d}q}{y} = -2(1+i)I, \qquad \int_B \omega = 2\kappa(1+i).$$

The exact period group is consequently

$$\Lambda = 4\kappa\mathbb{Z} + 2\kappa(1+i)\mathbb{Z}. \tag{417}$$

Its rectangular subgroup $4\kappa\mathbb{Z}+4i\kappa\mathbb{Z}$ has index two and is insufficient for the continuation argument.

To evaluate a critical value, continue above $q = 1$ to $q = R > 1$. Here $y = -i\sqrt{R^4-1}$ and

$$\int_0^R \frac{q^2\,\mathrm{d}q}{y} = I + i\int_1^R \frac{t^2\,\mathrm{d}t}{\sqrt{t^4-1}}, \qquad R(q,y) = 3iR + O(R^{-3}).$$

With $\varepsilon = 1/R$ and $Y(t) = \sqrt{1-t^4}$,

$$\int_1^R \frac{t^2\,\mathrm{d}t}{\sqrt{t^4-1}} = \int_\varepsilon^1 \frac{\mathrm{d}t}{t^2 Y(t)} = \frac{Y(\varepsilon)}{\varepsilon} - \int_\varepsilon^1 \frac{t^2\,\mathrm{d}t}{Y(t)}.$$

The constant after subtracting $R$ is $-I$. Formula (414) therefore gives $F(I_-) = i\kappa$. The conjugate path gives $F(I_+) = -i\kappa$. Adding all periods, the complete finite critical-value set is

$$\mathcal{V} = (i\kappa + \Lambda) \cup (-i\kappa + \Lambda) = 2\kappa\mathbb{Z} + i\kappa(2\mathbb{Z}+1). \tag{418}$$

In particular it misses $\mathcal{S}$. This calculation alone does not identify which boundary values are singularities of the chosen inverse, a question addressed below.

We also need to rule out finite asymptotic values of the lifted primitive. Uniformize the compact genus-one curve as $E = \mathbb{C}/\Omega$. Since the residues vanish, the pullback of $\omega$ has a single-valued meromorphic primitive $G$ on $\mathbb{C}$: integrals around its poles vanish, and the local primitive extends meromorphically through each pole. For $\lambda \in \Omega$,

$$G(z+\lambda) = G(z) + P(\lambda), \qquad P(\Omega) = \Lambda, \qquad |P(\lambda)| \geq c|\lambda|. \tag{419}$$

The last inequality holds because the images of a lattice basis, computed above, are real-linearly independent. The poles of $G$ are simple and form finitely many $\Omega$-periodic orbits. Choose a closed fundamental parallelogram with no poles on its boundary, and small open disks about its poles whose lattice translates are pairwise disjoint. On the compact remainder $|G| \le B$. Hence, outside these disks in the cell translated by $\lambda$, $|G| \ge |P(\lambda)| - B$. For each $M$, the set $\{|G| \le M\}$ outside a sufficiently large disk is therefore contained in the union of the disjoint translated pole disks. A continuous curve escaping to infinity with $G$ tending to a finite value would eventually lie in that union. Its connected tail would have to stay in one bounded pole disk, a contradiction. This rules out every finite asymptotic curve; the full rank of the period homomorphism is essential in this argument.

Start with the regular real inverse germ of $G$ corresponding to $0 < q < 1$. It continues along every path in $\mathcal{S}$. Indeed, consider a maximal lift approaching a finite path endpoint $u_* \in \mathcal{S}$. A finite cluster point of its lifted values cannot be a pole and cannot be critical, by (418). A regular inverse chart at that cluster point traps and extends the final part of the lift: choose two concentric target disks in the chart, and use that the remaining path stays in the smaller disk after the lift has entered it. If there is no finite cluster point, the lift escapes to infinity with $G \to u_*$, contrary to the preceding paragraph. Thus no endpoint obstructs continuation. The monodromy theorem on the simply connected strip supplies a single-valued holomorphic inverse $z(u)$ there.

Compose with the elliptic functions $q(z), y(z)$. Their poles occur only at the infinite points, whose primitive values lie in $\mathcal{V}$; their zeros occur at poles of the primitive. Neither event occurs on the strip inverse. Thus $q, y, X = 1/y$ are holomorphic and nonzero on $\mathcal{S}$. They have the claimed logarithms by simple connectivity, and their positive real normalization identifies the density powers. This proves the lemma. ∎

**Lemma 72** (The central hits and regular translated boundary terms)**.** *The inverse in Lemma 71 reaches $I_-$ at $u = i\kappa$ and $I_+$ at $u = -i\kappa$. For every nonzero integer $j$ it has a regular holomorphic continuation across $u = 2\kappa j \pm i\kappa$. For* $\operatorname{Re} w > 0$ *put*

$$\rho_w(u) = q(u)^6 \exp((w+1)\log y(u)). \tag{420}$$

*The sum of its noncentral translates $\sum_{j\neq 0} \rho_w(u + 2\kappa j)$ extends holomorphically to full neighborhoods of both $i\kappa$ and $-i\kappa$, with normal convergence locally uniformly in $w$.*

**Proof.** An explicit path is needed to identify the central critical value as an actual hit of the strip inverse. Take $r_0 = 2/3$ and $\phi = \pi/8$, and follow

$$q = r_0 e^{i\theta}\ (0 \le \theta \le \phi), \qquad q = r e^{i\phi}\ (r_0 \le r < \infty).$$

For $|q| < 1$, the origin expansion is the convergent series

$$F(q) = \kappa - 2/q + \sum_{n\ge 1} b_n q^{4n-1}, \qquad b_n = \frac{2(3/2)_n}{n!(4n-1)} > 0.$$

At $|q| = r_0$,

$$\sum_{n\ge1} b_n r_0^{4n-1} \le \frac{2}{3r_0}\big((1-r_0^4)^{-3/2}-1\big) = (81/65)^{3/2}-1 < \frac{2}{5}.$$

The last inequality follows by squaring, using $25\cdot 81^3 < 49\cdot 65^3$. As $\sin(\pi/8) < 2/5$, the circular arc has $|\operatorname{Im} F| < 8/5$. On the other hand

$$\kappa = 3\int_0^1 \frac{q^2\,\mathrm{d}q}{\sqrt{(1-q^2)(1+q^2)}} \ge \frac{3\pi}{4\sqrt{2}} > \frac{8}{5}. \tag{421}$$

Thus this arc lies strictly in the inverse strip. At its endpoint $\sin(\pi/8) > 3/8$ also yields $\operatorname{Im} F(r_0 e^{i\phi}) > 29/40$.

Along the ray, take $y = \sqrt{1-ir^4}$ with argument in $(-\pi/4, 0)$. Then

$$\frac{\mathrm{d}F}{\mathrm{d}r} = 2r^{-2}e^{-i\pi/8}(1-ir^4)^{-3/2}, \qquad \arg\frac{\mathrm{d}F}{\mathrm{d}r} = -\frac{\pi}{8} + \frac{3}{2}\arctan(r^4).$$

Its imaginary part changes sign at most once, at $r_* = (\tan(\pi/12))^{1/4} < 3/4$. Thus $H(r) = \operatorname{Im} F(re^{i\phi})$ decreases only up to $r_*$ and increases thereafter. The possible decrease from $r_0$ to $r_*$ is at most

$$2\sin(\pi/8)(1/r_0 - 1/r_*) \le \tfrac{1}{3}\sin(\pi/8) < \tfrac{2}{15}.$$

It follows that $H(r) > 71/120 > 0$. The ray tends to $I_-$ on the same primitive lift used in (418): connect it at large radius $R$ to the real ray above $q = 1$ by an arc in $0 < \arg q \le \pi/8$. Its differential is $O(R^{-8})\,\mathrm{d}q$, so the connecting integral tends to zero. The two paths are homotopic in this sector with no extra circuit about a pole or branch point. Hence their primitive limits agree, and $F(re^{i\phi}) \to i\kappa$. Its increasing tail lies below $\kappa$; the preceding decreasing part is bounded above by its initial value, already less than $8/5$. The full image path is consequently in $\mathcal{S}$. Uniqueness of continuation of the strip inverse forces it to follow this path all the way to $I_-$. Conjugation gives the lower hit at $I_+$.

We next construct explicit regular inverse charts for all noncentral translates. In the upper-turn chart, $y = 1/X$ and

$$F(y) = 1/y - E(y), \qquad E(y) = \sum_{n\ge1} \frac{(5/4)_n}{n!(2n-1)} y^{2n-1}.$$

On $|y| = 1/2$,

$$|E(y)| \le 2\big((4/3)^{5/4}-1\big) < 1,$$

since $(4/3)^5 < (3/2)^4$. The equation $F(y) = u$ becomes $1 - yE(y) - uy = 0$. For $|u| > 3$, on $|y| = 1/2$,

$$|uy| > 3/2 > |1 - yE(y)|.$$

Rouché's theorem gives exactly one root in $|y| < 1/2$, counted with multiplicity. It is nonzero and simple, also because $F_y = -y^{-2}(1-y^2)^{-5/4} \neq 0$ there. Uniqueness gives a holomorphic root on $|u| > 3$, with no monodromy ambiguity. Set $q = (1-y^2)^{1/4}$ on the branch near 1. At large positive real $u$ this is the real inverse, and therefore it matches that inverse throughout the right component of $\mathcal{S} \cap \{|u| > 3\}$. It continues the chosen branch across every point with $\operatorname{Re} u > 3$. The logarithm of $y$ on that simply connected half-plane is chosen to match its positive real value. Formula (420) is then holomorphic there. Since $2\kappa > 3$, this handles $2\kappa j + i\kappa$ for $j \geq 1$.

In the origin chart write

$$F(q) = \kappa - 2/q + G_0(q), \qquad G_0(q) = \sum_{n\geq 1} \frac{2(3/2)_n}{n!(4n-1)} q^{4n-1}.$$

On $|q| = 1/2$,

$$|G_0(q)| \leq \frac{4}{3}\big((16/15)^{3/2} - 1\big) < 1.$$

For $|u-\kappa| > 5$, Rouché's theorem applied to $-2 + qG_0(q) - (u-\kappa)q$ gives exactly one root in $|q| < 1/2$, since $|(u-\kappa)q| > 5/2 > |-2+qG_0(q)|$ on the circle. It is nonzero, simple, and holomorphic in $u$. Set $y = (1-q^4)^{1/2}$ near 1; its range has a holomorphic logarithm near 1, so (420) is defined throughout this inverse chart. It matches the real inverse at large negative real $u$, and hence on the left component of $\mathcal{S} \cap \{|u-\kappa| > 5\}$. For $j \leq -1$,

$$|2\kappa j + i\kappa - \kappa| = \kappa\sqrt{(2j-1)^2+1} \geq \kappa\sqrt{10} > 5.$$

Each such point is on the boundary of that matching left component: first move horizontally left and then to the negative real axis while staying outside the disk. This identifies the regular inverse as the chosen branch, rather than another preimage. Conjugation handles the lower boundary.

Choose a positive $\delta$ smaller than $2\kappa - 3$, $\kappa\sqrt{10} - 5$, and $\kappa$. For $|u - i\kappa| < \delta$, all positive translates lie in the right inverse half-plane and all negative translates in the origin inverse chart. The inversions give, uniformly on compact subsets of this disk,

$$y(u+2\kappa j) = (u+2\kappa j)^{-1} + O(j^{-3}) \quad (j \to +\infty),$$

$$q(u+2\kappa j) = -2/(u+2\kappa j - \kappa) + O(|j|^{-5}) \quad (j \to -\infty).$$

It follows that

$$\rho_w(u+2\kappa j) = \begin{cases} O_w(j^{-\operatorname{Re} w - 1}), & j \to +\infty, \\ O_w(|j|^{-6}), & j \to -\infty. \end{cases} \tag{422}$$

The bounds are locally uniform for $\operatorname{Re} w > 0$. Normal convergence proves the assertion about the holomorphic remainder. The same proof works in the lower disk. ∎

## H.5 No common Mellin zero

**Theorem 20** (No common zero in the Mellin strip)**.** *For every $w$ with $0 < \operatorname{Re} w < 1$, at least one positive integer $k$ satisfies $M_k(w) \neq 0$.*

**Proof.** The function $F$ in (398) is strictly increasing from $-\infty$ to $+\infty$. Changing variables $u = F(X)$ in (409) gives

$$M_k(w) = C(w)\left(\frac{4\pi k}{\kappa}\right)^{-w} \int_{\mathbb{R}} \rho_w(u)\sin(\pi k u/\kappa)\,\mathrm{d}u, \quad C(w) = \frac{2\sqrt{2}}{T_0}\Gamma(w)\sin\frac{\pi w}{2}. \quad (423)$$

The real density agrees with (420), since

$$\rho_w(u) = (1 - X(u)^{-2})^{3/2} X(u)^{-w-1}.$$

The coefficient $C(w)$ is nonzero in the stated strip. On the real line the endpoint expansions are

$$X(u) = u + O(u^{-1}) \quad (u \to +\infty), \qquad q(u) = -2/(u-\kappa) + O(|u|^{-5}) \quad (u \to -\infty).$$

They and their derivatives give, for every $j \geq 0$,

$$\rho_w^{(j)}(u) = \begin{cases} O_{w,j}(u^{-\operatorname{Re} w - 1 - j}), & u \to +\infty, \\ O_{w,j}(|u|^{-6-j}), & u \to -\infty. \end{cases} \quad (424)$$

Thus the complex periodization

$$\mathcal{P}_w(t) = \sum_{j \in \mathbb{Z}} \rho_w(t + 2\kappa j) \quad (425)$$

is smooth and $2\kappa$-periodic on the real line, with uniformly convergent derivative series on compact intervals. Absolute integrability permits unfolding each sine coefficient. Hence all $M_k(w)$ vanish if and only if all sine coefficients of $\mathcal{P}_w$ vanish, if and only if

$$\mathcal{P}_w(t) = \mathcal{P}_w(-t) \quad (t \in \mathbb{R}). \quad (426)$$

For the last equivalence, the odd continuous function $\mathcal{P}_w(t) - \mathcal{P}_w(-t)$ has zero constant and cosine coefficients automatically; its sine coefficients then all vanish. Uniqueness of Fourier coefficients, for instance by uniform convergence of Fejér means, applies separately to its real and imaginary parts. No reality assumption at nonreal $w$ is used.

The inverse charts in the preceding proofs also give the expansions (424) uniformly on closed narrower strips. Equivalently, on the right invert $F(X) = X + O(X^{-1})$; on the left invert (415), and identify both inverses by their real values. Thus (425) converges normally on compact subsets of $\mathcal{S}$ and is holomorphic there. Lemma 72 gives holomorphic remainders $A_+, A_-$ on full neighborhoods of the two central boundary points such that

$$\mathcal{P}_w(u) = \rho_w(u) + A_+(u) \quad (u \text{ near } i\kappa), \qquad \mathcal{P}_w(u) = \rho_w(u) + A_-(u) \quad (u \text{ near } -i\kappa). \quad (427)$$

Every noncentral translate is included in these remainders.

It remains to compare the central coefficients with consistent branches. Put $\zeta = u - i\kappa$. At $I_-$, with $t = 1/q$,

$$\zeta = \frac{2i}{7}t^7(1 + O(t^4)), \qquad y = -iq^2(1 + O(q^{-4})).$$

These are convergent local analytic expansions. Absorb the analytic unit into $t(1 + O(t^4))^{1/7}$; inversion is then a convergent Puiseux expansion. Let $\operatorname{Log}_- \zeta$ have argument in $(-\pi, 0)$, the half-disk inside the strip. On the explicit hit path of Lemma 72,

$$\arg q = \pi/8, \qquad \arg \zeta \longrightarrow -3\pi/8, \qquad \arg y \longrightarrow -\pi/4.$$

The continued logarithms therefore satisfy precisely

$$\begin{aligned} \log q &= \frac{\log(2/7) + i\pi/2 - \operatorname{Log}_- \zeta}{7} + O(\zeta^{4/7}), \\ \log y &= 2\log q - i\pi/2 + O(\zeta^{4/7}). \end{aligned} \tag{428}$$

There is no unspecified multiple of $2\pi i$. Along the circular arc and ray $y$ moves continuously from its positive real value to argument $-\pi/4$ without winding. Agreement of this Puiseux branch with the strip inverse on the hit path extends by uniqueness to a sufficiently small lower half-disk.

Put $\nu = (2w + 8)/7$. Substitution in (420) gives

$$\rho_w(i\kappa + \zeta) = C_+(w)e^{-\nu \operatorname{Log}_- \zeta}(1 + O_w(|\zeta|^{4/7})), \qquad C_+(w) = (2/7)^\nu e^{i\pi(\nu - w - 1)/2}. \tag{429}$$

At the lower point use $v = u + i\kappa$ and $\operatorname{Log}_+ v$ with argument in $(0, \pi)$. Conjugating the path, but leaving the complex parameter $w$ unchanged, gives by the same direct substitution

$$\rho_w(-i\kappa + v) = C_-(w)e^{-\nu \operatorname{Log}_+ v}(1 + O_w(|v|^{4/7})), \qquad C_-(w) = (2/7)^\nu e^{-i\pi(\nu - w - 1)/2}. \tag{430}$$

In particular this is not an asserted conjugacy of $\rho_w$ at fixed nonreal $w$.

For $\arg \zeta \in (-\pi, 0)$ one has the exact identity $\operatorname{Log}_+(-\zeta) = \operatorname{Log}_- \zeta + i\pi$. Equations (427)–(430) therefore yield, along any fixed ray in that half-disk,

$$\begin{aligned} &\lim_{\zeta \to 0} e^{\nu \operatorname{Log}_- \zeta}\big(\mathcal{P}_w(i\kappa + \zeta) - \mathcal{P}_w(-i\kappa - \zeta)\big) \\ &\qquad = D(w) = C_+(w) - e^{-i\pi\nu}C_-(w) = C_+(w)\left(1 - e^{-i\pi(9 - 3w)/7}\right). \end{aligned} \tag{431}$$

The Puiseux errors gain $4/7$ in their real exponent; the holomorphic remainders vanish after multiplication because $\operatorname{Re} \nu > 0$. This justifies the limit with the entire periodized density retained. Its coefficient is nonzero: $C_+(w) \neq 0$, and a zero of the bracket would require $(9 - 3w)/7$ to be an even integer, whereas its real part lies strictly between $6/7$ and $9/7$.

If all Mellin transforms vanished, (426) and the identity theorem would make $\mathcal{P}_w(u) - \mathcal{P}_w(-u)$ identically zero on $\mathcal{S}$. Approaching the central point inside that strip contradicts (431). This proves the theorem. ∎

**Corollary H.1** (Finite detection on a bounded frequency interval)**.** *Fix $0 \le \gamma < 1$ and $R < \infty$. There are a finite set $\mathcal{F} \subset \mathbb{N}$ and a constant $a_{\gamma,R} > 0$ such that*

$$\sum_{k\in\mathcal{F}} \frac{2k}{\kappa} \left| M_k\left(\frac{1+\gamma}{2} + i\eta\right)\right|^2 \ge a_{\gamma,R} \qquad (|\eta| \le R). \tag{432}$$

*For every $f \in L^2(\theta^{-\gamma}\,\mathrm{d}\theta)$,*

$$a_{\gamma,R} \int_{-R}^{R} |\widehat{F}_\gamma(\eta)|^2 \,\mathrm{d}\eta \le \sum_{k\in\mathcal{F}} \frac{2k}{\kappa} \|T_k f\|^2_{L^2(c^{-2-\gamma}\,\mathrm{d}c)}. \tag{433}$$

**Proof.** Each multiplier in (413) is continuous because its logarithmic kernel is integrable. At every $\eta \in [-R, R]$, Theorem 20 supplies a nonzero channel and an open nonvanishing neighborhood. Choose a finite subcover and take the positive minimum of the resulting finite square sum. Plancherel proves (433), including the factor $2k/\kappa$ from the actual row density. No assertion about an infinite unfiltered family is needed. ∎

## H.6 Exact weighted transfer on the entire shrinking half-line

A compact-scale limit does not by itself control inputs spread over infinitely many logarithmic scales. The complete-row tail bounds in Lemma 69 supply that additional control. Fix $k$ and take a monotone collar for $n_k(c) = 2k/(r(c)-2)$. For sufficiently large integers $n$ define its full inverse cells by

$$C_{k,n} = \{c : n - \tfrac{1}{2} \le n_k(c) < n + \tfrac{1}{2}\}, \qquad c_n = c_{k,n}, \qquad L_n = \log(1/c_n).$$

Their disjoint union over $n \ge N$ is $(0, d_{k,N})$, with $d_{k,N}$ the upper endpoint of $C_{k,N}$ and $d_{k,N} \downarrow 0$. For $0 \le \gamma < 1$ let

$$m_{n,\gamma} = \int_{C_{k,n}} c^{-2-\gamma}\,\mathrm{d}c, \qquad \beta_{n,\gamma} = \left(\frac{L_n c_n^{-\gamma}}{m_{n,\gamma}}\right)^{1/2}. \tag{434}$$

The notation suppresses the fixed channel in $m_{n,\gamma}$.

**Theorem 21** (Weighted fixed-channel half-line transfer)**.** *Fix $0 < \gamma < 1$, a sufficiently small horizontal cap $(0, \theta_0)$, and a positive integer $k$. On the actual cells in $(0, d_{k,N})$ define*

$$(\mathcal{A}_{k,N,\gamma} a)(c) = \beta_{n,\gamma} R_{k,n}[\mathbf{1}_{(0,\theta_0)} a] \quad (c \in C_{k,n},\ n \ge N). \tag{435}$$

*It extends to a bounded operator from $L^2((0,\theta_0), \theta^{-\gamma}\,\mathrm{d}\theta)$ to $L^2((0, d_{k,N}), c^{-2-\gamma}\,\mathrm{d}c)$, and*

$$\left\| \mathcal{A}_{k,N,\gamma} - \sqrt{\frac{2k}{\kappa}}\, \mathbf{1}_{(0,d_{k,N})} T_k \mathbf{1}_{(0,\theta_0)} \right\|_{\mathrm{op}} \longrightarrow 0 \qquad (N \to \infty). \tag{436}$$

*Its norm is normalized exactly by*

$$\|\mathcal{A}_{k,N,\gamma} a\|^2_{L^2(c^{-2-\gamma}\,\mathrm{d}c)} = \sum_{n\ge N} L_n c_n^{-\gamma} |R_{k,n}[\mathbf{1}_{(0,\theta_0)} a]|^2. \tag{437}$$

*All assertions also hold for $\gamma = 0$ and for a fixed finite Hilbert direct sum of channels, each with its own full cells.*

**Proof.** The norm identity follows immediately from the constant value on each exact cell. We prove boundedness and convergence without assuming beforehand that the infinite row sum is finite. The differentiable counting estimate (391) gives, uniformly for large $n$,

$$\begin{aligned} m_{n,0} &= \frac{\kappa}{2kL_n}(1+o(1)), \qquad \sup_{c\in C_{k,n}} |c/c_n - 1| = O(c_n/(kL_n)), \\ m_{n,\gamma} &= c_n^{-\gamma} m_{n,0}(1+O_\gamma(c_n/(kL_n))), \\ q_{n,\gamma} := \frac{\beta_{n,\gamma}}{\beta_{n,0}} &= 1+O_\gamma(c_n/(kL_n)), \qquad \beta_{n,0} = \sqrt{2k/\kappa}\, L_n(1+o(1)). \end{aligned} \tag{438}$$

These statements follow by integrating the exact unit $n_k$-mass of a cell and using $|n_k'(c)| = (2kL(c)/(\kappa c^2))(1+o(1))$. The errors tend to zero uniformly over $n \geq N$. In particular the constant-cell correction $q_{n,\gamma}$ is near one; it must not be replaced by the variable function $c^{-\gamma/2}$.

Apply the logarithmic isometries (411), extending the input by zero for $x \geq \log\theta_0$. For $e^y \in C_{k,n}$ the actual logarithmic kernel is

$$\mathcal{H}_{N,\gamma}(y,x) = e^{(1+\gamma)(x-y)/2}\beta_{n,\gamma} b_{k,n}(e^x) = q_{n,\gamma} e^{\gamma(x-y)/2}\mathcal{H}_{N,0}(y,x). \tag{439}$$

The boundary kernel is $\sqrt{2k/\kappa}\, H_{k,\gamma}(x-y)$, on the same input and output half-lines. This equality gives the precise weight conjugation, including the node-to-cell correction.

Write $t = x - y$. The complete actual-row estimates (407), the relative cell bound, and (438) imply, outside a fixed logarithmic-ratio interval,

$$|\mathcal{H}_{N,\gamma}(y,x)| \leq \begin{cases} C_{k,\gamma} e^{(1+\gamma)t/2}, & t < -R_0, \\ C_{k,\gamma} e^{-(1-\gamma)t/2}, & t > R_0. \end{cases} \tag{440}$$

For the boundary kernel the first decay is even stronger, and the second has only the extra factor $1+t$, by Lemma 70. Schur's test therefore gives, uniformly for large $N$,

$$\Big\| \mathbf{1}_{|x-y|>R}\big(\mathcal{H}_{N,\gamma} - \sqrt{2k/\kappa}\, H_{k,\gamma}(x-y)\big)\Big\|_{\mathrm{op}} \leq C_{k,\gamma}\big(e^{-(1+\gamma)R/2} + (1+R)e^{-(1-\gamma)R/2}\big). \tag{441}$$

The same one-variable integrable majorant bounds both Schur integrals. Restricting either physical half-line only decreases them. Thus this estimate controls infinitely many scales at once and does not count their number.

For $|x-y| \leq R$, partition each logarithmic line into unit intervals. Each output interval has at most $2\lceil R\rceil + 3$ input neighbors, and conversely. On an output block $y \in [j, j+1]$ take $h = e^j$. The scaled output $c/h$ and every neighboring input $\theta/h$ belong to fixed positive compact intervals depending only on $R$. Proposition H.1 applies on a fixed larger collar before restriction to the block. On this collar $L_n/\log(1/h) \to 1$ uniformly. Its exact compact-cell embedding and the present unweighted embedding differ, after the unitary rescalings, precisely by the factor $\sqrt{L_n/\log(1/h)}$, with the fixed density $2k/\kappa$ retained. Consequently their local unweighted kernels have the same Hilbert–Schmidt limit. For the

weighted kernels, the factor $e^{\gamma(x-y)/2}$ is a bounded smooth function on this bounded-ratio set, and $q_{n,\gamma} \to 1$ uniformly. The local Hilbert–Schmidt difference therefore still tends to zero. More explicitly, for some modulus $\eta_{k,\gamma,R}(h) \to 0$, every such block difference is bounded by $\eta_{k,\gamma,R}(e^j)$.

Full cells crossing block boundaries are first included in the fixed collar and then restricted. The same restrictions are made on both kernels, so their Hilbert–Schmidt difference can only decrease. This also covers the top output block cut at $d_{k,N}$. There is no false isometry assigned to a clipped cell. Replace the modulus by its decreasing supremum envelope. Since $e^j \le d_{k,N}$ for all retained output blocks, the finite-neighbor estimate gives

$$\left\| \mathbf{1}_{|x-y|\le R}\big(\mathcal{H}_{N,\gamma} - \sqrt{2k/\kappa}\, H_{k,\gamma}(x-y)\big) \right\|_{\mathrm{op}} \le C_R \sup_{0<h\le d_{k,N}} \eta_{k,\gamma,R}(h) \longrightarrow 0. \tag{442}$$

To see the finite-neighbor estimate directly, first take an input supported in finitely many unit intervals; apply the block operator bounds and Cauchy–Schwarz over the bounded number of neighbors on both sides. The resulting constant is independent of the number of intervals, so density proves the same estimate on the entire half-line.

For any tolerance choose $R$ large in (441), and then $N$ large in (442). This proves (436). Since the boundary operator is bounded by (412), these same estimates prove boundedness of the actual operator and justify its extension from compactly supported inputs. The cell norm identity then extends by completion. A finite sum of the channelwise squared errors proves the final assertion. ∎

**Lemma 73** (Compact outer output)**.** *For $0 \le \gamma < 1$, $d > 0$, and fixed $k$, the operator $\mathbf{1}_{[d,\infty)} T_k \mathbf{1}_{(0,\theta_0)}$ is Hilbert–Schmidt between the weighted spaces in* (411)*. Its squared Hilbert–Schmidt norm is*

$$\int_0^{\theta_0/d} |K_k(z)|^2 z^\gamma \log\left(\frac{\theta_0}{dz}\right)\, \mathrm{d}z < \infty. \tag{443}$$

**Proof.** The squared kernel integral relative to the input measure $\theta^{-\gamma}\,\mathrm{d}\theta$ and output measure $c^{-2-\gamma}\,\mathrm{d}c$ is

$$\int_d^\infty \int_0^{\theta_0} \theta^\gamma |K_k(\theta/c)|^2 c^{-2-\gamma}\, \mathrm{d}\theta\, \mathrm{d}c.$$

Set $\theta = cz$ and integrate $\mathrm{d}c/c$ over $d < c < \theta_0/z$ to obtain (443). At zero, Lemma 70 gives an integrable $O(z^{2+\gamma}|\log z|)$ bound. At the finite brake, a squared logarithm is integrable; the upper endpoint is harmless. This treats all $c \ge d$, not just a compact output interval. ∎

One useful sequential consequence makes the scope of the transfer explicit. Suppose $a_j$ is bounded in $L^2((0,\theta_0), \theta^{-\gamma}\,\mathrm{d}\theta)$, tends strongly to zero on each closed subinterval of $(0,\theta_0)$, and, for each fixed $k$, its actual weighted cap rows obey

$$\lim_{N\to\infty} \limsup_{j\to\infty} \sum_{n\ge N} L_n c_n^{-\gamma} |R_{k,n}[a_j]|^2 = 0.$$

The transfer theorem makes the boundary outputs small on the shrinking half-line. Local convergence and weighted boundedness imply weak convergence to zero in the full weighted input space, by density of tests supported away from zero. Lemma 73 makes the remaining output converge strongly to zero for every fixed $d$. Therefore $\|T_k a_j\|_{L^2(c^{-2-\gamma}\,\mathrm{d}c)} \to 0$ for every fixed $k$. Corollary H.1 then gives

$$\|F_{\gamma,j}\|_{H^{-1}(\mathbb{R})} \longrightarrow 0. \tag{444}$$

Indeed its Fourier mass on each fixed bounded interval tends to zero, while the remaining squared $H^{-1}$ norm is bounded by $(1+R^2)^{-1}\sup_j \|F_{\gamma,j}\|_2^2$.

In applying this consequence to a globally null input, the hypothesis on cap rows must come from the original full-row identity and a bound on the complementary angular input. No angular or frequency cutoff inherits nullity. The results of this section supply the fixed-channel weighted transfer and finite-frequency detection; they assert neither a uniform all-channel upper frame nor a numerical coercivity constant, and do not classify the original invariant-average kernel.

# I Interior regularity from the complete integer rows

The estimates in this appendix use the complete resonant rows from Section 5.2. A localization of the input is never assumed to have zero data. The result needed for compactness is a strictly positive local Sobolev gain; we also prove the stronger all-order estimate used in finite observation approximation.

**Theorem 22** (Interior estimates in the original norm)**.** *For every compact $K \Subset I$ and every finite $s \geq 0$,*

$$\|a\|_{H^s(K)} \leq C_{K,s}\|a\|_{L^2(I)}, \qquad a \in \mathcal{N}_0. \tag{445}$$

*The same estimate holds for the angular realization of $\mathcal{N}_E$. Constants can depend on $K$ and $s$; no estimate through either axis or uniformity as $s \to \infty$ is asserted.*

## I.1 The actual angular kernels, including both brakes

Fix a compact regular separation interval $I_c \Subset (0, c_*)$ inside a nonzero-twist chart for $r(c) = T_u(c)/T_v(c)$. The two brake angles $\theta_i(c) < \theta_o(c)$ and their derivatives are separated from zero and from the angular axes. Choose a target interval strictly between both brakes for every $c \in I_c$. Such a chart exists near any fixed angle in $I$: the small positive separation collar has $r'(c) > 0$, while the turning interval tends to $I$ as $c \downarrow 0$. Use only the actual indices

$$m \in 2\mathbb{Z}, \quad n \in \mathbb{N}, \qquad r(c) = m/n.$$

Unreduced pairs are retained, as harmonics of the primitive orbit.

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$$m \in 2\mathbb{Z}, \quad n \in \mathbb{N}, \qquad r(c) = m/n.$$

Unreduced pairs are retained, as harmonics of the primitive orbit.

**Lemma 74** (Uniform Abel and reflected kernels)**.** *There is a jointly analytic excursion kernel $\mathcal{K}_c(t,\theta)$ on the required closed triangles, with*

$$E_c[a](t) = \int_0^t \frac{\mathcal{K}_c(t,\theta)a(\theta)}{\sqrt{t-\theta}}\,\mathrm{d}\theta,$$
$$\mathcal{K}_c(t,t) = \frac{2\sqrt{2}\,\varrho_c(t)^3}{\sqrt{\partial_t Q_0(e_t)}} > 0, \qquad \varrho_c(t) = \left[\frac{c}{4\sin t(1-\sin^4 t)}\right]^{1/5}.$$

*Here $E_c$ is twice the physical excursion integral, including the $x$-reflected partner, without a period denominator. Put*

$$v_\theta(c,\theta) = \frac{\Omega_v(c)}{\cos\theta\sqrt{2(1-(c/b(\theta))^{4/5})}},$$
$$v(c,\theta_i) = 0, \qquad v(c,\theta_o) = \pi, \qquad b(\theta) = 4\sin\theta(1-\sin^4\theta)Q_0(e_\theta)^{-5/4}.$$

*Let $t_c$ be the inverse of $v(c,\cdot)$. The complete row density is $b_{mn} = P_{c,n} - N_{c,n}$, where*

$$P_{c,n}(\theta) = \frac{1}{\pi T_u \overline{S}} \int_{v(c,\max(\theta,\theta_i))}^{\pi} \frac{\mathcal{K}_c(t_c(z),\theta)\cos(nz)}{\sqrt{t_c(z)-\theta}}\,\mathrm{d}z \tag{446}$$

*for $\theta < \theta_o$, and $P_{c,n} = 0$ for $\theta > \theta_o$. For every finite $j, M$, $\|N_{c,n}\|_{C^j([0,\pi/2])} \le C_{j,M} n^{-M}$. All constants are uniform on $I_c$.*

**Proof.** At an angular critical point in the positive quadrant, $\ddot{\theta} = -\varrho^2\partial_\theta Q_0(e_\theta) < 0$. Each positive $u$ pulse therefore has one positive-quadrant excursion with one nondegenerate maximum. There are $n$ such pulses in the common auxiliary interval $nT_u$, and the negative $u$ pulses give their $x$ reflections. The state at a maximum is determined by $(c,t)$, with radial velocity $\pm\sqrt{2(1-Q_0(e_t)\varrho_c(t)^4)}$; the two signs are time reversals and have the same occupation measure. At the maximum an analytic Morse coordinate factors out $\sqrt{t-\theta}$. The sum of its two branches is analytic and gives the displayed diagonal coefficient. At a brake, temporarily allow the energy to vary and use the radial initial velocity $R$ as independent data. Time reversal makes the two-branch numerator even in $R$ and in the Morse variable. Substitution of $R^2 = 2(1-Q_0(e_t)\varrho_c(t)^4)$ gives an analytic extension through the brake. Away from the maximum, angular crossings are transverse; in particular $c > 0$ prevents an angular critical point at $\theta = 0$. Compactness provides uniform analytic neighborhoods and finite parameter-derivative bounds.

The simultaneous descending turning phase is $nv(c,t)$, because $m\Omega_u = n\Omega_v$. Folding all excursions, then integrating against the invariant phase character, changes the phase integral into $\int_0^\pi E_c[a](t_c(z))\cos(nz)\,\mathrm{d}z$ with coefficient $(\pi T_u\overline{S})^{-1}$. Fubini gives (446). For every even $m$ the same calculation can be made on the reduced $|u|$ oscillator, with Fourier pair $(m/2,n)$. Using a repeated auxiliary period repeats its excursion count and its period denominator together. This proves the formula for every unreduced row, without a primitive-parity restriction.

For the negative contribution use the complete phase torus. If $X$ is the auxiliary flow and $\Theta$ the physical polar angle, an angular stationary point would satisfy $c = 4\varrho^5 \sin\Theta(1-\sin^4\Theta)$. This is impossible on the closed negative half-plane for $c > 0$. The torus stays away from the origin, so $|X\Theta|$ is uniformly positive there. On a level circle of $\Theta$ take $T = \Theta_\beta\partial_\alpha - \Theta_\alpha\partial_\beta$. For the actual periodic character,

$$T(m\alpha - n\beta) = m\Theta_\beta + n\Theta_\alpha = nX\Theta/\Omega_u.$$

Integration by parts along the closed level circle gains $n^{-1}$. All density and vector-field derivatives are uniformly bounded in finitely many charts. Repetition, including additional integrations for angular derivatives, proves the reflected estimate through both axes. Periodicity of the integer character removes boundary terms; no noninteger continuation of this reflected row is used. ∎

The even turning expansions are

$$t_c(z) = \theta_i + \alpha_i z^2 + O(z^4), \qquad t_c(z) = \theta_o - \alpha_o(\pi - z)^2 + O((\pi - z)^4), \quad \alpha_i, \alpha_o > 0. \tag{447}$$

Watson's endpoint expansion in (446), on a fixed visible interior, gives

$$P_{c,n}(\theta) = n^{-1/2}A_0(c,\theta)\cos(nv(c,\theta) + \pi/4) + O(n^{-3/2}), \quad A_0 = \frac{\mathcal{K}_c(\theta,\theta)\sqrt{v_\theta}}{\sqrt{\pi}T_u\overline{S}} > 0. \tag{448}$$

There is a full finite classical expansion to any requested order. At $z = 0, \pi$ the relevant amplitudes are even, so integer cosine integration has no endpoint terms. The positive row is rapidly decreasing on fixed invisible intervals below the inner brake. Only the retained classical coefficients will be extended in continuous frequency below; these integer endpoint cancellations are not asserted for noninteger $n$.

**Lemma 75** (Actual brake estimates)**.** *Let $\Delta = |\theta - \theta_b(c)|$ denote distance to either brake. On the visible side the coefficient of $n^{-j-1/2}$ in* (448) *is $\Delta^{-1/4-j/2}$ times a smooth analytic coefficient. On a shell $r = r_b(\phi) \pm y/P^2$, with $y$ in a fixed positive compact interval, it is $P^{1/2+j}$ times a function with uniformly bounded finite $(\phi, y)$ derivatives. The exact kernel satisfies*

$$\int_{|\theta-\theta_b|<2\varepsilon} |b_{mn}(\theta)|^2 \,\mathrm{d}\theta \le C(n^{-2} + n^{-1}\varepsilon^{1/2}), \tag{449}$$

$$\left|R_{mn}\left[(\theta-\theta_b)^j\chi((\theta-\theta_b)/\varepsilon)\right]\right| \le C_{j,M}\varepsilon^{j+1}(n\sqrt{\varepsilon})^{-M} \tag{450}$$

*for fixed smooth compactly supported $\chi$, every integer $j \ge 0$ and $M \ge 1$. Both sides of the inner brake and the complete reflected row are included.*

**Proof.** If $z_0 = v(c,\theta)$ near the inner brake, factor $t_c(z) - t_c(z_0) = (z - z_0)(z + z_0)H(c, z^2, z_0^2)$, $H > 0$. The endpoint coefficients are therefore $z_0^{-j-1/2}F_j(c, z_0^2)$; $z_0^2$ is a positive analytic multiple of $\Delta$. The outer proof uses $\pi - z$ instead. Rescaling the endpoint distance gives

the parameter $n\sqrt{\Delta}$ and the stated conormal expansion, with an arbitrary additional inverse power of this parameter for the remainder. The substitution to moving-center shell coordinates proves the asserted derivatives; separate fixed-$r$ derivative bounds would not suffice for this step.

At the outer brake the local integral is a smooth even-amplitude version of

$$(-1)^n \int_0^{z_0} \frac{\cos(nz)}{\sqrt{z_0^2 - z^2}}\,\mathrm{d}z = (-1)^n \frac{\pi}{2} J_0(nz_0).$$

Its bound is $C(1 + nz_0)^{-1/2}$, from the substitution $z = z_0 \sin t$ and stationary phase at $t = \pi/2$. At the inner brake the denominators are respectively $\sqrt{z^2 - z_0^2}$ and $\sqrt{z^2 + z_0^2}$. Splitting at $z = 1/n$ gives $C(1 + |\log(nz_0)|)$ when $nz_0 \le 1$. For $nz_0 \ge 1$, the visible endpoint estimate is $C(nz_0)^{-1/2}$ and the invisible estimate is $C_M(nz_0)^{-M}$ by even cosine integration by parts. The remaining smooth integral has the endpoint cancellations in (447). Squaring and using $\Delta \asymp z_0^2$ gives (449); in particular $n^{-2}\int_0^1 x(1 + |\log x|)^2\,\mathrm{d}x$ is finite, with no $\log n$ loss.

For (450), the Abel response at $t = \theta_b + \varepsilon u$ is $\varepsilon^{j+1/2}$ times

$$\int_{-2}^{\min(u,2)} \frac{\mathcal{K}_c(\theta_b + \varepsilon u, \theta_b + \varepsilon v) v^j \chi(v)}{\sqrt{u - v}}\,\mathrm{d}v.$$

This smooth Abel convolution has bounded derivatives on bounded $u$ intervals, and its $k$th derivative is $O((1 + u)^{-k-1/2})$ for $u \ge 3$. After $t = t_c(z)$ its $M$th derivative has $L^1$ norm $C\varepsilon^{j+1-M/2}$. At the inner brake this includes the long tail $C\varepsilon^{j+1} z^{-M-1}$ for $z \gg \sqrt{\varepsilon}$; it does not presume an $O(\varepsilon^{j+1})$ undifferentiated $L^1$ bound there. Odd derivatives vanish at both phase endpoints. Integer cosine integration by parts proves the result. The reflected part is $O(\varepsilon^{j+1} n^{-M})$ by Lemma 74. ∎

## I.2 A compact-independent finite phase order

The following algebraic argument is included because repeated local regularity gains need one positive step independent of successive compact enlargements. Mere analyticity on each compact set would not supply that assertion.

**Lemma 76** (Uniform finite multiplicity)**.** *There is an integer $M \ge 2$ such that every zero of every nonidentically zero phase*

$$H(c;\theta,\phi) = \sigma v(c,\theta) - \sigma' v(c,\phi) - \pi p r(c) - 2\pi q, \qquad p, q \in \mathbb{Z}, \quad \sigma, \sigma' \in \{\pm 1\},$$

*has order at most $M$ in $c$, whenever both fixed angles are in the strict turning interior. The only identically zero case is $\sigma = \sigma'$, $\theta = \phi$, $p = q = 0$. On $H = 0$, the critical function $H_r(r, \theta, \Phi(r,\theta))$ has order at most $M - 1$, where $H(r,\theta,\Phi) = 0$.*

**Proof.** First, nonidentity follows from exact small-separation limits. Put $L = \log(1/c)$. Lemma 58 gives the scalar period estimates $T_v = L/\sqrt{2} + O(1)$ and $T_u - 2T_v = O(c)$. Integrating the exact $v_\theta$ gives

$$Lv(c,t) \longrightarrow 2\pi \log(\sec t + \tan t), \qquad L(r(c) - 2) \longrightarrow 0. \tag{451}$$

At the small lower turning angle, the change $z = b(t)/c$, with $b(t) = 4t + O(t^3)$, bounds the phase integral up to any fixed small angle by a constant times that angle. On the remaining interval ordinary dominated convergence proves (451). If $H$ vanishes on an interval, analytic continuation toward $c = 0$ first gives $q = -p$. Multiplication by $L$ then gives equality of the two signed strictly positive, strictly increasing functions $\log(\sec t + \tan t)$. Thus signs and angles agree, and the nonconstant $r$ forces $p = q = 0$.

To make the order uniform, use the actual scalar potential $U(x) = x^{10} - x^2$. For $s = \sin\theta$ put

$$\begin{aligned} D(s) &= 4s(1 - s^4), \qquad Q(s) = 1 + 10s^2 + 5s^4, \qquad R = (c/D(s))^{1/5}, \\ u &= \sqrt{R(1 + s)}, \qquad w = \sqrt{R(1 - s)}, \qquad k = \sqrt{2(1 - R^4 Q(s))}. \end{aligned}$$

Direct expansion gives $2(c - U(u)) = k^2u^2$ and $2(-c - U(w)) = k^2w^2$. On the curves $y^2 = 2f_e(x)$, $f_e(x) = e + x^2 - x^{10}$, let

$$J_u = \int_{a(c)}^{u} \frac{\mathrm{d}x}{\sqrt{2(c - U(x))}}, \qquad J_v = \int_{h(c)}^{w} \frac{\mathrm{d}x}{\sqrt{2(-c - U(x))}},$$

where $a$ is the upper crossing root and $h$ the upper positive-well root. Differentiating in $\theta$ gives

$$\frac{\mathrm{d}u}{ku} - \frac{\mathrm{d}w}{kw} = \frac{\mathrm{d}\theta}{k\cos\theta}, \qquad v = \frac{2\pi}{T_v}(J_u - J_v). \tag{452}$$

The integration constant is zero at the inner brake. Algebraically, with $P(s) = 5s^3 - 5s^2 + 5s - 1$,

$$\begin{aligned} 4D'Q - 5DQ' &= -16P(s)(5s^3 + 5s^2 + 5s + 1), \\ (1 + s)^4 - Q &= 4s(1 - s)(1 + s^2), \qquad 5(1 - s)^4 - Q = -4P(s). \end{aligned}$$

As $P$ is strictly increasing, at the inner brake $u = a, w = h$; at the outer one $u = a, w$ is the lower well root. This also verifies the endpoint phase $\pi$ in (452). All four upper endpoints from $\theta, \phi$ have $y > 0$ in the strict turning interior and obey $x_c = x/(10c)$.

Here is a rational differential system with no pole in that real interior. Let $\Delta(e) = e(3125e^4 - 256)$ and

$$\begin{aligned} A_e(x) &= \frac{3125e^4 - 2500e^3x^2 + 2000e^2x^4 - 1600ex^6 + 1280x^8 - 256}{\Delta(e)}, \\ B_e(x) &= -\frac{x(625e^4 - 500e^3x^2 + 400e^2x^4 - 320ex^6 + 256x^8 - 256)}{2\Delta(e)}. \end{aligned}$$

Multiplication verifies $A_e f_e + B_e f_e' = 1$. Hence

$$V = \partial_e - B_e \partial_x + (A_e/2) y \partial_y$$

is tangent to $y^2 = 2f_e$ and carries each lower branch point. It is regular there: $y$ is a local coordinate and $\mathrm{d}x/y = \mathrm{d}y/f_e'(x)$. The only excluded energies are $e = 0$ and $3125e^4 = 256$, so neither $e = c$ nor $e = -c$ has a pole for $0 < c < c_* = (4/5)5^{-1/4}$.

For $\omega_j = x^j \,\mathrm{d}x/y$, $0 \le j \le 8$,

$$\mathcal{L}_V \omega_j = W_j \,\mathrm{d}x/y, \quad W_j = -\partial_x(x^j B_e) - x^j A_e/2, \quad \mathrm{d}_x(Ry) = (2f_e R' + f_e' R)\,\mathrm{d}x/y.$$

The degree of $W_j$ is at most 16. The last polynomial with $R = x^m$ has leading term $-(2m+10)x^{m+9}$. Successively removing degrees 16 down to 9 therefore constructs polynomials $R_j$, of degree at most 7, and rational coefficients $M_{j\ell}(e)$ such that

$$W_j \,\mathrm{d}x/y = \sum_{\ell=0}^{8} M_{j\ell}(e)\omega_\ell + \mathrm{d}_x(R_j y). \tag{453}$$

Only nonzero rational numbers are divided out in this reduction; all coefficients belong to $\mathbb{C}[e, \Delta(e)^{-1}]$. For a period vector $P^\varepsilon$ and an incomplete vector $J^\varepsilon$ with upper endpoint $(x, y)$, (453) gives

$$\begin{aligned}
(P_j^\varepsilon)' &= \varepsilon \sum_\ell M_{j\ell}(\varepsilon c) P_\ell^\varepsilon, \\
(J_j^\varepsilon)' &= \varepsilon \sum_\ell M_{j\ell}(\varepsilon c) J_\ell^\varepsilon + \varepsilon R_j(\varepsilon c, x) y + \frac{x^j}{y}\left[\frac{x}{10c} + \varepsilon B_{\varepsilon c}(x)\right].
\end{aligned}$$

There is no lower-endpoint term: the horizontal field carries it and $R_j y$ vanishes there. For the physical cycles $P_0^+ = T_u$, $P_0^- = T_v$.

Take the polynomial ring generated by $c$, the four endpoint pairs, their 36 incomplete-integral coordinates, 18 period coordinates, and four constant coefficient parameters $S, T, P, Q$, localized by $c$, $3125c^4 - 256$, and the four upper ordinates $y_i$. It is Noetherian. The preceding system, together with

$$Dx_i = x_i/(10c), \quad Dy_i = [\varepsilon_i + f'_{\varepsilon_i c}(x_i)x_i/(10c)]/y_i, \quad DS = DT = DP = DQ = 0,$$

defines a derivation of this one ring. The fixed observable

$$g = 2S(J_{1,0} - J_{2,0}) - 2T(J_{3,0} - J_{4,0}) - PP_0^+ - 2QP_0^-$$

restricts, by (452), to $(T_v/\pi)H$. The ideals $(g, Dg, \ldots, D^j g)$ stabilize. At a stabilization index $M$, the ideal is $D$ invariant; thus vanishing of all jets through $M$ implies vanishing of every jet. Analyticity then gives an identically zero germ. This proves a single finite multiplicity bound for all evaluations and all integer $p, q$. No numerical value of $M$ or finite-dimensional approximation of an integral is used.

Finally $H_\phi \neq 0$ on $H = 0$. If $H(c, \theta, \Phi(c)) = 0$, then

$$H(c, \theta, \phi_0) = (\phi_0 - \Phi(c))A(c), \quad A(c_0) \neq 0, \qquad H_c(c, \theta, \Phi) = -H_\phi(c, \theta, \Phi)\Phi'(c).$$

Thus the order of the zero-set critical function is one less than the fixed-endpoint zero order. A regular analytic change $c \leftrightarrow r$ preserves this order; reversing $\theta, \phi$ gives the other direction. ∎

## I.3 The complete interior Poisson normal

Choose smooth frequency cutoffs $\chi \in C_c^\infty((1, 2))$ and $\eta \in C_c^\infty(I_r)$, where $I_r = r(I_c)$. For a fixed compact visible chart write

$$D_N f(m, n) = N^{-1/2}\chi(n/N)\eta(m/n) \sum_{\sigma=\pm} \int a_\sigma(m/n, \theta, n)e^{-i\sigma n v(m/n,\theta)} f(\theta)\, \mathrm{d}\theta.$$

The known amplitudes have uniform symbol bounds $|\partial_r^j \partial_\theta^k \partial_n^\ell a_\sigma| \leq C_{jk\ell} n^{-\ell}$ and principal coefficient $a_{\sigma,0} = (A_0/2)e^{-i\sigma\pi/4}$.

**Lemma 77** (All-dual normal estimate)**.** *The same-sign, zero-dual contribution near the angular diagonal is a bounded semiclassical pseudodifferential operator $P_N$. Every other contribution has $L^2$ operator norm $O(N^{-\delta_*})$, where one may take*

$$\delta_* = \min\{1/[12(M - 1)], 1/2\} > 0. \tag{454}$$

*Its constant depends on the compact geometry. The principal symbol on the sign-$\sigma$ frequency cone is*

$$p_0(\theta, \xi/N) = \frac{\pi|\xi|}{N} \int \frac{\eta(r)^2 |a_{\sigma,0}(r, \theta)|^2}{v_\theta(r, \theta)^2} \chi\left(\frac{|\xi|}{N v_\theta(r, \theta)}\right)^2 \mathrm{d}r. \tag{455}$$

*It is positive on a fixed nonempty frequency band after shrinking the target and ratio intervals.*

**Proof.** The lattice $2\mathbb{Z} \times \mathbb{Z}$ has dual $\pi\mathbb{Z} \times 2\pi\mathbb{Z}$ and density $1/2$. Poisson summation, with $n = Nt$, $m = nr$, gives each sign/dual kernel as

$$\frac{N}{2} \iint t\chi(t)^2 \eta(r)^2 A(r, \theta, \phi, Nt) e^{iNtH(r;\theta,\phi)}\, \mathrm{d}r\, \mathrm{d}t, \quad H = \sigma v(r, \theta) - \sigma' v(r, \phi) - \pi p r - 2\pi q. \tag{456}$$

For $|p| + |q|$ large, the gradient in $(m/N, n/N)$ is bounded below by $C(1 + |p| + |q|)$. Repeated two-dimensional integration by parts gives $C_L N^{1-L}(1 + |p| + |q|)^{-L}$, summable to arbitrary inverse order. This leaves finitely many duals, rather than only the zero dual.

On a nontrivial zero set, $H_\theta = \sigma v_\theta \neq 0$ and $H_\phi = -\sigma' v_\phi \neq 0$. Solve $H = 0$ for $\phi = \Phi(r, \theta)$. Lemma 76 gives, on finitely many compact charts,

$$\big|\{r : |H_r(r, \theta, \Phi(r, \theta))| < \varepsilon\}\big| \leq C\varepsilon^{1/(M-1)}, \tag{457}$$

uniformly in $\theta$, and similarly with $\theta, \phi$ reversed. This is the elementary one-variable finite-derivative sublevel estimate; the derivatives and lower bounds are chosen on a finite cover. The order is uniform even though the constants are not.

Near $|H_r| \le 2\varepsilon$, integrate (456) in $t$ to obtain $CN \int 1_{|H_r|\le 2\varepsilon}(1+N|H|)^{-L}\,\mathrm{d}r$. Change the source variable to $z = H$ in the first Schur integral. The critical function differs from that in (457) by $O(|z|)$. The portion $|z| \le \varepsilon$ is bounded by $C\varepsilon^{1/(M-1)}$; the remaining part is $C_L(N\varepsilon)^{-L}$ after increasing $L$. The reverse coordinate change gives the identical second Schur bound. Away from $|H_r| \le \varepsilon$, integrate twice in $r$. Derivatives of the smooth critical cutoff and of $H_r^{-1}$ cost at most $C\varepsilon^{-6}$, giving kernel bound $CN^{-1}\varepsilon^{-6}$. Taking $\varepsilon = N^{-1/12}$ proves the asserted gain.

For the retained diagonal write $H = \sigma(\theta-\phi)h_0$, with $h_0 = \int_0^1 v_\theta(r, \phi + s(\theta-\phi))\,\mathrm{d}s > 0$. The change $\xi = \sigma n h_0$ in (456) gives amplitude $|\xi|/(2Nh_0^2)$. In the convention $(2\pi)^{-1}\int e^{i(\theta-\phi)\xi}p\,\mathrm{d}\xi$, its diagonal is (455). Smooth expansion gives the usual bounded full symbol and an $O(N^{-1})$ first remainder. Positivity follows from $A_0 > 0$ and a ratio subinterval where both frequency cutoffs are positive. ∎

## I.4 The intermediate collars and their unbounded dual indices

**Lemma 78** (Intermediate brake layers)**.** *For a fixed interior target, let $M_N$ be the complete comparison operator from the visible collars $N^{-3/2} \lesssim \Delta \lesssim 1$ at both moving brakes. Then*

$$\begin{aligned} \|M_N\|_{L^2\to L^2} &\le CN^{-1/2}\log(2+N), \\ \|M_N f\|_2 &\le C_j N^{-j-1/2}\|f\|_{H^j(L)}, \qquad j \ge 1, \end{aligned}$$

*where $L \Subset I$ contains the fixed source collars. All actual expansion remainders, all Poisson duals, and the outer phase value $\pi$ are retained.*

**Proof.** On one dyadic layer $\Delta \asymp P^{-2}$, $P_0 \le P \le CN^{3/4}$, write $r = r_b(\phi) + \varepsilon y/P^2$. The source phase is $v(r,\phi) = e_b + \varepsilon_b A(r,\phi)\sqrt{\varepsilon(r - r_b(\phi))}$, where $e_b = 0$ or $\pi$ and $A > 0$. For the phase $H$ in (456), $H_{rr} \asymp P^3$ with fixed sign. A stationary point is possible only for one sign of $p$ with $|p| \asymp P$; it is unique in an enlarged layer. In the $y$ coordinate the nonconstant phase has size $P^{-1}$ and a nonzero uniformly bounded Hessian. Its parameter is $Nt/P$, not $Nt/P^2$. Lemma 75 supplies uniformly smooth moving-center amplitudes of size $P^{1/2}(P/N)^j$. Stationary phase therefore gives, for each stationary $p$ and every $q$,

$$N^{1/2}P^{-1}\int A_{N,P,p}(\theta,\phi,t)e^{iNt(h_p(\theta,\phi)-2\pi q)}\,\mathrm{d}t, \tag{458}$$

with bounded prescribed finite derivatives after the critical phase has been extracted. The critical point may lie outside the support; a smooth enlarged chart gives the same statement there. The exact envelope identities are

$$(h_p)_\theta = \sigma v_\theta(r_*,\theta) \asymp 1, \qquad (h_p)_\phi = -\sigma' v_\phi(r_*,\phi) \asymp P, \tag{459}$$

in absolute value, with fixed signs. After $t$ integration the absolute kernel is bounded by $C_L N^{1/2} P^{-1}(1 + N|h_p - 2\pi q|)^{-L}$. For fixed target, the source phase image crosses $O(P)$ periods of the $q$ lattice, while $\mathrm{d}\phi \le C\,\mathrm{d}h/P$. For fixed source, the target image crosses $O(1)$ periods and $\mathrm{d}\theta \le C\,\mathrm{d}h$. Both Schur integrals, with the sum over every $q$, are $O(N^{-1/2}/P)$. Summing the $O(P)$ stationary integers $p$ gives $CN^{-1/2}$ per layer and hence the logarithmic bound.

For nonstationary $p$, integration in $y$ gives any factor $(P/N)^L(1 + |p|/P)^{-L}$. Large $q$ are summed using integration in $t$; the remaining count is $O(1 + |p|)$. Thus all such kernels, stationary remainders, and discrete source remainders have the bound

$$C_L N P^{1/2}(P/N)^L. \tag{460}$$

For example $L = 12$ gives $N^{-13/8}$ before the logarithmic layer sum. This accounts for unbounded duals, rather than truncating them at $N$ or at $N^{3/4}$.

For source regularity, the reduced phase has expansion $h_p = -\pi p r_b(\phi) + h_0(\theta, \phi) + P^{-1}h_{1,P}$, with bounded finite derivatives. Thus all derivatives of $1/(h_p)_\phi$ are $O(P^{-1})$. Integrating (458) $j$ times in $\phi$ costs $(NP)^{-j}$ and differentiates only the input and the already smooth reduced amplitude. The fixed source cutoff removes boundary terms. The same two Schur integrals and the $p$ sum give $CN^{-j-1/2}P^{-j}$ per layer, whose sum is bounded for $j \ge 1$. In particular no fixed-$r$ moving cutoff is differentiated at a false cost $P^2$. For any requested $j$, choose $L \ge 4j + 12$ in (460); then its supremum bound is $O(N^{-j-13/8})$. The invisible inner rows are controlled by their $n\sqrt{\Delta} \ge cN^{1/4}$ rapid bound with one further integration order; the outer invisible positive kernel is zero. All these error estimates are absolute kernel bounds, so require no higher derivatives at an axis. ∎

## I.5 A full-input identity and the narrow $L^2$ frame

For a fixed target geometry put

$$\tau_{mn}(\theta) = n^{-1/2} \sum_{\sigma=\pm} \frac{A_0(r,\theta)}{2} e^{-i\sigma\pi/4} e^{-i\sigma n v(r,\theta)}, \quad w_{mn} = \chi(n/N)\eta(m/n)\sqrt{n/N}.$$

Let $\rho_1$ be a fixed target cutoff and define

$$S_N z = \rho_1 \sum_{m \ meven, n>0} w_{mn}\overline{\tau_{mn}} z_{mn}, \qquad Z_N a = S_N\{w_{mn} R_{mn}[a]\}.$$

Lemma 77 applied to these leading symbols gives $\|S_N\|_{\ell^2 \to L^2} \le C$. The physical normalizations are unchanged: $w_{mn} n^{-1/2} = N^{-1/2}\chi\eta$.

Complete nullity implies $R_{mn}[a] = 0$ already at original $C^1$ regularity. Indeed, on a rational torus the continuous function $Sg$ has zero auxiliary average on each closed orbit because $\mathrm{d}t = S\,\mathrm{d}\tau$. Averaging this identity against the invariant character $e^{-i(m\alpha - n\beta)}$ gives its torus Fourier coefficient zero by compact-domain Fubini. This needs no absolutely

convergent Fourier series or second angular derivative. Thus $Z_N a = 0$ for every real $N$ large enough; its rows still have integer indices.

Partition each complete row into fixed visible interiors, both narrow brake layers of width $\varepsilon = N^{-3/2}$, the intermediate collars, the invisible positive part, and the entire reflected part. This is a partition of the same integral. A finite product-chart cover of the fixed visible set allows Lemma 77 there; the cutoffs sum to one near the target diagonal. One obtains the exact comparison identity on a smaller target interval

$$0 = P_N a + E_N a + M_N a + T_N a + G_N a. \tag{461}$$

$P_N$ retains every needed lower-order same-sign diagonal symbol; $E_N$ contains the other fixed-interior terms; $G_N$ contains actual integer remainders and all reflected/invisible pieces. These last pieces satisfy $\|G_N\|_{L^2\to L^2} \le C_B N^{-B}$ at any prescribed finite order, by increasing expansion lengths. There are $O(N^2)$ rows and each target test is $O(N^{-1/2})$; the arbitrary angular supremum bounds therefore suffice. The same assertion for intermediate remainders follows from (460) with larger $L$.

**Lemma 79** (Narrow frame at every source order)**.** *With the actual narrow operator in* (461)*,*

$$\|T_N f\|_2 \le C_s N^{-3/8-3s/2}\|f\|_{H^s(L)}, \qquad s \ge 0, \tag{462}$$

*where $L$ is a fixed compact neighborhood of the moving source brakes. All nonprimitive harmonics are included.*

**Proof.** Write $T_N = S_N B_N$, where $(B_N f)_{mn} = w_{mn} R_{mn}[\chi_{\mathrm{end}} f]$. By (449), each narrow density has squared $L^2$ norm $CN^{-7/4}$. For fixed $n$, consecutive allowed numerators differ by two. Since the brake derivative in $r$ is bounded away from zero, the centers have spacing at least $c/n$. Their intervals of width $N^{-3/2}$ have bounded overlap for this fixed $n$. Cauchy–Schwarz in each interval, then summation in $m$, and finally over $O(N)$ denominators give

$$\|B_N\|^2_{L^2\to\ell^2} \le CN^{-3/4}.$$

One must not replace this argument by a bound on overlap across all rational ratios: repeated ratios at different denominators are real harmonics and have not been removed. Bounded $S_N$ proves the case $s = 0$.

For integer $j \ge 1$, let $p_W$ be the $L^2(W)$ projection of $f$ onto polynomials of degree $j-1$ on an enlarged narrow interval $W$. Rescaling its length $\asymp \varepsilon$ gives

$$\|f - p_W\|_{L^2(W)} \le C_j \varepsilon^j \|f^{(j)}\|_{L^2(W)}, \quad p_W = \sum_{k<j} \alpha_k (\theta - \theta_b)^k, \quad |\alpha_k| \le C_j \varepsilon^{-k-1/2}\|f\|_{L^2(W)}.$$

The remainder, summed by the same fixed-$n$ frame, has norm $CN^{-3/8}\varepsilon^j\|f\|_{H^j(L)}$. For the polynomial part, (450) gives each row at most $C\varepsilon^{1/2}(n\sqrt{\varepsilon})^{-M}\|f\|_{L^2(W)}$. Its frame norm is $C\sqrt{N\varepsilon}\,N^{-M/4}\|f\|_2 = CN^{-1/4-M/4}\|f\|_2$. Choose $M \ge 6j+1$. This is smaller

than the required remainder. The projection coefficients depend on the row but are never differentiated. Thus no Sobolev point trace has entered the proof. Interpolation between consecutive integer estimates, for the same linear $T_N$ with a fixed interval extension and source cutoff, proves (462) for all real $s \geq 0$. ∎

## I.6 Band inversion and finite-step regularity

Before making any $N$-dependent critical partition in (456), the source phase derivative is $H_\phi = -\sigma' v_\phi$, uniformly nonzero and independent of the dual indices. For integer $j$, integrate $j$ times in $\phi$ first. This produces $N^{-j}$ times operators with the same smooth amplitude class, acting on derivatives through order $j$. Lemma 77 applies with the same exponent $\delta_*$, and interpolation gives

$$\|E_N f\|_2 \leq C_s N^{-s-\delta_*} \|f\|_{H^s(L)}.$$

Lemma 78, with interpolation of its logarithmic $L^2$ endpoint for $0 \leq s \leq 1$, gives the convenient bound $\|M_N f\|_2 \leq C_s N^{-s-3/8} \|f\|_{H^s(L)}$. Indeed $\log(2+N) \leq C N^{1/8}$. Together with (462) and the arbitrary-order global $L^2$ remainder, (461) yields

$$\|P_N a\|_{L^2(J_1)} \leq C_s N^{-s-\eta_*} (\|a\|_{H^s(L)} + \|a\|_2), \quad \eta_* = \min(\delta_*, 3/8). \tag{463}$$

Here $L \Subset I$ contains every nonsmoothing source region of this fixed reconstruction. Its choice is independent of $a$.

The principal symbol (455) is bounded below on $A \leq |\xi|/N \leq B$ for some $0 < A < B$. Retain all lower-order diagonal terms to a requested finite order. Ordinary local semiclassical symbolic division then gives, for nested fixed spatial cutoffs and a frequency cutoff $\beta$ inside this band,

$$\rho_0 \beta(D/N) a = Q_N(\rho_1 P_N a) + \mathcal{E}_N a, \quad \|Q_N\|_{2\to 2} \leq C, \quad \|\mathcal{E}_N\|_{2\to 2} \leq C_M N^{-M}.$$

Explicitly start with symbol $\rho_0 \beta / p_0$ and solve each successive composition coefficient; the larger elliptic support keeps every division away from zero. Spatially separated terms $Q_N(1-\rho_1)P_N$ are rapidly decreasing by their smooth compact semiclassical frequency support. Thus this inversion does not require $P_N a$ small outside $J_1$.

To commute $\rho_0$ through the frequency projection, split the input with a fixed cutoff $\rho_2 = 1$ near $\operatorname{supp} \rho_0$. On $\rho_2 a$, the multiplier difference and rapid Fourier decay of $\rho_0$ give

$$\|[\beta(D/N), \rho_0](\rho_2 a)\|_2 \leq C_s N^{-s-1} \|\rho_2 a\|_{H^s}.$$

On comparable input frequencies this is the usual $N^{-1}$ commutator times $N^{-s}$; on non-comparable frequencies the fixed cutoff supplies arbitrary decay. The spatially separated input $(1-\rho_2)a$ costs only $C_M N^{-M} \|a\|_2$ by the Schwartz kernel. Combining with (463) gives the same bound for $\beta(D/N)(\rho_0 a)$.

A single dyadic sequence need not cover this narrow visible band. Choose an integer $L_0$ with $2^{1/L_0} < B/A$, with slack for the support of $\beta$, and use the finite shifted family $N = 2^{k+\ell/L_0}$, $0 \le \ell < L_0$. Its nonzero bands have bounded overlap and cover all sufficiently large frequencies. These real scale parameters only select actual integer rows. The square-function sum at every strict gain gives

$$\|a\|_{H^{s+\gamma_*}(K)} \le C_{K,s}(\|a\|_{H^s(L)} + \|a\|_2), \qquad \gamma_* = \eta_*/2 > 0. \tag{464}$$

Low frequencies are controlled by $L^2$.

To prove Theorem 22, choose $q$ with $q\gamma_* \ge s$. Starting from the final target compact, choose $q$ successively larger buffered compact source sets required by (464), in reverse order. Every finite enlargement stays inside $I$. Starting at order zero on the largest compact, iterate toward the target. The step $\gamma_*$ is independent of these choices by Lemma 76; only finitely many constants are multiplied. This proves (445) using global $L^2$, without an endpoint derivative bound. For $a \in \mathcal{N}_E$, approximate in $E$ by original $\mathcal{N}_0$ profiles. Coercivity and (445) applied to differences show convergence in each local $H^s$, agreeing with $Ja$. □

# J The EO energy space and compact angular observations

Throughout this appendix $I = (0, \pi/2)$, $\delta = \pi/2 - \theta$, and $\|\cdot\|_2$ denotes $L^2(I, \mathrm{d}\theta)$, without a reflection factor. An EO profile is extended evenly in $x$ and oddly in $y$. Let $\mathcal{W}_0$ be the original angular $C^1$ EO profiles with zero vertical value for which the finite physical-window quotient in Lemma 53 is continuous, bounded, and zero on the vertical orbit. Put

$$\mathcal{W}_P = \{a \in \mathcal{W}_0 : a(\pi/2 - \delta) = o(\delta^2)\}.$$

The original complete-null space with zero vertical value is denoted by $\mathcal{N}_0$. The original-input result of Theorem 18, followed by Lemma 53, gives $\mathcal{N}_0 \subset \mathcal{W}_P$. This entry uses neither completion nor finite dimensionality. Smooth profiles compactly supported in $I$, with their EO extension, also belong to $\mathcal{W}_P$.

For use below recall the positive representation from Proposition F.4:

$$E(a) = D(a) + P(a), \qquad P(a) = c_E \int_0^1 \mu(K)|I_a(K)|^2 \,\mathrm{d}K, \qquad c_E = \frac{32}{3\pi}. \tag{465}$$

$$D(a) = \frac{1}{2}\int_{I^2} \kappa(s,t) R_*(s) R_*(t) \left| \frac{a(s)}{R_*(s)} - \frac{a(t)}{R_*(t)} \right|^2 \mathrm{d}s\,\mathrm{d}t. \tag{466}$$

Here $R_*(\theta) = 20\cos\theta\sin\theta(1 + \sin^2\theta)$. Here $\kappa \ge 0$, $R_* > 0$ in $I$, and the kernel is finite almost everywhere off the diagonal. These formulas are initially identities for the original finite-window class. In particular the OO form used in $D$ is the legitimate sign flip of a zero-vertical EO force. No sign flip of a nonzero-vertical reference is used.

A single dyadic sequence need not cover this narrow visible band. Choose an integer $L_0$ with $2^{1/L_0} < B/A$, with slack for the support of $\beta$, and use the finite shifted family $N = 2^{k+\ell/L_0}$, $0 \le \ell < L_0$. Its nonzero bands have bounded overlap and cover all sufficiently large frequencies. These real scale parameters only select actual integer rows. The square-function sum at every strict gain gives

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## J.1 An explicit angular coercivity bound

**Lemma 80** (Ground-state gap)**.** *For every original $C^1$ OO quadrant profile $a$ there is a real number $t$ such that*

$$\|a - tR_*\|_2^2 \le 1\,643\,520\, D(a) < 2^{21} D(a)$$

*when $D(a) > 0$; the non-strict inequality with $2^{21}$ also includes $D(a) = 0$.*

**Proof.** We give the endpoint estimates as well as the interior argument. Write

$$b = \frac{\cos^2\theta}{1+\sin^2\theta}, \qquad q = 1 - b^2, \qquad \beta = \sqrt{1-K},$$
$$u = \frac{\beta^2}{b^2}, \qquad w = 1 - u, \qquad d_b = \sqrt{2-2b}.$$

The scalar differentiation and sandwich formulas for the Green kernel in (342)–(346) give, for $u \le 1/4$,

$$-F_q'(K) \ge G_{\mathrm{sc}} U b^4 \min\{P_0, wP_1/K\},$$

where

$$P_0 = 20w^2 + bu(14 - 11u) - b^2u(33u^2 - 56u + 20),$$
$$P_1 = 33b^3u^3 - 34b^3u^2 + 11b^2u^2 - 10b^2u - 42bu^2 + 44bu - 20u + 20,$$
$$G_{\mathrm{sc}} = K(1-K)^{3/2}(K-q)^{-3/4}/b^2, \qquad U = (d_b/2)K^{-1/4}[(1-K)(K-q)]^{-5/4}.$$

All these are exact algebraic quantities. For $0 \le b \le 1$ and $0 \le u \le 1/4$,

$$P_0 \ge 45/4, \qquad P_1 \ge 20w, \qquad wP_1/K \ge 45/4.$$

Indeed $P_0$ is concave in $b$. At $b = 0$ it is $20w^2$; at $b = 1$ it is $20 - 46u + 65u^2 - 33u^3$, decreasing on this interval and at least $771/64$. Also

$$P_1 - 20w = bu\{b^2u(33u - 34) + b(11u - 10) - 42u + 44\} \ge 0,$$

since the expression in braces is minimized at $b = 1$ and then exceeds $34 - 65/4$. Consequently

$$-F_q'(K) \ge \frac{45 d_b}{8} K^{3/4}\beta^{1/2} b^{-2} \quad (\beta \le b/2). \tag{467}$$

Let $c_0 = [32\,\mathrm{Beta}(5/4, 3/2)]^{-1}$. Jensen's inequality for the probability density $(3/2)(1-t)^{1/2}$ gives $\mathrm{Beta}(5/4, 3/2) \le 8/15$, so $c_0 \ge 15/256$. The exact positive series in the Green formula is

$$W_q(K) = \frac{3 b_{\mathrm{ph}} T_0 (1+b)^3}{32 b^{3/2}} \left( \sum_{j \ge 0} c_j \beta^j \right) F_q(K), \qquad b_{\mathrm{ph}} = 4\sqrt{2}, \quad c_j \ge 0.$$

Each summand has nonpositive $K$ derivative. Retaining $j=0$ and using the preceding inequality gives

$$-W_q'(K) \geq \frac{135 b_{\mathrm{ph}} T_0 c_0}{256}(1+b)^3 d_b K^{3/4}\beta^{1/2} b^{-7/2}.$$

For ordered angles $s<t$ put $b=b(s)>z=b(t)$. The exact occupation density and kernel are

$$G(K,t)=\frac{(1+z)^3}{2b_{\mathrm{ph}}z^{7/2}}\beta^{5/2}K^{-3/4}(z^2-\beta^2)^{1/4}, \qquad \kappa(s,t)=\frac{4}{T_0}\int_{q(t)}^{1} G(K,t)(-W'_{q(s)}(K))\,\mathrm{d}K.$$

Restrict this nonnegative integral to $0<\beta<z/2$. Since $(z^2-\beta^2)^{1/4}\geq (9/10)z^{1/2}$ and $2\int_0^{z/2}\beta^4\,\mathrm{d}\beta = z^5/80$, we obtain

$$\kappa(s,t)\geq \frac{1}{1440}(1+b)^3(1+z)^3\sqrt{2-2b}\,b^{-7/2}z^2. \tag{468}$$

The last rounding is $243(15/256)/20480 = 3645/5242880 > 1/1440$. There is no uniform positive lower bound at the vertical corner; its powers have been retained in (468).

Set $\psi(b)=a(\theta(b))/R_*(\theta(b))$. Direct differentiation gives

$$R_*\,\mathrm{d}\theta = 40(1+b)^{-3}|\,\mathrm{d}b|, \qquad \|a-tR_*\|_2^2=\int_0^1 \omega(b)|\psi(b)-t|^2\,\mathrm{d}b, \quad \omega(b)=\frac{1600\sqrt{2b(1-b)}}{(1+b)^5}.$$

Thus (466) and (468) imply

$$D(a)\geq \frac{10}{9}\int_{0<z<b<1} k(b,z)|\psi(b)-\psi(z)|^2\,\mathrm{d}z\,\mathrm{d}b, \quad k(b,z)=\sqrt{2-2b}\,b^{-7/2}z^2. \tag{469}$$

Use the symmetric extension of $k$ off the diagonal. For $A=[1/4,1]$, $J=[1/2,3/4]$, set $t=4\int_J\psi$. For $b\in A$, $z\in J$,

$$k(b,z)\geq \frac{125}{331776}\omega(b).$$

For $b\leq z$ this follows by minimizing $b^{3/2}(1+b)^5\sqrt{1-z}z^{-7/2}/[1600\sqrt{1-b}]$ at $b=1/4$, $z=3/4$. For $b\geq z$ the ratio is $z^2(1+b)^5/(1600b^4)\geq 1/200$. The factor $1/2$ from the symmetric form and Jensen's inequality yield

$$C_A:=\int_A \omega|\psi-t|^2\leq 19200 D(a).$$

For $h_j=2^{-j}$, $I_j=(h_j/2,h_j)$, let $m_j,v_j$ be the mean and variance of $\psi-t$ on $I_j$. Pairing $I_j$ with $I_{j+1}$ in (469) gives

$$H:=\sum_{j\geq 1} h_j^{1/2}[v_j+(m_j-m_{j+1})^2]\leq (576/5)D(a).$$

On $I_1$, $\omega\geq 128$, hence $m_1^2\leq C_A/32$. Geometric Cauchy–Schwarz and summation give

$$\sum_j h_j^{3/2}m_j^2\leq \frac{6}{5}m_1^2+5H, \qquad \sum_j h_j^{3/2}v_j\leq H/2.$$

Since $\omega(b) \le 2400\sqrt{b}$, it follows that

$$\int_0^{1/2} \omega |\psi - t|^2 \le 45 C_A + 6600 H.$$

Combining the overlapping cover $A \cup (0,1/2)$ gives

$$\|a - tR_*\|_2^2 \le [46(19200) + 6600(576/5)]D = 1\,643\,520 D.$$

All averages are legitimate for the original OO class, whose quotient by $R_*$ is bounded at the axes. ∎

**Lemma 81** (A compact rank average)**.** *There is an explicit angular $L^2$ functional $T$ with*

$$T(R_*) = 1, \qquad |T(f)| \le 16 \|f\|_2, \qquad |T(a)|^2 \le \frac{15}{16} E(a) \quad (a \in \mathcal{W}_0).$$

**Proof.** Write $\alpha = q^{-1}$, $\beta = \sqrt{1-K}$ and $c(K) = (1+\beta)^3/(16\sqrt{2}\sqrt{K}\beta)$. The exact rank formula (356) is

$$I_f(K) = -c(K) f(\alpha(K)) + \int_0^{\alpha(K)} \Lambda(K,s) f(s)\, \mathrm{d}s, \qquad \Lambda \ge 0,$$

and $I_{R_*}(K) = \tfrac{1}{2}\sqrt{(1+\beta)/\beta}$. Take $A = 1/4$, $B = 1/2$, and

$$\zeta(K) = 8\sqrt{\frac{\beta}{1+\beta}}\, 1_{[A,B]}(K), \qquad T(f) = \int_A^B \zeta(K) I_f(K)\, \mathrm{d}K.$$

Then $\zeta I_{R_*} = 4$ and $T(R_*) = 1$. Changing variables in the boundary term gives the density $j_T = k - m$, where

$$k(s) = 1_{q(s)<B} \int_{\max(A,q(s))}^{B} \zeta(K) \Lambda(K,s)\, \mathrm{d}K,$$
$$m(s) = 1_{A \le q(s) \le B}\, b(s)(1+b(s))^3.$$

Here the boundary subtraction has not been omitted. We record sufficient bounds on the exact kernel to verify the constants. Put $D = \operatorname{arctanh}\sqrt{K}$, $x = \operatorname{arctanh}\sqrt{q/K}$, and $M = \int_{D-x}^{D+x} \cosh(z)^{-3/2}\, \mathrm{d}z$. Its two pieces are

$$\Lambda_E = \frac{3}{64}\frac{(1+b)^{5/2}}{b^{9/2}} \sqrt{q}(1-K) K^{-3/2} [3(1-K^2) + b(K+3)],$$
$$\Lambda_M = -\frac{3(1+b)^3}{128\sqrt{2} b^{7/2}} \beta^{5/2} K^{-1/4} [3K + 2(1-q) - 3q/K] M (K-q)^{-3/4}.$$

On $A \le K \le B$, $q < K$, use $1/\sqrt{2} \le \beta \le \sqrt{3}/2 < 7/8$, $\beta \le b \le 1$, $M < 4$, and $\max\{-3K - 2(1-q) + 3q/K, 0\} \le 1 - K$. Direct substitution gives

$$0 \le \Lambda(K,s) \le \frac{1455}{256}\sqrt{q(s)}(1-K)K^{-3/2} + \frac{3}{4}(K - q(s))^{-3/4}. \tag{470}$$

Moreover $\zeta < 11/2$, $\alpha(A) \ge 1/4$, $\alpha(B) < 7/16$, and $\int_0^{\alpha(B)} q(s)\,\mathrm{d}s < 343/3072$. The last two follow from $q(s) \le 4s^2$ and $\sin(7/16) > 5/12 > \sqrt{2}-1$. The two integrals of the right side of (470) give nonnegative densities $k_E, k_M$ such that

$$k \le k_E + k_M, \quad \|k_E\|_2 < 8, \quad \|k_M\|_2 < 31/4, \quad \|m\|_2^2 < 27/4.$$

For completeness, use $\int_{1/4}^{1/2}(1-K)K^{-3/2}\,\mathrm{d}K = 5 - 3\sqrt{2} < 61/80$ to obtain $\|k_E\|_2^2 < 4359170778501/68719476736 < 64$. The fractional integral over any interval of length $1/4$ is at most $2\sqrt{2}$, giving $\|k_M\|_2 < 33\sqrt{14}/16 < 31/4$. Finally $0 \le m < 6$ on an interval of length $< 3/16$. As $k, m \ge 0$,

$$\|j_T\|_2^2 \le \|k\|_2^2 + \|m\|_2^2 < (8 + 31/4)^2 + 27/4 = 4077/16 < 256.$$

These bounds also justify absolute Fubini separately in both terms, so this is an actual $L^2$ density.

To bound $T$ by energy, the positive volume series gives

$$\mu(K) = b_{\mathrm{ph}} T_0 S(\beta) K^{3/4} \beta^{-3/2},$$

$$S(\beta) = \sum_{j \ge 0} \frac{a_j \beta^j}{32\,\mathrm{Beta}(5/4, 3/2 + j/2)}, \qquad \sum a_j z^j = (1 - z/2 - z^2/4)^{-3/2}.$$

The exact beta recurrence and period identity give $b_{\mathrm{ph}} T_0 c_0 = 21/16$. Thus

$$\mu(K) \ge \frac{21}{16}\left[\frac{1-\beta^2}{\beta^2(1-\beta/2-\beta^2/4)^2}\right]^{3/4} > 12/5 \quad (A \le K \le B).$$

To verify the last minimum without sampling, the bracket is decreasing in $\beta$: its logarithmic derivative has negative numerator

$$4 - 4\beta - 3\beta^2 + 2\beta^3 + 2\beta^4 = 4(1-\beta)^4 + 12\beta(1-\beta)^3 + 9\beta^2(1-\beta)^2 + \beta^4 > 0.$$

At $\beta = \sqrt{3}/2$ the bracket exceeds $9/4$, and $63\sqrt{6}/64 > 12/5$. Since $\zeta^2 < 30$ and $c_E > 10/3$, weighted Cauchy–Schwarz yields

$$|T(a)|^2 \le (25/8)\int_A^B \mu |I_a|^2 \le (15/16) E(a).$$

The use of $R_*$ is only through this compact trace; no full EO rank energy is assigned to $R_*$ or to $a - tR_*$. ∎

**Proposition J.1** (Angular coercivity)**.** *On $\mathcal{W}_0$,*

$$\|a\|_2 \le 2^{19}\sqrt{E(a)}. \tag{471}$$

**Proof.** Apply Lemma 80 to the genuine OO sign flip and write $g = a - tR_*$. Then $\|g\|_2 \le 2^{11}\sqrt{E}$. Lemma 81 gives $|t| = |T(a) - T(g)| \le \sqrt{E} + 16\|g\|_2$. Since $\|R_*\|_2^2 = 925\pi/16 < 196$,

$$\|a\|_2 \le 225\|g\|_2 + 14\sqrt{E} \le (225 \cdot 2^{11} + 14)\sqrt{E} < 2^{19}\sqrt{E}.$$

Only the $L^2$ density is applied to $g$. ∎

## J.2 Closability and the original Peano completion

**Proposition J.2** (Faithful completion)**.** *The completion $H_E$ of $\mathcal{W}_0$ in the norm $\sqrt{E}$ embeds injectively and continuously into $L^2(I)$. Its closed Dirichlet and rank maps retain* (465)*. Let $H_P$ be the closed subspace generated by $\mathcal{W}_P$, and let $\mathcal{N}_E$ be the $E$ closure of $\mathcal{N}_0$. Then $\mathcal{N}_E \subset H_P \subset H_E$.*

**Proof.** Coercivity produces a continuous realization $J : H_E \to L^2(I)$; injectivity requires a separate argument. Suppose $a_j$ is $E$ Cauchy and tends to zero in $L^2$. The functions

$$Ga_j(s,t) = \sqrt{\kappa(s,t)R_*(s)R_*(t)/2}\left[\frac{a_j(s)}{R_*(s)} - \frac{a_j(t)}{R_*(t)}\right]$$

are Cauchy in $L^2(I^2)$. A pointwise convergent subsequence of $a_j$ shows that their limit is zero almost everywhere and hence in $L^2$. The rank images are Cauchy in $L^2(\mu\,\mathrm{d}K)$. On every compact $K$ interval, Lemma 56 bounds their norm by a constant times $\|a_j\|_2$. Their limit is zero there, and a countable exhaustion gives zero on $(0,1)$. Thus $E(a_j) \to 0$. The closed extensions of $G$ and $I$ prove the stated identity. This proves a closure statement, without identifying $H_E$ with a maximal formal integral domain or assigning endpoint values to its elements. ∎

## J.3 The moving rank kernel and the vertical weight

**Lemma 82** (Weighted rank control)**.** *For $a \in H_P$,*

$$\int_I (1+\delta^{-4})|Ja|^2\,\mathrm{d}\theta \le C\{P(a) + \|Ja\|_2^2\}. \tag{472}$$

*Moreover for every $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that*

$$D(a) \le \varepsilon P(a) + C_\varepsilon \|Ja\|_2^2. \tag{473}$$

**Proof.** Use $b = \cos^2\theta/(1+\sin^2\theta)$ and $F(b) = a(\theta(b))$, extended by zero outside $(0,1)$. The exact rank trace, with its origin boundary term, becomes

$$I_a(1-\beta^2) = c(\beta)\mathcal{R}_\beta F(\beta), \qquad c(\beta) = \frac{(1+\beta)^3}{16\sqrt{2}\sqrt{1-\beta^2}\,\beta}, \tag{474}$$
$$\mathcal{R}_\beta F(\beta) = -F(\beta) + \int_1^\infty p_\beta(w)F(\beta w)\,\mathrm{d}w.$$

Here $p_\beta = 0$ for $w \ge 1/\beta$, so the full angular history is retained. On $1 < w < 1/\beta$, put $K = 1-\beta^2$, $b = \beta w$, $q = 1-b^2$, and use the $M$ integral in Lemma 81. Direct substitution in the displayed $\Lambda_E, \Lambda_M$ gives

$$p_E = \frac{3(1+\beta w)^2[(4-\beta^2)w + 6\beta - 3\beta^3]}{4K(1+\beta)^3 w^5},$$

$$p_M = -\frac{3\sqrt{2}}{16}\frac{\beta K^{-3/4}(1+\beta w)^2}{(1+\beta)^3}\frac{[(5-2\beta^2)w^2-6+3\beta^2]M}{w^4\sqrt{1-\beta w}(w^2-1)^{3/4}}, \qquad p_\beta = p_E + p_M.$$

In particular the moving-diagonal singularity has exponent $-3/4$, which is integrable.

Set $X = L^2((0,\infty), b^{-5/2}\,\mathrm{d}b)$. For original $a \in \mathcal{W}_P$, $F(b) = o(b)$, so $F \in X$ before any estimate is applied. The normal operator

$$\mathcal{R}_0 F(b) = 3\int_1^\infty w^{-4} F(bw)\,\mathrm{d}w - F(b)$$

is diagonalized by the unitary map $F \mapsto e^{-3x/4}F(e^x)$, followed by the Fourier transform with phase $e^{-i\eta x}$. Its multiplier is

$$m(\eta) = \frac{3/4 + i\eta}{9/4 - i\eta}, \qquad 1/9 \le |m(\eta)|^2 \le 1.$$

Consequently

$$\|F\|_X \le 3\|\mathcal{R}_0 F\|_X, \qquad \|\mathcal{R}_0 F\|_X \le \|F\|_X. \tag{475}$$

This full-line argument fixes the domain: the constant indicial mode is not in $X$. Equivalently, if $H = 3b^3\int_b^1 F(t)t^{-4}\,\mathrm{d}t$ and $g = H - F$, then $H' = 3g/b$, and the original Peano condition fixes $H(0) = 0$.

For $\beta \le 1/4$ the exact formulas imply

$$|p_E - 3w^{-4}| \le 64\beta w^{-3}, \qquad |p_M| \le 108\beta\frac{w^2+1}{w^4(w^2-1)^{3/4}}. \tag{476}$$

Here is a bound at the upper end of the moving interval which is sometimes important. Uniformly for $0 < b < 1$, $M/\sqrt{1-b} \le 15$: for $b \le 1/2$ use $M < 4$; for $b \ge 1/2$ put $t = \sqrt{q/K}$, use $t^2 \le 4/5$, $t^2 < (9/4)(1-b)$, and $M \le 2\operatorname{arctanh} t \le 10t$. This proves the second bound in (476) with the displayed constant. For the first write $p_E = 3w^{-4}AB$, where

$$A = \frac{(1+\beta w)^2}{K(1+\beta)^3}, \qquad B = 1 - \beta^2/4 + (6\beta - 3\beta^3)/(4w).$$

For $z = \beta w \le 1$, $|A-1| < 8z$, $|B| < 2$, and $|B-1| < 2z$, giving the claimed rounding.

Crucially take the supremum over the moving output parameter before integrating:

$$e(\beta_0) = \int_1^\infty w^{3/4}\sup_{0<\beta\le\beta_0}|p_\beta(w) - 3w^{-4}|\,\mathrm{d}w.$$

The zero extension contributes $3w^{-4}1_{w\ge 1/\beta_0}$. Since

$$\int_1^\infty \frac{w^{3/4}(w^2+1)}{w^4(w^2-1)^{3/4}}\,\mathrm{d}w < 22,$$

by the bounds $5(w-1)^{-3/4}$ on $(1,2)$ and $(5/2)w^{-11/4}$ on $(2,\infty)$, we have

$$e(\beta_0) \le (256/5)\beta_0 + 2376\beta_0 + (4/3)\beta_0^{9/4} < 2^{12}\beta_0. \tag{477}$$

Minkowski's inequality and $\|F(w\,\cdot)\|_X = w^{3/4}\|F\|_X$ therefore give

$$\|1_{\beta<\beta_0}(\mathcal{R}_\beta - \mathcal{R}_0)F\|_X \le e(\beta_0)\|F\|_X.$$

This is an operator-norm bound for the varying kernel, rather than pointwise convergence at a frozen $\beta$.

The exact physical density in (474) is comparable to $\beta^{-5/2}\,\mathrm{d}\beta$ on a fixed small interval. Thus its $\mathcal{R}_\beta F$ norm there is bounded by $C\sqrt{P}$. On the complementary output interval, triangularity gives

$$1_{\beta\ge\beta_0}\mathcal{R}_0 F = 1_{\beta\ge\beta_0}\mathcal{R}_0[1_{b\ge\beta_0}F], \qquad \|1_{b\ge\beta_0}F\|_X \le C_{\beta_0}\|a\|_2.$$

The latter follows using the bounded derivative $|\,\mathrm{d}b/\,\mathrm{d}\theta|$; no inverse horizontal Jacobian is used. Combining with (475), choose $3e(\beta_0) < 1/2$ and absorb. Then $\|F\|_X^2 \le C(P+\|a\|_2^2)$. Since $b = \delta^2/2+O(\delta^4)$ and $|\,\mathrm{d}b/\,\mathrm{d}\theta| = \delta(1+O(\delta^2))$, this is (472). Applying the estimate to differences of $E$-Cauchy original profiles proves it on $H_P$, with the same $L^2$ representative and no completed Peano trace.

Finally Lemma 55 gives $D(a) \le 2\int a_0|a|^2$ with $a_0(\theta) \le C(1+\delta^{-2})$. For $\delta < d$, use $\delta^{-2} \le d^2\delta^{-4}$; on the complement the coefficient is bounded. Thus $D \le Cd^2P + C_d\|a\|_2^2$, proving (473). This argument restricts a positive energy integral; it does not assume that a vertical input cutoff is an energy multiplier. ∎

**Proposition J.3** (A numerical vertical constant)**.** *The following squared-norm estimate holds on $H_P$:*

$$\int_I (1+\delta^{-4})|Ja|^2\,\mathrm{d}\theta \le 2^{87}E(a). \tag{478}$$

**Proof.** The exact positive series above has $S(\beta) \ge c_0 \ge 1/32$. Also

$$b_{\mathrm{ph}}T_0 = 4\int_{\mathbb{R}} \operatorname{sech}(z)^{1/2}\,\mathrm{d}z \ge 4.$$

Substituting $\mathrm{d}K = 2\beta\,\mathrm{d}\beta$ in the rank square gives

$$P(a) = c_E\int_0^1 \frac{b_{\mathrm{ph}}T_0 S(\beta)(1+\beta)^6}{256K^{1/4}}\beta^{-5/2}|\mathcal{R}_\beta F(\beta)|^2\,\mathrm{d}\beta.$$

The bracket is at least $2^{-11}$ and $c_E > 1$. Take $\beta_0 = 2^{-15}$. By (477), $3e(\beta_0) \le 3/8 < 1/2$. The small-output rank norm is at most $2^6\sqrt{P}$. Also $|b'(\theta)| \le 2$, so

$$\|1_{b\ge\beta_0}F\|_X \le \sqrt{2}\,\beta_0^{-5/4}\|a\|_2 < 2^{20}\|a\|_2.$$

Absorption gives $\|F\|_X \le 6(2^6\sqrt{P}+2^{20}\|a\|_2) \le 2^{42}\sqrt{E}$ by (471). For $0 < \delta \le \pi/4$, $b \le \delta^2$ and $\mathrm{d}b/\,\mathrm{d}\delta \ge \delta/4$. For $\delta \ge \pi/4$, $\delta^{-4} < 16$. Consequently

$$\int_I (1+\delta^{-4})|a|^2 \le 4\|F\|_X^2 + 17\|a\|_2^2 \le (2^{86}+17\cdot 2^{38})E(a) \le 2^{87}E(a).$$

Pass to $H_P$ on differences as before. ∎

**Theorem 23** (Weighted realization)**.** *Let $W = L^2(I, (1+\delta^{-4})\,\mathrm{d}\theta)$. The angular realization is an isomorphism $J : H_P \to W$: there are constants $C_1, C_2$ such that*

$$C_1\|Ja\|_W^2 \le E(a) \le C_2\|Ja\|_W^2, \qquad a \in H_P,$$

*and $J$ is onto. The first inequality may be taken with $C_1 = 2^{-87}$.*

**Proof.** Only the upper bound and surjectivity remain. On a fixed small $\beta$ interval the physical rank density is bounded above by a constant times $\beta^{-5/2}$, and the operator bound (477), together with (475), controls its rank square by $C\|F\|_X^2$. On the complementary $K$ interval use the full low-$K$ rank bound of Lemma 56. Thus $P \le C\|a\|_W^2$; the positive ground-row upper bound gives the same for $D$. Initially this holds on $\mathcal{W}_P$, and it passes to its closure. Every $f \in W$ can be approximated in $W$ by smooth profiles compactly supported in $I$. Their EO extensions belong to $\mathcal{W}_P$. The upper bound makes this approximation $E$ Cauchy. Its limit in $H_P$ has angular realization $f$, by the already proved injectivity and $L^2$ convergence. This proves surjectivity. It assigns no endpoint trace, original $C^1$ regularity, or nullity to an arbitrary element of $W$. ∎

**Corollary J.1** (Location of a possible energy defect)**.** *If $a_j \in H_P$ is $E$ bounded and $\|Ja_j\|_2 \to 0$, then $D(a_j) \to 0$. If in addition $E(a_j) \to e$, then, for every fixed $K_0 < 1$,*

$$c_E \int_{K_0}^1 \mu(K)|I_{a_j}(K)|^2\,\mathrm{d}K \longrightarrow e.$$

**Proof.** Apply (473) and let $\varepsilon \downarrow 0$ after the sequence limit. The complementary rank square tends to zero by the low-$K$ $L^2$ bound. This localizes a possible defect; it does not eliminate it. ∎

## J.4 Weighted horizontal compactness from actual null rows

Fix a sufficiently small horizontal collar $(0, \theta_0)$. The following theorem uses the complete all-integer row estimates of Appendix G and the boundary operators of Appendix H. We give the gluing proof, including its original-domain bootstrap and its infinite-scale summation.

**Theorem 24** (Subcritical horizontal weights)**.** *For every fixed $0 < \gamma < 1$ there is $G_\gamma < \infty$ such that*

$$\int_0^{\theta_0} \theta^{-\gamma}|a(\theta)|^2\,\mathrm{d}\theta \le G_\gamma E(a), \qquad a \in \mathcal{N}_0. \tag{479}$$

*The inclusion extends with the same angular realization to $\mathcal{N}_E$, and this inclusion into $L^2((0,\theta_0), \theta^{-\gamma}\,\mathrm{d}\theta)$ is compact. The constants need not remain bounded as $\gamma \uparrow 1$.*

**Proof.** Write $a_H = 1_{(0,\theta_0)}a$, $a_{\mathrm{off}} = a - a_H$ and use the isometry

$$F_\gamma(x) = e^{(1-\gamma)x/2}a(e^x), \qquad x < \log\theta_0,$$

extended by zero to $\mathbb{R}$. First consider a sequence of original complete-null profiles with bounded $E$ and bounded $\|F_\gamma\|_2$, whose angular $L^2$ norm tends to zero on each compact subset of $(0, \theta_0]$. The weighted norm is an assumption only for this intermediate sequential assertion.

**Bounded Mellin frequencies.** For each fixed positive integer $k$, the boundary operator has the form $T_k a(c) = \int K_k(\theta/c) a(\theta)\,\mathrm{d}\theta$. Lemma 70 gives $K_k(z) = O_k(z)$ at zero and $O_k(\log(2+z)/z)$ at infinity. After the unitary output conjugation for $L^2(c^{-2-\gamma}\,\mathrm{d}c)$, it is convolution with

$$H_{k,\gamma}(t) = e^{(1+\gamma)t/2} K_k(e^t), \qquad \widehat{H}_{k,\gamma}(\eta) = M_k((1+\gamma)/2 + i\eta). \tag{480}$$

The kernel is integrable precisely in the range used here. Theorem 20 and Corollary H.1 imply that on every bounded $\eta$ interval finitely many actual $k$ give a positive lower bound for the sum of squared multipliers. No all-frequency lower frame for finitely many $k$ is inferred.

Use the exact cells at the actual nodes $c_{k,n}$. If $m_{n,\gamma} = \int_{\text{cell } n} c^{-2-\gamma}\,\mathrm{d}c$, represent the row by the cell constant

$$\left(\frac{L_n c_{k,n}^{-\gamma}}{m_{n,\gamma}}\right)^{1/2} R_{k,n}[a_H], \qquad L_n = \log(1/c_{k,n}).$$

Its squared norm is exactly $\sum_n L_n c_{k,n}^{-\gamma} |R_{k,n}[a_H]|^2$. The fixed factor $\sqrt{2k/\kappa}$ in the boundary limit is the one in Theorem 21 and is harmless for a fixed finite set of $k$. The same lemma gives operator-norm transfer on the whole shrinking output half-line. It is not a sum of one transfer error per logarithmic block.

Full nullity says $R_{k,n}[a_H] = -R_{k,n}[a_{\text{off}}]$ at each actual row. The complete fixed-$k$ off-cap estimate in Lemma 67 therefore bounds the squared weighted data on $c < d$ by

$$C_k d^{6/5-\gamma} E(a). \tag{481}$$

The exact transfer error tends to zero uniformly on this half-line. On $c \geq d$ the boundary operator is Hilbert–Schmidt between the weighted spaces, since its squared kernel integral, apart from the fixed normalization, is

$$\int_0^{\theta_0/d} |K_k(z)|^2 z^\gamma \log\frac{\theta_0}{dz}\,\mathrm{d}z < \infty.$$

The bounded input sequence is weakly zero in the weighted space by its local convergence and density of compactly supported tests. Thus the outer output tends strongly to zero. First take the sequence limit, then $d \downarrow 0$ in (481) and in the transfer error. Every fixed-$k$ boundary output tends to zero. The finite multiplier cover now gives vanishing Fourier mass on every bounded interval. Splitting the Fourier integral at a large radius, using the bounded $L^2$ norm for the tail, proves

$$\|F_\gamma\|_{H^{-1}(\mathbb{R})} \longrightarrow 0. \tag{482}$$

**The common output space and exact IMS identity.** Choose a small logarithmic spacing $\ell > 0$ and smooth real translates $\chi_j(x) = \chi(x - j\ell)$ with bounded overlap and $\sum_j \chi_j^2 = 1$. Their supports lie in narrow intervals $j\ell + X$. Choose the same smooth output collar $\rho(c/h)$ and buffered target $e^X$ at every scale; in particular $\min e^X > \max \operatorname{supp}(\rho)/4$. Set $h_j = e^{j\ell}$, $L_j = \log(1/h_j)$ and

$$(B_j a)_{k,n} = \sqrt{L_j}\rho(c_{k,n}/h_j)R_{k,n}[a].$$

On a target-supported input this is a bounded all-integer analysis map by Lemma 61. For a general input it is initially used only in finite row sums and in the mixed limits justified by the separate source estimates. At each actual row only a bounded number of $j$ occur, and $L_j$ is comparable to $\log(1/c_{k,n})$.

For any finite row set and a real multiplier $\chi$, put $N = B^*B$. Then

$$\begin{aligned}\operatorname{Re}\langle Ba, B(\chi^2 a)\rangle - \|B(\chi a)\|^2 = \frac{1}{2}\iint (\chi(x) - \chi(y))^2 \\ \cdot N(x,y)a(y)\overline{a(x)}\,\mathrm{d}y\,\mathrm{d}x.\end{aligned} \tag{483}$$

The sign on the right is positive. Applying the full null row to $a = a_H + a_{\mathrm{off}}$ gives

$$\|B_j(\chi_j a_H)\|^2 = -\operatorname{IMS}_j(a_H) - \operatorname{Re}\langle B_j a_{\mathrm{off}}, B_j(\chi_j^2 a_H)\rangle. \tag{484}$$

In particular $a_H$ and $\chi_j a_H$ have not been assigned zero data. Multiply (484) by $h_j^{-\gamma}$. On the translated local chart the unweighted input isometry equals $h_j^{\gamma/2}p(x)F_\gamma(x + j\ell)$, with $p(x) = e^{\gamma x/2}$. Conjugation by this fixed positive smooth multiplier preserves the positive diagonal in Lemma 61. Its multiplication part commutes with $\chi_j$, and therefore vanishes in the IMS error.

**Compact defects and infinitely many scales.** The local normal is positive multiplication plus a norm-precompact family of compact operators and a remainder tending to zero. A compact family on fixed intervals has, for every $\varepsilon > 0$, the uniform form estimate

$$|\langle K_h u, v\rangle| \le \varepsilon\|u\|_2\|v\|_2 + C_\varepsilon\|u\|_{H^{-1}}\|v\|_2. \tag{485}$$

Approximation by finitely many smooth finite-rank kernels proves this statement. Lemmas 64 and 65 supply the same estimate on every fixed finite set of relative source-scale windows. The latter retains the whole reflected upper-time operator and its actual sampling estimate. Its constants are chosen in the order cap radius, high-frequency threshold, then $h$; no continuum estimate is substituted for an actual row bound.

For smooth translates $\zeta_j$ with bounded overlap,

$$\sum_j \|\zeta_j F\|_{H^{-1}(\mathbb{R})}^2 \le C\|F\|_{H^{-1}(\mathbb{R})}^2. \tag{486}$$

Indeed synthesis $\{g_j\} \mapsto \sum_j \zeta_j g_j$ is bounded from $\ell^2(H^1)$ to $H^1$ by the product rule and bounded overlap; duality gives (486). Thus all bounded-relative-scale compact errors sum by Cauchy–Schwarz, rather than by counting the infinitely many scales.

For completeness the far-scale matrix must also be summed before using a scalar error. The unweighted source bounds of Lemma 66, after multiplication by $e^{\gamma(i-j)\ell/2}$, are

$$\begin{aligned} t_{ji}^{\mathrm{small}} &\le C e^{(1+\gamma)(i-j)\ell/2}, \qquad i \ll j, \\ t_{ji}^{\mathrm{large}} &\le C\left[e^{-(3/5-\gamma/2)(i-j)\ell} + \frac{e^{(3/5-\gamma/2)j\ell+(2/5+\gamma/2)i\ell}}{1+|j|}\right], \qquad i \gg j. \end{aligned}$$

The second term comes from the scale-resolved upper input endpoint, not from replacing it prematurely by $\theta_0/h$. For $h_j \le d$, the Schur norm outside $|i-j|\ell \le R$ is bounded by

$$C\left[e^{-(1+\gamma)R/2} + e^{-(3/5-\gamma/2)R} + \frac{d^{3/5-\gamma/2}}{\log(1/d)}\right]. \tag{487}$$

For the cell term use its separable majorant $u_j v_i$; both geometric row and column sums converge. This pays the whole all-integer sampling error once. Summing an $O(h/L)$ error independently over all frequency bands would not prove (487). All exponents are positive for $0 < \gamma < 1$.

The complete off-cap mixed bound in Lemma 67 is $Ch_j^{3/5}\sqrt{E}\|v_j\|_2$. After weighting and summation it gives

$$\sum_{h_j \le d} h_j^{-\gamma} |\langle B_j a_{\mathrm{off}}, B_j(\chi_j^2 a_H)\rangle| \le C d^{3/5-\gamma/2}\sqrt{E}\,\|F_\gamma\|_2. \tag{488}$$

This uses the original vertical mass bound $\|1_{\delta<\nu} a\|_2 \le C\nu^2\sqrt{E}$ from Lemma 82; it gives no energy to the cutoff input.

Begin (484) with finitely many $j$ and a common smooth finite row exhaustion. First exhaust the rows. Target squares converge by the all-integer target upper bound; each mixed term converges by its local, separated, inner-brake, far-tail, or off-cap bound. Then exhaust $j \le J$, where $h_J \le d$. Equations (485), (486), (487), and (488) provide summable bounds for this second passage. No global all-row source-analysis map on $a_{\mathrm{off}}$ is presumed.

The local lower bound, summed in this order, consequently gives

$$\begin{aligned} \lambda \sum_{j \le J} \|\chi_j F_\gamma\|_2^2 \le\ & [C e^{-(1+\gamma)R/2} + C e^{-(3/5-\gamma/2)R} + C_R\varepsilon + \omega_R(d)]\|F_\gamma\|_2^2 \\ & + C_{R,\varepsilon}\|F_\gamma\|_{H^{-1}}\|F_\gamma\|_2 + C\|F_\gamma\|_{H^{-1}}^2 + C d^{3/5-\gamma/2}\sqrt{E}\,\|F_\gamma\|_2, \end{aligned} \tag{489}$$

where $\lambda > 0$ and $\omega_R(d) \to 0$ for fixed $R$. The latter includes (487)'s cell term and the uniform small-$h$ envelopes of the finite-relative-scale remainders. For the sequence under consideration, the omitted target region is bounded away from $\theta = 0$, so its norm tends to zero. Take a subsequence realizing the limsup of $\|F_\gamma\|_2$. Choose $R$ large, then $\varepsilon$ small, then $d$ small, so the square-norm coefficient is below $\lambda/2$. Use (482) and then let $d \downarrow 0$ in (488). This proves $\|F_\gamma\|_2 \to 0$ for the intermediate sequence.

**The original-domain bootstrap.** Every original $C^1$ EO input has $a(0) = 0$ and hence an individually finite horizontal weighted norm. If (479) failed on $\mathcal{N}_0$, normalize a violating sequence to have weighted norm one and $E \to 0$. Coercivity gives global angular $L^2 \to 0$, so all hypotheses of the intermediate assertion hold. Its conclusion contradicts the normalization. Thus the bound is proved before it is used for completion or compactness.

For a bounded $E$-weak-zero sequence of original null profiles, Theorem 22 and Rellich compactness give local angular $L^2 \to 0$. The newly proved bound supplies the weighted boundedness; the intermediate assertion gives strong convergence in that norm. Finally apply (479) to differences to extend the realization to $\mathcal{N}_E$. Approximate any completed weak-zero sequence by original null inputs within $1/j$ in $E$. Their difference is small in the weighted norm, and the original sequence argument proves compactness on $\mathcal{N}_E$. ∎

**Corollary J.2** (Compact angular realization)**.** *The map $J : \mathcal{N}_E \to L^2(I)$ is compact. For each fixed smooth horizontal cutoff $\chi_H$ and every $0 < \gamma < 1$, the map $a \mapsto \chi_H Ja$ into $L^2(\theta^{-\gamma}\,\mathrm{d}\theta)$ is compact.*

**Proof.** On a bounded energy set the vertical $L^2$ tail is at most $2^{87}\nu^4$ in squared norm by (478), and the horizontal squared tail is at most $G_\gamma \tau^\gamma$. On the retained compact interval use the positive interior Sobolev gain and Rellich. This proves angular compactness. The weighted conclusion is Theorem 24 with a bounded smooth multiplier. ∎

The proof of (479) is qualitative. It neither assigns a numerical value to $G_{3/4}$ nor bounds it by a norm chosen after finite dimensionality. The explicit principal coefficient in the row-calculus appendix controls one leading term. A numerical $G_{3/4}$ would additionally require numerical finite-frequency cover bounds and one quantified absorbable packet of the full actual-row IMS defects, with the same cutoffs and cells.

# K Exact ordinary endpoint operators and energy gluing

All numerators in this appendix are raw physical coarea integrals $T_g(c) = \int g\delta(H - 1)\delta(C - c)\,\mathrm{d}q\,\mathrm{d}W$. Dividing by a $c$-dependent torus clock would change the differentiated operator and is not done. Set $\mathsf{Q}_c = c\partial_c - 1$; an output restriction is always applied after this distributional derivative.

## K.1 The exact angular density

Let $s = \sin\theta$, $q(s) = 1 + 10s^2 + 5s^4$, $f_0(w) = w - w^5$ and

$$\lambda_\pm(s) = (1 \pm s)q(s)^{-1/4}, \qquad C_+(z,s) = f_0(\lambda_- z), \quad C_-(z,s) = -f_0(\lambda_+ z).$$

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Integrating the physical momentum direction on $H = 1$, and then setting the physical radius to $zq(s)^{-1/4}$, gives

$$\begin{aligned} K(c,s) &= 2q(s)^{-3/2}\int_0^1 \frac{z^5 \mathbf{1}_{C_-<c<C_+}}{\sqrt{(C_+-c)(c-C_-)}}\,\mathrm{d}z, \\ T_a(c) &= 2\int_I a(\theta)[K(c,\sin\theta)-K(c,-\sin\theta)]\,\mathrm{d}\theta. \end{aligned} \tag{490}$$

The momentum double covering contributes one full angular integral, not an additional factor two. The remaining factor two in the second line is physical $x$ reflection. In particular $K(c,-s) = K(-c,s)$ and $T_a(-c) = -T_a(c)$. Fix smooth source cutoffs $\chi_V, \chi_H$ at the vertical and horizontal axes, with disjoint supports and equal to one closer to those axes. Put $\chi_I = 1 - \chi_V - \chi_H$. These are restrictions of one original input; no cut input is asserted to be null.

## K.2 Vertical comparison in the exact brake coordinate

Put $r = \tan(\delta/2)$ and

$$y = c_b(r) = \frac{r^2(1-r^8)(1+r^2)^{1/4}}{(1+r^{10})^{5/4}}, \qquad F_a(y) = -\frac{(1+r(y)^2)^5}{16}\chi_V(\theta(y))a(\theta(y)). \tag{491}$$

The coordinate is increasing in a fixed small collar. Write $r^2 = y\rho(y)$, $a_b(y) = yr'(y)/r(y)$ and $t(y) = r(y)^8$; $\rho(0) = 1$, $a_b(0) = 1/2$. For $p_e(u) = \sqrt{2(e+u^2-u^{10})}$, let

$$\begin{aligned} J_-(c,\lambda,t) &= \int_{q_l(-c)}^{q_+(-c)} \frac{u^{11}\,\mathrm{d}u}{p_{-c}(u)\sqrt{2(u^2+\lambda-tu^{10})}}, \\ J_+(c,\lambda,t) &= \int_{q_l(-c)/r}^{q_+(c)} \frac{u^{11}\,\mathrm{d}u}{p_c(u)\sqrt{2(u^2-\lambda-tu^{10})}}, \end{aligned}$$

where the second integral is used only on its visible interval and $\lambda = c/r^2$, $t = r^8$. The exact positive visibility boundary is $c = c_b(r)$: the coincident roots obey $q_+(c)^8 = (1+r^2)/(1+r^{10})$. Consequently

$$\begin{aligned} T_V(c) &= 8B_{\mathrm{ex}}F_a(c), \qquad B_{\mathrm{ex}}F(c) = \int_0^\infty K_V(y,c/y)F(y)\frac{\mathrm{d}y}{y}, \\ K_V(y,z) &= a_b(y)\{J_-(yz,z/\rho(y),t(y)) - \mathbf{1}_{z<1}J_+(yz,z/\rho(y),t(y))\}. \end{aligned} \tag{492}$$

This identity includes the entire negative lower-turn history. Its normal is

$$B_0F(c) = \int K_0(c/y)F(y)\frac{\mathrm{d}y}{y}, \quad K_0(z) = \frac{1}{4}\int_0^1 \frac{u^{10}}{\sqrt{1-u^8}}\left[\frac{1}{\sqrt{u^2+z}} - \frac{\mathbf{1}_{u^2>z}}{\sqrt{u^2-z}}\right]\mathrm{d}u. \tag{493}$$

Thus $K_0(z) = K_V(0,z) = -4d(z)$ with $d$ from Appendix E.

**Lemma 83** (Fixed-brake comparison)**.** *Let $X = L^2((0,\infty), y^{-5/2}\,\mathrm{d}y)$, and use the same space in output coordinate $c$. For independently small $c_0, y_0 > 0$,*

$$\|\mathbf{1}_{c<c_0}\mathsf{Q}_c(B_{\mathrm{ex}} - B_0)F\|_X \le C(c_0 + y_0)\|F\|_X, \qquad \operatorname{supp} F \subset (0, y_0).$$

*The constant is uniform in the two collars and at every Mellin frequency.*

**Proof.** For small $z$ regularize the lower roots before differentiating. In the positive term use the inverse $Q_t$ of $w = u\sqrt{1-tu^8}$ and $w = \sqrt{v^2+\lambda}$. Its exact lower integrand is a smooth factor times

$$\frac{u^{11}Q_t'(w)}{w\sqrt{c+u^2-u^{10}}}.$$

It is $w^9$ times a uniformly smooth function of $w^2, t, c/w^2$. Here $c/w^2 \le Cy$; one $z$ derivative costs at most $w^{-2}$ and a $y$ derivative only bounded ratios. The mixed derivative is bounded by $Cw^7$. In the negative term use the inverse $Q_1$ of $w = u\sqrt{1-u^8}$ and $w = \sqrt{v^2+c}$. Its mixed derivatives are bounded by sums of

$$\frac{w^8}{\sqrt{w^2+\lambda}}, \quad \frac{w^{10}}{(w^2+\lambda)^{3/2}}, \quad \frac{cw^6}{\sqrt{w^2+\lambda}}, \quad \frac{zw^8}{(w^2+\lambda)^{3/2}}, \quad \frac{zw^{10}}{(w^2+\lambda)^{5/2}},$$

each at most $Cw^7$. The outer roots are simple and are regularized by a fixed square-root coordinate. These integrable majorants justify the mixed derivatives through $y = z = 0$. Since $K_V(y,0) = 0$ for every $y$,

$$|K_V - K_0| + |z\partial_z(K_V - K_0)| \le Cyz \quad (z \text{ small}).$$

Near $z = 1$ the two positive endpoints $l, A$ satisfy $A - l = (1-z)d(y,z)$, with $d$ smooth positive. Factor their denominators as $(A-u)A_1$ and $(u-l)B_1$, and put $u = l + (A-l)v$. The positive integral becomes

$$J_+ = \int_0^1 \frac{[l+(A-l)v]^{11}}{\sqrt{A_1(y,z,l+(A-l)v)B_1(y,z,l+(A-l)v)}} \frac{\mathrm{d}v}{\sqrt{v(1-v)}}.$$

This amplitude has a smooth one-sided continuation across $z = 1$; the physical indicator remains outside it. With $j(y,z) = a_b(y)J_+$, $j(0,1) = \pi/16$ and $j(y,1) - \pi/16 = O(y)$. Distributional differentiation therefore gives the diagonal multiplier $j(c,1)F(c)$, plus a regular kernel whose difference from the normal is $O(y)$ on each compact positive ratio interval. No shifted delta is approximated.

For $z \ge 2$ the positive branch is absent. The test $u^{11}/\sqrt{2(u^2 + z/\rho - tu^{10})}$ and its logarithmic $z$ derivative have weighted $C^1$ norm $O(z^{-1/2})$; their differences from $y = 0$ have norm $O(yz^{-1/2})$. The scalar measure estimates (315)–(316) then give

$$|K_V - K_0| + |z\partial_z(K_V - K_0)| \le C(c+y)z^{-1/2}, \quad c = yz \le c_0.$$

Conjugate by $UF(x) = e^{-3x/4}F(e^x)$. The regular kernel has the bound

$$C(c_0 + y_0)z^{-3/4}M(z),$$

where $M(z) = z$ near zero, is bounded on the middle interval and equals $z^{-1/2}$ near infinity. Its $\mathrm{d}z/z$ integral is finite and bounds both Schur integrals. The separate diagonal error has norm $Cy_0$. This proves the result first on compact smooth inputs and then by density, without differentiating an unknown input. ∎

## K.3 The scalar normal inverse and its sign

Write $\mathsf{A}_v = \mathsf{Q}_c B_0$. Under $U$ its multiplier, at $\alpha = 3/4 + i\eta$, is

$$m(\alpha) = -4(\alpha - 1)D(\alpha), \tag{494}$$

with $D$ in (321). The canceled pole at $\alpha = 1$ has nonzero value; there is no zero for $0 < \Re\alpha < 2$. The critical inverse measure in (322), with the factor $-1/2$ and the appropriate weight conjugation, is the inverse $\mathsf{S}_v = \mathsf{A}_v^{-1}$. It is the same convolution inverse in every weighted supremum class

$$\mathcal{B}_\alpha = \{F : \sup_{y>0} y^{-\alpha}|F(y)| < \infty\}, \qquad 0 < \alpha < 2. \tag{495}$$

Exponential moments justify compatibility on intersections of these classes, not merely nonvanishing on separate Mellin lines.

**Proposition K.1** (Positive normal inverse)**.** *The scalar inverse preserves nonnegativity and* $\|\mathsf{S}_v\|_{X\to X} < 24$. *There are finite positive-convolution approximants* $\mathsf{S}_v^{[N]}$ *such that*

$$\|\mathsf{S}_v - \mathsf{S}_v^{[N]}\|_{X\to X} < 24(4/5)^{N+1}.$$

*These are normal-operator estimates, distinct from the mechanical remainder contraction below.*

**Proof.** Let $k(z) = (z\partial_z - 1)K_0(z)$ off $z = 1$. We first prove $k(z) < 0$. Below the jump write $4K_0 = J_- - J_+$ with

$$J_- = \int_0^1 \frac{u^{10}\,\mathrm{d}u}{\sqrt{1-u^8}\sqrt{u^2+z}}, \qquad J_+ = \int_0^{\pi/2} h(z + (1-z)\sin^2\phi)\,\mathrm{d}\phi,$$
$$h(t) = \frac{t^{9/2}}{\sqrt{1+t+t^2+t^3}}.$$

For $p(t) = 1 + t + t^2 + t^3$, direct differentiation gives

$$8p^{7/2}t^{-3/2}h''' = 48t^9 + 168t^8 + 378t^7 + 693t^6 + 936t^5 + 1077t^4 + 1122t^3 + 1107t^2 + 756t + 315 > 0.$$

Thus $K_0''' < 0$. Twice integrating by parts against $t^{-1/2}(1-t)^{3/2}$ gives $J_+''(0) = J_-''(0) = \frac{3}{8}\int_0^1 t^2(1-t^4)^{-1/2}\,\mathrm{d}t$; all boundary terms vanish. Therefore $K_0''(0) = 0$, so $k'(z) = zK_0''(z) < 0$ and $k(0) = 0$. Above the jump,

$$k(z) = -\frac{1}{8}\int_0^1 \frac{u^{10}(2u^2+3z)}{\sqrt{1-u^8}(u^2+z)^{3/2}}\,\mathrm{d}u < 0.$$

The jump itself remains the positive atom $\pi/16$. Consequently $U\mathsf{A}_v U^{-1} = a_* \operatorname{Id} - T$, where $a_* = \pi/16$ and $T$ is convolution by the strictly positive $L^1$ kernel $-e^{-3x/4}k(e^x)$. Its mass is $a_* - m_0$, where

$$m_0 = m(3/4) = \frac{\pi}{128}\frac{\Gamma(17/16)}{\Gamma(25/16)}\frac{\Gamma(5/4)}{\Gamma(7/4)}(1+\sqrt{2}).$$

The beta integral shows that $\Gamma(t)/\Gamma(t+1/2)$ decreases in $t > 0$. Both arguments are below $3/2$, hence $m_0 \geq \pi^2(1+\sqrt{2})/512 > 27/640 > 1/24$. The Neumann series $a_*^{-1}\sum_{j\geq 0}(T/a_*)^j$ converges in operator norm and preserves positivity. The Fourier multiplier of $T$ has modulus at most its mass, with equality at zero frequency, so the inverse norm is exactly $1/m_0$. Finally $a_* < 11/56$ gives $\|T/a_*\| < 691/880 < 4/5$. Truncation proves the stated error. ∎

## K.4 The horizontal logarithm and the fixed interior

**Lemma 84** (Complete complementary components)**.** *For fixed sufficiently small horizontal source and output collars,*

$$\|\mathbf{1}_{c<c_H}\mathsf{Q}_c T_H[a]\|_X \leq C_H \|\chi_H a\|_{L^2(\theta^{-1/2}\,\mathrm{d}\theta)}.$$

*For every fixed compact interior source interval $J \Subset I$, there is $c_I > 0$ such that the corresponding operator is Hilbert–Schmidt from $L^2(J)$ to $L^2((0,c_I), c^{-5/2}\,\mathrm{d}c)$, and its output below $d \leq c_I$ has norm at most $C_J d^{9/4}\|a\|_2$.*

**Proof.** The common rest-point value in (490) at $z = 1$ is $b(s) = 4s(1-s^4)q(s)^{-5/4}$, with $b'(0) = 4$. Let $A(s) = \partial_z C_+(1,s)$ and $D_s = \partial_z C_-(1,s)$. At $s = 0$ they equal $-4, 4$. The exact upper-root factorization gives

$$K(c,s) = -L_0(s)\log|c - b(s)| + R(c,s), \qquad L_0(s) = \frac{2q(s)^{-3/2}}{\sqrt{(-A(s))D_s}}, \tag{496}$$

where $L_0$ is smooth even, $L_0(0) = 1/2$, and $|R_c(c,s)| \leq C(1 + |\log|c - b(s)||)$. Here are the uniform details. With $e = c - b(s)$ the two small upper radial roots satisfy $r_+ - r_- = eB(e,s)$, $B > 0$. Put $m = (r_+ + r_-)/2$, $d = (r_+ - r_-)/2$ and $r = m + \sqrt{\zeta^2 + d^2}$. The density is a smooth amplitude divided by $\sqrt{\zeta^2 + d^2}$. Subtract its value $L_0(s)$ at $\zeta = e = 0$. The remaining numerator is $O(\sqrt{\zeta^2 + d^2})$, with bounded first $e$ derivative; its differentiated integral is $O(1 + |\log|e||)$. The subtracted scalar integral has primitive $\operatorname{arsinh}(\zeta/|d|)$ and produces exactly the logarithm in (496). At the scalar origin set $v = C_+(z,s)$ and $v = c + t^2$. The exact $z^5$ numerator gives an integrand of size $(c+t^2)^{9/2}$ and a first derivative of size $(c+t^2)^{7/2}$. Thus the remainder bound is uniform also at $c = s = 0$. Reflection gives $R(c,-s) = R(-c,s)$.

Set $t = b(\sin\theta)$ and $g_H(t) = 2L_0(\sin\theta(t))\chi_H(\theta(t))a(\theta(t))|\theta'(t)|$. The principal signed horizontal numerator is

$$H_1 g_H(c) = \int_0^\infty \log\frac{c+t}{|c-t|} g_H(t)\,\mathrm{d}t.$$

Its Mellin multiplier on the line $\alpha = 3/4 + i\eta$, after applying $\mathsf{Q}_c$, is $\pi(\alpha-1)\tan(\pi\alpha/2)/\alpha$, which is uniformly bounded. The formula follows by integration by parts in the signed logarithm and the beta reflection integral; the two logarithmic boundary terms at $t = c$ cancel. The isometries are $g_H(e^x)e^{x/4}$ and $H_1 g_H(e^x)e^{-3x/4}$. Plancherel proves the asserted principal mapping. For the signed remainder $R(c,s) - R(-c,s)$, absolute continuity gives

$$|\mathsf{Q}_c[R(c,s) - R(-c,s)]| \le Cc(1 + |\log c| + |\log|c-t||).$$

Its squared angular integral is bounded by $Cc^2(1+|\log c|^2)$, so its squared Hilbert–Schmidt norm is bounded by $C\int_0^{c_H} c^{-1/2}(1+|\log c|^2)\,\mathrm{d}c < \infty$.

For $\theta \in J \Subset I$, choose $c_I$ below all positive rest-point values $b(\sin\theta)$. The upper radial endpoint is simple, below $z = 1$, and depends analytically on signed $c$; substituting its distance as a square removes the root. At the origin the small root changes its branch at $c = 0$. Removing a collar $z < \epsilon$ changes each one-sided $c$ jet of order $j \le 3$ by at most $C_J\epsilon^{5-j}$: after $z = u(c,s) + t^2$, the $j$th derivative is bounded by $C(c+t^2)^{9/2-j}$. The remaining integral is smooth through zero, so these one-sided jets match as $\epsilon \downarrow 0$. Thus $K(c,s)$ is uniformly $C^3$ in signed $c$ on this fixed interior. The complete signed kernel is odd and $\mathsf{Q}_c$ cancels its linear term. More precisely,

$$\mathsf{Q}_c[K(c,s) - K(-c,s)] = \int_{-c}^{c} v[K_{cc}(v,s) - K_{cc}(0,s)]\,\mathrm{d}v = O_J(c^3).$$

Its squared weighted kernel integral below $d$ is $O_J(d^{9/2})$, proving the stated Hilbert–Schmidt estimate. ∎

## K.5 Non-nested leakage and compact energy control

**Lemma 85** (Ordinary gluing)**.** *On the original zero-vertical complete-null class and its energy closure,*

$$E(a) \le C\left(\|Ja\|_2^2 + \|\chi_H Ja\|^2_{L^2(\theta^{-1/2}\,\mathrm{d}\theta)}\right). \tag{497}$$

**Proof.** The energy equivalence in Theorem 23 and the exact coordinate (491) give $E(a) \asymp \|Ja\|_2^2 + \|F_a\|_X^2$. In fact the density ratio of $\|F_a\|_X^2$ to $\delta^{-4}|\chi_V a|^2\,\mathrm{d}\theta$ tends to $1/16$. The normal satisfies $m_B\|F\|_X \le \|\mathsf{A}_v F\|_X$ for some $m_B > 0$ (one may take $m_B > 1/24$). For any fixed output threshold $d > 0$, the omitted output obeys

$$\|\mathbf{1}_{c\ge d}\mathsf{A}_v F_a\|_X \le C_d\|Ja\|_2. \tag{498}$$

Indeed split only within this estimate at $y = d/2$. On the smaller source interval the regular kernel is $O((c/y)^{-1/2})$, so the output is bounded by $Cc^{-1/2}\int |F_a(y)|y^{-1/2}\,\mathrm{d}y$. The latter moment is controlled by ordinary angular $L^2$ because $y^{-1/2}\,\mathrm{d}y$ is a bounded smooth multiple of $\mathrm{d}\theta$ in the exact coordinate. On $y \ge d/2$, the $X$ norm of $F_a$ is itself bounded by $C_d\|Ja\|_2$ and the full-line normal upper bound applies. This proves (498) without any nesting of the source and output collars.

First choose $y_0$ so small that the $Cy_0$ part of Lemma 83 is below $m_B/4$, and fix the three source cutoffs. Then choose $c_0 < \min(c_H, c_I)$ so small that $Cc_0 < m_B/4$. The exact original equation $8B_{\mathrm{ex}}F_a + T_H[a] + T_I[a] = 0$, after distributional differentiation, Lemma 84 and (498), gives

$$\tfrac{1}{2}m_B\|F_a\|_X \le C_H\|\chi_H Ja\|_{L^2(\theta^{-1/2}\,\mathrm{d}\theta)} + C'\|Ja\|_2.$$

This proves (497) on originals. Every displayed map has an already bounded realization on the energy closure, so approximation by original null inputs extends the inequality. No completed element was assigned an endpoint value before this passage. ∎

## K.6 An explicit supported mechanical contraction

Let $P_V$ be the coordinate-space support projection $\mathbf{1}_{y<y_0}$. The mechanical remainder is $R_V = \mathbf{1}_{c<d}\mathsf{Q}_c(B_{\mathrm{ex}} - B_0)P_V$. The explicit kernel bounds of Proposition L.1 are

$$\begin{aligned}
&|L(y,z)| \le 2^{12}yz && (z \le z_L),\\
&|L(y,z)| \le 2^{13}y && (z_L \le z \le 2),\\
&|L(y,z)| \le 2^{10}(c+y)z^{-1/2} && (z \ge 2),\\
&|j(c,1) - \pi/16| \le 2^7 c,
\end{aligned}$$

where $L$ is the regular differentiated difference, $c = yz$, $y_0 = 2^{-12}$, $d = 2^{-80}$ and $z_L = 1/16$. The diagonal is separate. The low-ratio log kernel is bounded by $2^{12}c^{1/4}y^{3/4}$. Its squared Hilbert–Schmidt integral over the larger rectangle is $(4/3)2^{24}d^{1/2}y_0^{3/2}$. For the middle and large pieces use $y = c/z \le d/z$ in the two Schur integrals. Including the diagonal gives

$$\|R_V\| \le \frac{2}{\sqrt{3}}2^{12}d^{1/4}y_0^{3/4} + C_*d, \quad C_* = 2^7 + \frac{4}{7}2^{13}(z_L^{-7/4} - 2^{-7/4}) + \frac{6}{5}2^{-5/4}2^{10}. \tag{499}$$

In particular $C_* < 2^{20}$, and Proposition K.1 implies

$$\|\mathsf{D}_v\| < 2^{-12}, \qquad \mathsf{D}_v = P_V\mathsf{S}_vR_V. \tag{500}$$

For example, $\sqrt{3} > 5/3$ bounds the leading term by $24(6/5)2^{-17} < 2^{-12}$, and the remaining $24\,2^{-60}$ is smaller than the resulting strict margin. This is an ambient supported-operator bound, so it controls every Neumann iterate. It is not the scalar positive-convolution ratio $4/5$, and does not assert positivity of the exact mechanical inverse. The same constants also give vertical-component contractions below $1/4$ at $d = 2^{-40}$ and below $1/2$ at $d = 2^{-36}$. Those larger output collars require separate compatibility with the fixed horizontal and interior estimates before they can be used for full reconstruction. Lemma 86 supplies the explicit ordinary-observation leakage bound $6/d$. Numerical observation-tail and separating-matrix bounds are separate matters.

# L Explicit vertical mechanical bounds

This appendix gives numerical bounds for the exact mechanical kernel, including its diagonal measure. The estimates concern the original coarea integrals and hold on ambient weighted spaces. They do not require an angularly localized input to remain in the invariant-average kernel.

## L.1 Exact coordinate and statement

Put

$$R = r^2, \qquad y = R\psi(R), \qquad \psi(R) = \frac{(1-R^4)(1+R)^{1/4}}{(1+R^5)^{5/4}}, \qquad a_v(y) = \frac{yr'(y)}{r(y)}. \tag{501}$$

The source collar is $0 < y \le y_0 = 2^{-12}$. Write $c = yz$, $\lambda = z\psi(R)$ and $t = R^4$. With $p_e(u) = \sqrt{2(e + u^2 - u^{10})}$, define the two exact radial integrals

$$J_-(c,\lambda,t) = \int_{u_l(c)}^{u_B(c)} \frac{u^{11}\,\mathrm{d}u}{p_{-c}(u)\sqrt{2(u^2+\lambda-tu^{10})}}, \tag{502}$$

$$J_+(c,\lambda,t) = \int_{u_+(\lambda,t)}^{u_A(c)} \frac{u^{11}\,\mathrm{d}u}{p_c(u)\sqrt{2(u^2-\lambda-tu^{10})}}. \tag{503}$$

The positive integral is present only when its lower endpoint is strictly below its upper endpoint; equality is treated by its one-sided amplitude. The outer-brake coordinate (501) makes this condition exactly $z < 1$. Thus the signed kernel is

$$K(y,z) = a_v(y)\big[J_-(yz, z\psi(R), R^4) - \mathbf{1}_{z<1}J_+(yz, z\psi(R), R^4)\big]. \tag{504}$$

Let $K_0(z) = K(0,z)$ and $\mathsf{Q}_z = z\partial_z - 1$. We use $L(y,z)$ for the regular part of $\mathsf{Q}_z(K(y,z) - K_0(z))$. The positive amplitude $j(y,z) = a_v(y)J_+(yz, z\psi(R), R^4)$ has a one-sided smooth extension to $z = 1$. The remaining diagonal multiplier is $j(c,1) - \pi/16$.

**Proposition L.1.** *The following bounds hold with the stated physical normalization:*

$$|L(y,z)| \le 2^{12}yz, \qquad 0 < y \le 2^{-12}, \quad 0 < z \le 1/16, \tag{505}$$

$$|L(y,z)| \le 2^{13}y, \qquad 0 < y \le 2^{-12}, \quad 1/16 \le z \le 2,\ z \ne 1, \tag{506}$$

$$|L(y,z)| \le 1024(c+y)z^{-1/2}, \qquad z \ge 2, \quad c = yz \le 2^{-20}, \quad y \le 2^{-12}, \tag{507}$$

$$|j(c,1) - \pi/16| \le 128c, \qquad 0 \le c \le 2^{-12}. \tag{508}$$

*At $z = 1$, the bounds for the regular kernel are understood one-sidedly; they do not include the separate diagonal measure. In particular, all four estimates apply simultaneously when $0 < c \le d \le 2^{-36}$.*

# L Explicit vertical mechanical bounds

This appendix gives numerical bounds for the exact mechanical kernel, including its diagonal measure. The estimates concern the original coarea integrals and hold on ambient weighted spaces. They do not require an angularly localized input to remain in the invariant-average kernel.

## L.1 Exact coordinate and statement

Put

$$R = r^2, \qquad y = R\psi(R), \qquad \psi(R) = \frac{(1-R^4)(1+R)^{1/4}}{(1+R^5)^{5/4}}, \qquad a_v(y) = \frac{yr'(y)}{r(y)}. \tag{501}$$

The source collar is $0 < y \le y_0 = 2^{-12}$. Write $c = yz$, $\lambda = z\psi(R)$ and $t = R^4$. With $p_e(u) = \sqrt{2(e + u^2 - u^{10})}$, define the two exact radial integrals

$$J_-(c,\lambda,t) = \int_{u_l(c)}^{u_B(c)} \frac{u^{11}\,\mathrm{d}u}{p_{-c}(u)\sqrt{2(u^2+\lambda-tu^{10})}}, \tag{502}$$

$$J_+(c,\lambda,t) = \int_{u_+(\lambda,t)}^{u_A(c)} \frac{u^{11}\,\mathrm{d}u}{p_c(u)\sqrt{2(u^2-\lambda-tu^{10})}}. \tag{503}$$

The positive integral is present only when its lower endpoint is strictly below its upper endpoint; equality is treated by its one-sided amplitude. The outer-brake coordinate (501) makes this condition exactly $z < 1$. Thus the signed kernel is

$$K(y,z) = a_v(y)\big[J_-(yz, z\psi(R), R^4) - \mathbf{1}_{z<1}J_+(yz, z\psi(R), R^4)\big]. \tag{504}$$

Let $K_0(z) = K(0,z)$ and $\mathsf{Q}_z = z\partial_z - 1$. We use $L(y,z)$ for the regular part of $\mathsf{Q}_z(K(y,z) - K_0(z))$. The positive amplitude $j(y,z) = a_v(y)J_+(yz, z\psi(R), R^4)$ has a one-sided smooth extension to $z = 1$. The remaining diagonal multiplier is $j(c,1) - \pi/16$.

**Proposition L.1.** *The following bounds hold with the stated physical normalization:*

$$|L(y,z)| \le 2^{12}yz, \qquad 0 < y \le 2^{-12}, \quad 0 < z \le 1/16, \tag{505}$$

$$|L(y,z)| \le 2^{13}y, \qquad 0 < y \le 2^{-12}, \quad 1/16 \le z \le 2,\ z \ne 1, \tag{506}$$

$$|L(y,z)| \le 1024(c+y)z^{-1/2}, \qquad z \ge 2, \quad c = yz \le 2^{-20}, \quad y \le 2^{-12}, \tag{507}$$

$$|j(c,1) - \pi/16| \le 128c, \qquad 0 \le c \le 2^{-12}. \tag{508}$$

*At $z = 1$, the bounds for the regular kernel are understood one-sidedly; they do not include the separate diagonal measure. In particular, all four estimates apply simultaneously when $0 < c \le d \le 2^{-36}$.*

## L.2 Coordinate bounds and fixed-domain formulas

Let $b = \psi_R/\psi$. Direct differentiation gives

$$b = -\frac{4R^3}{1-R^4} + \frac{1}{4(1+R)} - \frac{25R^4}{4(1+R^5)}.$$

Initially $R \le 1/256$ suffices to see $0 < b \le 1/4$: the positive term exceeds $1/8$, whereas the sum of the negative terms is below $1/16$. Thus $\psi \ge 1$ and the actual inverse collar satisfies $R \le y \le 2^{-12}$. On this sharper interval,

$$1 \le \psi \le 1 + R/4 < 9/8, \qquad |b_R| \le \tfrac{1}{4} + 24R^2 + 64R^6 + 25R^3 + 32R^8 < \tfrac{1}{2}.$$

Consequently

$$R_y = \frac{1}{\psi(1+Rb)} \le 1, \quad \psi_y = \frac{b}{1+Rb} \le \tfrac{1}{4}, \quad a_v = \frac{1}{2(1+Rb)}, \quad |(a_v)_y| \le \tfrac{1}{4},$$
$$0 \le t \le 2^{-48}, \quad 0 \le t_y \le 2^{-34}, \quad \lambda \ge z, \quad \lambda_z \le 9/8, \quad \lambda_y \le z/4, \quad \lambda_{yz} \le 1/4. \tag{509}$$

These inequalities also prove that the chart is increasing. Its value at $R = 1/256$ exceeds $2^{-12}$, so the entire declared collar is covered.

All roots in the rest of this appendix are *squared* radii. Put $F(w) = w - w^5$, and let

$$F(\ell) = F(B) = c, \qquad A^5 - A = c,$$

where $\ell$ is the small root, $B$ the root near one, and $A \ge 1$. For $0 \le c \le 2^{-11}$, explicit root brackets and inverse derivatives yield

$$0 \le \ell \le 2c \le 2^{-10}, \quad 31/32 \le B \le 1, \quad 1 \le A \le 1 + c/4 < 9/8,$$
$$|\ell_c| \le 2, \quad |B_c| \le 1/3, \quad 0 < A_c \le 1/4, \qquad |\ell_{cc}|, |B_{cc}|, |A_{cc}| \le 1. \tag{510}$$

Indeed $F(2c) \ge c$, $F(31/32) > 2^{-11}$, and $5(31/32)^4 - 1 > 3$. The second inverse derivatives have absolute values $20w^3/|1-5w^4|^3$. These bounds give $|\ell_{cc}| \le 160(2^{-10})^3 < 1$ and $|B_{cc}| \le 20/27 < 1$; the bound for $A$ is identical with the appropriate sign.

For $0 \le z \le 1$, let $\ell_+$ be the small root of $\ell_+ - t\ell_+^5 = \lambda$. The derivative $1 - 5tw^4$ is at least $15/16$ on $[0, 9/8]$. To verify visibility without an unspecified smallness condition, fix $R > 0$ and note that

$$(A - tA^5) - c/R = A(1-R^4) - c(R^4 + 1/R).$$

Its derivative in $c$ is negative. At $c = y$ it vanishes, since $A(y)^4 = (1+R)/(1+R^5)$. Therefore $\lambda \le \ell_+ \le A$ for $c \le y$. The limiting definitions at $R = 0$ are $\ell_+ = z$, $A = 1$. Implicit differentiation gives, with $D = 1 - 5t\ell_+^4$,

$$(\ell_+)_y = \frac{\lambda_y + t_y\ell_+^5}{D}, \qquad (\ell_+)_z = \frac{\lambda_z}{D}, \qquad (\ell_+)_{yz} = \frac{\lambda_{yz} + 5t_y\ell_+^4(\ell_+)_z + 20t\ell_+^3(\ell_+)_y(\ell_+)_z}{D}.$$

Using $\ell_+ \le 2\lambda \le 9z/4$ and $\ell_+ \le 9/8$ proves

$$0 \le (\ell_+)_y \le z/3, \qquad 0 < (\ell_+)_z \le 6/5 < 5/4, \qquad |(\ell_+)_{yz}| \le 1/3. \tag{511}$$

For example $t_y \ell_+^5 \le (9/2)2^{-34} z < z/1024$; the numerator of the mixed derivative is at most $1/4 + 12\,2^{-34} + 12\,2^{-48}$. Also $|A_y| \le z/4$, $|A_z| \le y/4$, and $|A_{yz}| < 1/3$.

Now substitute $w = u^2$ into (502)–(503). The factor is $1/4$, since $u^{11}\,\mathrm{d}u = w^5\,\mathrm{d}w/2$ and the two mechanical square roots contribute another factor two. For the negative branch put $w = \ell + (B - \ell)\sin^2\phi$ and set

$$\begin{aligned} \mathscr{C} &= w^3 + (B+\ell)w^2 + (B^2 + B\ell + \ell^2)w + (B^3 + B^2\ell + B\ell^2 + \ell^3), \\ M &= w + \lambda - t w^5. \end{aligned}$$

The exact identity $F(w) - c = (w - \ell)(B - w)\mathscr{C}$ gives

$$J_- = \frac{1}{2}\int_0^{\pi/2} w^5 \mathscr{C}^{-1/2} M^{-1/2}\,\mathrm{d}\phi. \tag{512}$$

Both scalar endpoints are retained. In particular $\mathscr{C} \ge B^3 > 1/2$.

For the visible positive branch put $w = \ell_+ + (A - \ell_+)\sin^2\phi$ and set

$$\mathscr{P} = c/A + A^3 w + A^2 w^2 + A w^3 + w^4, \qquad T = \sum_{j=0}^{4} w^{4-j}\ell_+^j, \qquad \mathscr{H} = 1 - tT.$$

Here $c + w - w^5 = (A - w)\mathscr{P}$ and $w - \lambda - t w^5 = (w - \ell_+)\mathscr{H}$. Hence

$$J_+ = \frac{1}{2}\int_0^{\pi/2} w^5 \mathscr{P}^{-1/2} \mathscr{H}^{-1/2}\,\mathrm{d}\phi, \qquad \mathscr{P} \ge w, \quad \mathscr{H} \ge 15/16. \tag{513}$$

The factor $1/2$ in both formulas includes the Jacobian of the sine substitution. Formula (513) extends to the coalescing endpoint $z = 1$ as an amplitude; the visibility indicator in (504) remains external.

## L.3 Uniform regular derivatives on the low and middle rectangles

We give derivative bounds on the whole rectangle $0 \le y \le 2^{-12}$, $0 \le z \le 2$; the positive branch is used only for $z \le 1$. This stronger statement makes the later ratio split independent of a limiting argument.

On the negative branch,

$$c \le \ell \le w \le 1, \quad |w_y| \le 2z, \quad |w_z| \le 2y, \quad |w_{yz}| \le 3, \quad |w_y w_z| \le 4c \le 4w.$$

The ten positive cubic monomials in $\mathscr{C}$ give

$$\tfrac{1}{2} < \mathscr{C} \le 10, \quad |\mathscr{C}_y| \le 60z, \quad |\mathscr{C}_z| \le 60y, \quad |\mathscr{C}_{yz}| \le 100.$$

For the last bound each monomial contributes at most $9 + 24c$, so the sum is below 100. With $h = \mathscr{C}^{-1/2}$ this yields

$$|h| \le 2, \quad |h_y| \le 90z, \quad |h_z| \le 90y, \quad |h_{yz}| \le 160.$$

Indeed the last reciprocal derivative is at most $10800\sqrt{2}\,c + 100\sqrt{2} < 160$. Moreover

$$M \ge (w+z)/2, \quad |M_y| \le 8M, \quad |M_z| \le 3, \quad |M_{yz}| \le 5.$$

These follow from the exact identities

$$\begin{aligned} M_y &= (1 - 5tw^4)w_y + \lambda_y - t_y w^5, \\ M_z &= (1 - 5tw^4)w_z + \lambda_z, \\ M_{yz} &= (1 - 5tw^4)w_{yz} + \lambda_{yz} - 5t_y w^4 w_z - 20tw^3 w_y w_z. \end{aligned}$$

Writing $b_- = M^{-1/2}$ and $N = w^5$, we obtain

$$\begin{gathered} |(b_-)_y| \le 4M^{-1/2}, \quad |(b_-)_z| \le \tfrac{3}{2}M^{-3/2}, \quad |(b_-)_{yz}| \le 21M^{-3/2}, \\ |N_y| \le 10zw^4, \quad |N_z| \le 10yw^4, \quad |N_{yz}| \le 95w^4. \end{gathered}$$

In the last estimate it is important to use $c \le w$ before replacing powers of $w$ by their upper bound.

For clarity, the nine terms in $(Nhb_-)_{yz}$, ordered as

$$\begin{aligned} &N_{yz}hb_-, \quad N_y h_z b_-, \quad N_y h(b_-)_z, \\ &N_z h_y b_-, \quad N h_{yz} b_-, \quad N h_y (b_-)_z, \\ &N_z h (b_-)_y, \quad N h_z (b_-)_y, \quad N h (b_-)_{yz}, \end{aligned}$$

are bounded respectively by

$$285, \quad 1, \quad 90, \quad 1, \quad 240, \quad 405, \quad 1, \quad 1, \quad 126.$$

For example $M \ge w/2$, $z \le 2M$, and $900\sqrt{2}\,c < 1$ control the mixed terms. Their sum is $1150 < 2048$. The same product rule gives

$$|Nhb_-| < 3, \quad |(Nhb_-)_y| < 512, \quad |(Nhb_-)_z| < 16, \quad |(Nhb_-)_{yz}| < 2048. \tag{514}$$

For the positive branch, $z \le \ell_+ \le w \le A \le 9/8$ and

$$|w_y| \le z/3, \qquad |w_z| \le 5/4, \qquad |w_{yz}| \le 1/3.$$

Differentiating the stable expression for $\mathscr{P}$ gives

$$|\mathscr{P}_y| \le 9z \le 9\mathscr{P}, \quad |\mathscr{P}_z| \le 20, \quad |\mathscr{P}_{yz}| \le 40.$$

To check the mixed bound, the four degree-four monomials contribute at most $8+(80/3)z$. The remaining term satisfies $|(c/A)_{yz}| < 2$, by differentiating $c/A$ as a scalar function of $c$. Thus, for $p = \mathscr{P}^{-1/2}$,

$$|p| \le w^{-1/2}, \quad |p_y| \le \tfrac{9}{2} w^{-1/2}, \quad |p_z| \le 10 w^{-3/2}, \quad |p_{yz}| \le 155 w^{-3/2}.$$

The five monomials of $T$ give $|T| \le 10$, $|T_y| \le 10z$, $|T_z| \le 38$, $|T_{yz}| \le 50$. It follows that

$$|\mathscr{H}_y| \le 2^{-30}, \quad |\mathscr{H}_z| \le 2^{-42}, \quad |\mathscr{H}_{yz}| \le 2^{-28}.$$

For $h_+ = \mathscr{H}^{-1/2}$, its value is at most two and its three listed derivatives are at most one. Finally

$$|N_y| \le \tfrac{5}{3} w^5, \quad |N_z| \le 7w^4, \quad |N_{yz}| \le 10 w^4.$$

The nine terms in $(Nph_+)_{yz}$, in the same order as above, are bounded by

$$40, \quad 67, \quad 4, \quad 126, \quad 620, \quad 9, \quad 14, \quad 20, \quad 2.$$

Their sum is $1002 < 1024$. Using $w^{7/2}, w^{9/2} < 2$ gives

$$\begin{aligned} |Nph_+| &< 4, \qquad |(Nph_+)_y| < 32, \\ |(Nph_+)_z| &< 128, \quad |(Nph_+)_{yz}| < 1024. \end{aligned} \tag{515}$$

The estimates retain positive powers of $w$ at a zero endpoint, so the integrands and the differentiated integrands extend by zero there. They justify differentiation under the fixed-domain integrals, including one-sided parameter derivatives. Since each integral has factor $1/2$ and interval length $\pi/2$, the same displayed bounds hold for $J_\pm$. Together with (L.2), they imply

$$|K_y| \le 2^9, \qquad |K_{yz}| \le 2^{11} \tag{516}$$

on each regular branch. At $z = 0$ the two original integrals agree for every $y$, so $K(y,0) = K_0(0) = 0$. Integrating $K_{yz}$ first in $y$ and then in $z$ proves $|K - K_0| \le 2^{11} yz$ and $|K_z - (K_0)_z| \le 2^{11} y$ for $z < 1$. This gives (505). On the whole middle rectangle, (516) instead gives $|\mathsf{Q}_z(K - K_0)| \le (2^9 + 2\,2^{11})y < 2^{13} y$, proving (506) without crossing the visibility jump.

## L.4 The exact jump and the unbounded ratio tail

At coalescence let $U$ denote the common radius in the original $u$ variable, and write $V = U^8 = (1+R)/(1+R^5)$. The endpoint derivatives of the two vanishing squared momenta are $4U(5U^8 - 1)$ and $4U(1 - 5R^4U^8)$. The beta integral of mass $\pi$ therefore gives

$$j(y,1) = \frac{\pi}{4} \frac{a_v(y) V^{5/4}}{\sqrt{(5V-1)(1-5R^4V)}}. \tag{517}$$

It equals $\pi/16$ at zero. On $R \le 1/256$ one may use $1 \le V \le 2$, $|V_R| \le 2$, $V^{5/4} \le 3$, $|(V^{5/4})_R| \le 5$. The denominator product is at least two and its derivative is below twenty. On $R \le 1/256$ the same rational bound above gives $|b_R| < 1/2$. The identity

$$(a_v)_R = -\frac{b + Rb_R}{2(1 + Rb)^2}$$

therefore gives the convenient bounds $a_v \le 1$, $|(a_v)_R| \le 8$ and $R_y \le 2$. Hence $|j_R| \le 8 \cdot 3 + 5 + 3 \cdot 10 = 59$, and $|j_y| \le 118 < 128$. This proves (508) by integration from zero. The diagonal is evaluated at $y = c$ when it acts on an input.

For $z \ge 2$ the positive branch is absent. Retain (512), now using independent parameters

$$0 \le c \le 2^{-20}, \qquad z/2 \le \lambda \le 2z, \qquad 0 \le t \le 1/256. \tag{518}$$

The scalar endpoints satisfy $0 \le \ell \le 2c$, $127/128 \le B \le 1$, $|\ell_c| \le 2$, $|B_c| \le 1/3$. On the fixed sine domain, $|w_c| \le 2$, $1/2 < \mathscr{C} \le 10$, and $|\mathscr{C}_c| \le 10(2 + 1 + 2) = 50$. With $\mathfrak{a} = w^5 \mathscr{C}^{-1/2}$ this gives

$$|\mathfrak{a}| \le 2, \qquad |\mathfrak{a}_c| \le 120.$$

Moreover $M = w + \lambda - t w^5 \ge \lambda \ge 1$ and $|M_c| \le 4$. Direct differentiation of (512) therefore proves

$$|(J_-)_c| \le 124 \lambda^{-1/2} \le 248 z^{-1/2}.$$

Define $H = \lambda (J_-)_\lambda - J_-$. In terms of the same fixed domain,

$$H = -\tfrac{1}{2} \int_0^{\pi/2} \mathfrak{a} \left( M^{-1/2} + \tfrac{\lambda}{2} M^{-3/2} \right) \mathrm{d}\phi.$$

The auxiliary expression has absolute value at most $3\lambda^{-1/2}/2$ and $M$ derivative at most $5\lambda^{-3/2}/4$. Its total $\lambda$ derivative, at fixed $c, t$, is exactly $-3\lambda M^{-5/2}/4$. Thus

$$|H| \le 6 z^{-1/2}, \qquad |H_c| \le 380 z^{-1/2},$$
$$|H_\lambda| \le 6 z^{-3/2}, \qquad |H_t| \le 8 z^{-3/2}.$$

These inequalities hold throughout the independent box (518), so they can be used on interpolation segments. The chart also gives $|a_v - 1/2| \le 4y$, $|\lambda - z| \le 4yz$, $t \le 4y$, and $a_v \le 1$. Since $z \partial_z = c \partial_c + \lambda \partial_\lambda$ at fixed $y$,

$$\mathsf{Q}_z K = a_v \big[ c (J_-)_c + H(c, \lambda, t) \big], \qquad \mathsf{Q}_z K_0 = \tfrac{1}{2} H(0, z, 0).$$

Subtracting and interpolating successively in $c$, $\lambda$, $t$ and $a_v$ bounds the five terms by

$$248 c z^{-1/2}, \quad 380 c z^{-1/2}, \quad 24 y z^{-1/2}, \quad 16 y z^{-1/2}, \quad 24 y z^{-1/2}.$$

Their sum is at most $(628c + 64y) z^{-1/2}$, proving (507). This argument keeps the entire negative lower history even as $z$ tends to infinity. Proposition L.1 is proved.

## L.5 A whole-observation leakage bound

Here is a complementary bound which explains the cost of making the output cutoff small. Put

$$X = L^2((0,\infty), y^{-5/2}\,\mathrm{d}y), \qquad \Phi(g)(y) = -\frac{(1+R)^5}{16}\chi_V(\theta(y))g(\theta(y)), \quad |\chi_V| \le 1,$$

where $\theta = \pi/2 - 2\arctan r$. Let $\mathsf{A}_v = \mathsf{Q}_c B_0$ be the scalar normal operator.

**Lemma 86.** *For $0 < d < 2y_0$, the map*

$$\mathsf{L}_d g = \mathbf{1}_{c\ge d}\mathsf{A}_v\Phi(g)$$

*has a bounded extension from $L^2(\,\mathrm{d}\theta)$ into $X$ with $\|\mathsf{L}_d\| \le 6/d$. The formula is interpreted by this extension when $\Phi(g) \notin X$.*

**Proof.** Let $g_v = \psi + R\psi_R$. The exact measure is $|\,\mathrm{d}\theta/\,\mathrm{d}y| = [(1+R)\sqrt{R}\,g_v]^{-1}$. On the collar, $\psi \ge 1/2$, $g_v \le 2$, and $(1+R)^{11} < 2$. Consequently

$$\frac{|\Phi(g)|^2 y^{-1/2}\,\mathrm{d}y}{|g|^2\,\mathrm{d}\theta} \le \frac{(1+R)^{11} g_v}{256\sqrt{\psi}} \le \frac{1}{32} < \frac{1}{16}, \qquad \|\Phi(g)\|_{L^2(y^{-1/2}\,\mathrm{d}y)} \le \tfrac{1}{4}\|g\|_2.$$

Split $F = \Phi(g)$ at $y = d/2$. For the upper part, $\|F\mathbf{1}_{y\ge d/2}\|_X \le \|g\|_2/(2d)$. The full scalar-normal bound $\|\mathsf{A}_v\| \le 10$ therefore gives $5\|g\|_2/d$, including its diagonal atom. For the lower part and $c \ge d$, the ratio is at least two and the complete regular normal kernel satisfies $|k_0(c/y)| \le (3/32)(c/y)^{-1/2}$. There is no atom in this region. Its output is bounded by

$$\frac{3}{32}c^{-1/2}\int_0^{d/2} |F(y)|y^{-1/2}\,\mathrm{d}y.$$

Cauchy–Schwarz and $\int_d^\infty c^{-7/2}\,\mathrm{d}c = (2/5)d^{-5/2}$ give

$$\|\mathsf{L}_{d,\mathrm{low}} g\|_X^2 \le \frac{9}{20480\sqrt{2}}\, d^{-2}\|g\|_2^2 < \frac{1}{(32d)^2}\|g\|_2^2.$$

The sum is bounded by $(5+1/32)\|g\|_2/d < 6\|g\|_2/d$. Both pieces are defined for every $L^2$ input and agree with the exact formula on the dense class supported away from the vertical endpoint. This gives the asserted extension without applying $\mathsf{A}_v$ outside its original domain. ∎

The tail bound used in this last proof can also be read directly from the scalar kernel. For $z \ge 2$ only the positive-denominator integral remains in $K_0$, and applying $\mathsf{Q}_z$ gives

$$|k_0(z)| \le \frac{3}{8}z^{-1/2}\int_0^1 \frac{u^{10}}{\sqrt{1-u^8}}\,\mathrm{d}u = \frac{3}{64}B(11/8,1/2)z^{-1/2} \le \frac{3}{32}z^{-1/2}.$$

The global bound $\|\mathsf{A}_v\| \le 10$ follows, in particular, from the scalar normal decomposition on the logarithmic energy line: $\mathsf{A}_v = (\pi/16)I - \mathsf{T}_v$, where $\mathsf{T}_v$ is positive convolution of mass $\pi/16 - m(3/4)$ and the Mellin value $m(3/4)$ is positive. Thus even $\|\mathsf{A}_v\| \le \pi/8 < 10$ holds. This is a bound for the entire normal operator, including its atom, not just the large-ratio tail. Neither Lemma 86 nor Proposition L.1 estimates the horizontal observation tail or identifies the dimension of the full invariant-average kernel.

# M The quartic vertical value trace

The results in this appendix use ordinary nullity and the original angular $C^1$ norm. They do not use complete resonant rows or finite dimensionality. Retain the exact coordinate $y = c_b(\tan(\delta/2))$, $F_a$ in (491), and $\mathsf{A}_v = \mathsf{Q}_c B_0$.

## M.1 Subcritical improvement from the exact equation

**Lemma 87** (Subquartic values)**.** *For every original $C^1$ EO ordinary-null profile with $a(\pi/2) = 0$, and each $0 < \alpha < 2$,*

$$|F_a(y)| \le C_\alpha \|a\|_{C^1} y^\alpha$$

*on a fixed sufficiently small collar. Consequently $a(\pi/2 - \delta) = o(\delta^{4-\epsilon})$ for every $\epsilon > 0$. The constants may depend on the fixed ordinary-null interval and on $\alpha$.*

**Proof.** Two signed cancellations improve the forcing and the error in Lemma 83. For compact $C^1$ horizontal input $g_H$ with $g_H(0) = 0$, integration by parts gives

$$\frac{\mathsf{Q}_c H_1 g_H(c)}{c^2} = \int_0^\infty A(s) g_H'(cs)\,\mathrm{d}s, \qquad A(s) = s \log \frac{s+1}{|s-1|} - 2.$$

Here $A$ is integrable, and its primitive

$$\frac{s^2-1}{2} \log \frac{s+1}{|s-1|} - s$$

has equal zero limits at zero and infinity. Hence $\int A = 0$. Dominated convergence gives $\mathsf{Q}_c H_1 g_H = o(c^2)$, with an $O(c^2)\|g_H'\|_\infty$ bound. Only continuity of $g_H'$ is used.

The exact remainder $R$ of (496) also satisfies $|R_{ccc}(c,s)| \le C(1 + |c - b(s)|^{-2})$ off its graph. To check the additional derivatives, in the upper-root coordinates of Lemma 84 write $v = (\zeta^2 + d^2)^{1/2}$ and let $F_1$ be the amplitude after subtracting its leading constant. For $j \le 3$, $|\partial_e^j v| \le C_j v^{1-j}$ and $|\partial_e^j F_1| \le C_j v^{1-j}$. Three derivatives of $F_1/v$ are therefore bounded by $Cv^{-3}$, whose integral is $O(|e|^{-2})$. The explicitly subtracted scalar logarithm has a smooth remainder. At the scalar origin the $z^5$ numerator gives the integrable majorants $(c + t^2)^{9/2-j}$ for $j \le 3$; removing a radial collar changes the jets by $O(\epsilon^{5-j})$ and proves matching through zero. For $t = b(s) > 2c$, odd cancellation now gives

$$|\mathsf{Q}_c[R(c,s) - R(-c,s)]| \le Cc^3(1 + t^{-2}).$$

For $t < 2c$ use the earlier bound $Cc(1 + |\log c| + |\log|c - t||)$. Since the original odd horizontal input is $O(t)\|a\|_{C^1}$, integration gives

$$\mathsf{Q}_c T_{H,\mathrm{rem}}[a] = O(c^3(1 + |\log c|))\|a\|_{C^1}. \tag{519}$$

Thus the complete horizontal forcing is $o(c^2)$ and uniformly $O(c^2)\|a\|_{C^1}$.

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*on a fixed sufficiently small collar. Consequently $a(\pi/2 - \delta) = o(\delta^{4-\epsilon})$ for every $\epsilon > 0$. The constants may depend on the fixed ordinary-null interval and on $\alpha$.*

**Proof.** Two signed cancellations improve the forcing and the error in Lemma 83. For compact $C^1$ horizontal input $g_H$ with $g_H(0) = 0$, integration by parts gives

$$\frac{\mathsf{Q}_c H_1 g_H(c)}{c^2} = \int_0^\infty A(s) g_H'(cs)\,\mathrm{d}s, \qquad A(s) = s \log \frac{s+1}{|s-1|} - 2.$$

Here $A$ is integrable, and its primitive

$$\frac{s^2-1}{2} \log \frac{s+1}{|s-1|} - s$$

has equal zero limits at zero and infinity. Hence $\int A = 0$. Dominated convergence gives $\mathsf{Q}_c H_1 g_H = o(c^2)$, with an $O(c^2)\|g_H'\|_\infty$ bound. Only continuity of $g_H'$ is used.

The exact remainder $R$ of (496) also satisfies $|R_{ccc}(c,s)| \le C(1 + |c - b(s)|^{-2})$ off its graph. To check the additional derivatives, in the upper-root coordinates of Lemma 84 write $v = (\zeta^2 + d^2)^{1/2}$ and let $F_1$ be the amplitude after subtracting its leading constant. For $j \le 3$, $|\partial_e^j v| \le C_j v^{1-j}$ and $|\partial_e^j F_1| \le C_j v^{1-j}$. Three derivatives of $F_1/v$ are therefore bounded by $Cv^{-3}$, whose integral is $O(|e|^{-2})$. The explicitly subtracted scalar logarithm has a smooth remainder. At the scalar origin the $z^5$ numerator gives the integrable majorants $(c + t^2)^{9/2-j}$ for $j \le 3$; removing a radial collar changes the jets by $O(\epsilon^{5-j})$ and proves matching through zero. For $t = b(s) > 2c$, odd cancellation now gives

$$|\mathsf{Q}_c[R(c,s) - R(-c,s)]| \le Cc^3(1 + t^{-2}).$$

For $t < 2c$ use the earlier bound $Cc(1 + |\log c| + |\log|c - t||)$. Since the original odd horizontal input is $O(t)\|a\|_{C^1}$, integration gives

$$\mathsf{Q}_c T_{H,\mathrm{rem}}[a] = O(c^3(1 + |\log c|))\|a\|_{C^1}. \tag{519}$$

Thus the complete horizontal forcing is $o(c^2)$ and uniformly $O(c^2)\|a\|_{C^1}$.

For the exact vertical kernel, signed $c$ reflection interchanges the two scalar allowed intervals. The complete kernel is odd in $c$. The lower-root regularizations in the proof of Lemma 83 permit three $z$ derivatives and one $y$ derivative at the corner: before differentiation the density has order $w^9$, and each parameter derivative costs at most two powers. The worst resulting majorant is $Cw$, still integrable. Removing a lower radial collar makes the signed jets match, with error $O(\epsilon^{8-2j})$, $j \le 3$. Therefore $K_V(y,0) = \partial_z^2 K_V(y,0) = 0$ and

$$|(z\partial_z - 1)(K_V(y,z) - K_0(z))| \le Cyz^3 \quad (z \text{ small}). \tag{520}$$

The fixed-jump and large-ratio bounds are unchanged. The differentiated error is consequently bounded, in both $\mathcal{B}_\alpha$ and $L^2(y^{-2\alpha-1}\,\mathrm{d}y)$, by $C_\alpha(c_0 + y_0)$ for $-1/2 < \alpha < 3$: its Mellin majorant is $z^3$ at zero, bounded in the middle and $z^{-1/2}$ at infinity. The integral $\int z^{-\alpha} M(z)\,\mathrm{d}z/z$ gives precisely this strip.

Theorem 18 gives $F_a = o(y)$, so the original input belongs to $\mathcal{B}_1$ before this argument. Fix $1 < \alpha < 2$. Choose the vertical source collar and then the output collar so that $\mathsf{D}_v = P_V \mathsf{S}_v R_V$ is a contraction simultaneously in $\mathcal{B}_1$ and $\mathcal{B}_\alpha$. These are qualitative choices; they need not equal the explicit energy collars used in (500). The exact ordinary equation gives

$$(\mathrm{Id} + \mathsf{D}_v)F_a = P_V \mathsf{S}_v G_a, \quad G_a = \mathbf{1}_{c \ge c_0} \mathsf{A}_v F_a - \tfrac{1}{8}\mathbf{1}_{c<c_0}(\mathsf{Q}_c T_H[a] + \mathsf{Q}_c T_I[a]).$$

The low-output forcing has order $c^2$, the fixed-interior term order $c^3$, and the high-output normal tail is $O(c^{-1/2})$. Its bounded middle-output part is controlled by the original $C^1$ norm and the fixed diagonal/regular kernels. Thus $G_a$ belongs to both weighted supremum classes, with bound $C_\alpha \|a\|_{C^1}$. The inverses in (495) are the same finite-measure formula. The Neumann solution in $\mathcal{B}_\alpha$ has compact upper support, hence also belongs to $\mathcal{B}_1$; uniqueness there identifies it with the original $F_a$. This proves the asserted gain for $1 < \alpha < 2$, and smaller exponents follow. Choosing a slightly larger exponent gives the little-oh statement. Equality almost everywhere yields the pointwise bound because the original profile is continuous. ∎

## M.2 The convergent critical forcing primitive

**Lemma 88** (Critical forcing integral)**.** *On the same original zero-vertical ordinary-null class the improper integral*

$$I(a) = \lim_{d \downarrow 0} \int_d^\infty (\mathsf{A}_v F_a)(c) \frac{\mathrm{d}c}{c^3} \tag{521}$$

*exists and $|I(a)| \le C\|a\|_{C^1}$. It need not be absolutely convergent at its lower endpoint.*

**Proof.** The only initially conditional part is the signed horizontal model. Let $K(u) = e^u A(e^u)$, with $A$ from the previous proof, and put $B(u) = \int_{-\infty}^u K(v)\,\mathrm{d}v$. The displayed

primitive shows

$$B(\log s) = \frac{s^2-1}{2}\log\frac{s+1}{|s-1|} - s < 0, \qquad \int_{\mathbb{R}} B(u)\,\mathrm{d}u = -\frac{\pi^2}{4}.$$

It decays exponentially at both ends. For the mass, integration by parts reduces it to $-\int_0^\infty \log(s)A(s)\,\mathrm{d}s$; expanding the logarithm below and above one gives $2+4\sum_{n\geq 0}[(2n+1)^2(2n+3)^2]^{-1} = \pi^2/4$. The equality follows by partial fractions and the odd reciprocal-square sum. With $V(x) = g_H'(e^x)$, distributional integration by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\int B(u)V(x+u)\,\mathrm{d}u = -e^{-2x}\mathsf{Q}_c H_1 g_H(e^x).$$

No derivative of $V$ is taken. Bounded dominated convergence therefore proves the exact improper identity

$$\int_0^\infty \mathsf{Q}_c H_1 g_H(c)\frac{\mathrm{d}c}{c^3} = -\frac{\pi^2}{4}g_H'(0). \tag{522}$$

Choose any fixed $1 < \alpha < 2$. By Lemma 87, $|F_a(y)| \leq C_\alpha\|a\|_{C^1}y^\alpha$. The exact vertical error $E_V = \mathsf{Q}_c(B_{\mathrm{ex}} - B_0)F_a$ obeys $|E_V(c)| \leq C_\alpha\|a\|_{C^1}c^{\alpha+1}$. Indeed (520) gives the integrable ratio power $z^{2-\alpha}\,\mathrm{d}z/z$ at zero after factoring $c^{\alpha+1}$; the middle and diagonal terms have the same order, and the large ratio bound gives $(1+z^{-1})z^{-\alpha-1/2}\,\mathrm{d}z/z$. Its $c^{-3}$ integral is absolutely convergent since $\alpha > 1$. The horizontal remainder (519) and the interior $O(c^3)$ term have the same absolute-integrability property. The full null equation is

$$\mathsf{A}_v F_a = -\tfrac{1}{8}(\mathsf{Q}_c H_1 g_H + \mathsf{Q}_c T_{H,\mathrm{rem}} + \mathsf{Q}_c T_I) - E_V \quad (0 < c < c_0).$$

Moreover $g_H'(0) = a'(0)/16$, since $b'(0) = 4$ and $L_0(0) = 1/2$. Combining this identity with (522) yields

$$\begin{aligned} I(a) = &\int_{c_0}^\infty \mathsf{A}_v F_a\,\frac{\mathrm{d}c}{c^3} + \frac{\pi^2}{32}g_H'(0) + \frac{1}{8}\int_{c_0}^\infty \mathsf{Q}_c H_1 g_H\,\frac{\mathrm{d}c}{c^3} \\ &- \frac{1}{8}\int_0^{c_0}(\mathsf{Q}_c T_{H,\mathrm{rem}} + \mathsf{Q}_c T_I)\frac{\mathrm{d}c}{c^3} - \int_0^{c_0} E_V\frac{\mathrm{d}c}{c^3}. \end{aligned} \tag{523}$$

Every integral on this right side is absolutely convergent. At infinity the normal is $O(c^{-1/2})$ and $\mathsf{Q}_c H_1 g_H = O(c^{-1})$. All pieces involve the same original input and have the stated $C^1$ bound. No Dini continuity assumption on $a'$ is required. ∎

## M.3 Removing the critical Mellin zero

**Lemma 89** (Critical value inverse)**.** *Suppose $F \in C((0,\infty))$ has compact upper support and $F(y) = O_\alpha(y^\alpha)$ for every $\alpha < 2$ sufficiently close to two. Assume $c^{-2}\mathsf{A}_v F(c) \to 0$ as $c \downarrow 0$ and that $\int_0^\infty \mathsf{A}_v F(c)c^{-3}\,\mathrm{d}c$ converges improperly. Then*

$$\lim_{y\downarrow 0}\frac{F(y)}{y^2} = -\frac{1}{m'(2)}\int_0^\infty \mathsf{A}_v F(c)\frac{\mathrm{d}c}{c^3}, \qquad m'(2) = -\frac{3\pi^2}{512}B(3/4,1/2) < 0. \tag{524}$$

**Proof.** Put $U(x) = e^{-2x}F(e^x)$ and $f(x) = e^{-2x}\mathsf{A}_v F(e^x)$. The actual differentiated kernel, including its atom, gives $f = \mu_2 * U$, where

$$\mu_2 = (\pi/16)\delta_0 + k_2(t)\,\mathrm{d}t, \qquad k_2(t) = e^{-2t}(z\partial_z - 1)K_0(z)\big|_{z=e^t}.$$

The regular density is $O(e^t)$ at $-\infty$ and $O(e^{-5t/2})$ at $+\infty$. The first bound follows from the cubic cancellation of the signed kernel, and the second from its full negative branch. The bilateral transform is $m(2+\zeta)$, initially by integration in the convergent strip and then by analytic continuation. Its total mass is $m(2) = 0$, and differentiating (494) gives the coefficient in (524). Let $p(t) = \mu_2((-\infty, t])$. Zero total mass gives the same exponential tails for $p$ at both ends; $p' = \mu_2$ in distributions and $\widehat{p}(\zeta) = m(2+\zeta)/\zeta$, with the removable value $\int p = m'(2) =: a_2$. The finite measure $N = \mu_2 + p(t)\,\mathrm{d}t$ has transform

$$n(\zeta) = m(2+\zeta)\frac{\zeta+1}{\zeta}, \qquad n(0) = a_2.$$

It is zero-free for $|\Re\zeta| < 1$. In this strip the gamma factors have no zeros or poles, and the only tangent zero is canceled at $\zeta = 0$; the neighboring tangent poles lie on the boundary. On each smaller strip $n = \pi/16 + O(|\Im\zeta|^{-1})$ and $n' = O(|\Im\zeta|^{-2})$, by the beta-integral argument in Appendix E. The same Fourier inversion and contour shift therefore construct

$$\Omega = (16/\pi)\delta_0 + r(t)\,\mathrm{d}t, \quad N * \Omega = \delta_0, \quad \Omega(\mathbb{R}) = 1/a_2, \quad \int e^{\eta|t|}\,\mathrm{d}|\Omega|(t) < \infty \quad (\eta < 1).$$

The assumed subcritical bounds put $U$ in every bilateral subexponential growth class. All the preceding convolutions and associativity identities are thus justified by the exponential moments. In particular $(p * U)' = f$. Let $h(x) = \int_{-\infty}^{x} f(s)\,\mathrm{d}s$, interpreted as an improper integral. It exists, tends to zero at $-\infty$, and $p*U = h+C$ for a constant $C$. Hence $N*U = f+h+C$. This right side tends to $C$ at $-\infty$ and retains subexponential growth. Applying $\Omega$ and dominated convergence shows that $U(x) \to C/a_2$. The dominance uses boundedness on a left half-line and the exponential moments on the right; boundedness of $U$ was not assumed. Finally upper compact support of $U$ gives $f(x), p*U(x) = O(e^{-5x/2})$ at $+\infty$. Taking that limit in $p * U = h + C$ yields $C = -\int_{\mathbb{R}} f$. This proves (524). The homogeneous constant is retained until the upper support condition fixes it. ∎

**Theorem 25** (Quartic vertical Peano trace)**.** *For an original $C^1$ EO ordinary-null profile with zero vertical value, there is a finite scalar $\lambda(a)$ such that*

$$a(\pi/2 - \delta) = \lambda(a)\delta^4 + o(\delta^4), \qquad \lambda(a) = I(a)/m'(2). \tag{525}$$

*The reflected statement follows by parity, and $|\lambda(a)| \le C\|a\|_{C^1}$ on each fixed ordinary-null interval.*

**Proof.** Lemmas 87 and 88 give every hypothesis of Lemma 89. In particular the exact ordinary forcing is $o(c^2)$; its improper primitive is the coupled expression (523). Since

$y \sim \delta^2/4$ and $a = -16(1+r^2)^{-5}F_a$ where the cutoff is one, the physical conversion has the minus sign $\lambda = -\lim F_a/y^2 = I(a)/m'(2)$. Different admissible cutoffs agree near the axis, so their actual limits agree. The trace formula then proves cutoff independence of the complete functional $I(a)$, although its individual terms change. ∎

A fourth-order Peano value expansion is not a $C^4$ endpoint theorem. No derivative quotient has been assigned a limit. Nor does (525) prove $\lambda(a) = 0$ or injectivity of this trace on the complete-null space. Even vanishing of its coefficient would leave higher-order or flat profiles to be examined. The nonzero vertical-trace affine problem retains its reference numerator and is not covered by this theorem.

# N Exact reconstruction and finite-dimensional comparison spaces

Let $\mathcal{H}_P$ be the completed Peano energy space of Theorem 23, with injective angular realization $J$, and let $\mathcal{N}_0$ be the original zero-vertical complete-null space. This appendix constructs an ambient operator which reconstructs members of $\mathcal{N}_0$ from compact observations. It does not parametrize the nullspace by arbitrary observation data.

## N.1 A supported inverse and its physical lift

Use fixed cutoffs from Appendix K. Write $H = (0, \theta_H)$, where $0 < \theta_H < \pi/2$ and $\operatorname{supp}\chi_H \subset [0, \theta_H)$. Set

$$\mathcal{Z} = L^2(I,\, \mathrm{d}\theta) \oplus L^2((0,\theta_H), \theta^{-1/2}\,\mathrm{d}\theta), \qquad \mathcal{K}a = (Ja, \chi_H Ja).$$

The second component is already cut off; the horizontal operator below does not apply $\chi_H$ again. Set $X_V = P_V X$, with $P_V$ the support projection in the exact $y$ coordinate. Write

$$R_V = \mathbf{1}_{c<c_0}\mathsf{Q}_c(B_{\mathrm{ex}} - B_0)|_{X_V}, \quad \mathsf{D}_v = P_V \mathsf{S}_v R_V, \quad \|\mathsf{D}_v\| \le q < 1.$$

Qualitative small-collar choices suffice. The explicit compatible vertical collars of (500) give $q < 2^{-12}$; horizontal and interior output restrictions may be decreased further without worsening that estimate.

Let $\Phi(g)$ be the coordinate/amplitude formula for $F_a$ in (491), with $Ja$ replaced by $g$. Define the bounded observation maps

$$\begin{aligned} Lg &= \mathbf{1}_{c\ge c_0}\mathsf{A}_v\Phi(g), \\ Hh &= \mathbf{1}_{c<c_0}\mathsf{Q}_c T[h], \\ I_0 g &= \mathbf{1}_{c<c_0}\mathsf{Q}_c T[\chi_I g], \\ \mathcal{B}(g,h) &= Lg - (Hh + I_0 g)/8. \end{aligned}$$

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For arbitrary $g \in L^2$, $\Phi(g)$ need not belong to $X$. The first line means the bounded leakage-kernel extension proved in (498), not application of a full-line operator outside its domain. If $\|L\| \le l$, $\|H\| \le h_0$, $\|I_0\| \le i_0$, then

$$\|\mathcal{B}\| \le b = \left[(l + i_0/8)^2 + (h_0/8)^2\right]^{1/2}.$$

On an original null input the exact ordinary equation is

$$\mathsf{A}_v F_a = \mathcal{B}\mathcal{K}a - R_V F_a.$$

After applying $P_V \mathsf{S}_v$ this gives the supported identity

$$F_a = (\mathrm{Id} + \mathsf{D}_v)^{-1} P_V \mathsf{S}_v \mathcal{B}\mathcal{K}a. \tag{526}$$

The support projection is taken after the full-line inverse.

Define $U_V : X_V \to \mathcal{H}_P$ by the physical angular realization

$$(JU_V F)(\theta(y)) = -\frac{16F(y)}{(1 + r(y)^2)^5}$$

on the coordinate collar and zero elsewhere. There is no division by $\chi_V$. Define $U_0 : L^2(I) \to \mathcal{H}_P$ by $JU_0 g = (1 - \chi_V)g$. The exact density ratio and Theorem 23 prove both maps bounded, with norms $v, u$, respectively. Thus

$$\mathcal{R}(g, h) = U_0 g + U_V (\mathrm{Id} + \mathsf{D}_v)^{-1} P_V \mathsf{S}_v \mathcal{B}(g, h) \tag{527}$$

is defined on all $\mathcal{Z}$ and satisfies

$$\mathcal{R}\mathcal{K}a = a \quad (a \in \mathcal{N}_0), \qquad \|\mathcal{R}\| \le C_R = u + \frac{vsb}{1 - q}, \quad s \ge \|\mathsf{S}_v\|. \tag{528}$$

Indeed its two angular pieces are $(1 - \chi_V)a$ and $\chi_V a$ by (526). The identity extends to the energy closure by boundedness. One may use $s = 24$; the other constants remain the norms of explicitly specified operators.

## N.2 A genuine ambient approximation bound

Let $\Pi_m$ be a specified finite-rank orthogonal projection on $\mathcal{Z}$, and suppose

$$\|(\mathrm{Id} - \Pi_m)\mathcal{K}a\|_{\mathcal{Z}} \le \eta_m \|a\|_E \qquad (a \in \mathcal{N}_0). \tag{529}$$

Let $P_m$ be the energy-orthogonal projection onto $V_m = \mathcal{R}\,\mathrm{Ran}\,\Pi_m \subset \mathcal{H}_P$. Best approximation and (528) give

$$\|(\mathrm{Id} - P_m)a\|_E \le \|a - \mathcal{R}\Pi_m \mathcal{K}a\|_E \le C_R \eta_m \|a\|_E. \tag{530}$$

The vectors spanning $V_m$ are images of known observation basis vectors under the exact operator $\mathcal{R}$. No estimate restricted to null inputs has been applied to $(\mathrm{Id} - P_m)a$.

The infinite inverse can also be truncated. Set

$$\mathcal{R}_M(g,h) = U_0 g + U_V \sum_{j=0}^{M} (-\mathsf{D}_v)^j P_V \mathsf{S}_v \mathcal{B}(g,h), \qquad \rho_M = \frac{vsbq^{M+1}}{1-q}.$$

Then $\|\mathcal{R} - \mathcal{R}_M\| \le \rho_M$. If $\|\mathcal{K}\|_{\mathcal{N}_0 \to \mathcal{Z}} \le \kappa$ and $P_{m,M}$ projects onto $\mathcal{R}_M \operatorname{Ran} \Pi_m$, then

$$\|(\mathrm{Id} - P_{m,M})a\|_E \le (C_R \eta_m + \rho_M \kappa)\|a\|_E. \tag{531}$$

Any further finite representation error of the spanning vectors must also be included. The small mechanical $q$ controls this Neumann error; it does not bound the observation error $\eta_m$.

Here is an explicit observation approximation formula. Choose a fixed $1/2 < \gamma < 1$, with

$$\int_H \theta^{-\gamma} |a|^2 \, \mathrm{d}\theta \le G_\gamma E(a), \quad \int_I \delta^{-4} |a|^2 \, \mathrm{d}\theta \le V_4 E(a).$$

Both estimates are proved in Appendix J; one may take $V_4 = 2^{87}$, while no numerical $G_{3/4}$ is asserted. Choose $0 < \tau < \min\{1, \theta_H\}$ and $0 < \nu < \pi/2 - \theta_H$, so the endpoint collars are disjoint. Discard $\theta < \tau$ and $\delta < \nu$ from the first observation and $\theta < \tau$ from the second. On the remaining compact intervals project to piecewise constants on cells of length at most $\Delta_m$, using the appropriate weighted averages for the second component. Let $B_{\tau,\nu}\|a\|_E$ bound the interior $L^2$ derivatives of both $a$ and $\chi_H a$, as supplied by the local regularity theorem. Weight comparison at the endpoints and the interval Poincaré inequality give

$$\eta_m^2 \le 2G_\gamma \tau^{\gamma - 1/2} + V_4 \nu^4 + \frac{1 + \tau^{-1/2}}{\pi^2} \Delta_m^2 B_{\tau,\nu}^2. \tag{532}$$

For the weighted average its error is no larger than that of the ordinary average, and $\theta^{-1/2} \le \tau^{-1/2}$ on retained cells. The factor two covers the two horizontal observations. All constants are independent of the particular original null input. In particular this is not an estimate obtained from an unknown basis-dependent $C^1$ norm after finite dimensionality.

## N.3 The exact residual and a sufficient separation test

For arbitrary $z = (g,h) \in \mathcal{Z}$, put $F_z = (\mathrm{Id} + \mathsf{D}_v)^{-1} P_V \mathsf{S}_v \mathcal{B} z$. The supported equation gives

$$\begin{aligned} G_z &= \mathsf{A}_v F_z + R_V F_z - \mathcal{B} z, \qquad P_V \mathsf{S}_v G_z = 0, \\ G_z &= -\mathsf{A}_v (\mathrm{Id} - P_V) \mathsf{S}_v (\mathcal{B} z - R_V F_z). \end{aligned} \tag{533}$$

Nothing makes the last omitted-output term zero for arbitrary data. The complete physical ordinary residual is

$$T[J\mathcal{R}z] = T[(1 - \chi_V)g] + 8B_V^{\mathrm{ex}} F_z, \qquad B_V^{\mathrm{ex}} F = \tfrac{1}{8} T[J U_V F],$$

where the last definition retains the exact mechanical vertical source on the whole regular interval. Thus $\mathcal{R}\mathcal{K} = \mathrm{Id}$ on known null inputs does not imply either $\mathcal{K}\mathcal{R} = \mathrm{Id}$ or ordinary nullity of $\mathcal{R}z$. Even an identity $\mathsf{Q}_c T = 0$ on a small interval would leave $T(c) = bc$ there and require an undifferentiated anchor.

A rigorous finite comparison can instead use the undifferentiated ordinary output on a fixed $J_c = [c_-, c_+] \Subset (0, c_*)$. The exact angular kernel (490) is bounded away from its inner logarithmic graph and its bounded outer jump. Both branches stay away from the axes and the fold on this fixed interval. The logarithmic root normal form therefore gives

$$|t(c,\theta)| \le C_J(1 + |\log|c - b(\sin\theta)||)$$

on finitely many inner charts, with a bounded complementary kernel. Hence $M_J^2 = \int_{J_c}\int_I |t(c,\theta)|^2 \,\mathrm{d}\theta\,\mathrm{d}c < \infty$, and $T_J J : \mathcal{H}_P \to L^2(J_c)$ is bounded by $2^{19} M_J$. This is an ambient bound, so it applies also to an approximation remainder that need not be null. If desired add finitely many actual rows $\ell_i$, with verified ambient bounds $|\ell_i(a)| \le b_i \|a\|_E$, and weights $w_i \ge 0$. Define

$$\mathcal{O}a = (T_J J a, (\sqrt{w_i}\ell_i(a))_i), \qquad B^2 = (2^{19} M_J)^2 + \sum_i w_i b_i^2.$$

Then $\|\mathcal{O}\| \le B$ and $\mathcal{O}a = 0$ on $\mathcal{N}_0$.

**Proposition N.1** (Finite-carrier separation criterion)**.** *Let $P$ be the exact energy-orthogonal projection onto a specified finite space $V \subset \mathcal{H}_P$. Suppose $\|(\mathrm{Id} - P)a\|_E \le \epsilon\|a\|_E$ on $\mathcal{N}_0$, with $0 \le \epsilon < 1$, and $\|\mathcal{O}v\|^2 \ge s_*^2\|v\|_E^2$ for all $v \in V$. If*

$$s_*^2(1 - \epsilon^2) > B^2\epsilon^2,$$

*then $\mathcal{N}_0 = \{0\}$.*

**Proof.** For a null input, $\mathcal{O}Pa = -\mathcal{O}(\mathrm{Id} - P)a$. Orthogonality gives $\|Pa\|_E^2 \ge (1-\epsilon^2)\|a\|_E^2$. The two operator estimates would then give the reverse of the strict displayed inequality unless $a = 0$. ∎

The proposition is a sufficient criterion, not a reported successful separation. The quantitative high-frequency horizontal error terms, a numerical $G_{3/4}$, retained-interval Sobolev constants and an enclosed continuous-output Gram margin have not all been supplied. In particular a finite matrix rank calculation without the ambient tail comparison does not identify the complete kernel. The exact reconstruction removes the logical difficulty of choosing an ambient comparison space; it does not remove these quantitative requirements.

# O A resonant certificate for the binary-quartic kernel

This appendix supplies the quantitative step in Proposition 5.1. Throughout, the potential is

$$P_0(x,y) = x^4 + 12x^2y^2 + 16y^4, \qquad H = \tfrac{1}{2}(W_x^2 + W_y^2) + P_0(x,y).$$

The kernel $\mathcal{K}$ consists of forces whose integral vanishes under every invariant probability on $H = 1$. We shall use one regular resonant torus, including all of its relative phases.

**Lemma 90** (A nonzero resonant coefficient)**.** *There is a parameter $c_{94}$ in the rational interval*

$$[c_L, c_U], \qquad c_L = 0.42391564641519901180613 34, \qquad c_U = c_L + 10^{-25}, \tag{534}$$

*for which $T_u/T_v = 9/4$. With the scalar phases based at the upper turning points, the physical-clock observable for $g = x^3y$ has double cosine coefficient*

$$\frac{-93006752507873712}{10^{23}} \;\le\; A_{9,4}[x^3y] \;\le\; \frac{-93006752505655829}{10^{23}} \;< -\frac{9}{10^7}. \tag{535}$$

*The fourth cosine coefficient of the physical invariant means over the relative-phase family is consequently less than $-9/(4\cdot 10^7)$.*

## O.1 The actual periodic family and its phase normalization

In canonical parabolic coordinates,

$$x = uv, \qquad y = (u^2 - v^2)/2, \qquad S = u^2 + v^2, \qquad p_u = vW_x + uW_y, \quad p_v = uW_x - vW_y,$$

the unit-energy equation is the zero level of

$$S(H-1) = \tfrac{1}{2}p_u^2 + U(u) + \tfrac{1}{2}p_v^2 + U(v), \qquad U(q) = q^{10} - q^2, \qquad \mathrm{d}t = S\,\mathrm{d}\tau. \tag{536}$$

Assign scalar energies $c$ and $-c$ to $u$ and $v$, respectively, and take $v > 0$. The squared turning points satisfy

$$a^5 - a = c, \qquad l^5 - l = h^5 - h = -c, \qquad a > 1, \quad 0 < l < h < 1. \tag{537}$$

For every $c$ in (534), elementary root brackets give

$$\frac{108}{100} < a < \frac{109}{100}, \qquad \frac{44}{100} < l < \frac{45}{100}, \qquad \frac{83}{100} < h < \frac{84}{100}. \tag{538}$$

All three roots are simple. The branches $a, l$ increase with $c$, while $h$ decreases. The two scalar periods have the endpoint-regularized forms

$$T_u = 4\int_0^{\pi/2} \left[2\left\{a^4 \sum_{j=0}^{4} \sin^{2j}\psi - 1\right\}\right]^{-1/2} \mathrm{d}\psi, \tag{539}$$

$$T_v = 2\int_0^{\pi/2} [2rQ(r)]^{-1/2}\,\mathrm{d}\psi, \qquad r = l + (h-l)\sin^2\psi, \tag{540}$$
$$Q(r) = r^3 + (l+h)r^2 + (l^2+lh+h^2)r + l^3 + l^2h + lh^2 + h^3.$$

The first formula follows from $u = \sqrt{a}\sin\psi$. For the second, use

$$r^5 - r + c = (r-l)(r-h)Q(r)$$

and then the displayed sine-squared substitution. Both formulas include both momentum branches of the scalar motion.

At a parameter with $T_u/T_v = 9/4$, write $u = u(\alpha)$ and $v = v(\beta)$, where each phase has period $2\pi$. They satisfy

$$u(-\alpha) = u(\alpha), \quad u(\alpha+\pi) = -u(\alpha), \qquad v(-\beta) = v(\beta).$$

Every relative phase $\phi$ gives an auxiliary periodic orbit $(\alpha,\beta) = (\alpha, 9\alpha/4+\phi)$, with $0 \le \alpha \le 8\pi$. Since $v^2 \ge l > 44/100$, the chart and time change in (536) are nonsingular on the whole torus. These are therefore genuine periodic orbits of the original physical Hamiltonian. Their common physical period is

$$(4T_u)\overline{S}, \qquad \overline{S} = \langle u^2\rangle + \langle v^2\rangle, \qquad 0 < \overline{S} < a + h < 2.$$

The independence of $\overline{S}$ from $\phi$ follows because $S$ is the sum of two one-variable functions and both oscillators traverse whole periods.

For $g = x^3y$, the clock-weighted observable is exactly

$$F(\alpha,\beta) = Sg(uv, (u^2-v^2)/2) = \tfrac{1}{2}(u^7v^3 - u^3v^7). \tag{541}$$

Define scalar cosine coefficients by

$$U_j = \frac{1}{\pi}\int_0^{2\pi} u(\alpha)^j\cos(9\alpha)\,\mathrm{d}\alpha, \qquad V_j = \frac{1}{\pi}\int_0^{2\pi} v(\beta)^j\cos(4\beta)\,\mathrm{d}\beta. \tag{542}$$

Product integration then gives

$$A_{9,4} := \frac{4}{(2\pi)^2}\int_0^{2\pi}\int_0^{2\pi} F(\alpha,\beta)\cos(9\alpha)\cos(4\beta)\,\mathrm{d}\alpha\,\mathrm{d}\beta = \tfrac{1}{2}(U_7V_3 - U_3V_7). \tag{543}$$

If $M(\phi)$ is the normalized physical mean of $g$ on the phase orbit, then

$$M(\phi) = \frac{1}{8\pi\overline{S}}\int_0^{8\pi} F(\alpha, 9\alpha/4+\phi)\,\mathrm{d}\alpha.$$

Integrating this identity against $\cos(4\phi)/\pi$ gives

$$\frac{1}{\pi}\int_0^{2\pi} M(\phi)\cos(4\phi)\,\mathrm{d}\phi = \frac{A_{9,4}}{2\overline{S}}. \tag{544}$$

Indeed the phase substitution produces $\cos(4\beta - 9\alpha)$; its sine product integrates to zero by separate phase evenness. This projection uses the whole family of actual means, so the possible presence of higher resonant multiples does not affect it.

## O.2 Rational enclosure of the resonant parameter

We give the finite arithmetic and its analytic error bounds explicitly. For a $2\pi$-periodic function $f$, analytic in $|\operatorname{Im} z| \le b$ and bounded there by $M$, contour displacement gives $|\widehat{f}(k)| \le Me^{-b|k|}$. Consequently its normalized periodic trapezoidal average with $N$ points differs from its integral average by at most

$$\frac{2M}{e^{bN}-1}. \tag{545}$$

This follows by summing the Fourier coefficients at the nonzero multiples of $N$.

The integrands in (539)–(540) are even and $\pi$-periodic in $\psi$. They extend analytically to $|\operatorname{Im}\psi| \le 1/200$, with a common bound 4, uniformly on (538). Here are explicit branch and denominator bounds. For the first integrand put $D = a^4 \sum_{j=0}^{4} \sin^{2j}\psi - 1$. On the real axis, $D > (108/100)^4 - 1 > 36/100$. On the indicated strip,

$$|D'| \le 40(109/100)^4 \cosh(1/200)^8 < 57.$$

For example $\cosh(1/200) \le 80000/79999$ proves the last inequality by rational arithmetic. Thus $\operatorname{Re} D > 3/40$, and the positive real inverse square root continues analytically with modulus less than 3. For the second integrand, the real value of $rQ(r)$ is larger than $(44/100)(83/100)^3 > 1/4$. Along the same strip,

$$|r - r_{\mathbb{R}}| < 0.00201, \qquad |r| < 0.843, \qquad |(rQ(r))_r| < 9.$$

The first inequality uses $r' = (h-l)\sin(2\psi)$, and the last follows from the positive coefficients of $Q$. Hence $\operatorname{Re}(rQ(r)) > 0.23$, and the second inverse square root has modulus less than 2. Both branches remain in a right half-plane.

In the variable $2\psi$ the strip width is $1/100$. Take $N_Q = 8192$ and write

$$q_Q = (3/8)^{80}, \qquad E_Q = \frac{56q_Q}{1-q_Q}. \tag{546}$$

Since $e > 8/3$ and $N_Q/100 > 80$, (545), including the period factors $2\pi$ and $\pi$, is bounded by $E_Q$ for either period.

For completeness, the arithmetic conventions determining the enclosures below are as follows. All interval endpoints lie on the dyadic grid $\delta\mathbb{Z}$, where $\delta = 2^{-180}$. A real number is enclosed by $[\delta\lfloor x/\delta\rfloor, \delta\lceil x/\delta\rceil]$. Addition and subtraction use their endpoint formulas. Products take the minimum and maximum of the four endpoint products and round outward to this grid. Positive reciprocals and integer divisions are rounded outward in the same way. For a nonnegative interval $[a,b]$, use the square-root interval

$$\left[\delta\left\lfloor\frac{\sqrt{a}}{\delta}\right\rfloor,\ \delta\left(1+\left\lfloor\frac{\sqrt{b}}{\delta}\right\rfloor\right)\right].$$

The integer square-root characterization makes these endpoints exact. Integer powers are formed by repeated squaring. Rational constants are first enclosed on the same grid.

To enclose $\pi$, let

$$s_m(x) = \sum_{j=0}^{m} \frac{(-1)^j x^{2j+1}}{2j+1}, \qquad I_m(x) = \operatorname{hull}\left\{ s_m(x), s_m(x) + \frac{(-1)^{m+1} x^{2m+3}}{2m+3} \right\}.$$

The alternating-series bound and Machin's identity give

$$\pi \in 16 I_{55}(1/5) - 4 I_{18}(1/239).$$

After exact quadrant reduction, every trigonometric node is enclosed by Horner evaluation of $\sum_{j=0}^{30} (-1)^j x^{2j}/(2j)!$ and an outward addition of $(8/5)^{61}/61!$, since $0 \le x \le \pi/2 < 8/5$.

For each endpoint $c_L, c_U$, take 175 exact rational bisections of (537), starting from $[1, 11/10]$, $[2/5, 1/2]$, and $[4/5, 9/10]$, on their respective monotone branches. Enclose the resulting root intervals on the dyadic grid. With $z_j = (1 - \cos(2\pi j/N_Q))/2$, evaluate the integrands in (539)–(540) for $0 \le j \le N_Q/2$. The polynomials in $z_j$ and $r$ are evaluated in Horner form. The weights are 1 at the two endpoints and 2 elsewhere. Divide the weighted sums by $N_Q$, multiply them by $2\pi$ and $\pi$, respectively, and enlarge both intervals by $E_Q$. This gives the rational enclosures

$$\begin{aligned} 4T_u(c_L) - 9T_v(c_L) &\in [-160428, -160426], 10^{-30}, \\ 4T_u(c_U) - 9T_v(c_U) &\in [279345, 279346], 10^{-30}. \end{aligned} \tag{547}$$

These finite bounds include the period quadrature remainder. Continuity of the regularized periods therefore gives a $c_{94} \in (c_L, c_U)$ with $4T_u(c_{94}) = 9T_v(c_{94})$. No monotonicity or uniqueness of the period ratio is required.

For the subsequent uniform enclosures, use the root boxes

$$a \in [a(c_L), a(c_U)], \quad l \in [l(c_L), l(c_U)], \quad h \in [h(c_U), h(c_L)],$$

with their bisection endpoints, and evaluate both period sums again on these whole boxes. This retains the full uncertainty in $c$ without assuming monotonicity of either period. In particular it gives $T_u < 4$ and $T_v < 2$.

## O.3 Analytic scalar orbits and validated node remainders

Both scalar motions satisfy

$$q'' = 2q - 10q^9. \tag{548}$$

Their real energies and (538) imply $|q| < 21/20$ and $|q'| < 7/5$. For the latter bound, use $c < 17/40$ and $\min U = -(4/5)5^{-1/4} > -27/50$, so $|q'|^2 < 2(17/40 + 27/50) < (7/5)^2$. At every real time the solution extends to a complex time disk of radius $1/50$, on which

$$|q| < 11/10, \qquad |q'| < 2. \tag{549}$$

Indeed, on this tube $|2q - 10q^9| \le 2(11/10) + 10(11/10)^9 < 26$. Along a complex segment of length $1/50$, the possible changes in $q$ and $q'$ are less than 0.04 and 0.52, respectively. Both leave strict room within (549). Local analytic existence and bounded continuation prove the uniform disk assertion.

Put $N = 2048$. The time increments $T_u/N, T_v/N$ are less than $1/500$, hence less than one tenth of the disk radius. If $q(\tau+s) = \sum_{n\ge0} a_n s^n$, its Taylor coefficients are determined by

$$a_0 = q(\tau), \quad a_1 = q'(\tau), \qquad a_{n+2} = \frac{2a_n - 10\sum_{i_1+\cdots+i_9=n} a_{i_1}\cdots a_{i_9}}{(n+1)(n+2)}. \tag{550}$$

Compute the convolution in this formula by successive squaring to obtain $q^2, q^4, q^8$, then multiplying by $q$, retaining at stage $n$ only the coefficients through $n$. For $0 \le n \le 24$, this gives the Taylor polynomial of $q$ through degree 26 and its derivative through degree 25. The Cauchy bounds from (549) give the respective omitted tails

$$\frac{11}{9}10^{-27} < 2\cdot10^{-27}, \qquad \frac{20}{9}10^{-26} < 3\cdot10^{-26}. \tag{551}$$

Use the preceding outward interval rules in (550) and Horner evaluation at the full interval time increment, adding the bounds on the right of (551) at every step. Start at $(\sqrt{a}, 0)$ for $u$ and $(\sqrt{h}, 0)$ for $v$. Take $N/4$ steps for $u$ and $N/2$ steps for $v$. Their final values are exactly 0 and $\sqrt{l}$, by the scalar reflection identities. The node intervals so obtained, with those exact endpoint enclosures, satisfy

$$\max_j \operatorname{width}(u_j) < 3130\cdot10^{-26}, \qquad \max_j \operatorname{width}(v_j) < 7689\cdot10^{-26}. \tag{552}$$

The preceding root and period intervals are used throughout this recurrence. Thus (552) retains their uncertainty and all propagated Taylor and rounding errors.

The normalized phase orbits are analytic in $|\operatorname{Im}\alpha| \le 1/50$ and $|\operatorname{Im}\beta| \le 1/50$: the corresponding imaginary times are less than $1/50$ because $T_u < 4$, $T_v < 2$ and $\pi > 3$. For $j, k \le 9$ their scalar Fourier integrands obey

$$|2q(\theta)^j \cos(k\theta)| < 3(11/10)^9 < 10 \qquad (|\operatorname{Im}\theta| \le 1/50).$$

Here $|\cos(k\theta)| \le e^{9/50} < 3/2$. Applying (545) gives the following common scalar quadrature error:

$$E_1 = \frac{20q_1}{1-q_1}, \qquad q_1 = (3/8)^{40}, \qquad \frac{20}{e^{N/50}-1} < E_1. \tag{553}$$

For $U_j$, the folded sum over $0 \le m \le N/4$ has weights 2 at its endpoints and 4 in its interior. For $V_j$, the sum over $0 \le m \le N/2$ has weights 1 at its endpoints and 2 in its interior. Each weight sum is $N$. The first folding uses that both $j$ and the harmonic 9 are odd; the second uses phase evenness. Multiply each weighted sum of $q_m^j \cos(2\pi km/N)$ by

$2/N$ and enlarge it by $E_1$. For the four coefficients needed in (543), the result is

| | enclosing endpoints, in units of $10^{-19}$ | |
|---|---|---|
| $U_3$ | 263072461211406715 | 263072461211410375 |
| $U_7$ | 876221133814187419 | 876221133814191078 |
| $V_3$ | 16725248538726098 | 16725248538729758 |
| $V_7$ | 56414227755351767 | 56414227755355426 |

(554)

All displayed endpoints are rational. Forming $\frac{1}{2}(U_7V_3-U_3V_7)$ in the dyadic intervals before the final rounding of (554) gives (535). Thus every scalar quadrature error is included before interval multiplication. Equations (544) and $0 < \overline{S} < 2$ prove the final assertion of Lemma 90.

## O.4 The polynomial kernel consequence

We conclude the proof of Proposition 5.1. Let

$$g = Ax^4 + Bx^3y + Cx^2y^2 + Dxy^3 + Ey^4 \in \mathcal{K}.$$

The unit-energy straight-orbit means on the horizontal and vertical axes are $A/3$ and $E/48$, respectively, by the quartic virial identity. Hence $A = E = 0$. The microcanonical invariant probability is separately reflection-even in $x$ and $y$, and its mean of $x^2y^2$ is strictly positive. Its mean of the two remaining odd monomials is zero, so $C = 0$.

The rotation force

$$g_{\mathrm{rot}} = 20xy(x^2 + 2y^2) = 20x^3y + 40xy^3$$

is centered under every invariant probability: for the bounded angular momentum $\Lambda = xW_y - yW_x$ on the compact energy shell,

$$\frac{\mathrm{d}\Lambda}{\mathrm{d}t} = -g_{\mathrm{rot}}.$$

Applying the resonant coefficient functional to this exact coboundary gives $20A_{9,4}[x^3y] + 40A_{9,4}[xy^3] = 0$. The complete invariant-mean condition for $g$ implies that its whole relative-phase mean vanishes, hence its fourth cosine coefficient vanishes. Lemma 90 therefore yields

$$0 = (B - D/2)A_{9,4}[x^3y], \qquad D = 2B.$$

Conversely, every multiple of $g_{\mathrm{rot}}$ belongs to $\mathcal{K}$ by the same coboundary identity. We have proved

$$\mathcal{K} \cap \mathcal{V}_4 = \mathrm{span}\{20xy(x^2 + 2y^2)\}.$$

This is a first-order statement at the specified base potential. Its proof uses actual invariant probabilities of that base and makes no integrability assumption on a perturbed family.

# P Rational action certificates

The strict numerical inequalities used in the proofs are rational interval comparisons. The interval operations below use exact rational endpoints and outward bounds for every transcendental quantity. The coefficient bounds in Lemmas 7 and 13, the cubic coefficient bound, and the sextic detecting integrals are specified analytically in the relevant proofs. The full-annulus twist certificate and the finite-phase and tail certificates for the sextic scalar response are specified in the preceding appendices. This appendix gives the curved-brake certificate.

## P.1 Curved-brake interval certificate

Use $N = 4096$ equal angular cells and outward rounding to multiples of $10^{-12}$ after interval arithmetic operations. Square roots of positive rational numbers are enclosed by integer square roots: if $x = A/B > 0$ and $S = 10^{12}$, take

$$k = \left\lfloor \sqrt{\lfloor AS^2/B \rfloor} \right\rfloor, \qquad k/S \le \sqrt{x} \le (k+1)/S.$$

These enclosures satisfy the squared endpoint inequalities. The starting bounds are

$$3141592653589/10^{12} < \pi < 3141592653590/10^{12},$$

certified by $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and consecutive alternating partial sums (indices through 22 and 8, respectively). On $[0, \pi/2]$, the sine Taylor polynomials through degrees 23 and 21 bound sine below and above, and the cosine polynomials through degrees 22 and 20 do the same. Monotonicity transports endpoint intervals into enclosures of $\sin^2\psi$ over each cell. The final cell endpoints are set to exactly zero or one where appropriate.

The coefficient $c$ in (59) is computed first. With $t = 2\tan\theta$, $s = \sin^2\theta$, $z_0 = s(1-s)$ and $D_0 = 1 + z_0$, the three integrals become

$$I_4 = \int_0^{\pi/2} \frac{1 - 2z_0}{D_0^{3/2}}\,\mathrm{d}\theta, \quad I_{22} = \frac{1}{2}\int_0^{\pi/2} \frac{z_0}{D_0^{3/2}}\,\mathrm{d}\theta, \quad J_* = \frac{1}{4}\int_0^{\pi/2} \frac{z_0(1-2z_0)}{D_0^{5/2}}\,\mathrm{d}\theta.$$

On each cell, the exact range of $s(1-s)$ includes $1/4$ if the cell crosses $1/2$. Summing interval ranges times interval cell lengths encloses the integrals, and the lower bound for $I_{22}$ is positive.

The cubic factor in (62) is strictly increasing, since its derivative is $3z^2 - 4z + 3 > 0$. Its values at $8/5$ and $17/10$ have opposite signs. Fifty exact bisections, followed by outward rounding, enclose its root. For each orbit cell take $w = \cos^2\psi$ and $R = zw$. The denominator in (63) is increasing in both positive variables $z, w$ and has a certified positive lower endpoint. Its reciprocal square-root interval gives the cell duration. Sum these durations to obtain $[Q_-, Q_+]$.

Evaluate the algebraic expression (68) minus (67) on the interval $R$ of each cell, multiply by its duration, and add. To bound the transverse projection, maintain cumulative time bounds $[t_-, t_+]$. Throughout the next cell, the phase $\pi\tau/(2Q)$ lies in

$$\left[\frac{\pi_- t_-}{2Q_+}, \frac{\pi_+(t_+ + \Delta t_+)}{2Q_-}\right].$$

Intersect its cosine range with $[0, 1]$, use the alternating Taylor bounds, and multiply by the force interval from (69) and the duration. This encloses the full projection without sampling its integrand at isolated points.

The following coarser rational bounds are consequences of those operations. All entries in the two numeric columns are integers divided by $10^9$.

| Quantity | Lower | Upper |
|---|---|---|
| $c$ | $-264300268$ | $-262819123$ |
| $Q$ | 909971128 | 910253649 |
| $v_0^{-2}\int_0^Q (SB_{\mathrm{even}} - B_{\mathrm{aux}})\,\mathrm{d}\tau$ | 2127830 | 2184651 |
| $v_0^{-1}\int_0^Q f\cos(\pi\tau/(2Q))\,\mathrm{d}\tau$ | $-91331979$ | $-90969906$ |

The finer rational endpoints verify $2 < 8Q/\pi < 3$ and

$$16 - (\pi/(2Q))^2 > 13, \qquad (3\pi/(2Q))^2 - 16 > 0.$$

Writing $P_{\mathrm{abs}}$ for the maximum absolute projection endpoint divided by $v_0$, the interval evaluation gives

$$T_- = -\frac{2P_{\mathrm{abs}}^2}{Q_-\,(16 - (\pi_+/(2Q_-))^2)} > -\frac{1408089}{10^9}.$$

The interval sum is at least $2127830/10^9$, and hence the total corrected curvature is greater than

$$\frac{2127830 - 1408089}{10^9} = \frac{719741}{10^9} > \frac{1}{2500}.$$

These estimates prove Lemma 9. An independently organized check with 8192 cells and outward scale $10^{11}$ gives a lower bound exceeding 0.0007368543; that decimal is a summary, not an input to the proof.

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